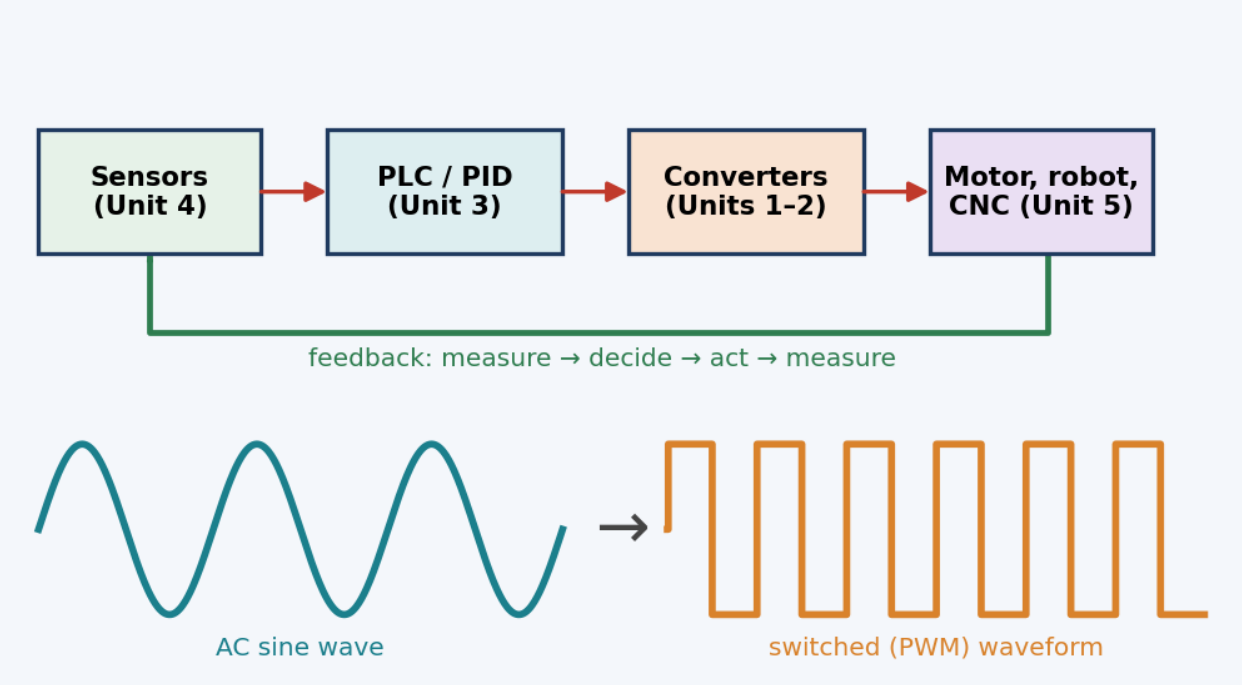
INDUSTRIAL ELECTRONICS & CONTROL

A Complete Visual Textbook, from Absolute Zero to the End-Semester Question Bank



Foundations (Chapters A–E) · Unit 1: Power switches, power factor, power quality · Unit 2: Converters, inverters, rectifiers · Unit 3: PLC, PID, SCADA, DCS, VFD · Unit 4: Sensors and data acquisition · Unit 5: Automation, communication, Industry 4.0

Prepared for: Dr. D. Y. Patil Institute of Technology, Pimpri, Pune-18 — Unit-wise Question Bank for Endsem Examination

Every figure in this book is drawn by program from the circuit or equation it shows; the waveforms are computed, not hand-sketched. Illustrative numbers are labelled as such.

How to Use This Book

This book is written for a reader who has never studied electricity or electronics. It has three parts: **Foundations** (Chapters A–E) teach what every later chapter assumes; **Units 1–5** follow your question bank in its original order and wording; **Reference sections** at the end collect glossaries, formulas, diagrams and a coverage checklist.

Box	Meaning
Exam question (orange)	the exact question from the uploaded question bank
Note on the source wording (yellow)	an ambiguity or error in the source, with the likely intended meaning
Key idea (blue)	the one-sentence takeaway
Worked example (green)	a complete solved calculation; every step is shown
Common misconception (red)	a likely mistake and why it is wrong

Suggested way to study

1. Read Foundations A–E once, doing the self-study questions. If you already know a chapter, skim it.
2. For each unit question: read the explanation, redraw the figure from memory, redo the worked example with different numbers, then check against the revision pack.
3. After each unit, use the end-of-unit pack: summary, formula sheet, diagram-revision sheet, glossary, review questions (with answers), and the examination-answer guide.
4. Before the exam, use the final revision guide and the coverage checklist at the back.

What is original source and what is added

The question wording comes from your question bank. Everything else (background, derivations, examples, diagrams, applications) is supporting material added to teach the question. Numerical values in worked examples are *assumed or illustrative* unless a standard defines them; no experimental measurement is claimed anywhere. Component ratings and typical figures are indicative, so check datasheets and standards for design work.

> **Safety**
> Everything here is theory and simulation. Never experiment directly on mains (230 V / 415 V) circuits or capacitors storing charge. For practice, use simulation software or low-voltage (≤ 12 V), current-limited bench supplies under supervision, and use properly rated, isolated test equipment.

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Foundation Chapter A: Understanding Industrial Electronics & Control

Where you are in the book. Before the five units of your question bank, this foundation section (Chapters A–E) teaches everything you need to read them. Nothing here is skipped over: if a word looks technical, it is defined the first time it appears.

A.1 What is engineering, and what is automation?

Engineering is the use of science and mathematics to design things that solve practical problems: a bridge that does not fall, a motor that turns at the right speed, a factory that fills 10,000 bottles an hour without anyone pressing a button. **Automation** means making a machine or process run with little or no human effort, by letting equipment *measure* what is happening, *decide* what to do, and *act* on that decision.

A.2 The five areas inside this subject

Area	Plain-English meaning	Where in this book
Electrical engineering	Generating, moving and using electrical energy (power stations, cables, motors).	Chapters B, C; Unit 1 (power factor)
Electronics	Using small voltages/currents and devices such as diodes and transistors to process signals and to switch.	Chapter D
Power electronics	Using electronic *switches* to convert and control large amounts of electrical power (AC↔DC, change of voltage, change of frequency).	Units 1 and 2
Instrumentation	Measuring physical quantities (pressure, temperature, speed…) and turning them into electrical signals.	Unit 4
Control	Using the measurements to automatically decide and act: PLCs, PID controllers, SCADA, drives, networks.	Units 3 and 5

A.3 Industrial Electronics & Control (IE&C)

Industrial electronics is the use of electronic devices and circuits to run industrial equipment, mainly by controlling how electrical power reaches machines. **Industrial control** is the use of sensors, controllers and actuators (devices that make something move or change) so that machines and processes behave as intended, automatically. IE&C is the combination: power that can be shaped, plus intelligence that decides how to shape it.

A.4 One complete example: an automatic water tank

A factory has a rooftop tank. The tank must stay between 40% and 90% full, with nobody watching it. Think about what has to happen:

1. A **sensor** measures the water level (Unit 4).
2. The sensor’s small electrical signal is cleaned and converted to a number by **signal conditioning** and **data acquisition** (Unit 4).
3. A **controller**, here a PLC, compares the level with the wanted range and decides ‘pump ON’ or ‘pump OFF’, or ‘pump speed 60%’ (Unit 3).
4. **Power electronics** (a motor starter, or a variable frequency drive, VFD) delivers the right electrical power to the **motor** (Units 1–3). Inside a VFD are rectifiers and inverters (Unit 2).
5. The **motor** turns the **pump**; water flows; the level changes. The motor and pump are **actuators**.
6. The sensor measures the new level, and the loop repeats. This repeated ‘measure → decide → act → measure’ is called **feedback control** (Unit 3).

7. A **network** sends the data to a control-room screen (SCADA) and to cloud software that watches motor vibration to warn of a failing bearing (Unit 5: IoT, predictive maintenance, Industry 4.0).

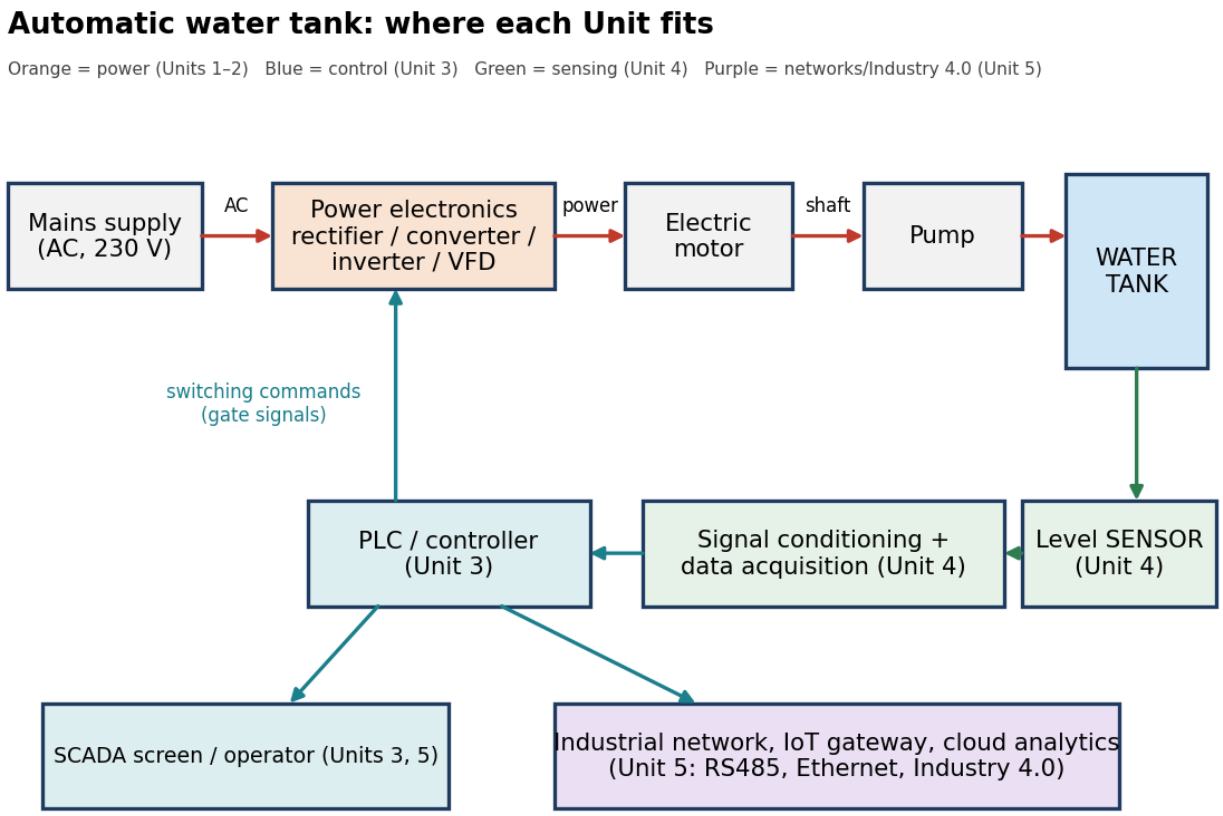


Figure 1. *Block diagram of the automatic water tank, colour-coded by the Unit that studies each block. Arrows show power (orange path) and information (blue/green paths).*

Key idea
Every unit of the syllabus is one part of this single loop: **power** (Units 1–2), **decision** (Unit 3), **measurement** (Unit 4), **communication and the smart factory** (Unit 5).

Foundation Chapter B: Basic Electricity and Circuits

B.1 Electric charge, electrons and the three kinds of material

Everyday idea. All matter is built from tiny particles. Some carry an electrical property called **charge**. There are two kinds: positive and negative. Opposite charges attract; like charges repel, just as the two ends of magnets behave. Charge is measured in **coulombs (symbol C)**.

An **electron** is a very small particle with a negative charge. In metals, a few electrons per atom are only loosely attached; these **free electrons** can wander through the material. When something pushes them in one overall direction, we have an electric current.

Material	Free electrons?	Behaviour	Examples / role in this book
Conductor	Very many	Current flows easily	Copper, aluminium wires and busbars
Insulator	Almost none	Current almost cannot flow	Plastic cable cover, rubber, air, glass
Semiconductor	Few, but *controllable*	Can be switched between conducting and blocking	Silicon: diodes, transistors, SCRs, IGBTs (Chapter D, Units 1–2)

B.2 Voltage, current and resistance

Imagine two water tanks joined by a pipe. If one tank is higher, water flows through the pipe. Three ideas appear:

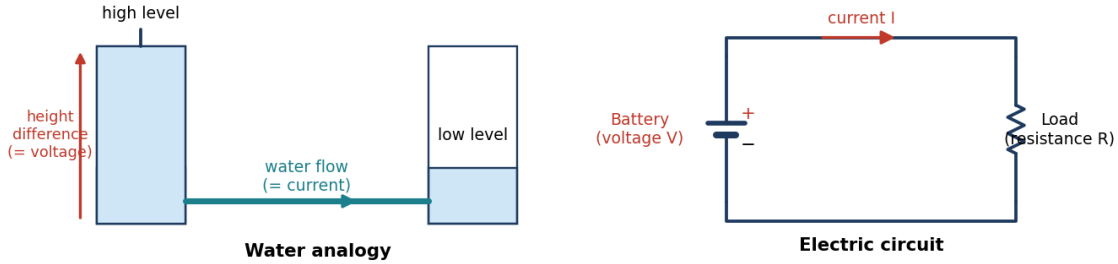


Figure 2. *The water analogy (left) and the matching electric circuit (right). Height difference ↔ voltage; flow of water ↔ current; narrow pipe ↔ resistance.*

- **Voltage (V)**, also called **potential difference**: the electrical ‘push’ between two points. Formally, the energy given to each coulomb of charge moving between them. Unit: **volt (V)**. Voltage is always *between two points*.
- **Current (I)**: the rate at which charge flows past a point. Unit: **ampere (A)**; 1 A = 1 coulomb per second. Formula: $I = Q/t$.
- **Resistance (R)**: how strongly a component opposes current. Unit: **ohm (Ω)**. A thin long wire has more resistance than a thick short one.
- **Conventional current** is drawn from the battery’s (+) terminal, through the circuit, to (−). Real electrons move the opposite way. Engineers use the conventional direction everywhere, and so does this book.

Worked example: charge and current
A motor winding passes 10 C of charge in 5 s. Find I.
Known: Q = 10 C, t = 5 s. **Unknown:** I.
$I = Q/t$ = 10 / 5 = **2 A**. Unit check: coulomb/second = ampere. ✔

B.3 A complete circuit: source, load, switch, ground

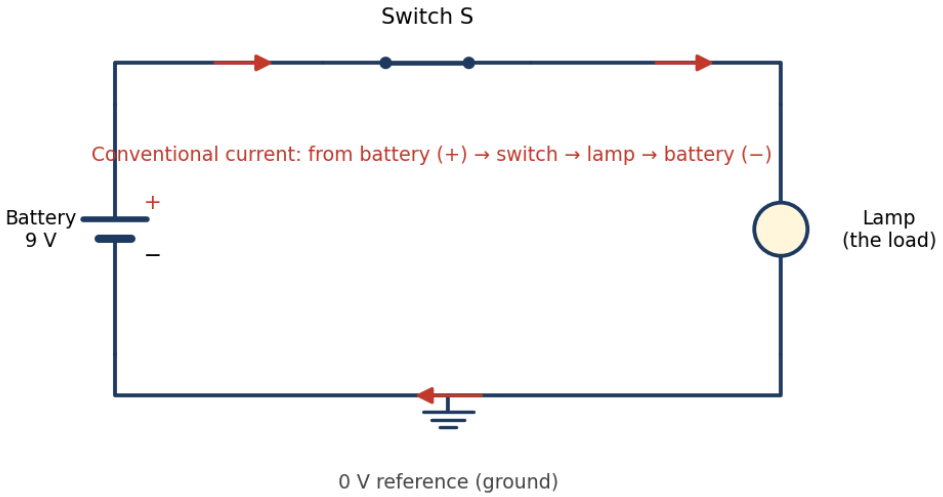


Figure 3. *A simple complete circuit. Conventional current leaves the battery's (+) terminal, passes the closed switch and the lamp, and returns to (−).*

- **Source**: supplies the voltage (battery, mains socket, power supply).
- **Load**: the part that uses the energy (lamp, motor, heater).
- **Switch**: opens or closes the path.
- **Ground / reference**: the point chosen as 0 V, from which other voltages are measured. (It is often, but not always, physically connected to earth for safety.)
- A circuit must be a **closed loop** because charge cannot vanish: whatever leaves one terminal must return to the other. The battery acts like a pump that keeps pushing charge around the loop.

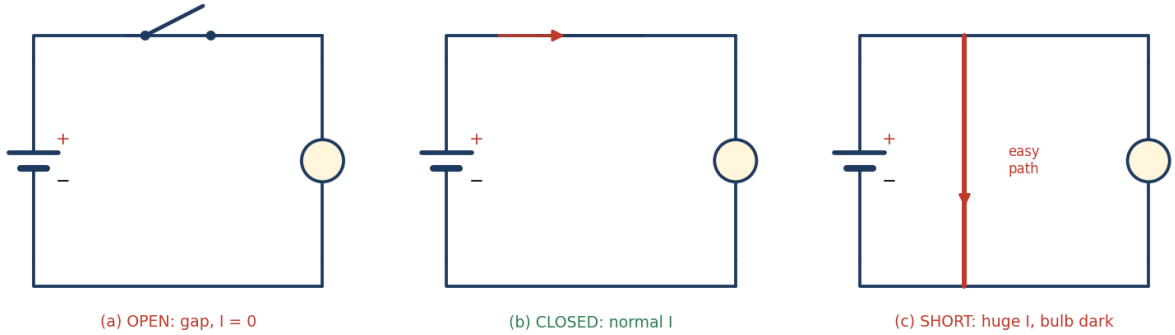


Figure 4. *Three states of the same circuit: (a) open circuit, switch open, no current; (b) closed circuit, normal current; (c) short circuit, a wire bypasses the load, current becomes dangerously large.*

Short circuit. A very low-resistance path joins the source terminals. Since current = voltage/resistance, a tiny resistance means an enormous current, overheating wires and causing fires. **Fuses** (a thin wire that melts) and **circuit breakers** (a switch that trips) are protective devices that interrupt the circuit when the current is too high.

> **Common misconception**
> 'The lamp uses up the current.' No: the same current returns to the battery. What the lamp uses up is **energy**, converted to light and heat.

B.4 Ohm’s law

Idea. For many materials, pushing harder (more voltage) gives proportionally more current, and a larger resistance gives less current for the same push.

$$V = I \times R \quad \Longleftrightarrow \quad I = \frac{V}{R} \quad \Longleftrightarrow \quad R = \frac{V}{I}$$

Symbol	Meaning	Unit
V	voltage across the component	volt (V)
I	current through the component	ampere (A)
R	resistance of the component	ohm (Ω)

Why it works (basic level). Voltage is the push, resistance is the opposition. Doubling the push doubles the flow; doubling the opposition halves the flow. **When it applies:** to resistors and to wires and heaters at constant temperature. It does *not* describe diodes, transistors or thyristors, whose current is not proportional to voltage; these appear from Chapter D onwards.

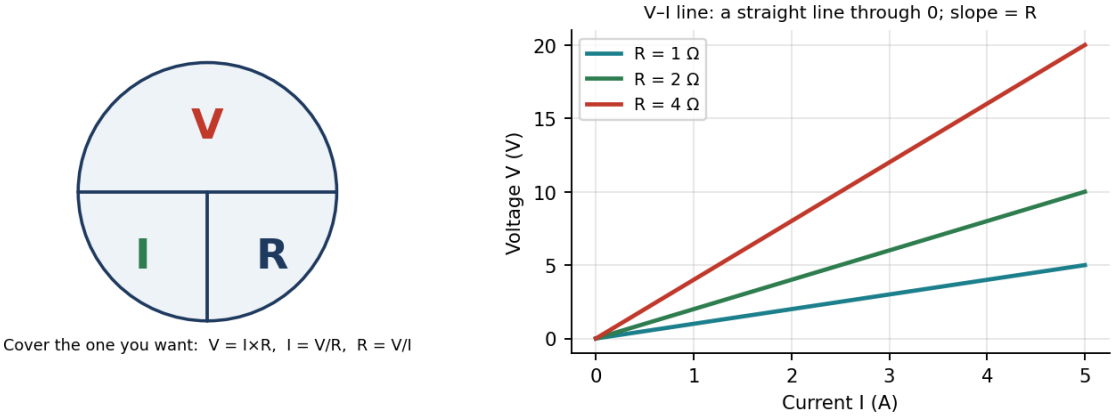


Figure 5. *Left: memory triangle for Ohm’s law. Right: for a resistor, the V–I graph is a straight line through the origin, and its slope equals R.*

Worked example: Ohm’s law
A 12 V battery is connected across a 4 Ω heater. Find the current.
Known: V = 12 V, R = 4 Ω. **Unknown:** I. **Equation:** I = V/R.
I = 12 / 4 = **3 A**. Unit check: volt/ohm = ampere. ✔
Reverse question: what voltage is needed to push 5 A through 4 Ω? V = I×R = 5 × 4 = **20 V**.

B.5 Electrical power and energy

Power is how fast energy is converted. A 100 W lamp converts 100 joules of electrical energy every second. **Energy** is the total amount converted over time.

$$P = V \times I \qquad P = I^2 R \qquad P = \frac{V^2}{R} \qquad E = P \times t$$

Symbol	Meaning	Unit
P	power	watt (W)
E	energy	joule (J), or kilowatt-hour (kWh) for electricity bills
t	time	second (s) or hour (h)

Derivation of the other two forms. Start from $P = VI$. Substitute $V = IR$ from Ohm’s law: $P = (IR)I = I^2R$. Instead substitute $I = V/R$: $P = V(V/R) = V^2/R$.

Worked example: power and energy
The 4 Ω heater of the previous example draws 3 A from 12 V.
P = V×I = 12 × 3 = **36 W**. Check: I²R = 9 × 4 = 36 W ✔.
Run for 2 hours: E = P×t = 36 W × 2 h = 72 Wh = **0.072 kWh** (1 kWh = 1000 Wh).

B.6 Series and parallel circuits

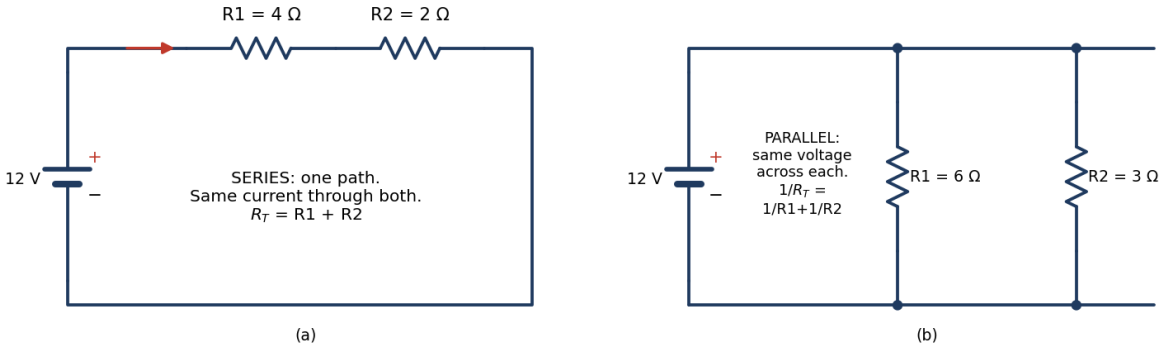


Figure 6. *(a) Series: components share one path. (b) Parallel: components share the same two nodes (connection points).*

	Series	Parallel
Current	same in every component	splits between branches; total = sum of branch currents
Voltage	splits between components; total = sum	same across every branch
Total resistance	$R_T = R_1 + R_2 + \ldots$	$1/R_T = 1/R_1 + 1/R_2 + \ldots$ (R_T is smaller than the smallest branch)
If one component opens	whole circuit stops	other branches keep working

Worked example: series
Figure 6(a): 12 V across R1 = 4 Ω and R2 = 2 Ω in series.
R_T = 4 + 2 = 6 Ω. I = V/R_T = 12/6 = **2 A** (the same in both).
V across R1 = 2×4 = 8 V; across R2 = 2×2 = 4 V. Check: 8 + 4 = 12 V ✔.

Worked example: parallel
Figure 6(b): 12 V across R1 = 6 Ω and R2 = 3 Ω in parallel.
Each branch has 12 V. I_1 = 12/6 = 2 A; I_2 = 12/3 = 4 A. Total I = 2 + 4 = **6 A**.
R_T = V/I = 12/6 = 2 Ω. Check by formula: 1/R_T = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2, so R_T = 2 Ω ✔.

B.7 Circuit symbols and how to read a circuit diagram

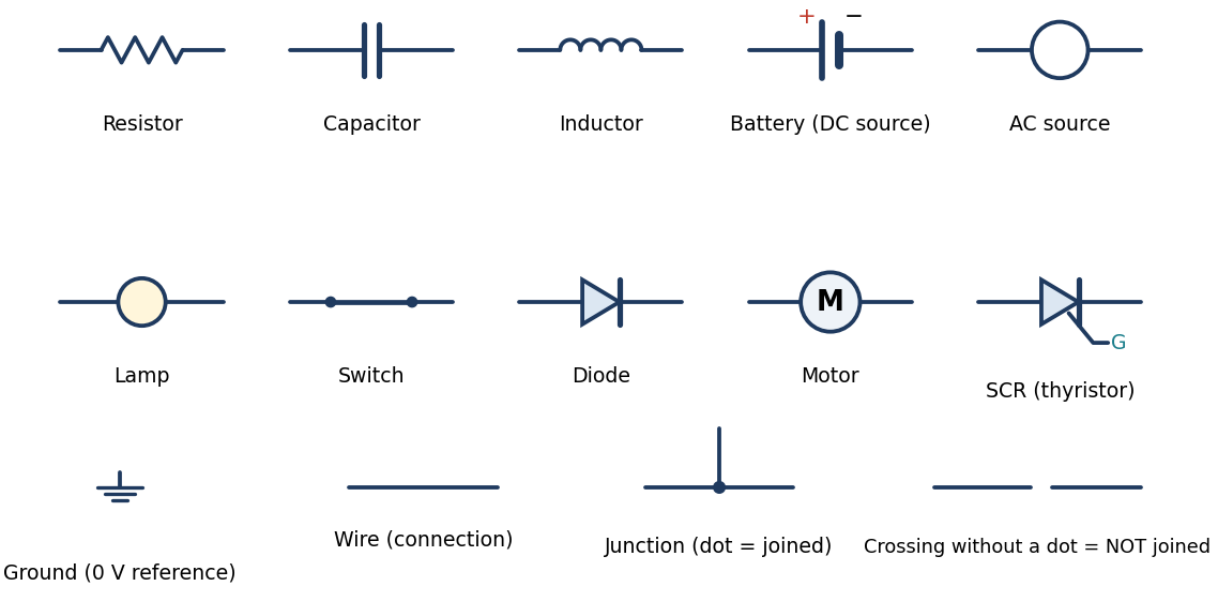


Figure 7. *Common circuit symbols used throughout this book. Component names and reference labels (R1, S1, D1…) are written beside each symbol.*

How to read a diagram. (1) Find the source. (2) Trace the loop from its (+) terminal, through every component, back to (−). (3) Identify the load. (4) Note which switches or devices can open the path. (5) Mark where current can and cannot flow in each state. We follow exactly this routine in every later circuit.

B.8 Self-study questions, Chapter B

1. A 230 V supply drives 5 A through a heater. Find its resistance and power.
2. Three resistors 10 Ω, 20 Ω and 30 Ω are connected in series across 12 V. Find the current and the voltage across the 20 Ω resistor.
3. Two 6 Ω resistors are in parallel. Find the total resistance.
4. Why does a short circuit produce a very large current?

Answers
1. R = V/I = 230/5 = **46 Ω**; P = VI = 230×5 = **1150 W**.
2. R_T = 60 Ω; I = 12/60 = **0.2 A**; V_{20} = 0.2×20 = **4 V**.
3. $1/R_T = 1/6 + 1/6 = 1/3$, so R_T = **3 Ω**.
4. Almost zero resistance in the path, so I = V/R becomes extremely large.

Foundation Chapter C: AC, DC and Waveforms

In Chapter B all voltages were steady. Industrial equipment also uses voltages that *change with time*. To describe them we draw a **waveform**: a graph with **time on the horizontal axis** and **voltage (or current) on the vertical axis**. Reading these graphs is the single most important skill for Units 1 and 2.

C.1 Direct current (DC) and alternating current (AC)

- **DC (direct current)**: the voltage keeps one polarity and (ideally) a constant value. Batteries, solar panels and the outputs of power supplies are DC.
- **AC (alternating current)**: the voltage and current repeatedly reverse direction, rising to a positive peak, falling through zero, reaching a negative peak, and returning. Mains electricity is AC.

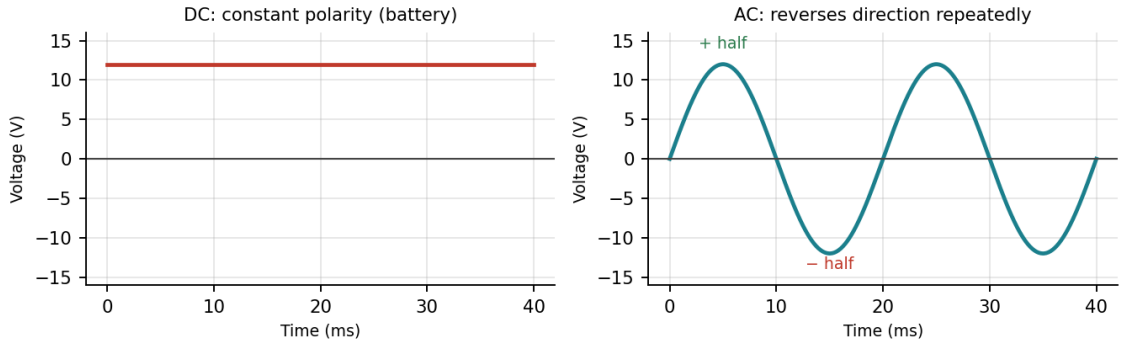


Figure 8. *Left: DC is a flat line. Right: AC swings above and below zero; the '−' half means the polarity is reversed.*

Why is mains AC? AC voltage can be raised or lowered easily by a **transformer** (a device that changes voltage levels). High voltage lets power travel long distances with low loss; it is then stepped down for use. Batteries produce DC because of how their chemistry works. Many industrial devices need DC (electronics, DC drives), so converting between the two, the subject of Unit 2, is a major task.

C.2 The sine wave and its vocabulary

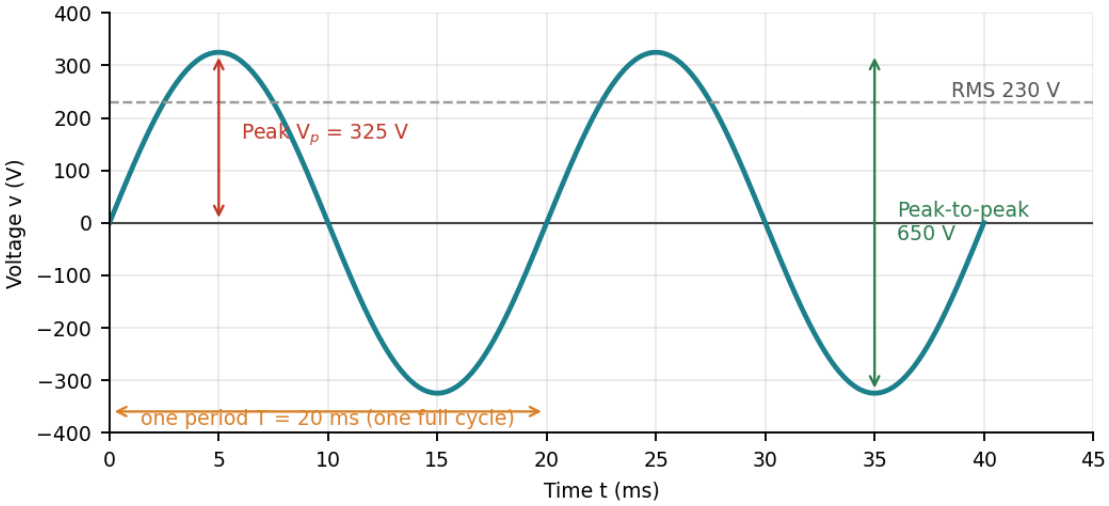


Figure 9. *A 50 Hz mains sine wave with 230 V RMS: peak 325 V, peak-to-peak 650 V, period 20 ms.*

$$v(t) = V_p \sin(2\pi f t) = V_p \sin(\omega t) \qquad f = \frac{1}{T} \qquad \omega = 2\pi f$$

Term	Meaning	Unit
v(t)	instantaneous voltage at time t	V
V_p	**peak** (amplitude): maximum value	V
Peak-to-peak	from negative peak to positive peak = $2V_p$	V

Term	Meaning	Unit
T	**period**: time for one full cycle	s
f	**frequency**: cycles per second = 1/T	hertz (Hz)
ω	**angular frequency**: how fast the angle advances, 2πf (one cycle = 2π radians = 360°)	rad/s
Phase	how far a wave is shifted in time/angle relative to another	degrees or radians

Why 2π? A sine function repeats every 2π radians (360°). To repeat *f* times per second, the angle must advance 2π*f* radians per second. A **radian** is simply another unit of angle: 180° = π rad, so 1 rad ≈ 57.3°.

Worked example: Indian mains
Mains: 50 Hz, 230 V RMS (RMS is explained in C.3).
T = 1/f = 1/50 = **0.02 s = 20 ms**. ω = 2πf = 2 × 3.1416 × 50 = **314.2 rad/s**.
V_p = √2 × 230 = 1.4142 × 230 = **325.3 V**. Peak-to-peak = 650.6 V.
Voltage at t = 5 ms: ωt = 314.2 × 0.005 = 1.571 rad (= 90°), so v = 325.3 × sin(1.571) = **325.3 V** (the positive peak, as in Figure 9).

C.3 Average and RMS values

A wave that goes up and down has several ‘sizes’. Which one matters depends on what you want to know:

- **Peak**: needed to choose insulation and components that survive the highest voltage.
- **Average value**: the mean of the waveform over time. For a symmetric sine over a full cycle it is **0** (positive and negative areas cancel), so averages are quoted for *rectified* waveforms (Units 1–2), and for the output of DC converters.
- **RMS (root-mean-square) value**: the DC value that would deliver the *same heating power* to a resistor. It is the number printed on equipment (‘230 V’).

Why ‘root-mean-square’? Power in a resistor is P = v²/R. Squaring makes every value positive, so negative half-cycles also deliver power. Take the **Mean** of the **Square** of the voltage, then take the **Root**. For a sine, the mean of sin² is ½ (Figure 10), hence:

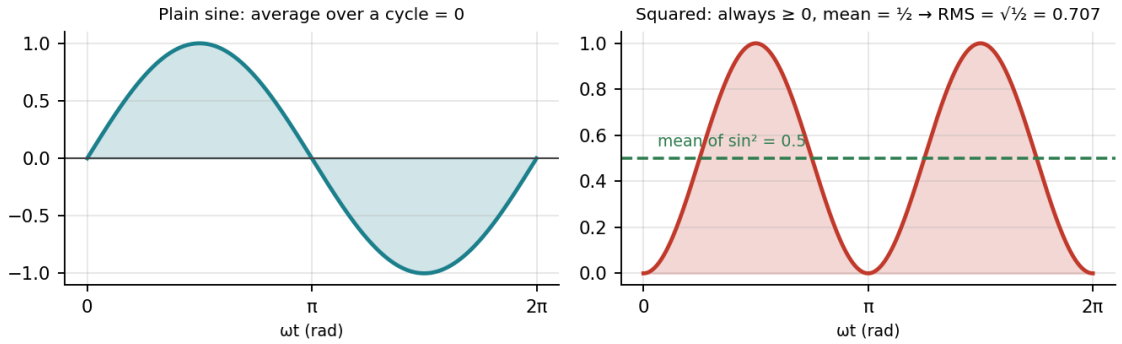


Figure 10. *Left: the average of a plain sine over a full cycle is zero. Right: squaring makes it non-negative; its mean is ½, so RMS = V_p×√½.*

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2\,dt} = \frac{V_p}{\sqrt{2}} \approx 0.707\,V_p \qquad V_{avg,\ half\text{-}wave\ rectified} = \frac{V_p}{\pi}, \qquad V_{avg,\ full\text{-}wave\ rectified} = \frac{2V_p}{\pi}$$

The averages come from integration, shown step by step in Chapter E.4. For a full-wave rectified sine, V_{avg} = 2V_p/π ≈ 0.637 V_p; for half-wave it is half of that, V_p/π ≈ 0.318 V_p.

Worked example: RMS and average
A sine has V_p = 100 V. RMS = 100/√2 = 100 × 0.7071 = **70.7 V**.
Full-wave rectified average: 2×100/π = 200/3.1416 = **63.7 V**. Half-wave rectified average: 100/π = **31.8 V**.

Common misconception
'RMS is the average.' No. For a sine the average is 0 and the RMS is 0.707 V_p. RMS measures heating effect; average measures the DC component.

C.4 Phase

Two waves with the same frequency may peak at different moments. The time (or angle) difference is the **phase difference**. In an inductive load such as a motor, the current peaks *after* the voltage: the current **lags**. This lag is the cause of the poor power factor that Unit 1 studies.

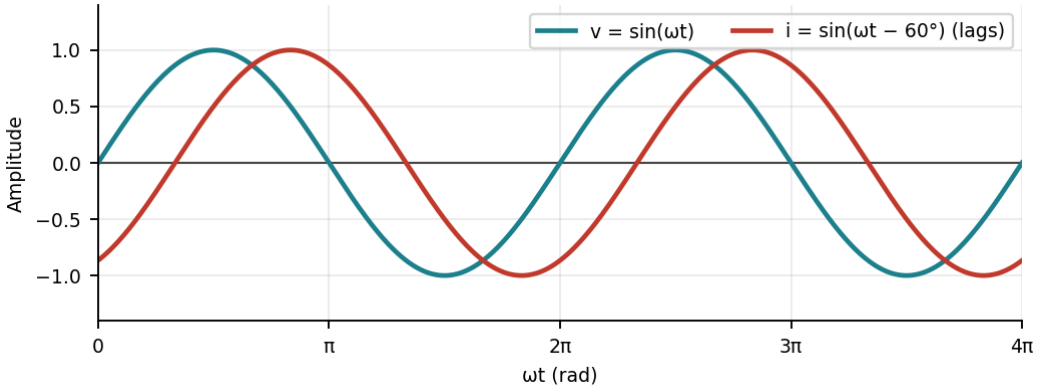


Figure 11. *Current lagging voltage by 60°. The two waves have the same frequency but the red wave reaches each peak 60° later.*

C.5 Switching and waveforms

A **switch** that opens and closes quickly turns a steady DC voltage into a **pulse train**. The load sees 12 V while the switch is ON and 0 V while OFF. The **duty cycle** D is the fraction of time the switch is ON, and the load's average voltage equals $D \times V_{in}$. This idea underlies choppers, DC–DC converters, PWM inverters and VFDs.

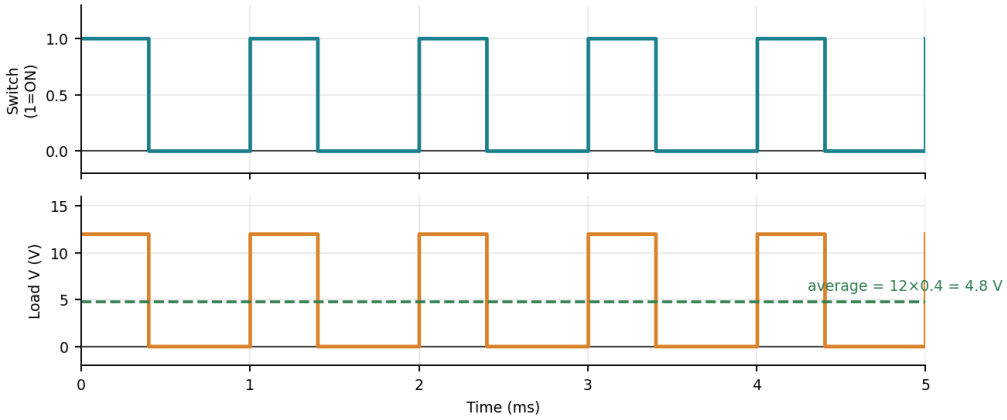


Figure 12. *A 12 V supply switched ON for 0.4 ms of each 1 ms period (duty cycle D = 0.4). Average = 12 × 0.4 = 4.8 V.*

C.6 Self-study questions, Chapter C

1. A sine wave has f = 400 Hz. Find T and ω.
2. A sine has V_p = 311 V. Find V_{rms}.
3. A 24 V supply is switched with D = 0.25. Find the average voltage.
4. State what 'current lags voltage' means.

Answers
1. T = 1/400 = **2.5 ms**; ω = 2π×400 = **2513 rad/s**.
2. 311/1.4142 = **220 V**.
3. 24 × 0.25 = **6 V**.
4. The current waveform reaches its peaks and zero-crossings later than the voltage waveform.

Foundation Chapter D: Semiconductors and Switching

D.1 Why semiconductors?

A **semiconductor** (silicon is the common one) normally conducts poorly, but we can change its behaviour in two ways: by **doping** it, adding trace impurities to give extra free electrons (**N-type**) or 'holes' where electrons are missing (**P-type**); and by applying voltages to control the flow. A **PN junction** is a P-type region touching an N-type region. It conducts in one direction only; this is the diode. Combining junctions in different patterns gives transistors and thyristors.

D.2 The diode, forward bias and reverse bias

A **diode** is a one-way valve for current. It has two terminals, the **anode (A)** and the **cathode (K)**.

- **Forward bias**: anode more positive than cathode. After the voltage exceeds about 0.7 V (silicon), the diode conducts easily.
- **Reverse bias**: anode more negative. The diode blocks; only a tiny **leakage current** flows. If the reverse voltage is too large, it reaches **breakdown** and may be destroyed (except for special Zener diodes).

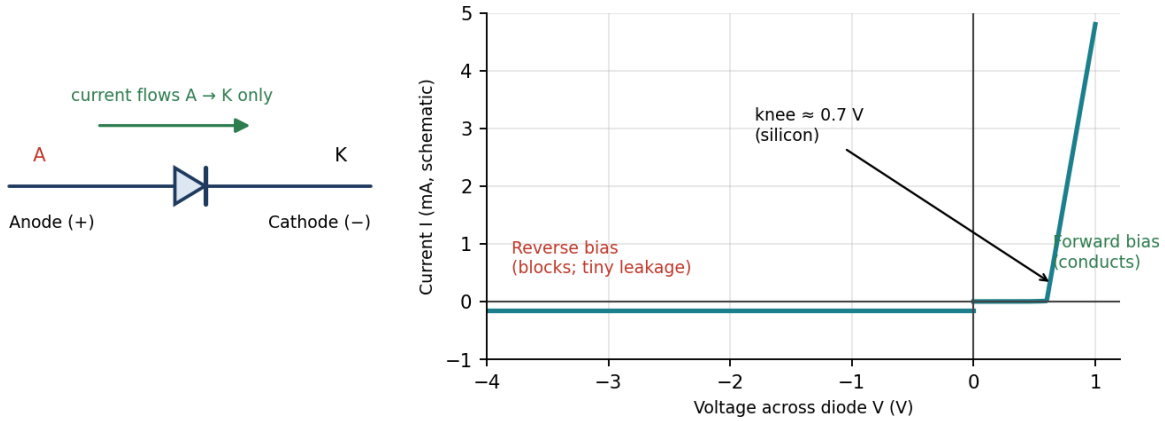


Figure 13. *Diode symbol with the allowed current direction (left) and its V–I characteristic (right, schematic, not to scale).*

Example: half-wave rectifier. A diode in series with a resistor and an AC source passes only the positive half-cycle, converting AC into pulsating DC.

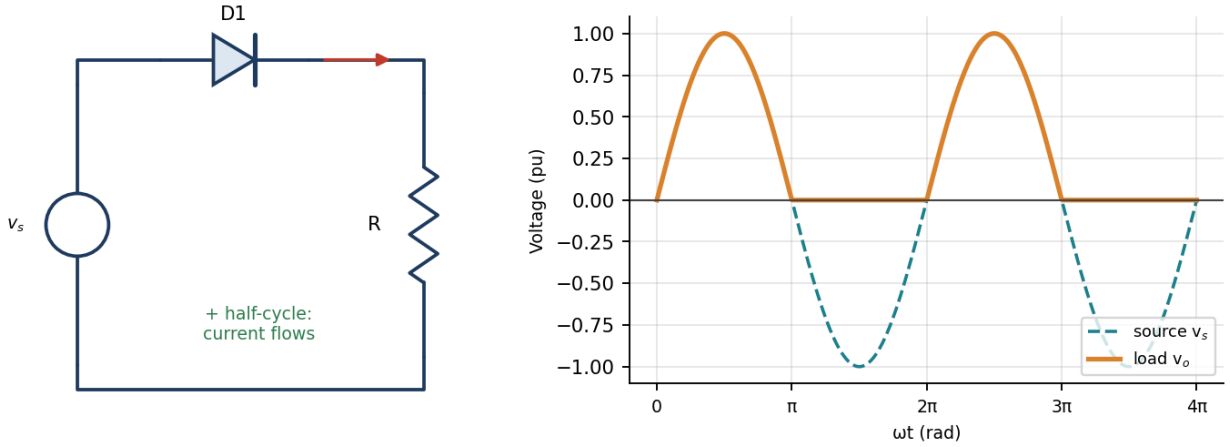


Figure 14. *Half-wave rectifier (ideal diode): current flows only while the source is positive, so the load voltage is the positive half-waves only.*

This simple circuit is the ancestor of every rectifier in Unit 2. Replacing the diode with an SCR gives *controlled* rectification.

D.3 The transistor: amplifier and switch

A **transistor** is a three-terminal semiconductor device in which a small signal at one terminal controls a larger current between the other two. Used in two very different ways:

- **As an amplifier**: operating in the middle (‘active’) region so the output current is proportional to the input; used in signal circuits.
- **As a switch**: driven fully ON (saturation: very low voltage across it) or fully OFF (cut-off: almost no current). Because either V≈0 or I≈0, the power lost in the device (P = V×I) is small. **All power electronics uses transistors and thyristors as switches.**

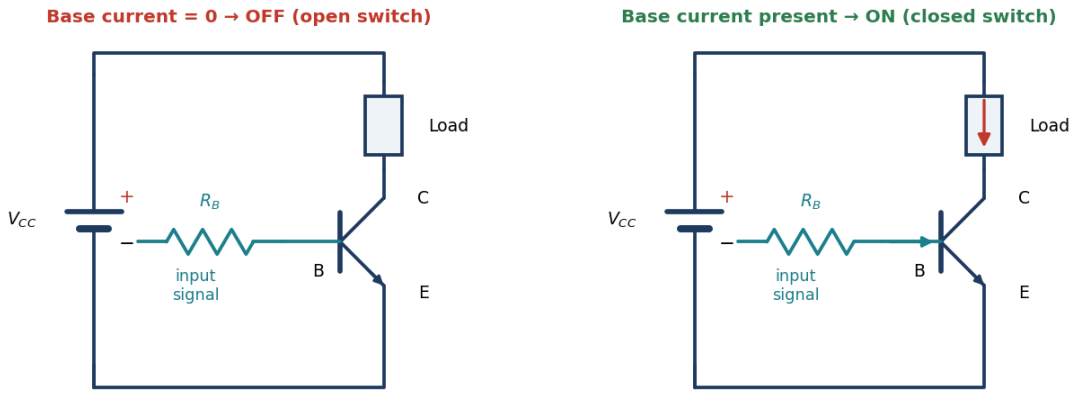


Figure 15. *An NPN transistor used as a switch. With no base current the collector–emitter path is open and the load is off (left). Base current closes it and load current flows (right).*

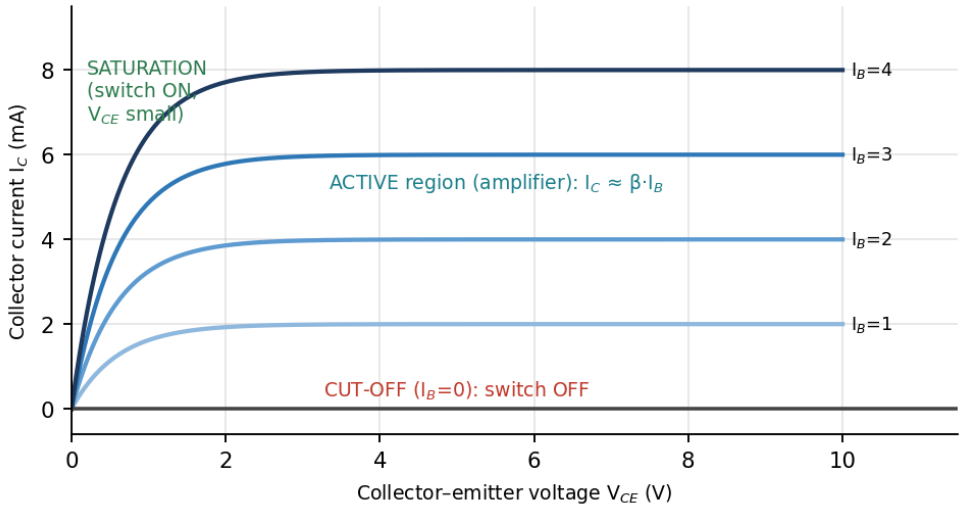


Figure 16. *Output characteristics of a BJT. Cut-off and saturation are the OFF and ON states used when switching; the active region is used for amplification.*

D.4 Why electronic switching controls power efficiently

Suppose a 12 V supply must give a motor an average of 6 V. **Method 1:** put a series resistor (or a transistor in the active region) to ‘drop’ the extra 6 V. With, say, 2 A flowing, the dropper wastes 6 V × 2 A = **12 W as heat**. **Method 2:** switch the supply fully ON and OFF at 50% duty. When ON, the switch has almost 0 V across it; when OFF, almost 0 A through it, so its loss is nearly zero. The motor sees an average of 6 V. This is the principle of the **power electronic converter**: an electronic circuit that uses switches to change the form of electrical power with little loss.

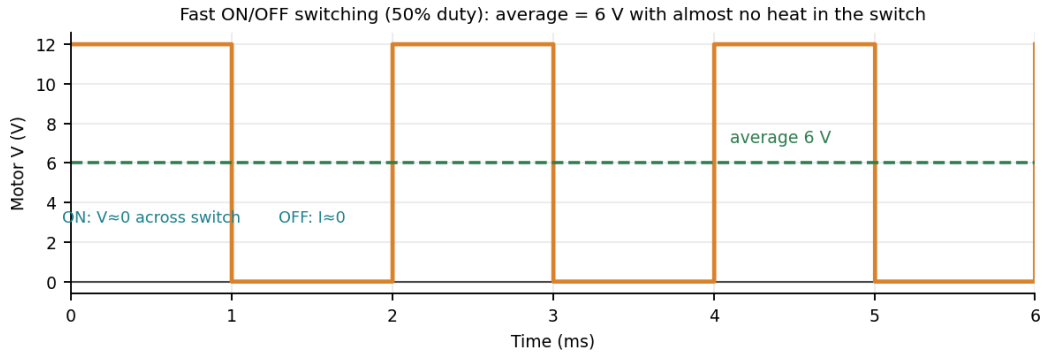


Figure 17. *Switching gives the same average voltage as a dropping resistor without the wasted heat.*

D.5 Meet the devices (first look)

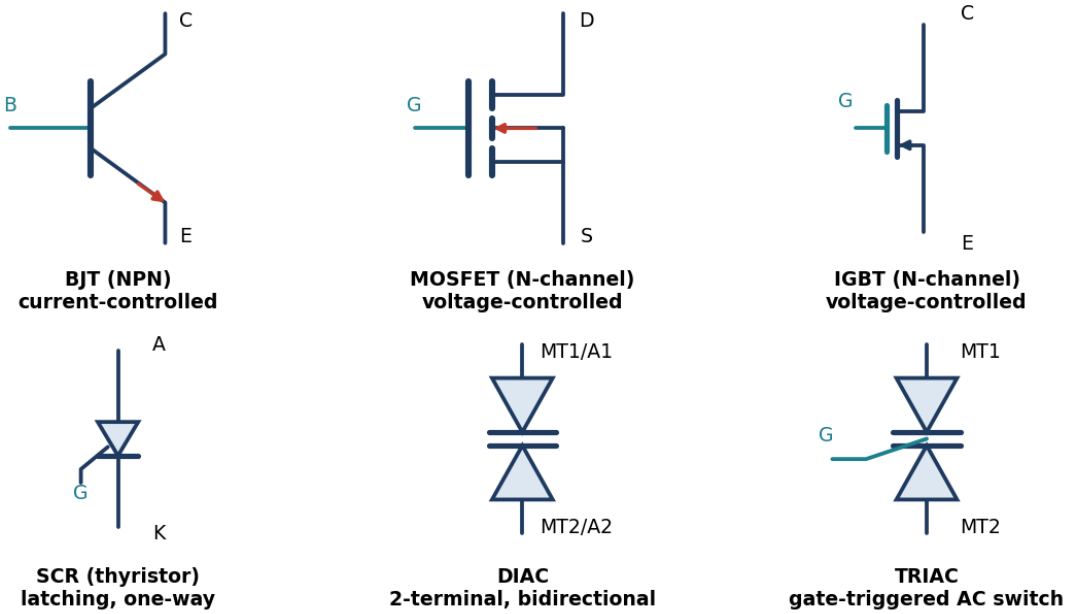


Figure 18. *Symbols of the six power switches in your question bank. Each is studied in detail in Unit 1.*

Device	Full name	Controlled by	One-line idea
BJT	Bipolar Junction Transistor	base *current*	Small base current controls a larger collector current.
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor	gate *voltage*	Voltage on an insulated gate opens a conducting channel; very fast.
SCR	Silicon Controlled Rectifier (thyristor)	gate pulse to turn ON	A diode that you can switch on; once on it stays on until current falls to zero.
IGBT	Insulated Gate Bipolar Transistor	gate *voltage*	MOSFET-style input with BJT-style high-current output.
DIAC	Diode for Alternating Current	none (voltage)	Two-terminal trigger that conducts both ways once its breakover voltage is exceeded.
TRIAC	Triode for Alternating Current	gate pulse	An AC switch: two SCRs back-to-back in one package.

D.6 Self-study questions, Chapter D

1. Which terminal of a diode must be more positive for conduction?
2. Why is a transistor operated as a switch more efficient than as a variable resistor?
3. Why are two names, ‘cut-off’ and ‘saturation’, used for the two states of a switching transistor?
4. A 24 V supply is chopped at 75% duty. What is the average output voltage?

Answers
1. The anode (by more than about 0.7 V for silicon).
2. In ON or OFF states, either V or I is almost zero, so P = V×I is small; as a variable resistor, both V and I are large.
3. Cut-off: no base current, the switch is OFF; saturation: transistor fully ON with minimum V_{CE}.
4. 24 × 0.75 = **18 V**.

Foundation Chapter E: Essential Engineering Mathematics

Only the mathematics used in this book is taught here, with examples. Return to this chapter whenever a later calculation looks unfamiliar.

E.1 Fractions, ratios, percentages

A **fraction** a/b means 'a divided by b'. A **ratio** compares two quantities (a transformer ratio 10:1 means one side is ten times the other). A **percentage** is a fraction of 100: 25% = 25/100 = 0.25. Duty cycle 40% means D = 0.4. Efficiency η = (output power/input power) × 100%.

> **Worked example: percentage**
> A motor takes 1000 W and delivers 850 W. Efficiency = 850/1000 = 0.85 = **85%**. The loss is 150 W.

E.2 Rearranging equations, substitution, units

Rule: **whatever you do to one side of an equation, do to the other.** Start from V = IR and want I: divide both sides by R: V/R = IR/R, so I = V/R. Want R: divide both sides by I: R = V/I.

> **Worked example: rearranging**
> Given P = V²/R, find V. Multiply both sides by R: PR = V². Take the square root: V = √(PR). For P = 100 W, R = 25 Ω: V = √(100×25) = √2500 = **50 V**.

Prefix	Symbol	Factor	Example
kilo	k	1000	2 kΩ = 2000 Ω
mega	M	1,000,000	3 MW = 3×10^{6} W
milli	m	0.001	5 mA = 0.005 A
micro	μ	0.000001	10 μF = 10×10^{-6} F

Always convert to base units (V, A, Ω, W, s, F, H) before substituting, and carry units through every step: an answer with the wrong unit signals an error.

E.3 Graphs and trigonometry

A graph shows how the vertical quantity (y) changes as the horizontal quantity (x) changes. For a V–I graph, x = I and y = V; the slope (rise ÷ run) = ΔV/ΔI = R. For a waveform, x is time or angle.

Sine function. For a right triangle, sin θ = opposite ÷ hypotenuse. As the angle goes 0° → 90° → 180° → 270° → 360°, sin θ goes 0 → 1 → 0 → −1 → 0. That is the sine wave. Key values are in Figure 19. Always set your calculator to **degrees** or **radians** to match your angle.

θ (degrees)	0	30	45	60	90	120	180	270	360
sin θ	0	0.5	0.707	0.866	1	0.866	0	−1	0
cos θ	1	0.866	0.707	0.5	0	−0.5	−1	0	1

Useful identity: $\sin^2\theta = (1 - \cos 2\theta)/2$, which is why the mean of sin² is ½ (the cos 2θ term averages to zero).

E.4 Differentiation and integration, only what control needs

Derivative = rate of change = slope of a graph. If position x changes with time, the derivative dx/dt is speed. A large derivative means the quantity is changing fast. A PID controller's D action uses the slope of the error. Rules needed: d(c)/dt = 0 for constant c; d(kt)/dt = k; d(sin θ)/dθ = cos θ.

Integral = accumulated total = area under a graph. If speed is graphed against time, the area under it is the distance travelled. The I action in PID accumulates error over time. Rules: $\int k\ dt = kt$; $\int \sin\theta\ d\theta = -\cos\theta$.

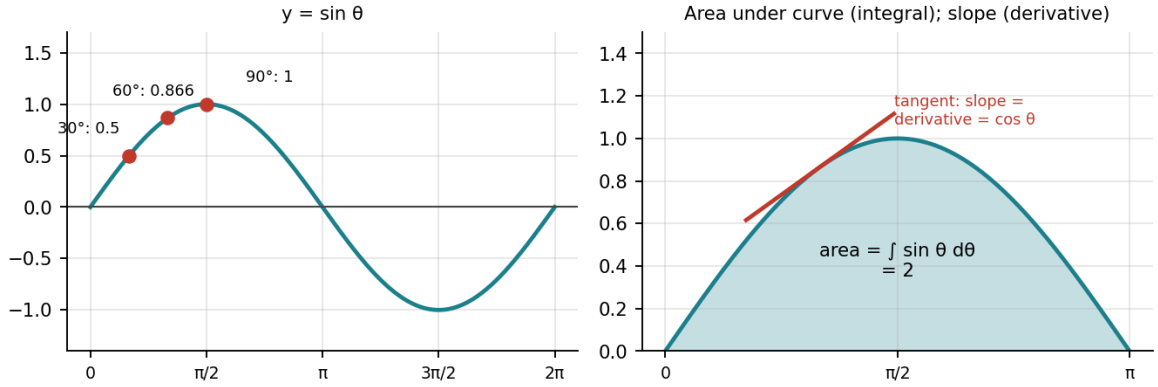


Figure 19. *Left: key sine values. Right: the area under half a sine (integral) and the slope (derivative) at a point.*

Worked example: average of a full-wave rectified sine (used in Unit 2)
Average over one half-cycle (0 to π) of $V_p \sin\theta = (1/\pi) \int_0^{\pi} V_p \sin\theta\ d\theta$.
$\int_0^{\pi} \sin\theta\ d\theta = [-\cos\theta]_0^{\pi} = (-\cos\pi) - (-\cos 0) = (-(-1)) - (-1) = 1 + 1 = 2$.
So $V_{avg} = (V_p/\pi) \times 2 = \mathbf{2V_p/\pi \approx 0.637\ V_p}$. For a half-wave rectified sine, the average is taken over a full 2π cycle but the area is the same $2V_p$, so $V_{avg} = 2V_p/2\pi = V_p/\pi$.

E.5 Binary numbers and Boolean algebra

Computers and PLCs use **binary**: only two digits, 0 and 1, matching 'off/on' switches. Each position is worth twice the previous one: ...16, 8, 4, 2, 1. So $1011_2 = 1\times8 + 0\times4 + 1\times2 + 1\times1 = \mathbf{11}$. To convert decimal 13 to binary: $13 = 8 + 4 + 1 \rightarrow 1101_2$.

Boolean algebra deals with variables that are only TRUE (1) or FALSE (0). Three basic operations: **AND** (A·B, true only if both true), **OR** (A+B, true if at least one true), **NOT** ($\bar{A}$, the opposite). Useful laws: A·0 = 0, A·1 = A, A+0 = A, A+1 = 1, A·A = A, A+$\bar{A}$ = 1, A·$\bar{A}$ = 0, and De Morgan's: NOT(A·B) = $\bar{A} + \bar{B}$ and NOT(A+B) = $\bar{A}\cdot\bar{B}$. Gate-by-gate truth tables appear in Unit 3.

E.6 Self-study questions, Chapter E

1. Rearrange $P = I^2R$ to find I.
2. Convert 4.7 kΩ to ohms and 250 mA to amperes.
3. Convert binary 10110 to decimal.
4. Find $\int_0^{\pi} 3 \sin\theta\ d\theta$.

Answers
1. $I = \sqrt{(P/R)}$.
2. 4700 Ω; 0.25 A.
3. 16 + 4 + 2 = **22**.
4. 3 × 2 = **6**.

UNIT 1: Power Switches, Power Factor, Power Quality, EMI/EMC and Relays

This unit asks eight questions. We follow your question bank's order exactly. Each question below is shown in an orange box with its original wording, then taught from the ground up, then converted into an exam answer in the revision pack at the end of the unit.

Unit 1, Question 1: Power switches and comparison of BJT, MOSFET, SCR, IGBT, DIAC, TRIAC

Exam question – Unit 1 Q1
Define Power Switches. Compare the working and characteristics BJTs, MOSFETs, SCR, IGBTs, DIAC, and TRIAC based on switching characteristics, operating parameters, applications etc. with neat diagrams.

Note on the source wording
The source sentence omits 'of' ('characteristics of BJTs…'). The intended meaning is to compare these six devices. The wording above is kept unchanged.

A. The problem and its purpose

A factory motor may need 50 A at 400 V; a lamp dimmer needs to change the power into a 1 kW lamp; an electric-vehicle inverter must switch hundreds of amperes thousands of times per second. A small control signal (milliwatts) must be able to command a huge flow of power (kilowatts), and do so without wasting much of it as heat. A **power switch** is the semiconductor device that does this.

B–D. Prerequisites, intuition and formal definition

Beginner view. A power switch is an electronic light-switch with no moving parts. Instead of a hand flipping a lever, a small electrical signal makes it 'closed' (current flows) or 'open' (no current).

Formal definition. A **power switch** is a semiconductor device operated in only two states, a fully **ON (conducting)** state with very low voltage drop and a fully **OFF (blocking)** state with very low leakage current, used to control the flow of electrical power in power electronic circuits.

Property	Ideal switch	Real device
ON voltage drop	0 V	≈ 0.7–3 V (conduction loss = V×I)
OFF leakage current	0 A	very small but not zero
Switching time	instant	finite: switching loss occurs while V and I are both non-zero
Voltage / current limits	unlimited	rated maximum V and I (exceeding destroys it)
Control power	none	drive circuit needed (gate/base)

Two kinds of loss. *Conduction loss* = V_{ON} × I while ON. *Switching loss* occurs during each transition, when the device briefly has both high voltage and high current. Faster switching (higher frequency) increases switching loss.

Classification by how the device is controlled:

- **Uncontrolled**: power diode (switches by itself, depending on circuit polarity).
- **Latching, turn-ON controlled**: SCR, TRIAC (a pulse turns them on; they turn off only when the current falls to zero).
- **Fully controlled (turn-ON and turn-OFF by the control terminal)**: BJT, MOSFET, IGBT.
- **Trigger device**: DIAC (no control terminal; breaks over at a fixed voltage).

E. Visual explanation: symbols and V–I characteristics

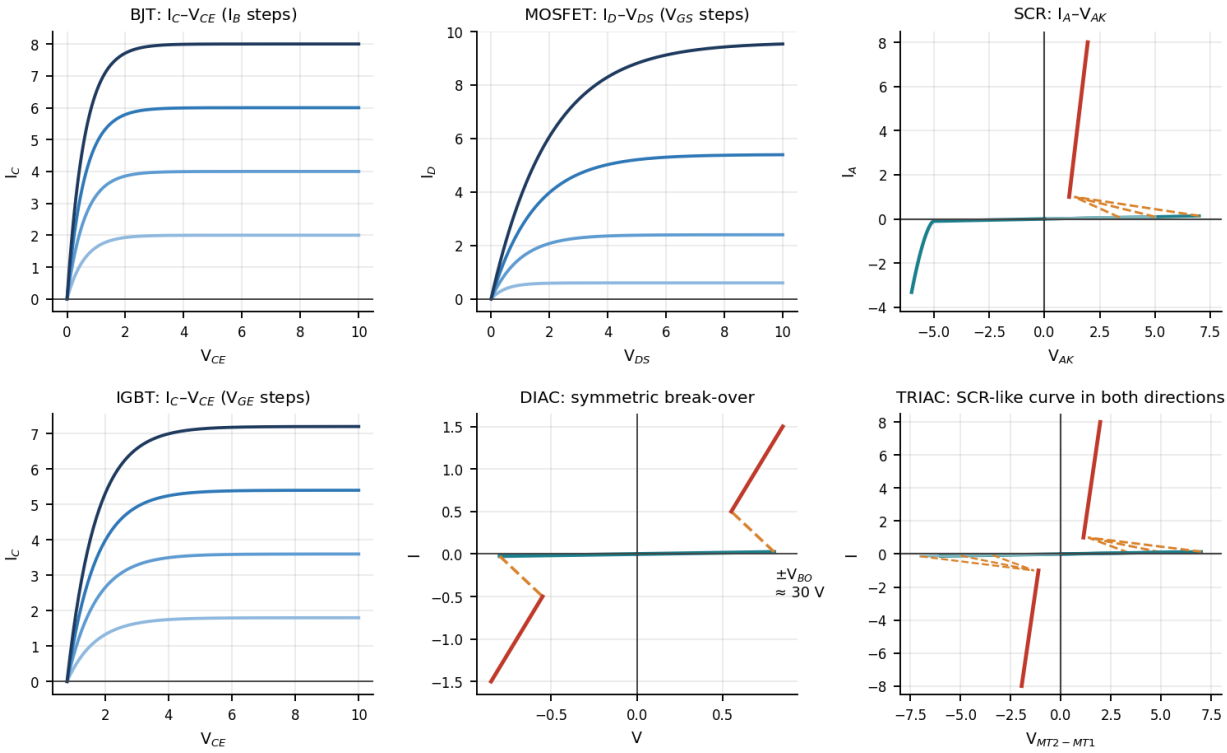


Figure 20. *Schematic V–I characteristics of the six devices. Curves are drawn for understanding, not to scale.*

F. Step-by-step working of each device

1. BJT (Bipolar Junction Transistor). Three layers (NPN or PNP) with terminals Base, Collector, Emitter. *Working:* a base current I_B flows into the base–emitter junction; this allows a much larger collector current $I_C \approx \beta \cdot I_B$ (β = current gain, typically 10–100 in power BJTs at switching). For switching, enough I_B is supplied to drive it into saturation ($V_{CE(sat)} \approx 1$–2 V); with $I_B = 0$ it is OFF. *Weakness:* needs continuous base current (a substantial drive circuit), is relatively slow, and can suffer 'second breakdown'. Largely replaced by MOSFETs and IGBTs.

2. MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor). Terminals Gate, Drain, Source. The gate is insulated from the channel by a thin oxide layer. *Working:* a gate–source voltage above the threshold $V_{GS(th)}$ attracts electrons to form a conducting channel between drain and source. No steady gate current flows (only a charging current to the gate capacitance), so it is voltage-controlled and very fast. ON state behaves like a small resistance $R_{DS(on)}$, so conduction loss = $I^2 \cdot R_{DS(on)}$. High-voltage MOSFETs have larger $R_{DS(on)}$, so they are best at low to medium voltage and high switching frequencies.

3. SCR (Silicon Controlled Rectifier, thyristor). Four layers PNPN, terminals Anode, Cathode, Gate. A short gate pulse turns it on when it is forward biased; after that the gate loses control and it stays on until the anode current falls below the holding current (details in Question 3).

4. IGBT (Insulated Gate Bipolar Transistor). MOSFET-like insulated gate (easy voltage drive) plus a bipolar output (low ON-state voltage even at high voltage ratings). Terminals Gate, Collector, Emitter (details in Question 6).

5. DIAC (Diode for Alternating Current). A two-terminal, three-layer device with no gate. It blocks in both directions until the applied voltage magnitude exceeds its **break-over voltage V_{BO}** (typically about 28–36 V for common types). Then it suddenly conducts, giving a current pulse. Because its trigger point is sharp and symmetrical, it is used to fire a TRIAC.

6. TRIAC (Triode for Alternating Current). Equivalent to two SCRs connected in inverse parallel, sharing one gate. Terminals MT1, MT2, Gate. A gate pulse (of either polarity) turns it on in either direction of MT2–MT1 voltage, so it can control AC power in both half-cycles. It turns off when the current passes through zero (every half-cycle of the AC).

G–H. Comparison table with operating parameters

Typical orders of magnitude are given; exact ratings vary by manufacturer and device generation, so always consult the datasheet. They are for comparison, not design.

Parameter	BJT	MOSFET	SCR	IGBT	DIAC	TRIAC
Control signal	base current	gate voltage	gate current pulse	gate voltage	none (break-over voltage)	gate current pulse
Turn-OFF by control?	yes	yes	no (needs current zero / forced commutation)	yes	self-resets when current falls	no (current zero each half-cycle)
Conduction direction	one	one (body diode reverse)	one	one	both	both
Typical voltage rating	up to ~1 kV	~20 V–1 kV (SiC higher)	up to several kV	~600 V–6.5 kV	~30 V trigger	~200–1000 V
Typical current rating	tens–hundreds A	few A–hundreds A	up to several kA	tens A–kA (modules)	small (tens of mA–1 A peak pulse)	few A–tens A
Switching speed / frequency	slow–medium, up to ~10–20 kHz	fastest, 100 kHz to MHz	slowest, line frequency up to few kHz	medium, ~1–30 kHz typical	–	line frequency (50/60 Hz)
ON-state drop	~1–2 V	$I \cdot R_{DS(on)}$	~1–2 V	~1.5–3 V	–	~1–2 V
Drive power	high (continuous I_B)	very low	low pulse	very low	none	low pulse
Power-handling class	medium	low–medium	very high	medium–high	trigger only	low–medium (AC)
Typical applications	older amplifiers/low-cost switching	SMPS, DC–DC converters, small motor drives	rectifiers, HVDC, soft starters, large AC controllers	VFDs, inverters, EV traction, UPS, welders	triggering TRIACs in dimmers	lamp dimmers, heater controllers, fan regulators, small AC motors

I. How engineers choose

- **Very high power at line frequency (rectifiers, HVDC):** SCR, the highest voltage/current ratings and robust.
- **Medium/high power with PWM and variable speed (VFDs, inverters):** IGBT.
- **Low voltage, very high frequency (SMPS, DC–DC):** MOSFET, small, efficient, needs little drive.
- **AC dimmer or heater control at home:** TRIAC triggered through a DIAC.
- **BJT** is mostly met in legacy and low-cost circuits.

Worked example: conduction loss comparison
A switch carries 20 A. Compare conduction loss for a MOSFET with $R_{DS(on)} = 0.05\ \Omega$ and an IGBT with $V_{CE(sat)} = 2$ V.
MOSFET: $P = I^2R = 20^2 \times 0.05 = 400 \times 0.05 =$ **20 W** (equivalent drop $20 \times 0.05 = 1$ V).
IGBT: $P = V \times I = 2 \times 20 =$ **40 W**.
At 20 A the MOSFET wins; at 100 A the MOSFET loss is $100^2 \times 0.05 = 500$ W, whereas the IGBT is $2 \times 100 = 200$ W. The crossover shows why IGBTs are chosen at high current and high voltage.

Common misconception
‘An SCR can be turned off by removing the gate signal.’ No. Once latched, the gate has no control; the anode current must fall below the holding current. ‘A MOSFET has no loss.’ It has $I^2R_{DS(on)}$ and switching losses.

M. Self-study questions

1. Define a power switch and name two losses in a real switch.
2. Which devices can be turned OFF through their control terminal?
3. Why is a DIAC used with a TRIAC?
4. Why is the IGBT preferred in a 400 V, 100 A motor drive over a MOSFET?

Answers
1. A semiconductor device used in only ON/OFF states to control power. Losses: conduction and switching.
2. BJT, MOSFET, IGBT (SCR/TRIAC cannot).
3. The DIAC gives a sharp, symmetrical trigger pulse at a fixed voltage, making TRIAC firing consistent.
4. At high voltage and current its ON-state drop (≈ 2 V) is lower than the MOSFET's I^2R loss, and it is still easy to drive.

Unit 1, Question 2: Impact of poor power factor and correction techniques

Exam question – Unit 1 Q2
What impact does a poor power factor have on industrial power system performance, and which techniques are commonly applied to correct it explain in details.

A. The problem

A factory pays for power that does not do useful work, and its cables and transformers are loaded with current that produces no motion or heat. Industrial plants have many induction motors, transformers and welders, all **inductive** loads, which create exactly this problem.

B–D. Prerequisites and the three kinds of power

In Chapter C we saw that in an inductive load the current **lags** the voltage by an angle φ. With φ ≠ 0, part of the power goes back and forth between the source and the load's magnetic fields rather than being used.

Power	Symbol / unit	Meaning	Formula (single-phase)
Active (real) power	P, watt (W) or kW	does useful work (shaft power, heat, light)	P = V I cos φ
Reactive power	Q, volt-ampere reactive (VAr)	builds magnetic fields; swings back to the source	Q = V I sin φ
Apparent power	S, volt-ampere (VA) or kVA	what the source/cables must supply	S = V I

$$S^2 = P^2 + Q^2 \qquad PF = \cos\varphi = \frac{P}{S}$$

The **power factor (PF)** is the ratio of useful power to apparent power. It ranges from 0 to 1; PF = 1 means all supplied current is doing useful work. **Lagging** PF: current lags voltage (inductive loads). **Leading** PF: current leads (capacitive loads).

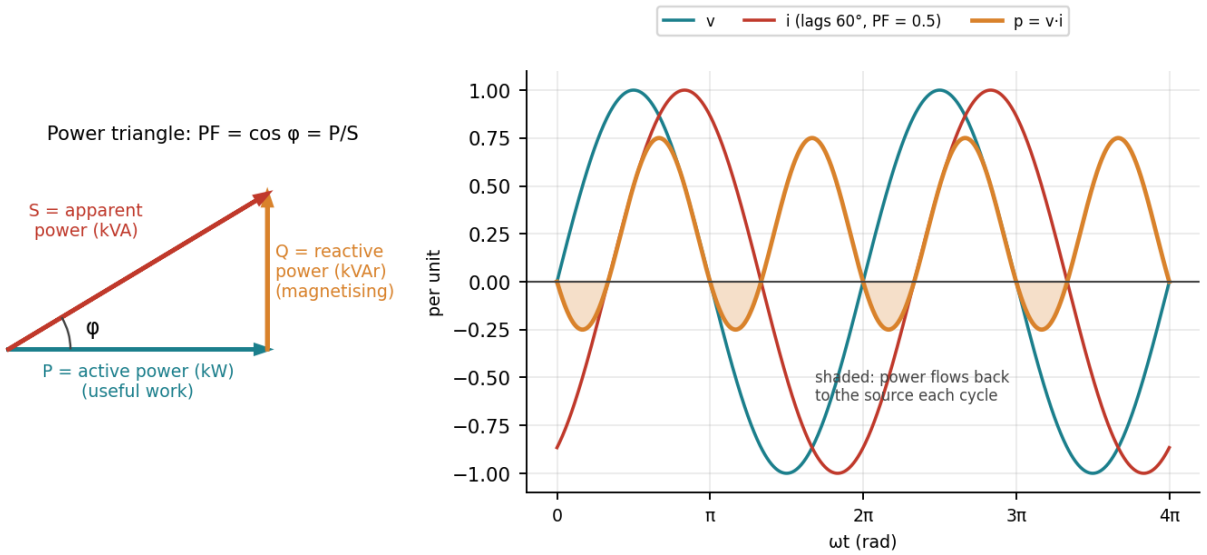


Figure 21. *Left: the power triangle. Right: with a 60° lag (PF = 0.5), instantaneous power p = v·i goes negative for part of each cycle: energy is returned to the source rather than used.*

E–G. Why poor PF matters: derivation

For the same useful power P at the same voltage V, the current is I = P/(V cos φ). A lower PF means a larger current. Since cable and transformer heating loss is I²R:

$$I = \frac{P}{V\cos\varphi} \qquad P_{loss} = I^2R = \frac{P^2R}{V^2\cos^2\varphi}$$

So the loss is proportional to 1/cos²φ. Improving PF from 0.7 to 0.95 reduces loss to (0.7/0.95)² = 0.543, i.e. by about 46%.

Effect of poor power factor	Explanation
Higher current for the same useful power	I = P/(V cos φ)
Larger I²R losses in cables, transformers, generators	loss ∝ 1/cos²φ → wasted energy and heating
Poor voltage regulation / larger voltage drop	the extra reactive current drops more voltage along the lines
Reduced capacity (kVA) of generators, transformers, cables	they are rated in kVA, so reactive current uses up capacity that could supply useful kW
Larger conductors and switchgear needed	to carry the higher current
Utility penalties / lower tariff benefits	utilities charge on kVA demand or penalise low PF (in India, DISCOMs commonly set PF-linked tariff provisions; check the current tariff order)

H. Correction techniques, in detail

The principle: reactive power of a capacitor is opposite in sign to that of an inductor. A capacitor placed near the inductive load *supplies* the reactive power the load needs, so the source only has to supply the useful (active) part.

Technique	How it works	Where used / notes
1. Static capacitor banks (fixed)	Capacitors in parallel with the load supply leading reactive current.	Simple, cheap. Fixed banks at individual large motors. Risk of over-correction at light load.
2. Automatic power factor correction (APFC) panel	A controller measures PF and switches capacitor steps in/out via contactors or thyristor switches.	Whole plants with varying load.

Technique	How it works	Where used / notes
3. Synchronous condenser / over-excited synchronous motor	A synchronous machine run over-excited acts like a variable capacitor, leading PF.	Large plants, can adjust continuously; costly and has maintenance.
4. Static VAr compensators (SVC) / STATCOM	Power-electronic (thyristor- or IGBT-based) devices that inject or absorb reactive power rapidly.	Fast-varying loads (arc furnaces, rolling mills); also helps voltage control and flicker.
5. Detuned (reactor-series) capacitor banks and active filters	A series reactor avoids resonance with harmonics; active filters also correct harmonics.	Plants with many VFDs/rectifiers (see Question 5).
6. Operating practices	Avoid lightly loaded motors; right-size motors; use high-efficiency motors.	Reduces the source of low PF.

The calculation of the capacitor size is in Question 8, which designs the circuit in detail.

K–L. Misconceptions and summary

Common misconception
'Low power factor wastes the kW the customer pays for.' Not exactly. The kW measured is real work; what low PF wastes is *current capacity* and causes extra I^2R loss and voltage drop. Also, PF correction by capacitor does **not** change the useful power delivered to the load; it only reduces the current drawn from the source.

Key idea
Poor PF ⇒ more current for the same work ⇒ more loss, less capacity, worse voltage, higher cost. Fix: put reactive-power sources (capacitors, synchronous condensers, SVC/STATCOM) near the loads.

Unit 1, Question 3: SCR / Thyristor working principle with V–I characteristics

Exam question – Unit 1 Q3
Explain the working principle of SCR/Thyristor with VI characteristics.

A. The problem

A diode conducts whenever it is forward biased; we cannot choose *when*. To control how much power reaches a load (the speed of a DC motor, the heat of a furnace) we need a diode that we can 'switch on' at a moment of our choice. That device is the SCR.

D. Construction

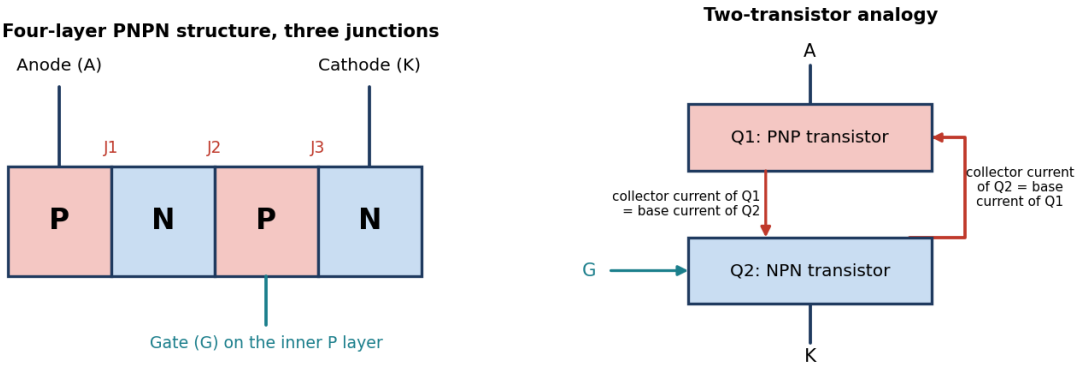


Figure 22. *Left: four-layer PNPN structure with three junctions J1, J2, J3. Right: the same device behaves like a PNP and an NPN transistor, each supplying the other's base current.*

SCR (Silicon Controlled Rectifier) has terminals Anode (A), Cathode (K), Gate (G). It is a **unidirectional** device: current flows only from A to K.

F. Working, step by step (cause and effect)

1. **Reverse blocking:** V_{AK} negative. Junctions J1 and J3 are reverse biased, so only a tiny leakage current flows. If V_{AK} reverses beyond the **reverse breakdown voltage V_{BR}**, avalanche breakdown can destroy the device.
2. **Forward blocking (OFF state):** V_{AK} positive, gate not triggered. J1 and J3 are forward biased, but J2 is reverse biased and blocks the voltage. Only leakage flows.
3. **Turn-ON by gate:** a small positive gate current injects carriers across the inner P–N region (the NPN transistor's base). Q2 starts to conduct, which supplies base current to Q1. Q1 then conducts more, feeding more base current back to Q2. This **regenerative (positive-feedback) action** drives both into saturation in microseconds; the SCR turns fully ON.
4. **Turn-ON by excessive forward voltage (break-over):** if V_{AK} exceeds the forward break-over voltage V_{BO} with no gate current, J2 avalanches and the SCR turns on. This is normally avoided; a higher I_G makes the SCR fire at a lower V_{BO}.
5. **Conduction (ON state):** voltage drop about 1–2 V, current limited by the external load. **The gate now loses control**: removing I_G does not turn it off, provided the anode current exceeds the **latching current I_L** when the gate pulse ends.
6. **Turn-OFF (commutation):** the anode current must be reduced below the **holding current I_H** (and reverse voltage applied briefly) for a time called the turn-off time t_q. In AC circuits this happens naturally every cycle (**natural commutation**). In DC circuits extra circuits force it (**forced commutation**).

E. The V–I characteristic

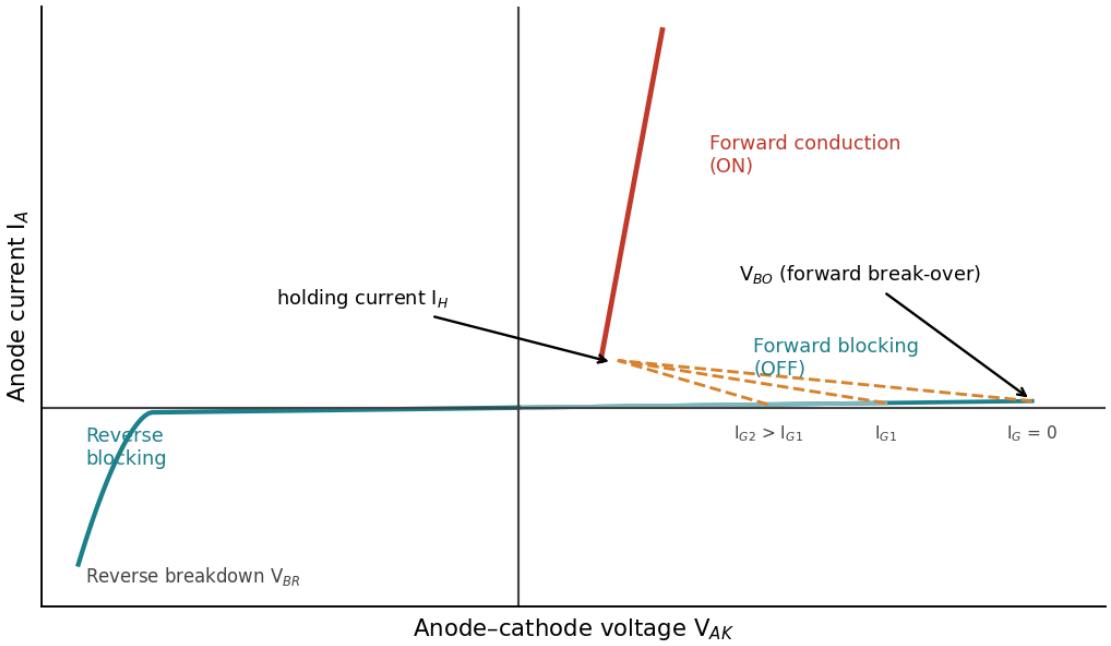


Figure 23. *V–I characteristics of an SCR (schematic). Increasing gate current reduces the break-over voltage; the dashed segment is the unstable negative-resistance transition.*

Region / parameter	Meaning
Reverse blocking	$V_{AK} < 0$: OFF; tiny leakage
Reverse breakdown V_{BR}	reverse avalanche; destructive if sustained
Forward blocking	$V_{AK} > 0$ and below V_{BO}, no gate pulse: OFF
Forward break-over V_{BO}	voltage at which it turns on without gate current
Forward conduction	ON state, V ≈ 1–2 V
Holding current I_H	minimum anode current to remain ON; below it the SCR turns off

Region / parameter	Meaning
Latching current I_L	minimum anode current needed at the end of the gate pulse for the SCR to stay on; $I_L > I_H$ (roughly 2–3 times)

G. Equation: phase-controlled half-wave rectifier with SCR

The SCR's main use is **phase control**. The gate pulse is delayed by the **firing (delay) angle α** after the instant the supply voltage becomes positive, so the SCR conducts only from α to π (resistive load).

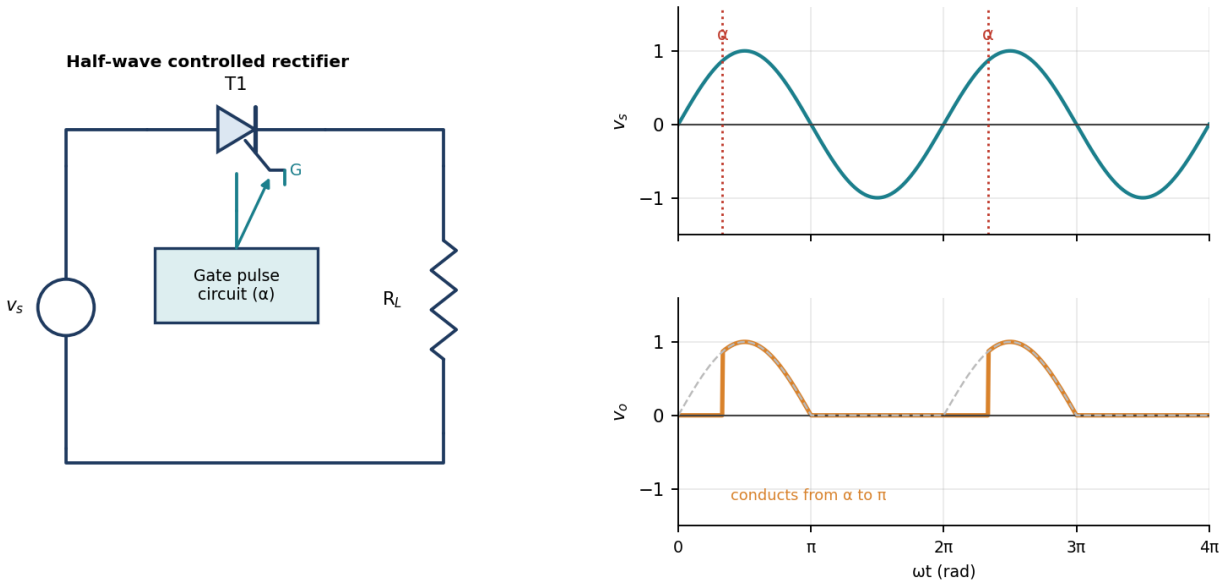


Figure 24. *Half-wave SCR circuit with resistive load R_L, and its waveforms. The SCR conducts from α to π; by changing α the average output voltage is adjusted.*

Let $v_s = V_m \sin \omega t$ (V_m = peak). The average output over one cycle (2π):

$$V_{avg} = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin\theta \, d\theta = \frac{V_m}{2\pi}[-\cos\theta]_{\alpha}^{\pi} = \frac{V_m}{2\pi}(-\cos\pi + \cos\alpha) = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

Check the limits: α = 0 gives V_m/π (an ordinary half-wave rectifier, from Chapter C); α = 180° gives 0 (SCR never conducts).

Worked example: firing angle control
Mains 230 V RMS: $V_m = \sqrt{2} \times 230 = 325.3$ V. α = 60°. Find V_{avg}.
$V_{avg} = (325.3 / 2\pi)(1 + \cos 60°) = (325.3 / 6.2832)(1 + 0.5) = 51.77 \times 1.5 =$ **77.7 V**.
For α = 0: 51.77 × 2 = 103.5 V; for α = 90°: 51.77 × 1 = 51.8 V; so raising α lowers the DC output.

I. Applications, J. Advantages, limitations

- **Applications:** controlled rectifiers for DC motor drives, battery chargers, HVDC converters, AC voltage controllers (heaters, lighting), soft starters for large induction motors, static switches and crowbar protection.
- **Advantages:** very high voltage/current ratings, rugged, low conduction loss, low-cost gate drive.
- **Limitations:** cannot be turned off by the gate; limited switching frequency; needs commutation circuits in DC applications; sensitive to rapid rise of voltage (dv/dt) and current (di/dt), so snubber circuits are used.

Common misconception
Mixing up holding and latching current: latching current is needed *just after turn-on*; holding current is needed to *stay on*. Latching ≥ holding.

Unit 1, Question 4: EMI/EMC considerations in industrial circuits

Exam question – Unit 1 Q4
Explain EMI/EMC considerations in industrial circuits with suitable examples.

A–D. Concepts

EMI (Electromagnetic Interference) is unwanted electrical noise from one device that disturbs another. **EMC (Electromagnetic Compatibility)** is the ability of equipment to work correctly in its electromagnetic environment *without* causing unacceptable interference to others and while tolerating the interference that exists. In short: EMI is the problem; EMC is the goal.

Why it arises in power electronics. Switches that change current rapidly (large di/dt) and voltage rapidly (large dv/dt) generate high-frequency energy, because a fast edge contains many high-frequency components. An inverter switching 400 V in tens of nanoseconds is a strong noise source.

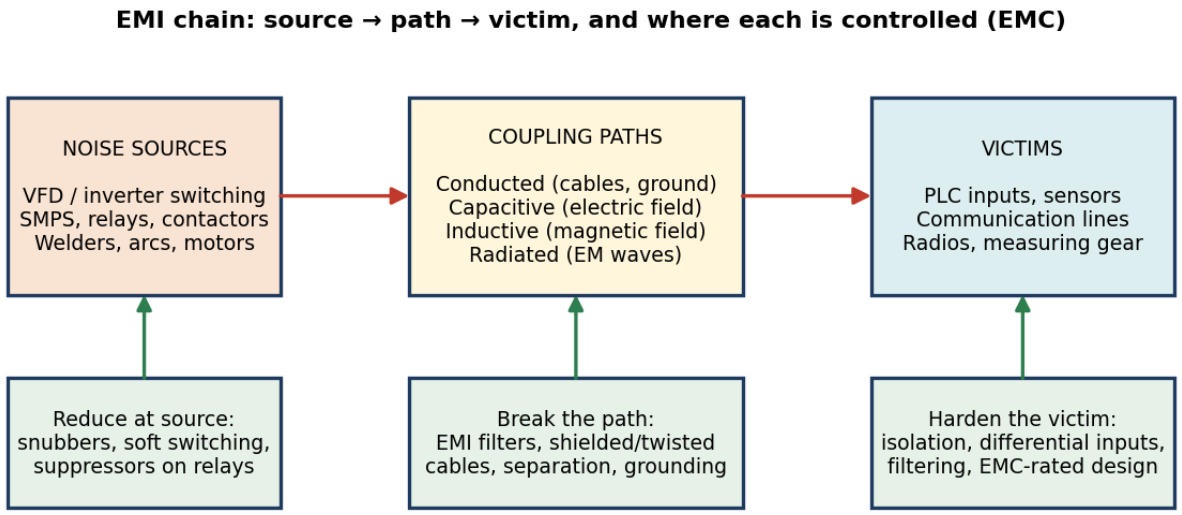


Figure 25. *EMI chain: every EMI problem has a source, a coupling path and a victim; EMC engineering acts on one or more of them.*

E–F. Coupling mechanisms

Coupling	How noise travels	Example
Conducted	through wires, power lines, ground connections	VFD switching noise on the mains upsetting other loads
Capacitive (electric field)	stray capacitance between nearby conductors couples dv/dt	an unshielded sensor cable beside a motor cable
Inductive (magnetic field)	changing current in one loop induces voltage in another loop (di/dt)	a long loop in a signal cable near a contactor coil
Radiated	electromagnetic waves	arc welding disturbing a wireless link
Common-impedance (ground)	two circuits share a return path; one's current creates voltage drop in the other	noisy motor ground shared with analogue sensor ground

G–I. Mitigation techniques with examples

Technique	How it helps	Industrial example
EMI filter (LC / ferrite)	blocks high-frequency noise from entering or leaving	line filter at VFD input
Shielded and twisted-pair cable	shield intercepts fields; twisting cancels induced noise	analogue 4–20 mA signals, RS485 lines

Technique	How it helps	Industrial example
Separation and routing	keep power and signal cables apart; cross at 90°	separate trays for motor cables and sensor cables
Proper grounding and bonding	gives noise a low-impedance path; avoids ground loops	single-point ground for analogue signals; short bonding straps
Snubbers and suppressors	limit voltage spikes at the source	RC snubber or flyback diode across a relay coil or contactor
Slower edges / soft switching	reduce dv/dt and di/dt	adjusting gate resistance of IGBT in a drive
Metal enclosures / cable glands	Faraday-cage effect	VFD in an earthed metal cabinet
Optical or galvanic isolation	breaks ground loops and common-mode paths	optocouplers on PLC inputs, isolated communication

EMC standards. Equipment is tested for *emissions* (how much noise it produces) and *immunity* (how much it tolerates). Examples: the IEC 61000 series, and CISPR limits for emission; equipment is marked (e.g. CE) after passing these.

Worked example: EMI troubleshooting
A VFD-driven pump causes a nearby level sensor's 4–20 mA reading to jitter by ±0.8 mA.
Likely path: switching noise couples capacitively/inductively into the unshielded sensor cable that runs 5 m parallel to the motor cable.
Fixes (in order): (1) re-route sensor cable away; (2) use shielded twisted-pair with shield grounded at one end; (3) fit a VFD output reactor/line filter; (4) check grounding. Each step addresses source, path or victim.

Common misconception
'Grounding everywhere solves noise.' Multiple grounds create loops through which noise current flows; use a deliberate grounding scheme.

Unit 1, Question 5: Power quality issues, harmonics and voltage sags

Exam question – Unit 1 Q5
What are the major power quality issues in industrial systems? Explain harmonics and voltage sags in detail

A–D. Power quality

Power quality is the degree to which the supplied voltage is a clean sine wave at the right magnitude and frequency, available without interruption. Poor power quality makes sensitive machines misbehave, trip, overheat or fail.

Issue	What it is	Typical causes	Effects
Voltage sag (dip)	short drop of RMS voltage to 10–90% for half a cycle to 1 minute (IEEE 1159)	faults on the grid, motor starting, large load switching	contactor drop-out, PLC resets, VFD trips, process stoppage
Voltage swell	short rise above 110%	load switch-off, single-line-to-ground fault	insulation stress, equipment damage
Interruption	voltage below 10% briefly or for a longer time	breaker operation, supply failure	complete stoppage
Transients / surges	very short high-voltage spikes	lightning, capacitor switching, inductive switching	insulation breakdown, damage to electronics
Harmonics	voltage/current components at multiples of 50 Hz	non-linear loads: rectifiers, VFDs, SMPS, arc furnaces	overheating, nuisance trips, capacitor failure, extra losses

Issue	What it is	Typical causes	Effects
Voltage unbalance	unequal voltages on the three phases	unequal single-phase loads	motor overheating and torque pulsation
Flicker	repetitive voltage fluctuations	arc furnaces, welders, large motor cycling	lights flicker, annoyance
Frequency variation / noise	drift or high-frequency noise	weak grids, generators; switching devices	timing errors, interference

Harmonics in detail

Beginner explanation. A pure mains voltage is a perfect 50 Hz sine. A **non-linear load** (a load whose current is *not* proportional to the voltage, such as a rectifier that only draws current in short pulses) draws a distorted current. Any repeating distorted wave can be written as the sum of sine waves: a **fundamental** at 50 Hz plus **harmonics** at integer multiples (100 Hz is the 2nd harmonic, 150 Hz the 3rd, and so on).

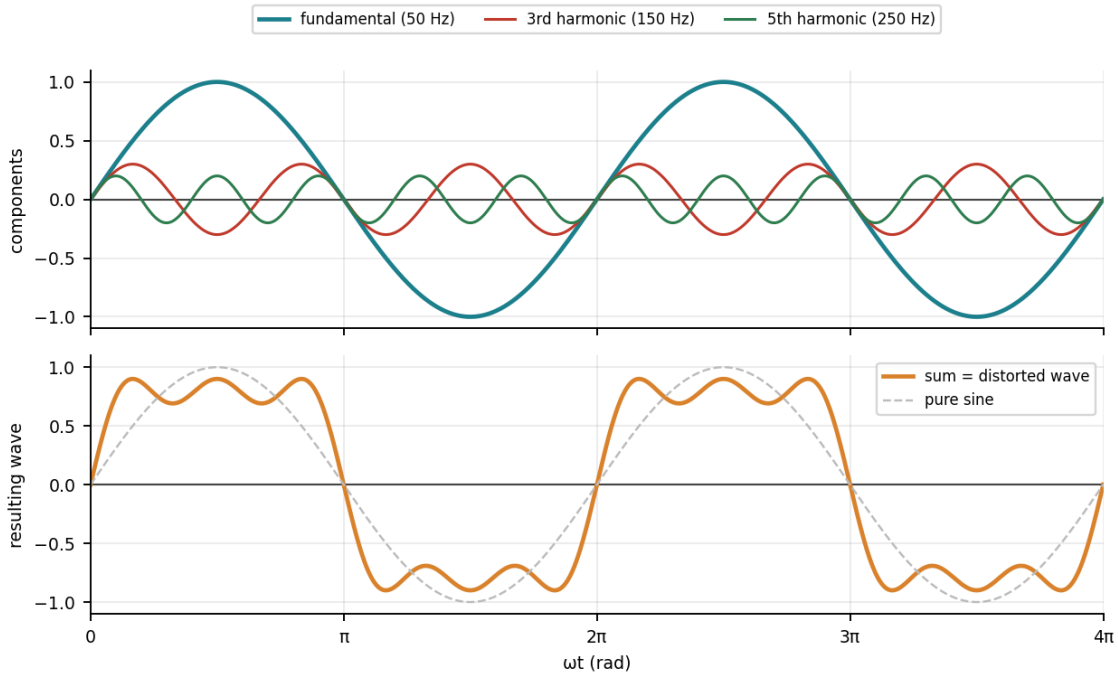


Figure 26. *A fundamental plus a 3rd and a 5th harmonic produces a distorted wave. The harmonics are the 'extra' sine waves hidden in the distortion.*

Which harmonics appear? A three-phase six-pulse rectifier (as in a VFD) produces harmonics of order n = 6k ± 1: 5th, 7th, 11th, 13th,... Single-phase SMPS loads produce odd harmonics, especially the 3rd; the 3rd, 9th ('triplen') harmonics add in the neutral conductor of a three-phase four-wire system instead of cancelling, which can overload the neutral.

$$THD_V = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots}}{V_1} \times 100\%$$

V_1 is the RMS fundamental; V_2, V_3,... are RMS values of the harmonics. THD (Total Harmonic Distortion) is a single number measuring distortion. The same formula, with currents, gives THD_I.

Worked example: THD calculation
V_1 = 230 V, V_3 = 20 V, V_5 = 12 V, V_7 = 8 V. Find THD.
Squares: 20^2 = 400; 12^2 = 144; 8^2 = 64. Sum = 608. √608 = 24.66 V.
THD = 24.66/230 × 100% = **10.7%**. This is well above the few-percent levels that standards such as IEEE 519 typically allow for voltage at the point of common coupling, so mitigation would be considered.

- **Effects of harmonics:** extra heating in transformers, motors and cables (skin effect and eddy losses rise with frequency); torque pulsation in motors; malfunction of electronic controls; resonance with capacitor banks (dangerous over-currents); neutral overheating; metering errors.
- **Mitigation:** line reactors and DC-link chokes; passive harmonic filters (tuned LC); active power filters; 12- or 18-pulse rectifiers; active front-end converters; detuned capacitor banks.

Voltage sags in detail

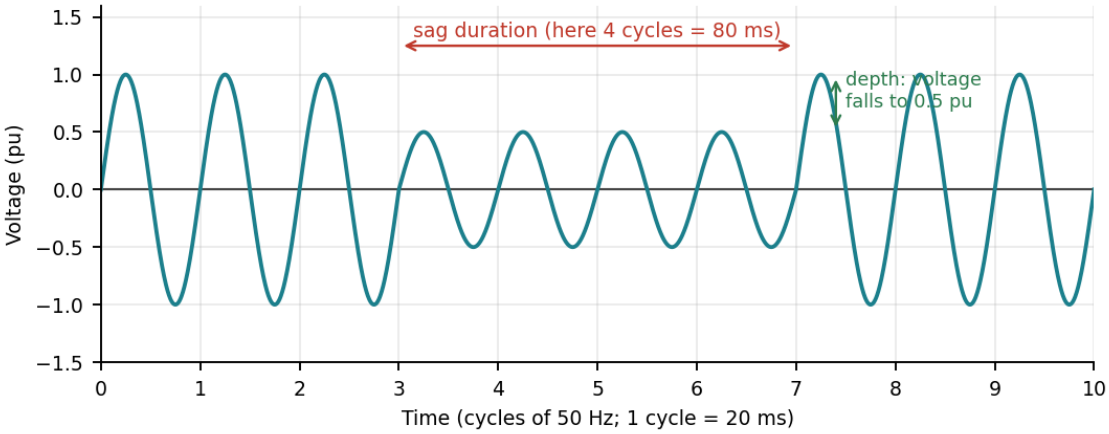


Figure 27. *A voltage sag: the amplitude falls to 0.5 per unit for four cycles, then recovers. Depth and duration characterise the sag.*

A sag is defined by its **depth** (how low the RMS voltage falls, e.g. to 50%) and **duration** (how long). The most common sources are short-circuit faults on the network (the voltage recovers when protection clears the fault) and the starting of large induction motors, whose starting current of about 5–7 times full-load current drags the supply voltage down.

- **Effects:** contactors and relays drop out; PLC and computers reset; VFDs trip on undervoltage; discharge lamps extinguish; production processes (semiconductor fabs, plastics, paper) lose batches.
- **Mitigation:** UPS (uninterruptible power supply) for control equipment; DVR (Dynamic Voltage Restorer) injecting the missing voltage; constant-voltage transformers; ride-through settings in drives; capacitor/energy-storage ride-through; soft starters or VFDs to limit motor starting current; PLC with battery backup; better system design to reduce the fault's impact.

Worked example: voltage-sag depth
A 400 V line sags to 280 V during a fault. Depth in per-unit: 280/400 = **0.70 pu** (70%), so a 30% sag. If cleared in 100 ms (5 cycles of 50 Hz), duration = 5 cycles. Many contactors drop out at about 50–70%, so this sag is borderline for them.

Unit 1, Question 6: IGBT working principle with V–I characteristics

Exam question – Unit 1 Q6
Explain the working principle of IGBT with V-I characteristics.

A–D. The problem and the idea

A MOSFET is easy to drive (voltage control, low gate power) but at high voltage its ON-state resistance becomes large, wasting power. A BJT has low ON-state drop at high voltage but needs large base current. The **IGBT (Insulated Gate Bipolar Transistor)** combines the two: MOSFET-style gate input and BJT-style current-carrying output.

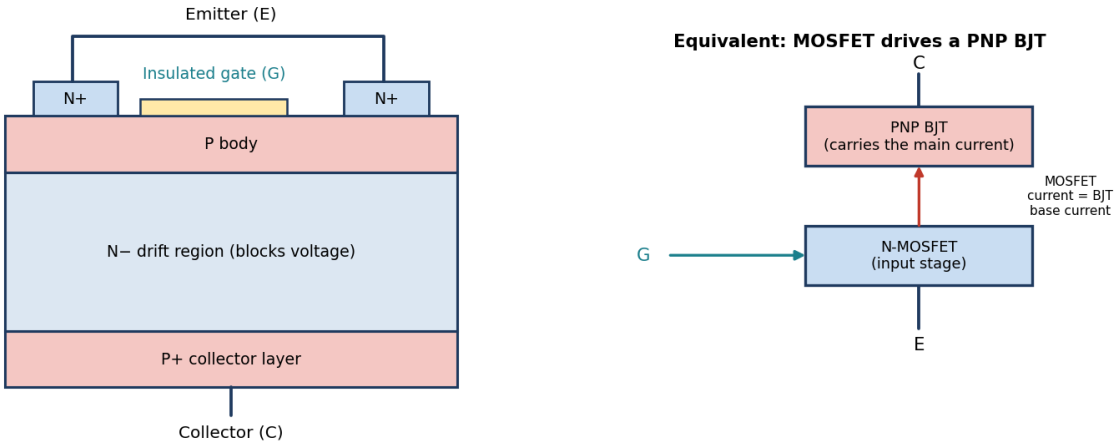


Figure 28. *IGBT: simplified cross-section (left) and equivalent circuit of a MOSFET driving a PNP transistor (right).*

Structure. It has four layers P+ collector, N− drift, P body, N+ emitter, with an insulated gate over the P body. The P+ layer at the collector is what makes it different from a MOSFET: it injects holes into the N− region, lowering resistance greatly (*conductivity modulation*).

F. Working, step by step

1. **OFF state:** V_{GE} below the threshold (about 4–6 V). No channel forms. The collector–emitter voltage is blocked across the N− drift region; only a small leakage flows.
2. **Turn-ON:** raise V_{GE} above threshold (typically +15 V in drives). A channel forms under the gate and electrons flow from the emitter, acting as base current for the internal PNP transistor.
3. **Conduction:** the P+ collector injects holes, flooding the N− region with carriers, which lowers its resistance. The device is ON with $V_{CE(sat)} \approx 1.5$–3 V even at high voltage ratings.
4. **Turn-OFF:** V_{GE} is reduced to 0 V (often −5 to −15 V to improve noise immunity). The channel disappears, electron current stops, but stored holes in the drift region take time to recombine, leaving a **tail current** that adds to turn-off loss and limits the switching frequency to roughly tens of kHz.

E. V–I characteristics

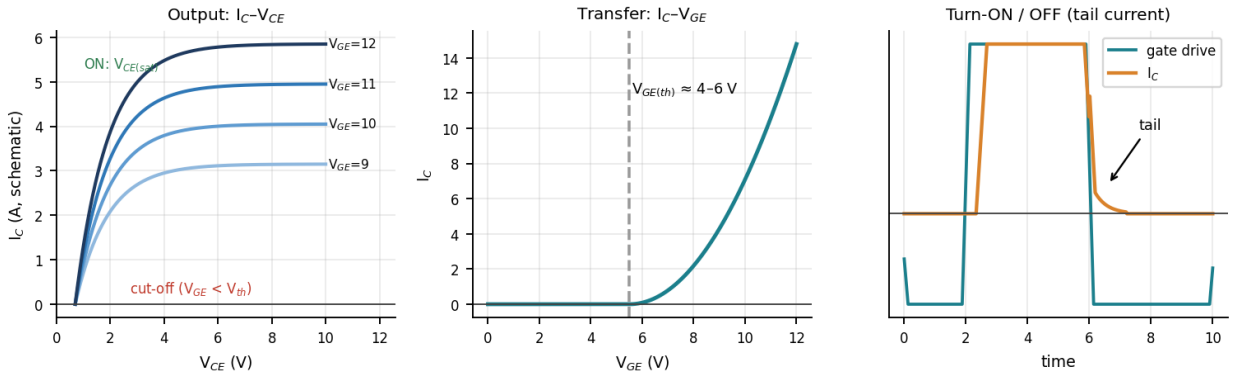


Figure 29. *IGBT characteristics (schematic): output characteristic with V_{GE} steps, transfer characteristic showing the threshold, and a switching sketch with turn-off tail.*

- **Output characteristic (I_C versus V_{CE}):** for V_{GE} below threshold the device is cut-off; for larger V_{GE} the current rises up to a level set by V_{GE} (active region); in the ON state $V_{CE(sat)}$ is small. Note the small knee voltage (~0.7 V) because of the internal PN junction.
- **Transfer characteristic (I_C versus V_{GE}):** no current until $V_{GE(th)}$, then I_C increases with V_{GE}.
- **Reverse direction:** an IGBT does not block much reverse voltage, so inverter modules include an antiparallel (free-wheeling) diode.

J. Advantages, limitations, applications

- **Advantages:** voltage-driven (low drive power), low ON-state drop at high voltage and current, rugged, short-circuit withstand, easy paralleling.
- **Limitations:** slower than MOSFETs (tail current), switching frequency up to tens of kHz, latch-up risk in older designs, only forward-voltage blocking.
- **Applications:** VFDs and servo drives, solar and wind inverters, UPS, traction drives and EV motor controllers, induction heating, welding.

Common misconception
Calling the IGBT a ‘better MOSFET’ in all respects. It beats the MOSFET at high voltage and current, but loses at high frequency and low voltage.

Unit 1, Question 7: Electromechanical relays and limit switches

Exam question – Unit 1 Q7
Explain the operation of electromechanical relays and limit switches used in automation with suitable diagram.

A–D. The problem and concept

A PLC output pin can give only a small current at 24 V; a motor needs 10 A at 230 V. A **relay** is an electrically operated switch: a small current in its **coil** moves a mechanical contact that carries the large current in a separate circuit. This gives (1) amplification of power, and (2) **electrical isolation** between control and load circuits.

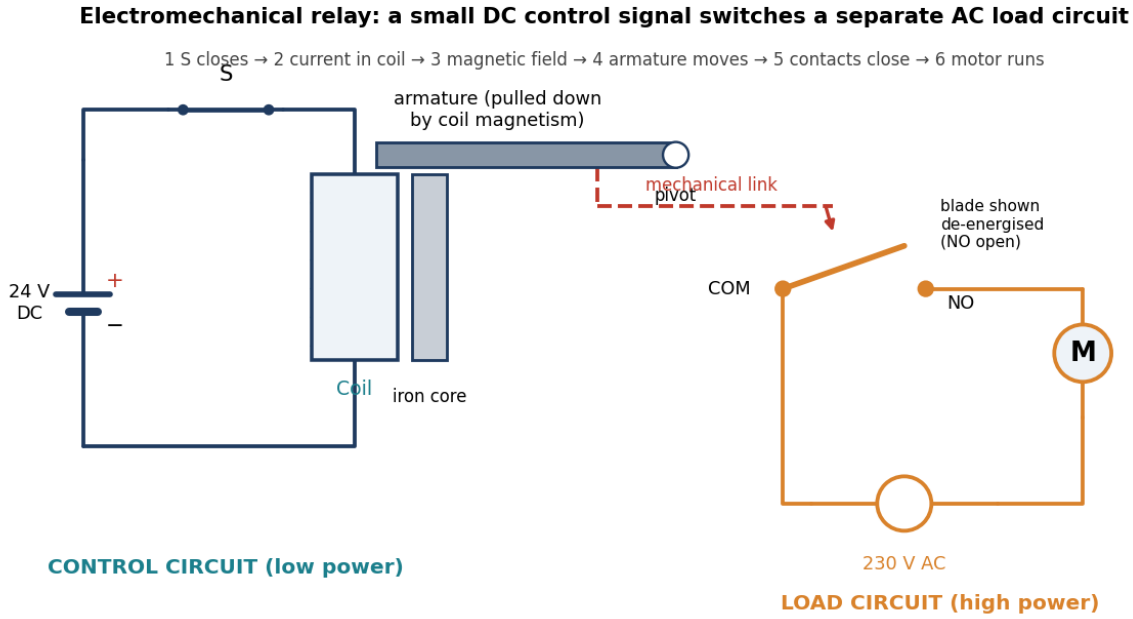


Figure 30. *Electromechanical relay: the coil circuit (left) is electrically separate from the load circuit (right).*

F. Working, step by step

1. With switch S open, no current flows in the coil. A return spring holds the armature away from the core; the **NO (Normally Open)** contact is open and the **NC (Normally Closed)** contact (if provided) is closed. The motor is OFF.
2. S closes. Current flows through the coil and creates a magnetic field in the iron core (electromagnet).
3. The magnetic force pulls the armature towards the core against the spring.
4. The armature motion moves the contact blade: NO closes, NC opens. The motor circuit is now complete; the motor runs.

5. When S opens, the coil current falls, the field collapses, and the spring returns the armature; contacts return to the normal state.

Terms. COM (common) is the moving contact's terminal. **NO** and **NC** describe the contact condition with the coil de-energised. **Coil voltage** (24 V DC, 230 V AC, etc.) and **contact rating** (e.g. 10 A at 250 V AC) are the key ratings. A **contactor** is a heavy-duty relay for motors.

Practical points. (1) The coil is inductive: when switched off it produces a voltage spike, so a **flyback (free-wheeling) diode** across a DC coil, or an RC snubber across an AC coil, protects the driving transistor and reduces EMI (Question 4). (2) Contacts arc and wear: choose a relay whose rating exceeds the load's inrush current. (3) Relays have limited mechanical life and slower switching (milliseconds) compared with solid-state relays, which use TRIACs or MOSFETs without moving parts.

Limit switches

A **limit switch** is a mechanically operated switch that detects the presence or position of an object when the object (or machine part) touches its actuator (lever, roller, plunger). It converts a mechanical position into an electrical signal (contact open/close).

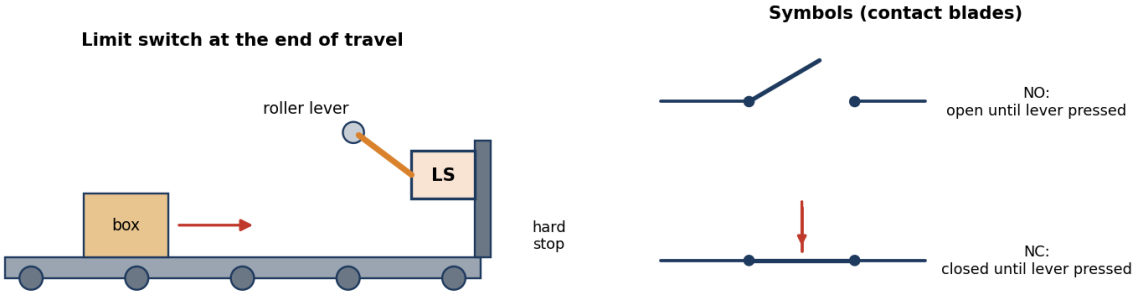


Figure 31. Limit switch at the end of a conveyor (left) and the contact states of NO and NC types (right).

- **Working:** the box moves, presses the roller lever, the lever moves an internal plunger that operates the contact: an NO limit switch closes, an NC opens. The PLC input sees this change.
- **Uses:** end-of-travel stops on conveyors, machine tools, lifts and gates; door interlocks and safety guards; counting objects; position confirmation in robots and presses.
- **Advantages:** simple, robust, cheap, high current-handling, unaffected by electrical noise. **Limitations:** physical contact causes wear, slower, needs mechanical alignment; proximity sensors (Unit 4) replace them where non-contact detection is needed.

Worked example: application scenario
Conveyor stop: a PLC output energises relay K1, whose contacts run the motor. A limit switch LS1 (NC) at the end of the conveyor is in the coil circuit: when the box presses LS1 it opens, de-energising K1 and stopping the motor. Using an NC contact is safer: a broken wire also stops the conveyor (fail-safe).

Unit 1, Question 8: Design of a capacitor-based power factor improvement circuit

Exam question – Unit 1 Q8
Design and explain a simple circuit for power factor improvement using capacitors.

Note on the source wording
The question bank gives no load data, so a unique numerical design is not possible. A clearly labelled **illustrative example** is designed below; the method is general.

A–F. Circuit and principle

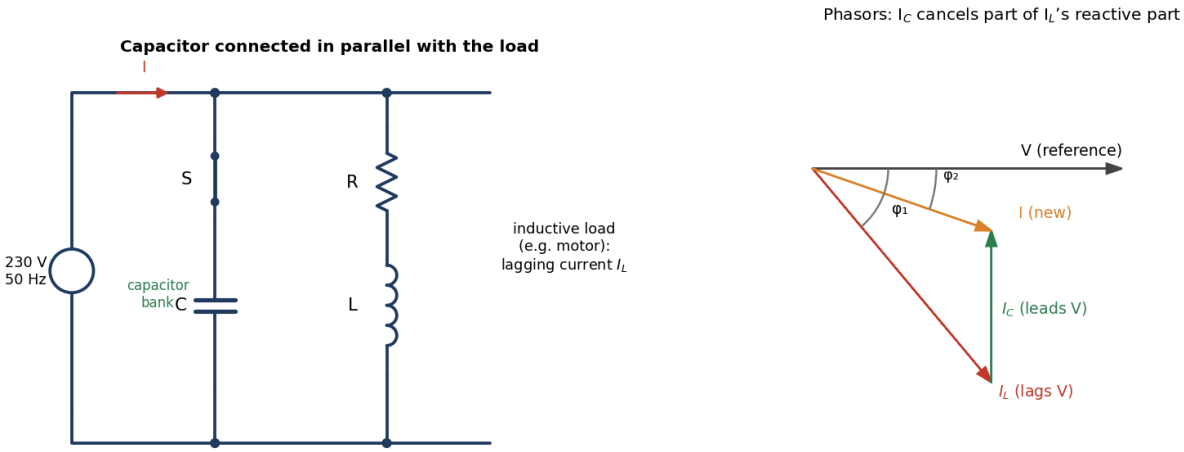


Figure 32. Left: capacitor connected in parallel with an inductive load. Right: phasor diagram; I_C cancels part of the lagging reactive current so the supply current I and angle φ_2 become smaller.

Circuit: AC source, inductive load (modelled as R in series with L), a parallel capacitor C with a switch/contactor S. **Why parallel?** The load keeps its full voltage and current unchanged; the capacitor simply supplies its own current, which leads the voltage by 90°. This leading current cancels part of the lagging reactive current of the load.

G. Design equations

Given load active power P, original PF cos φ_1, target PF cos φ_2. The load's reactive power is $Q_1 = P \tan \varphi_1$. After correction we want $Q_2 = P \tan \varphi_2$. The capacitor must supply the difference:

$$Q_C = Q_1 - Q_2 = P(\tan \varphi_1 - \tan \varphi_2)$$

A capacitor C at voltage V (RMS) and frequency f has reactive impedance $X_C = 1/(2\pi fC)$, takes current $I_C = V/X_C = 2\pi fCV$, and supplies $Q_C = V I_C$. Substituting:

$$Q_C = V I_C = V(2\pi fCV) = 2\pi fCV^2 \quad \Longrightarrow \quad C = \frac{Q_C}{2\pi fV^2}$$

Symbol	Meaning	Unit
P	load active power	W
Q_C	reactive power needed from the capacitor	VAr
V	RMS supply voltage across the capacitor	V
f	supply frequency	Hz
C	capacitance required	farad (F)

H. Fully worked illustrative example (assumed data)

Worked example: design (assumed, single-phase)
Assumptions: load 5 kW at PF 0.7 lagging, 230 V, 50 Hz; target PF 0.95.
Step 1. $\varphi_1 = \cos^{-1}(0.7) = 45.57°$, $\tan \varphi_1 = 1.0202$. $\varphi_2 = \cos^{-1}(0.95) = 18.19°$, $\tan \varphi_2 = 0.3287$.
Step 2. $Q_C = 5000 \times (1.0202 - 0.3287) = 5000 \times 0.6915 =$ **3457 VAr**.
Step 3. $C = Q_C / (2\pi fV^2) = 3457 / (314.16 \times 230^2) = 3457 / (314.16 \times 52\,900) = 3457 / 16\,619\,064 = 2.08 \times 10^{-4}$ F = **208 µF**.
Step 4 (check the benefit). Before: $I = P/(V \cos \varphi) = 5000 / (230 \times 0.7) = 31.06$ A. After: $I = 5000 / (230 \times 0.95) = 22.88$ A. Supply current falls by 26%, and I^2R loss falls to $(22.88/31.06)^2 = 0.543$, a 46% reduction.
Step 5 (units check). VAr / (rad/s · V²) = F ✔. **Practical choice:** use a standard 200–220 µF AC (power-factor) capacitor rated well above 230 V RMS (e.g. 440 V AC) with a discharge resistor.

Three-phase note. For a three-phase load, P is the total power and the formula $Q_C = P(\tan \varphi_1 - \tan \varphi_2)$ still holds for the total capacitor bank rating; per-phase capacitance in delta is $C = Q_C/(3 \cdot 2\pi f V_L^2)$ using line voltage V_L.

J–K. Practical issues and misconceptions

- **Avoid over-correction:** too much capacitance at light load makes the PF leading, can raise the voltage and cause resonance. Use APFC panels with several steps.
- **Harmonics:** capacitors form resonant circuits with system inductance; with VFDs present, use detuned (reactor-in-series) banks.
- **Safety:** capacitors store charge after switch-off, so fit discharge resistors and wait before touching; use fuses/breakers and contactors rated for capacitor inrush current.

Common misconception
Placing the capacitor in series with the load to correct PF. A series capacitor changes the voltage across the load; PF correction uses a **parallel** capacitor so that the load operation is unchanged.

Unit 1: End-of-Unit Revision Pack

Unit summary

- A power switch is a semiconductor operated as ON/OFF; losses are conduction loss and switching loss.
- BJT: current-controlled; MOSFET: voltage-controlled and fast; SCR: latching, high power; IGBT: MOSFET gate with BJT output; DIAC: bidirectional trigger; TRIAC: AC switch.
- Poor power factor (PF = P/S) raises current, loss, voltage drop, and needed kVA; corrected by parallel capacitors, APFC, synchronous condensers, SVC/STATCOM.
- SCR has four layers and three junctions; turns on by gate pulse through regenerative action, turns off when current falls below I_H; V–I curve has reverse blocking, forward blocking, break-over, conduction.
- EMI chain: source → path → victim; EMC solutions: filters, shielding, grounding, snubbers, isolation.
- Power quality problems: sags, swells, interruptions, transients, harmonics, unbalance, flicker. Harmonics come from non-linear loads; THD quantifies them. Sags are characterised by depth and duration.
- IGBT: insulated gate + PNP output, conductivity modulation, tail current; V_{GE} controls it.
- Relays isolate and amplify control signals; limit switches detect mechanical position.

Formula and equation sheet

Power triangle and PF

$$S^2 = P^2 + Q^2, \quad PF = \cos\varphi = \frac{P}{S}$$

Line loss versus PF

$$P_{loss} \propto \frac{1}{\cos^2\varphi}$$

SCR half-wave average (resistive)

$$V_{avg} = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

THD

$$THD = \frac{\sqrt{\sum_{n\geq 2} V_n^2}}{V_1} \times 100\%$$

Capacitor reactive power

$$Q_C = P(\tan\varphi_1 - \tan\varphi_2)$$

Capacitance for PF correction

$$C = \frac{Q_C}{2\pi f V^2}$$

Diagram-revision sheet

Practise drawing each of these from memory, labelling every component, then compare with the figure.

Figure	What to draw and label
Fig. 20	Q1: symbols of six devices and their V–I curves
see Q3	Q3: SCR structure, two-transistor analogy, V–I curve, half-wave circuit with waveforms
see Q2	Q2: power triangle; Q8: capacitor in parallel with load + phasors
see Q4	Q4: EMI chain (source–path–victim)
see Q5	Q5: harmonic waveform; voltage sag waveform
see Q6	Q6: IGBT cross-section and equivalent; characteristics
see Q7	Q7: relay and limit-switch diagrams

Glossary of important terms

Term	Meaning
Power switch	Semiconductor operated ON/OFF to control power
Conduction loss	Loss $V_{ON} \times I$ while the switch is ON
Switching loss	Loss during turn-on/off transitions
SCR / thyristor	Four-layer latching switch triggered by gate
Holding current	Minimum anode current to keep SCR ON
Latching current	Minimum anode current at end of gate pulse to keep SCR on
Firing angle α	Delay of gate pulse after zero crossing
Natural commutation	SCR turns off when AC current falls to zero
IGBT	Insulated-gate bipolar transistor
Power factor	P/S = cos φ
Reactive power	Q = V I sin φ, in VAr
APFC	Automatic power factor correction panel
EMI / EMC	Electromagnetic interference / compatibility
Harmonic	Component at an integer multiple of fundamental frequency
THD	Total harmonic distortion
Voltage sag	Short fall of RMS voltage to 10–90%
Relay	Electrically operated mechanical switch
Limit switch	Mechanically actuated position switch

Conceptual and numerical review questions

1. Calculate PF correction capacitance for a 10 kW single-phase load at 0.6 lagging, corrected to 0.9, at 230 V, 50 Hz.
2. An SCR is fired at α = 90° from a 325 V peak supply (half-wave, resistive). Find V_{avg}.
3. Why can’t the gate switch an SCR off?
4. A signal cable picks up noise from a nearby VFD cable. Name three fixes.
5. Voltage 230 V, harmonics 3rd = 15 V, 5th = 10 V. Find THD.
6. Why does an IGBT have lower ON-state drop than a high-voltage MOSFET?

Fully explained answers
1. $\varphi_1 = \cos^{-1}0.6 = 53.13°$, tan = 1.3333. $\varphi_2 = \cos^{-1}0.9 = 25.84°$, tan = 0.4843. $Q_C = 10\,000 \times (1.3333 - 0.4843) = 8490$ VAr. $C = 8490 / (314.16 \times 52\,900) = 8490 / 16\,619\,064 = 5.11 \times 10^{-4}$ F ≈ **511 µF**.
2. $V_{avg} = (325/2\pi)(1 + \cos 90°) = 51.7 \times 1 =$ **51.7 V**.
3. After regeneration the two internal transistors keep each other on without gate current; only a fall of anode current below I_H breaks the feedback.
4. Shielded twisted pair with proper grounding, physical separation/routing at 90°, filter/reactor at the VFD output.
5. √(225 + 100) = √325 = 18.03; 18.03/230 = **7.8%**.
6. Conductivity modulation: injected holes from the P+ collector flood the drift region and lower its resistance.

Examination-answer guide for the original questions

This guide complements the teaching chapters. It shows how to organise each original question as a complete exam answer.

Question	Structure of a strong answer
Q1	Definition (2–3 lines) → classification (uncontrolled / semi-controlled / fully controlled) → symbols (all six) → short working of each → V–I curves → **comparison table** with ≥ 7 parameters (control, rating, frequency, ON drop, drive, turn-off, application) → selection note.
Q2	Define PF and power triangle → derive $I = P/(V \cos\varphi)$ and loss ∝ $1/\cos^2\varphi$ → list effects → list correction methods (capacitors, APFC, synchronous condenser, SVC/STATCOM) → one worked number → conclusion.
Q3	Symbol and four-layer structure → two-transistor analogy → states: reverse blocking, forward blocking, conduction → V–I curve with V_{BO}, I_H, I_L, V_{BR} → turn-on/turn-off methods → applications.
Q4	Define EMI, EMC → source–path–victim diagram → coupling types → mitigation table → two examples (VFD, relay coil with diode).
Q5	List issues in a table → harmonics: cause, waveform, THD formula, effects, mitigation → voltage sag: definition, cause, waveform, effects, mitigation → conclusion.
Q6	Symbol → structure and equivalent circuit → working (off, on, conduction, off) → output and transfer characteristics → advantages, limitations, applications.
Q7	Relay diagram with coil, armature, NO/NC → step-wise working → ratings, protection diode → limit switch diagram → applications.
Q8	Circuit diagram + phasor/power triangle → state assumptions → formulas $Q_C = P(\tan\varphi_1 - \tan\varphi_2)$, $C = Q_C/(2\pi f V^2)$ → numerical example with steps → practical notes (APFC, discharge resistor).

UNIT 2: Power Converters, Inverters and Rectifiers

Unit 2 is the heart of power electronics. Ten questions follow, in your question bank's order. Before starting, learn this vocabulary, which organises everything in the unit:

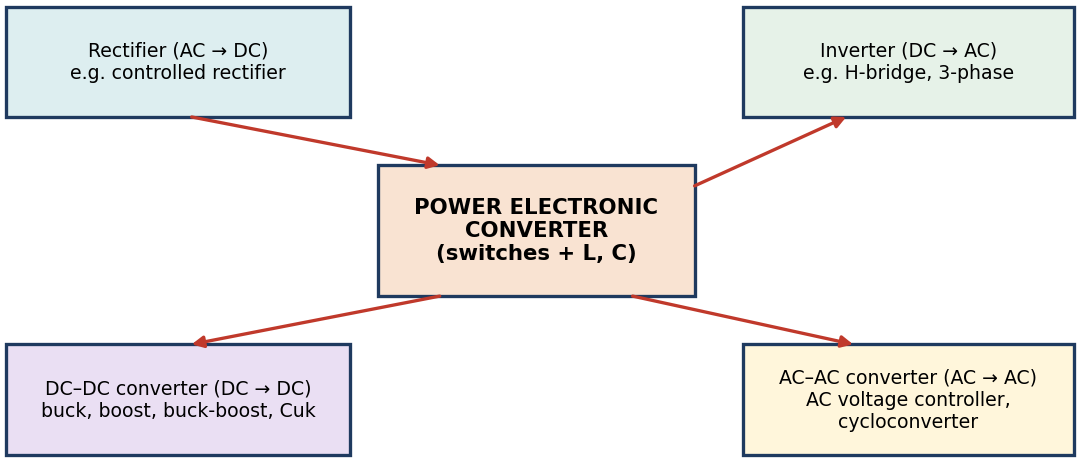


Figure 33. *The four basic kinds of power conversion. The word* ***converter*** *is the general name; 'rectifier', 'inverter', 'chopper/DC–DC converter' and 'AC controller/cycloconverter' are specific kinds.*

Common method. For every circuit we state the *input quantity*, the *switching arrangement*, the *output quantity*, the *load assumption* and the *purpose*. For every circuit we then (1) draw it, (2) trace the current in each switching state, (3) draw the waveforms, (4) derive the output formula, (5) check with a number. Switches are assumed **ideal** unless stated (zero drop, instantaneous), and real losses are discussed at the end.

Unit 2, Question 1: Inverter and the single-phase H-bridge inverter

> **Exam question – Unit 2 Q1**
> What is an inverter? Explain single-phase H-bridge inverter working principle with neat diagram and waveforms.

A. The problem

Batteries, solar panels and the DC link of a drive give DC, but AC motors, mains appliances and the grid need AC. A **solar inverter** turns panel DC into grid AC; a **UPS** turns battery DC into mains-like AC; a **VFD** creates AC of variable frequency to set motor speed.

B–D. Definition

Inverter: a power electronic circuit that converts **DC input to AC output** of controllable magnitude and frequency, using switches that connect the load to the DC source with alternating polarity. Input: DC voltage V_{dc}. Output: AC voltage v_o (square, stepped or PWM) at frequency f. Intuition: if you could flip a battery around the load over and over, the load would see alternating polarity; the H-bridge is a set of four electronic switches that does the flipping.

E–F. Circuit and working

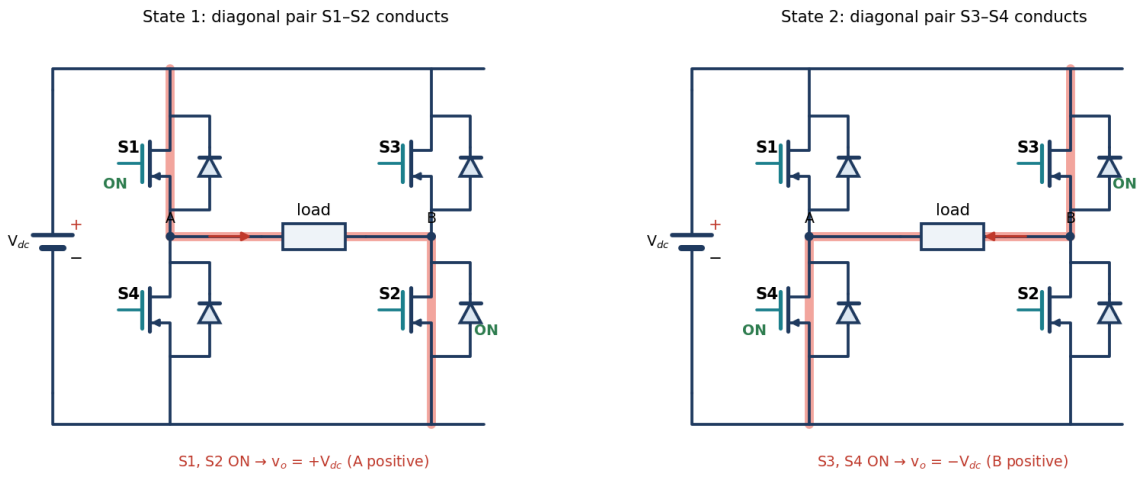


Figure 34. *Single-phase full-bridge (H-bridge) inverter. Each switch (S1–S4: IGBT or MOSFET) has an anti-parallel diode. The red path shows the current in each state.*

Components: DC source V_{dc}, four controlled switches S1–S4 (two legs: S1/S4 on the left leg, S3/S2 on the right leg), four feedback (anti-parallel) diodes, load between mid-points A and B. **Operating rule:** switches on the same leg (S1 & S4, or S3 & S2) must **never** be ON together, because that is a short circuit across V_{dc} (called *shoot-through*). A short **dead time** in which both are OFF is inserted between turning one OFF and its partner ON.

1. **0 to T/2:** S1 and S2 ON. Current flows $V_{dc}(+) \rightarrow$ S1 → A → load → B → S2 → $V_{dc}(-)$. Load voltage $v_o = +V_{dc}$.
2. **T/2 to T:** S3 and S4 ON. Current flows $V_{dc}(+) \rightarrow$ S3 → B → load → A → S4 → $V_{dc}(-)$. The load current direction is reversed, so $v_o = -V_{dc}$.
3. Repeating this every period T gives a **square-wave AC** of frequency f = 1/T. Changing the switching rate changes f; this is how a VFD sets the speed.

With an inductive load (R–L). The load current cannot reverse instantly. Just after S1,S2 turn OFF (and S3,S4 turn ON), the old current continues to flow, now backwards through the **feedback diodes** D3 and D4 (or D1/D2 in the other half), returning energy to the source. After the current crosses zero, S3,S4 conduct. This is why every switch has an anti-parallel diode (Figure 35 below).

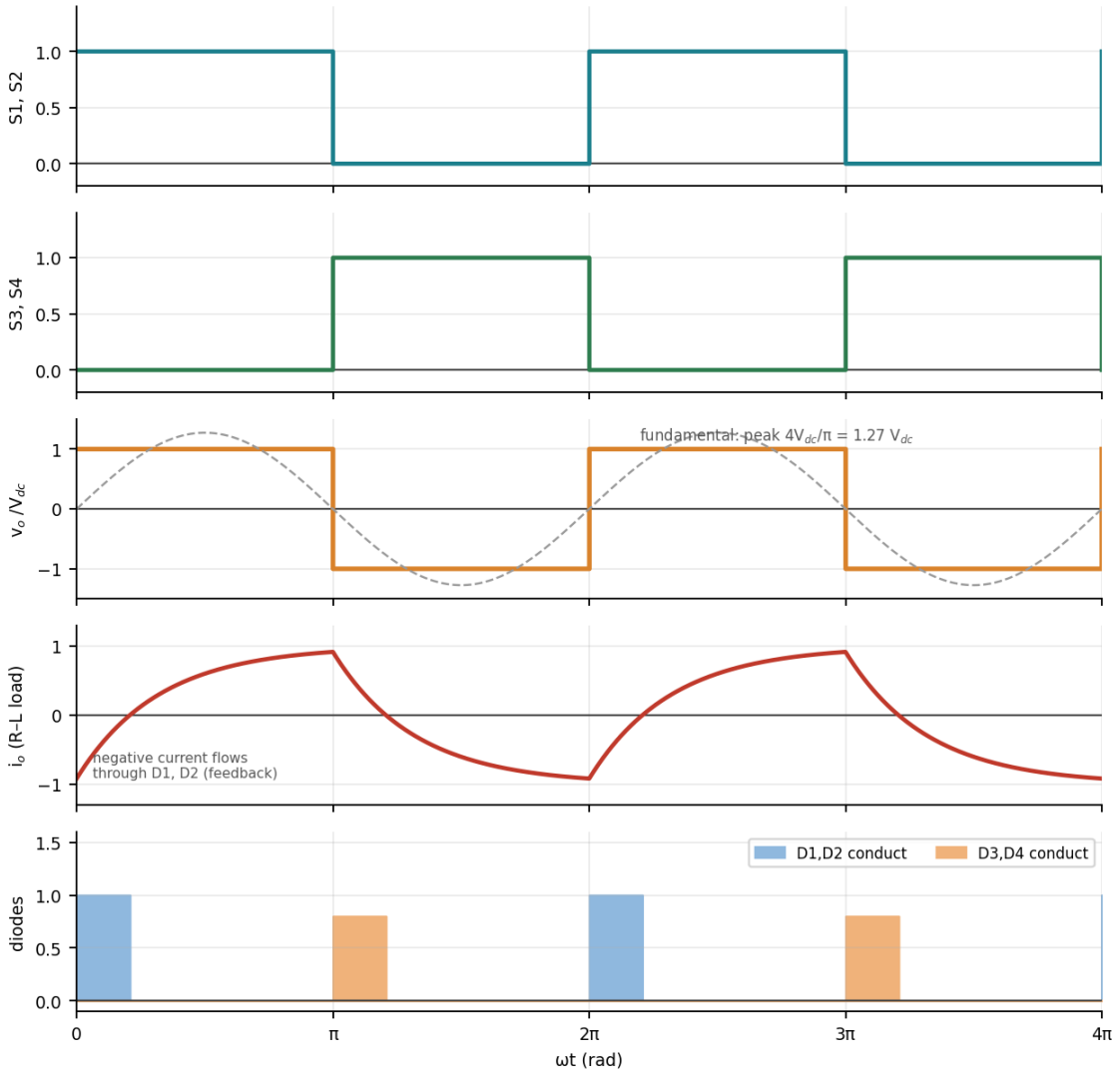


Figure 35. *Waveforms of the H-bridge: gate signals, output voltage (square wave; dashed = its fundamental), load current for an R–L load, and intervals in which the feedback diodes conduct.*

G. Equations

$$v_o(t) = \sum_{n=1,3,5,\ \ldots} \frac{4V_{dc}}{n\pi}\sin(n\omega t) \qquad V_{o,rms} = V_{dc} \qquad V_{1,rms} = \frac{4V_{dc}}{\sqrt{2}\,\pi} = 0.9\,V_{dc}$$

Here V_{dc} = supply (V), ω = 2πf, n = harmonic order (only odd). The square wave's RMS is simply V_{dc}, because $|v_o| = V_{dc}$ all the time. The fundamental is its 50 Hz sine part; the other terms are harmonics (see Unit 1 Q5). Distortion: THD = $\sqrt{[(V_{o,rms}/V_{1,rms})^2 - 1]} = \sqrt{[(1/0.9)^2 - 1]}$ = **48.3%**, which is why PWM (below) is used when a clean sine is needed.

Worked example: H-bridge inverter
V_{dc} = 48 V, R = 10 Ω load, f = 50 Hz.
$V_{o,rms}$ = 48 V; I_{rms} = 48/10 = 4.8 A; P = V^2/R = $48^2/10$ = **230.4 W**.
Fundamental: $V_{1,peak}$ = 4×48/π = 61.1 V, $V_{1,rms}$ = 43.2 V. Power in the fundamental = $43.2^2/10$ = 186.6 W; the remainder (≈ 44 W) is harmonic power, which only heats the load.
Switching: each switch is ON for T/2 = 10 ms, so two diagonal pairs alternate at 50 Hz.

Voltage and frequency control. The frequency is set by the switching rate. The output voltage can be controlled by **PWM (pulse-width modulation)**: each half-cycle is chopped into many pulses whose widths follow a sine pattern (*sinusoidal PWM*), producing a near-sinusoidal current with small harmonics at high frequency that are easily filtered. **Practical:** IGBTs for medium/high power, MOSFETs at low voltage; gate-driver isolation; dead time; snubbers; filter at the output for a pure sine (UPS, grid

inverters).

I–K. Applications and misconceptions

- Uses: UPS, solar and wind inverters, VFDs for motors, induction heating, electric vehicle drives, HVDC (conversion stations).
- Advantages: compact, efficient, controllable V and f. Limitations: square output has harmonics; needs filters and careful dead-time and protection.

Common misconception
'The inverter creates energy.' No: it only changes the form of power from DC to AC; output power ≤ input power. 'Anti-parallel diodes are optional.' With R–L loads they are essential: without them the inductive current has no path and the switches are destroyed by voltage spikes.

Unit 2, Question 2: AC voltage controllers and voltage regulation

Exam question – Unit 2 Q2
What is the need of AC voltage controllers? Explain the principle of voltage regulation in industrial systems.

A–D. The problem and the idea

A resistance heater, a lamp bank or an induction-motor fan sometimes needs *less* than full mains voltage. A series resistor wastes power as heat; a transformer with taps changes voltage only in steps and is bulky. An **AC voltage controller (AC regulator)** is a power electronic circuit that converts fixed AC to **variable-RMS AC at the same frequency**, using thyristors (or a TRIAC) to *connect the supply to the load for only part of each cycle*.

- **Needs:** heater/furnace temperature control, lighting dimmers, fan and pump speed (single-phase induction motors), soft start of induction motors (reducing starting current), transformer tap-changing, and voltage regulation in industrial supplies.
- **Compared with the alternatives:** efficient (no resistor loss), smooth, fast, compact, no moving parts.

E–F. Phase-angle control, step by step

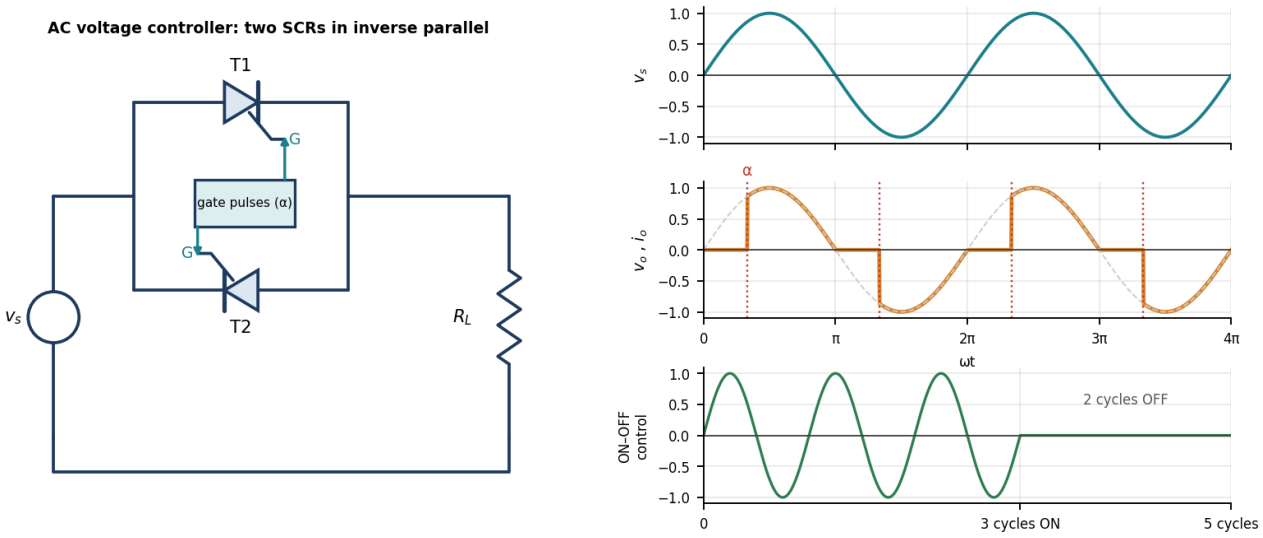


Figure 36. *Single-phase AC voltage controller with resistive load: two SCRs T1, T2 in inverse parallel (or one TRIAC). Right: phase-angle waveforms (middle) and integral-cycle (ON–OFF) control (bottom).*

1. Supply $v_s = V_m \sin \omega t$. In the positive half-cycle T1 is forward biased, but stays OFF until a gate pulse arrives at the **firing angle α**.

2. From α to π, T1 conducts and the load sees the source voltage: $v_o = v_s$. At π, the current falls to zero and T1 turns OFF by natural commutation.
3. In the negative half-cycle T2 is fired at π + α and conducts to 2π.
4. Increasing α shortens the conduction time, so the RMS output falls. α = 0 gives full voltage; α = 180° gives zero.

G. Derivation of the RMS output voltage (resistive load)

RMS is the square root of the mean of the square (Chapter C). Because the output pattern repeats every half-cycle (π), we may average over π:

$$V_{o,rms}^2 = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2\theta \, d\theta$$

Use $\sin^2\theta = (1 - \cos 2\theta)/2$ (Chapter E):

$$V_{o,rms}^2 = \frac{V_m^2}{2\pi}\int_{\alpha}^{\pi}(1-\cos 2\theta)\, d\theta = \frac{V_m^2}{2\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_{\alpha}^{\pi} = \frac{V_m^2}{2\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)$$

(the term sin 2π/2 is 0, and subtracting the lower limit gives −α + sin 2α/2.) With $V_m = \sqrt{2}\, V_s$ (V_s = supply RMS) so that $V_m^2 = 2V_s^2$:

$$V_{o,rms} = V_s\sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

Checks: α = 0 ⇒ √(π/π) = 1 ⇒ $V_o = V_s$ ✔. α = π ⇒ √0 = 0 ✔. Power factor of the input for a resistive load = $V_{o,rms}/V_s$.

Worked example: phase control
V_s = 230 V, R = 20 Ω, α = 90° (π/2 rad).
π − α + sin 2α/2 = π − π/2 + sin 180°/2 = 1.5708 + 0 = 1.5708. Divide by π: 0.5. √0.5 = 0.7071.
$V_{o,rms}$ = 230 × 0.7071 = **162.6 V**; I_{rms} = 162.6/20 = 8.13 A; P = V²/R = 162.6²/20 = **1322 W**; input PF = V_o/V_s = **0.707**.
Another point: α = 60°: π − 1.0472 + sin120°/2 = 2.0944 + 0.4330 = 2.5274; /π = 0.8045; √ = 0.897; V_o = **206.3 V**.

Integral-cycle (ON–OFF) control

Instead of cutting each cycle, switch whole cycles: the load receives *n* complete cycles then none for *m* cycles. Switching at the zero crossings produces little EMI.

$$V_{o,rms} = V_s\sqrt{\frac{n}{n+m}}$$

Example: n = 3, m = 2 ⇒ V_o = 230 × √0.6 = **178.2 V**. It suits loads with a large thermal time constant (furnaces) but causes flicker or torque pulsation in lamps and motors; it also creates sub-harmonics.

Principle of voltage regulation in industrial systems

Voltage regulation means keeping the voltage at the load close to its set value despite changes in supply or load. In a closed loop, a sensor measures the load voltage; the controller compares it with the reference and adjusts the firing angle (Unit 3 feedback). If the supply voltage falls, α is reduced so the load still gets the correct RMS value. Other regulation methods: tap-changing transformers, servo stabilisers, constant-voltage transformers, and the DC-link/inverter regulation of VFDs.

- **Limitations of phase control:** distorted current (harmonics, EMI), lagging PF at large α, and for R–L loads the control range is φ < α < π (see Question 6).
- **TRIAC dimmers**: a TRIAC with a DIAC-triggered RC network implements the same idea for household lamps.

Common misconception
'The output frequency changes with α.' No: the frequency remains 50 Hz; only the RMS value changes. Frequency changing is the job of cycloconverters (Question 10) and VFD inverters.

Unit 2, Question 3: Buck converter and boost converter

Exam question – Unit 2 Q3
Describe buck converter & boost converter (separately) with circuit diagram and waveform.

A–D. DC–DC conversion, the shared idea

A **DC–DC converter (chopper)** changes one DC voltage into another. The method is the one from Chapter D: switch the supply ON and OFF quickly with duty cycle $D = t_{ON}/T$, and use an **inductor** and a **capacitor** to smooth the result. The inductor stores energy as magnetic field while current flows, and releases it when the switch turns off; the capacitor holds the output voltage steady. Its key property: an inductor cannot change its current instantly, and in steady state the **average inductor voltage over one period is zero** (volt-second balance). All formulas below follow from that single fact. Assumptions: ideal switch and diode, steady state, **continuous conduction mode (CCM)** meaning the inductor current never falls to zero.

(a) Buck (step-down) converter

Input: V_{in} (DC). **Output:** V_o lower than V_{in}, same polarity. **Arrangement:** series switch S, then freewheeling diode D from ground, then the series inductor L, shunt capacitor C and load R.

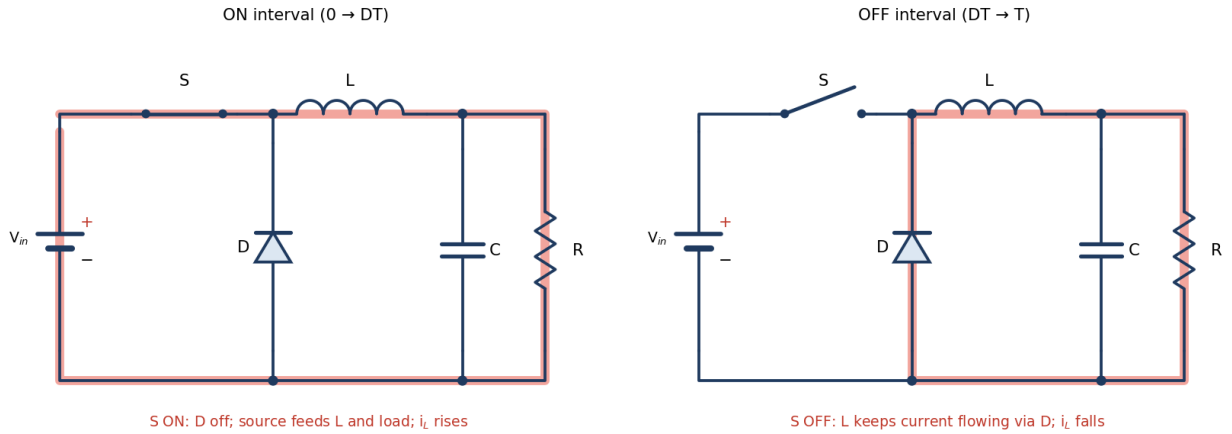


Figure 37. Buck converter in its two states. Red = current path. In the OFF interval the inductor current continues through the diode.

1. **ON (0 → DT):** S closed, D reverse biased. Voltage across L = $V_{in} - V_o > 0$, so i_L rises linearly. Energy goes to the load and to L.
2. **OFF (DT → T):** S open; L opposes the fall of current, its voltage reverses, D becomes forward biased and carries i_L. Voltage across L = $-V_o$, so i_L falls linearly.
3. C smooths the pulsating inductor current, so v_o has only a small ripple.

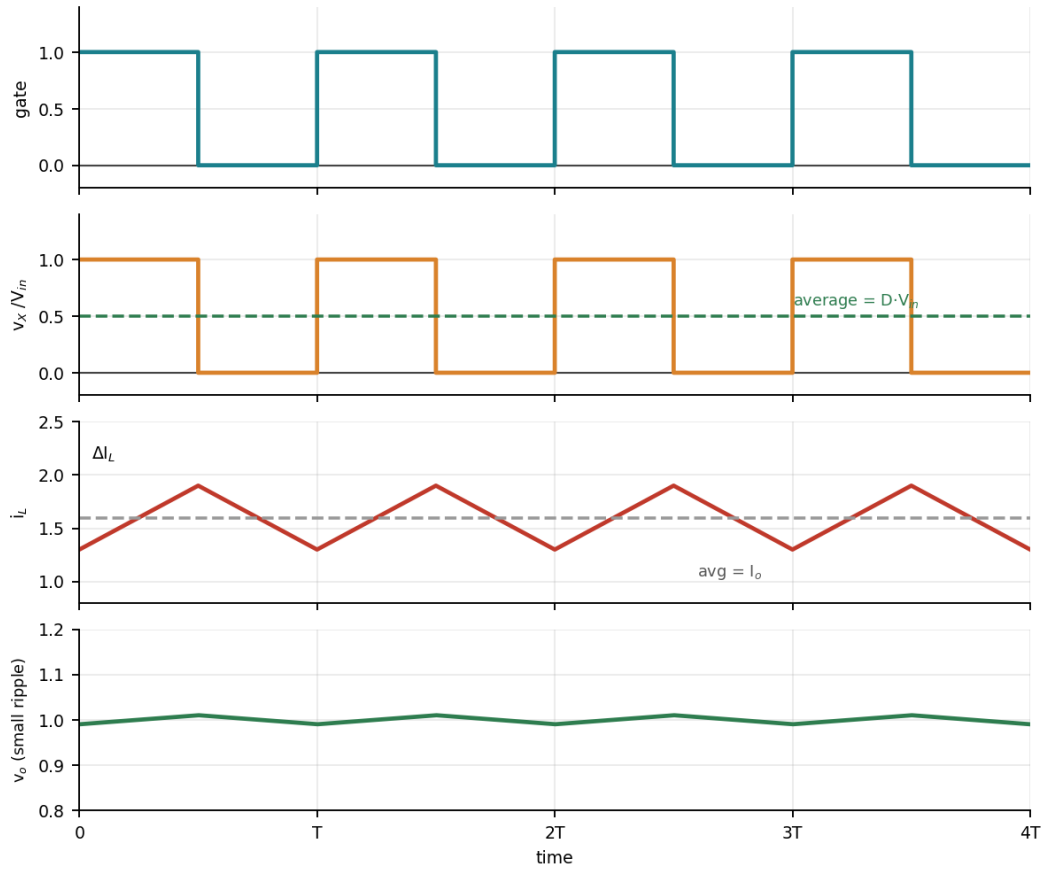


Figure 38. *Buck converter waveforms for D = 0.5 (schematic ripple exaggerated): gate signal, switch-node voltage v_X, inductor current i_L and output voltage.*

Derivation of the voltage ratio. Average inductor voltage = 0 over one period:

$$(V_{in} - V_o)\,DT - V_o\,(1-D)T = 0 \;\Rightarrow\; V_{in}D - V_oD - V_o + V_oD = 0 \;\Rightarrow\; V_o = D\,V_{in}$$

$$\Delta I_L = \frac{V_o(1-D)}{fL} \qquad \Delta V_o = \frac{\Delta I_L}{8fC} \qquad L_{crit} = \frac{(1-D)R}{2f}$$

Symbols: f = 1/T switching frequency (Hz); ΔI_L = peak-to-peak inductor ripple current (A); ΔV_o = output ripple (V); L_{crit} = boundary of CCM. ΔI_L comes from the OFF interval: the inductor current falls by $(V_o/L) \times (1-D)T$.

Worked example: buck converter design
Requirements: V_{in} = 24 V, V_o = 12 V, I_o = 2 A (R = 6 Ω), f = 20 kHz, ΔI_L = 20% of 2 A = 0.4 A, ΔV_o = 50 mV.
D = V_o/V_{in} = 12/24 = **0.5**.
L = $V_o(1-D)/(f\,\Delta I_L)$ = 12 × 0.5 / (20 000 × 0.4) = 6/8000 = 0.75 mH → **750 µH**.
C = $\Delta I_L/(8\,f\,\Delta V_o)$ = 0.4 / (8 × 20 000 × 0.05) = 0.4/8000 = 50 µF → **50 µF** (choose ≥ 100 µF low-ESR in practice).
Check CCM: L_{crit} = (1−0.5) × 6 / (2 × 20 000) = 3/40 000 = 75 µH; since 750 µH > 75 µH ✔.

(b) Boost (step-up) converter

Input: V_{in} (DC). **Output:** V_o *higher* than V_{in}, same polarity. **Arrangement:** inductor L in series with the source, switch S to ground after L, diode D in series to the output, capacitor C and load R.

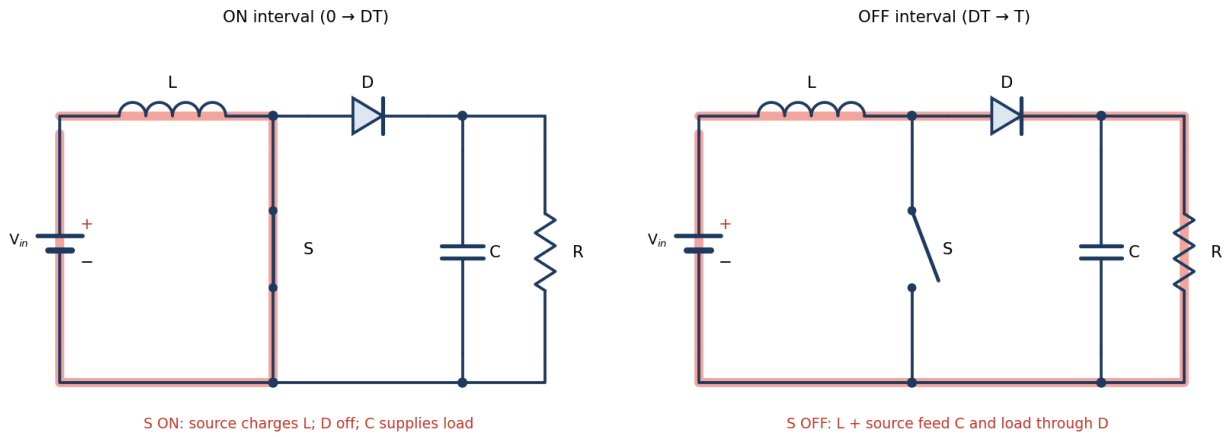


Figure 39. *Boost converter in its two states.*

1. **ON (0 → DT):** S closed. The source is directly across L, so i_L rises (voltage across L = V_{in}) and energy is stored in L. D is reverse biased (the output is higher than the switch node) so C alone supplies the load.
2. **OFF (DT → T):** S open. L forces its current to continue, so its voltage *adds* to the source: the node X rises above V_{in}, D conducts, and L + source feed C and the load. Voltage across L = $V_{in} - V_o$ (negative), so i_L falls.

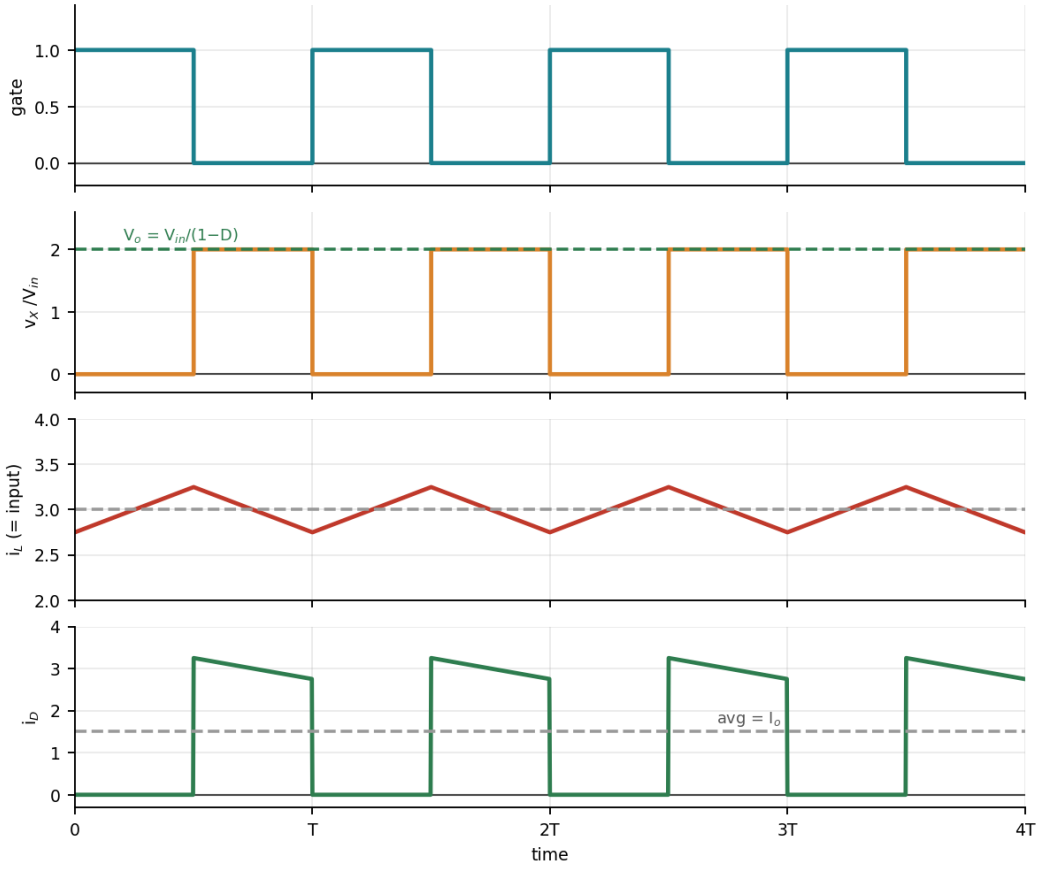


Figure 40. *Boost converter waveforms for D = 0.5: gate, switch-node voltage (0 while ON, V_o while OFF), inductor (input) current, and diode current.*

$$V_{in}\,DT + (V_{in} - V_o)(1-D)T = 0 \;\Rightarrow\; V_{in} = V_o(1-D) \;\Rightarrow\; V_o = \frac{V_{in}}{1-D}$$

$$\Delta I_L = \frac{V_{in}D}{fL} \qquad \Delta V_o = \frac{I_o D}{fC} \qquad I_{in} = \frac{I_o}{1-D} \qquad L_{crit} = \frac{D(1-D)^2 R}{2f}$$

Worked example: boost converter design
Requirements: V_{in} = 12 V to V_o = 24 V, R = 24 Ω (I_o = 1 A), f = 25 kHz, ΔI_L = 0.4 A, ΔV_o = 0.1 V.
$D = 1 - V_{in}/V_o = 1 - 12/24 =$ **0.5**. Input current (ideal) $I_{in} = I_o/(1-D)$ = 2 A (power: 12 × 2 = 24 W = 24 × 1 ✔).
$L = V_{in}D/(f\Delta I_L) = 12 \times 0.5/(25\,000 \times 0.4) = 6/10\,000 = 0.6$ mH → **600 µH**.
$C = I_oD/(f\Delta V_o) = 1 \times 0.5/(25\,000 \times 0.1) = 0.5/2500 = 200$ µF → **200 µF**.
$L_{crit} = 0.5 \times 0.25 \times 24/(2 \times 25\,000) = 3/50\,000 = 60$ µH ⇒ CCM ✔.

	Buck	Boost
Function	step-down	step-up
V_o/V_{in}	D (0 to 1)	1/(1−D) (≥ 1)
Inductor position	in series with the output	in series with the input
Input current	pulsating	continuous (smooth)
Output current	continuous	pulsating (from the diode)
Applications	point-of-load regulators, battery chargers, LED drivers, 24 V → 5 V	PV panels (MPPT), battery-powered systems, power-factor correction, EV auxiliary supplies

Common misconception
'A boost converter gives more power.' No. Power is conserved: voltage up means current down (I_o = (1−D) I_{in}). In practice, as D approaches 1 the real gain falls because of losses.

Unit 2, Question 4: Single-phase fully controlled full-wave bridge rectifier (resistive load)

Exam question – Unit 2 Q4
Explain the working of single-phase fully controlled full wave bridge rectifier with resistive load controlled with waveforms.

Note on the source wording
'resistive load controlled with waveforms' is awkward English. The intended task: explain the controlled bridge rectifier with a resistive load, with waveforms drawn for a chosen firing angle α.

A-D. Purpose and definition

A **rectifier** converts AC to DC. With diodes, the DC output is fixed. If we replace the diodes by SCRs and delay their firing, we get a **controlled rectifier** with adjustable DC output, used for DC-motor drives, battery charging and electrolysis. 'Fully controlled' means **all four** bridge devices are SCRs (a 'half-controlled' bridge mixes two SCRs and two diodes).

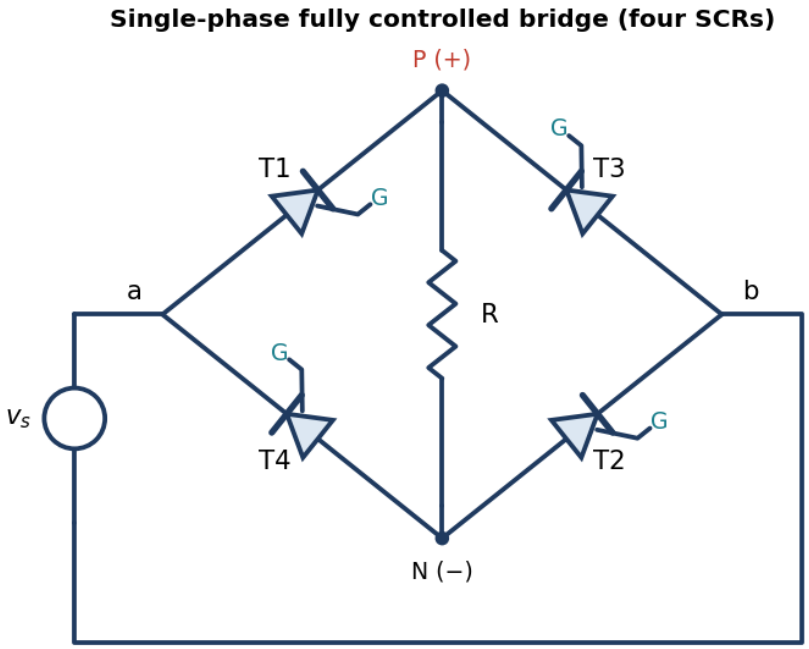


Figure 41. *Single-phase fully controlled bridge: SCRs T1–T4 between the AC source and a resistive load R. Current direction: T1, T2 conduct in the positive half-cycle; T3, T4 conduct in the negative half-cycle.*

F. Working, step by step

1. **Positive half-cycle** (terminal a is positive): T1 and T2 are forward biased but OFF. A gate pulse at ωt = α triggers both.
2. Current path: a → T1 → P → load R → N → T2 → b. The load sees $v_o = v_s$ (positive).
3. At ωt = π the supply voltage and current (resistive load) reach zero, T1, T2 turn OFF naturally. From π to π + α no device conducts: $v_o = 0$.
4. **Negative half-cycle:** at ωt = π + α, T3 and T4 are triggered. Path: b → T3 → P → R → N → T4 → a. The load voltage is again *positive* ($v_o = -v_s$), because the bridge reverses the connection.
5. Changing α changes the average output; the load current $i_o = v_o/R$ has the same shape as v_o.

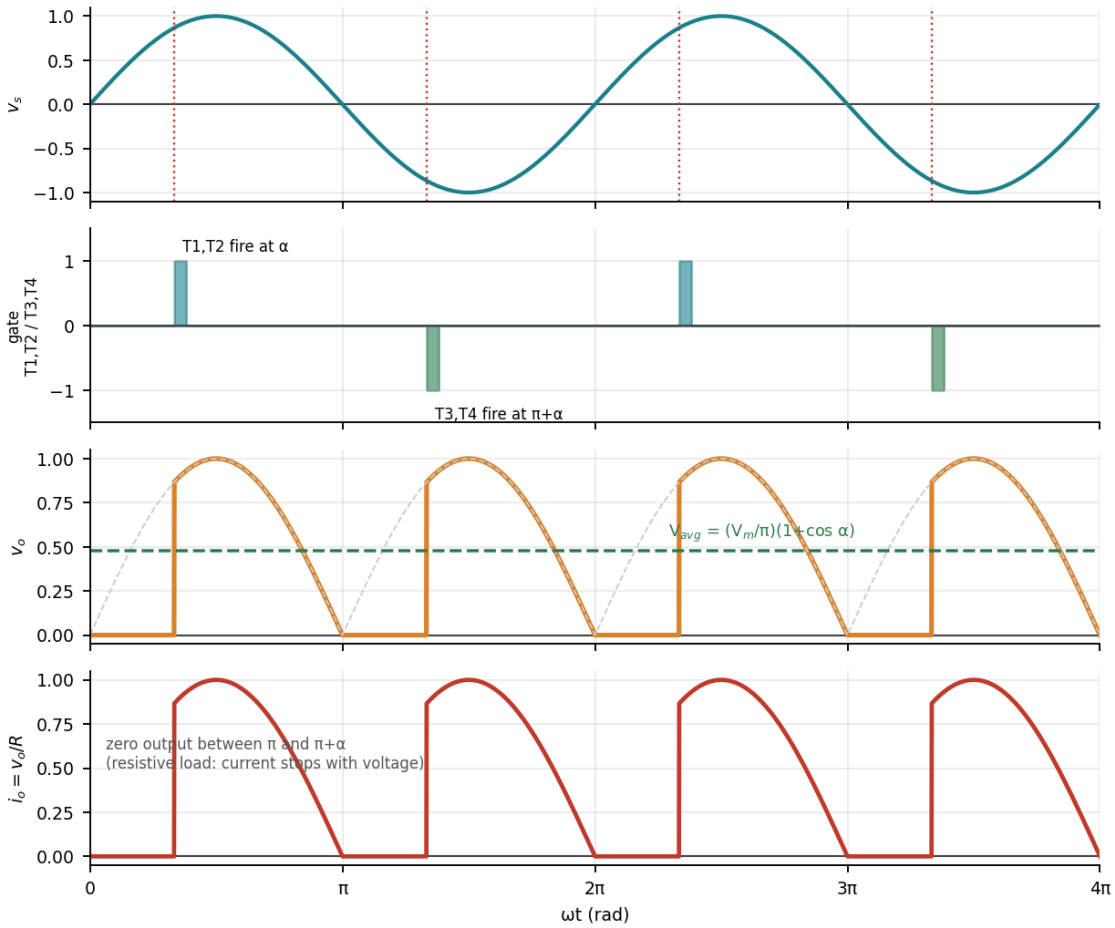


Figure 42. *Waveforms for α = 60°: supply, gate pulses, load voltage (shaded region between α and π, then π+α to 2π), and load current.*

G. Derivation

Because the pattern repeats every π, average over π:

$$V_{avg} = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin\theta\, d\theta = \frac{V_m}{\pi}[-\cos\theta]_{\alpha}^{\pi} = \frac{V_m}{\pi}(1+\cos\alpha)$$

$$V_{rms} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2\theta\, d\theta} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

(The second uses the same steps as in Question 2.) Checks: α = 0 ⇒ $V_{avg} = 2V_m/\pi = 0.637\ V_m$ (an uncontrolled full-wave bridge, Chapter E) and $V_{rms} = V_m/\sqrt{2}$ ✔. α = 180° ⇒ $V_{avg} = 0$ ✔. For a resistive load, the output cannot be negative, so the useful range is 0 < α < 180°.

Worked example: controlled bridge
V_s = 230 V RMS ⇒ V_m = 325.3 V; R = 10 Ω; α = 60°.
V_{avg} = (325.3/π)(1 + cos 60°) = 103.5 × 1.5 = **155.3 V**; I_{avg} = 15.5 A.
V_{rms} = (325.3/√2) × √[(1/π)(π − π/3 + sin120°/2)] = 230 × √(2.5274/π) = 230 × 0.897 = **206.3 V**; I_{rms} = 20.6 A; load power = 206.3²/10 = 4256 W.
Observe $I_{rms} > I_{avg}$: the output is pulsating DC, with ripple.

Practical notes: SCRs need isolated gate pulses synchronised to the supply zero crossing; snubbers limit dv/dt; with an inductive load or a DC motor, the current continues past π and the output can go negative at large α (the bridge then operates as an inverter, ‘regeneration’, for α > 90°).

Common misconception
Mixing up the firing angle: α is measured from the instant the SCR *would* start conducting if it were a diode (the zero crossing for a single-phase bridge), not from time zero of the experiment.

Unit 2, Question 5: DC–DC converters and the buck-boost converter

Exam question – Unit 2 Q5
What are DC-DC converters? Describe buck-boost DC–DC converters with circuit diagram and waveform.

A–D. DC–DC converters

DC–DC converters convert a DC voltage of one level to a DC voltage of another level (higher, lower, or either) with high efficiency by high-frequency switching of an 'inductor + capacitor' network, as in Question 3. Main types: buck (step-down), boost (step-up), **buck-boost** (step-up or step-down, inverted output), Cuk, SEPIC, and isolated forms (flyback, forward). Uses: SMPS, battery systems, solar MPPT, motor choppers, EV auxiliary supplies.

Buck-boost converter: the output can be **larger or smaller** than the input depending on D, but its polarity is **inverted** (negative with respect to the common ground). It is useful when the input may go above or below the required output (a battery that discharges, for instance).

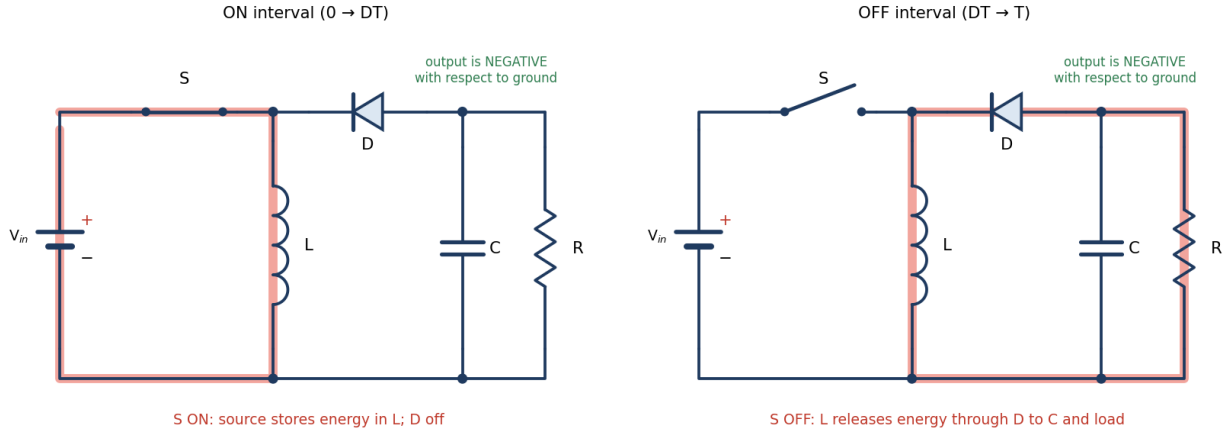


Figure 43. *Buck-boost converter (inverting). Inductor L is across the source when S is ON; when S is OFF, L pushes its current into C and the load through D.*

F. Working, step by step

1. **ON (0 → DT):** S closes. The source is across L, so $v_L = V_{in}$ and i_L rises. D is reverse biased (the output node is negative, the switch node is positive). C alone supplies the load.
2. **OFF (DT → T):** S opens. i_L must continue in the same direction, so the voltage across L reverses. The switch node swings negative, D conducts and L delivers energy to C and the load, with polarity negative at the output. $v_L = V_o$ (a negative number).

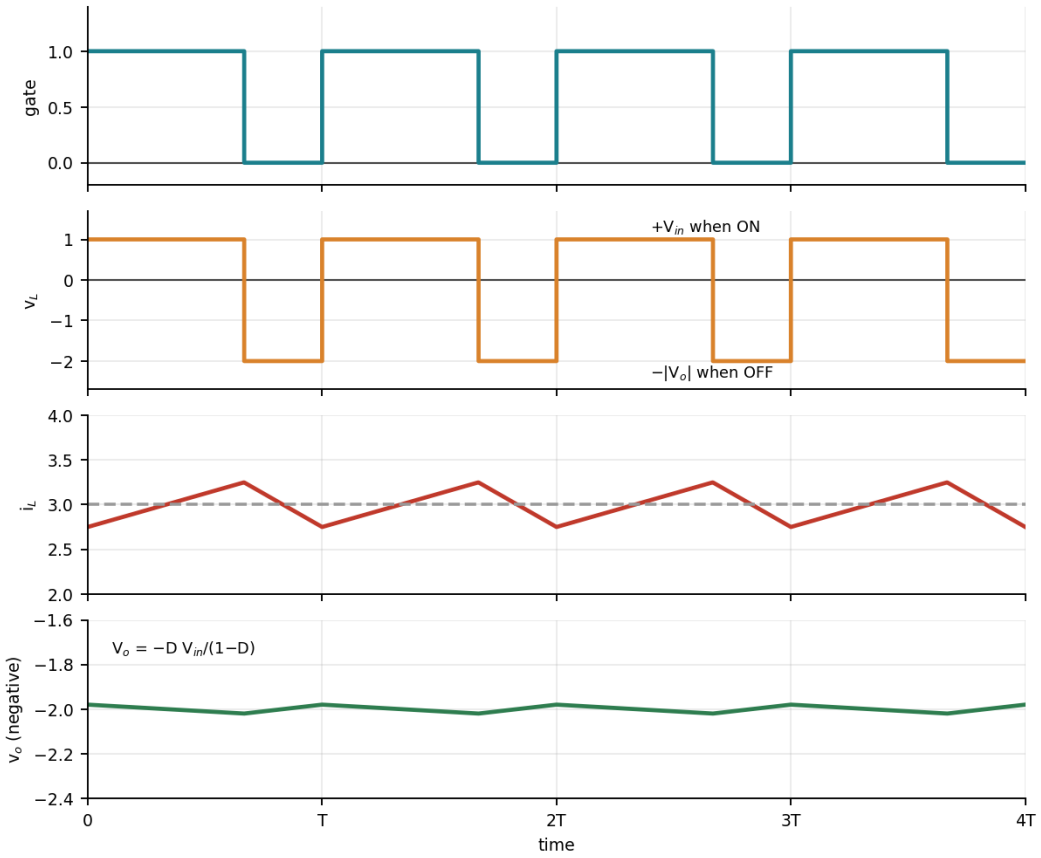


Figure 44. *Buck-boost waveforms for D = 2/3 ($V_o = -2V_{in}$): gate, inductor voltage, inductor current, output voltage.*

G. Equations

$$V_{in}\,DT + V_o(1-D)T = 0 \;\Rightarrow\; V_o = -\frac{D}{1-D}\,V_{in}$$

$$\Delta I_L = \frac{V_{in}D}{fL} \qquad \Delta V_o = \frac{|I_o|D}{fC} \qquad I_L = \frac{I_o}{1-D} \qquad L_{crit} = \frac{(1-D)^2 R}{2f}$$

D	0.25	0.5	0.75
$\lvert V_o\rvert/V_{in} = D/(1-D)$	0.333 (step-down)	1 (same magnitude)	3 (step-up)

Worked example: buck-boost design
V_{in} = 12 V, V_o = −24 V, R = 24 Ω ⇒ I_o = 1 A. f = 25 kHz.
$D/(1-D) = 24/12 = 2 \Rightarrow D = 2/3 =$ **0.667**.
I_L (average) = $I_o/(1-D) = 1/(1/3) = 3$ A. With ΔI_L = 20% × 3 = 0.6 A: $L = V_{in}D/(f\Delta I_L) = 12 \times 0.667/(25\,000 \times 0.6) = 8/15\,000 =$ **0.53 mH**.
For ΔV_o = 1% of 24 V = 0.24 V: $C = I_oD/(f\Delta V_o) = 1 \times 0.667/(25\,000 \times 0.24) =$ **111 µF**.
$L_{crit} = (1/3)^2 \times 24/(2 \times 25\,000) = 2.667/50\,000 = 53$ µH ⇒ CCM ✔. Note the switch must withstand V_{in} + $|V_o|$ = 36 V.

Advantages: one inductor, wide range of voltages. **Limitations:** inverted output, pulsating input and output currents (needs filtering), switch and diode must withstand V_{in} + $|V_o|$, higher ripple than Cuk.

Common misconception
'Buck-boost output is positive.' In this basic circuit it is negative with respect to the input common. (Non-inverting forms exist, such as the four-switch buck-boost, but are not the circuit of this question.)

Unit 2, Question 6: Single-phase AC to AC converters

Exam question – Unit 2 Q6
Write short note on single phase AC to AC converters with neat diagram and waveforms.

A–D. Concept and types

An **AC–AC converter** converts AC at one voltage and/or frequency into AC at another. Three routes exist:

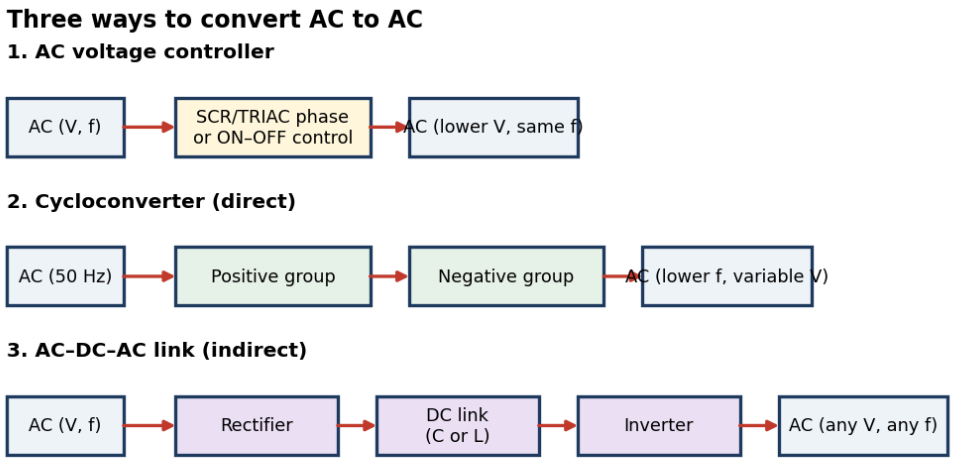


Figure 45. *Three ways to convert single-phase AC to AC.*

Type	Frequency	Voltage	Power flow	Examples
AC voltage controller (regulator)	same	variable	one-way	dimmers, heater control, soft starters, fan control
Cycloconverter	lower (variable)	variable	can reverse (regeneration)	gearless mill drives, ship propulsion (Q10)
AC–DC–AC (indirect)	any	any	depends	VFDs, UPS

E–G. The single-phase AC voltage controller

The basic circuit, waveforms and formula (phase-angle and ON–OFF control, R-load) were explained in Question 2 (Figure above). For an **R–L load** (such as a motor), the current lags the voltage by the load angle $\varphi = \tan^{-1}(\omega L/R)$, so after the supply voltage passes π, the thyristor keeps conducting until the current falls to zero at the **extinction angle β** ($\beta > \pi$).

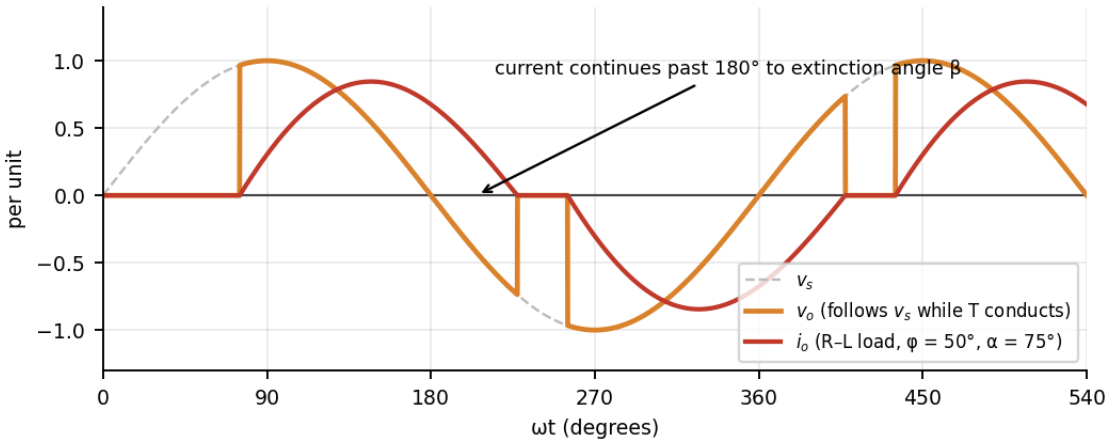


Figure 46. *AC voltage controller with R–L load: α = 75°, φ = 50°. The SCR keeps conducting beyond 180° until the current reaches zero (extinction angle β). During conduction $v_o = v_s$.*

$$i_o(\theta) = \frac{V_m}{Z}\left[\sin(\theta-\varphi) - \sin(\alpha-\varphi)\, e^{-(\theta-\alpha)/\tan\varphi}\right], \quad Z = \sqrt{R^2 + (\omega L)^2}, \quad \varphi = \tan^{-1}\frac{\omega L}{R}$$

This holds for $\alpha \le \theta \le \beta$, where β is found by solving $i_o(\beta) = 0$ (numerically). Symbols: Z = load impedance (Ω), φ = load angle, ωL = inductive reactance (Ω). Two key results: (1) the **control range** is $\varphi < \alpha < \pi$; if $\alpha \le \varphi$ the current is continuous and the output equals the full sine wave (no control); (2) the SCR is fired late, but conducts for a time $\pi - \alpha + (\beta - \pi)$.

Worked example: R-L load numbers
R = 10 Ω, ωL = 12 Ω ⇒ φ = $\tan^{-1}(1.2)$ = **50.2°**, Z = √(100 + 144) = 15.6 Ω. For control to be possible, α must exceed 50.2°; α = 75° is acceptable.
At full conduction ($\alpha \le \varphi$) $I_m = V_m/Z$ = 325.3/15.6 = 20.9 A (peak).

I–J. Where each is used; limitations

- **Single-phase AC controller:** lighting, resistive heaters, small fans and pumps. Limitations: harmonics, poor PF at high α, EMI (use filters).
- **Cycloconverter:** next question. **AC–DC–AC:** VFDs and UPS (a rectifier and an inverter back-to-back; see Units 2 Q1 and Q4 and Unit 3 Q8).

Unit 2, Question 7: Three-phase inverter

Exam question – Unit 2 Q7
Explain three-phase inverter circuit with neat diagram and waveforms.

A–D. Purpose

Industrial motors are three-phase. A **three-phase inverter** converts a DC link into three AC voltages 120° apart. It is the output stage of a VFD (Unit 3 Q8) and of grid-tied solar and wind converters. It needs three legs, one for each phase.

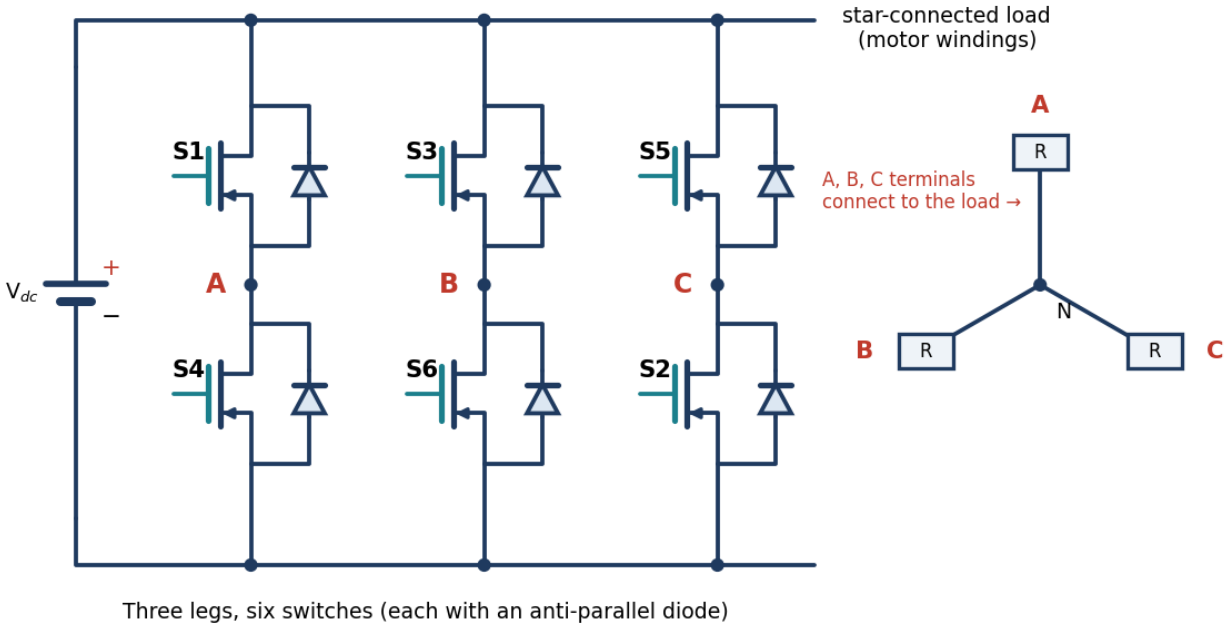


Figure 47. *Three-phase bridge inverter: six switches S1–S6 (IGBT with anti-parallel diode) in three legs (A: S1/S4, B: S3/S6, C: S5/S2), supplied from V_{dc}. The star-connected load (motor windings) is connected to A, B, C.*

F. Working: 180° conduction mode

In the **180° mode** each switch is ON for half a cycle (180°), the upper and lower switches of one leg are complementary, and the three legs are shifted by 120°. At every instant three switches (one per leg) conduct. The cycle divides into **six intervals of 60°**, giving six distinct circuit states:

Interval (ωt)	ON switches	v_{AN}	v_{BN}	v_{CN}	v_{AB}
0°–60°	S1, S6, S5	+1/3	−2/3	+1/3	+1
60°–120°	S1, S6, S2	+2/3	−1/3	−1/3	+1
120°–180°	S1, S2, S3	+1/3	+1/3	−2/3	0
180°–240°	S2, S3, S4	−1/3	+2/3	−1/3	−1
240°–300°	S3, S4, S5	−2/3	+1/3	+1/3	−1
300°–360°	S4, S5, S6	−1/3	−1/3	+2/3	0

(Voltages as multiples of V_{dc}; N is the load star point.) **Example of one state (0°–60°):** S1 connects A to +, S5 connects C to +, S6 connects B to −. Load phases A and C are in parallel (R/2) and in series with B (R), so the star point is at $2V_{dc}/3$, giving $v_{AN} = V_{dc} - 2V_{dc}/3 = V_{dc}/3$, $v_{BN} = -2V_{dc}/3$, $v_{CN} = V_{dc}/3$, matching the table.

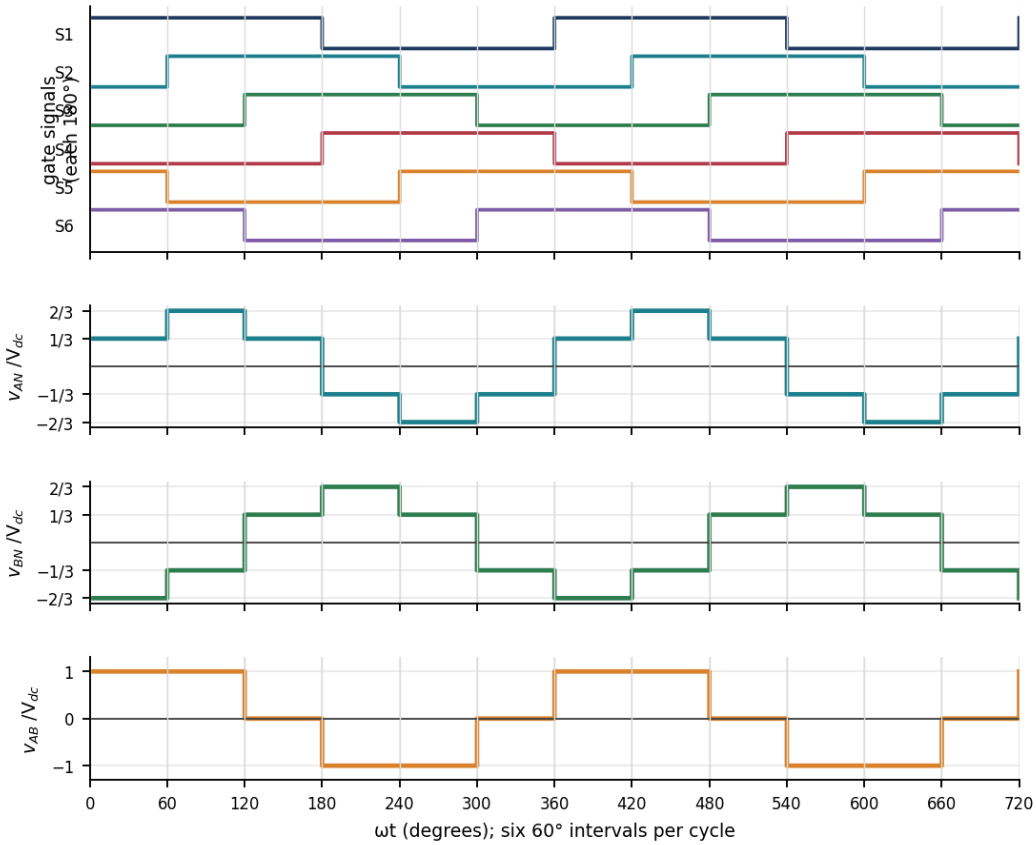


Figure 48. *Three-phase 180° inverter waveforms: the six gate signals, phase voltages v_{AN} and v_{BN} (six-step, levels $\pm V_{dc}/3$ and $\pm 2V_{dc}/3$), and line voltage v_{AB} (quasi-square, $\pm V_{dc}$ and 0).*

G. Equations

$$V_{LL,rms} = \sqrt{\frac{2}{3}}\,V_{dc} = 0.816\,V_{dc} \qquad V_{ph,rms} = \frac{\sqrt{2}}{3}V_{dc} = 0.471\,V_{dc} \qquad V_{LL1,rms} = \frac{\sqrt{6}}{\pi}V_{dc} = 0.78\,V_{dc}$$

Derivation of $V_{ph,rms}$: the phase voltage takes six values per cycle (±1/3, ±2/3, ±1/3, in units of V_{dc}), each for one-sixth of the cycle: $V^2 = (1/6)[\,(1/3)^2 + (2/3)^2 + (1/3)^2 + (1/3)^2 + (2/3)^2 + (1/3)^2\,]V_{dc}^2 = (1/6)(1/9 + 4/9 + 1/9 + 1/9 + 4/9 + 1/9)V_{dc}^2 = (1/6)(12/9)V_{dc}^2 = (2/9)V_{dc}^2$. So $V_{ph,rms} = (\sqrt{2}/3)V_{dc} = 0.471V_{dc}$. The line voltage is √3 times that: $0.816V_{dc}$. The line-voltage fundamental has a peak of $(2\sqrt{3}/\pi)V_{dc}$, i.e. an RMS of $0.78V_{dc}$. The line voltage contains no triplen harmonics (3rd, 9th…); the

dominant harmonics are the 5th and 7th.

Worked example: three-phase inverter
A VFD's DC link is 540 V (≈ 1.35 × 400 V line, the rectified level of a 400 V supply). Find the inverter's output.
$V_{LL,rms}$ = 0.816 × 540 = **441 V** (six-step total); fundamental $V_{LL1,rms}$ = 0.78 × 540 = **421 V**; $V_{ph,rms}$ = 0.471 × 540 = 254 V.
In practice, a VFD uses sinusoidal PWM, so the fundamental line voltage can be set from 0 up to about 0.612 × V_{dc} × √2… more simply, it is controlled to the motor's required V/f ratio (Unit 3 Q8).

- **120° mode:** each switch conducts 120°; only two switches conduct at any time, with natural dead time between switching events; lower utilisation, but safer against shoot-through.
- **Practical:** dead time, gate-drive isolation, snubbers, output filter or dV/dt filter, current sensing and protection; the diodes carry reactive current from the motor.
- **Applications:** VFDs, servo drives, UPS, grid-connected solar inverters, EV traction.

Common misconception
'Switches S1 and S4 can be ON together for a moment.' Never: it short-circuits V_{dc}. Each leg's two switches are complementary with a dead time.

Unit 2, Question 8: Inverters versus converters; the Cuk converter

Exam question – Unit 2 Q8
Differentiate between inverters and converters (any two). Write short notes on Cuk converter.(with circuit diagram and waveform)

Note on the source wording
'(any two)' is ambiguous. It most likely means 'give any two points of difference'. A fuller table of six points is given so that any two can be chosen. Also note the wording 'converter': in power electronics the term can mean *any* power converter (including inverters); here it is used in the narrower sense of an AC–DC converter (rectifier) or DC–DC converter, as in your unit's own list.

Point	Inverter	Converter (rectifier / DC–DC)
Conversion	DC → AC	AC → DC (rectifier) or DC → DC (chopper)
Input and output	DC input, AC output (single- or three-phase)	AC input, DC output; or DC at one level to DC at another
Output frequency	set by the switching (variable)	DC has zero frequency (rectifier output ripple at 2f or 6f)
Commutation	forced (controlled switches IGBT/MOSFET; SCRs need forced turn-off)	line/natural commutation for rectifiers; switch-mode for DC–DC
Typical devices	IGBT, MOSFET with anti-parallel diodes	diodes, SCRs (rectifiers); MOSFET, IGBT, diode (DC–DC)
Applications	UPS, VFD, solar/wind inverters, induction heating	battery chargers, DC-motor drives, SMPS, DC supplies, MPPT

Cuk converter

The **Cuk converter** (named after Slobodan Ćuk) is a DC–DC converter that gives an output voltage **higher or lower** than the input, with **inverted** polarity, like the buck-boost, but with **inductors at both input and output**. A capacitor C_1 transfers the energy from input side to output side.

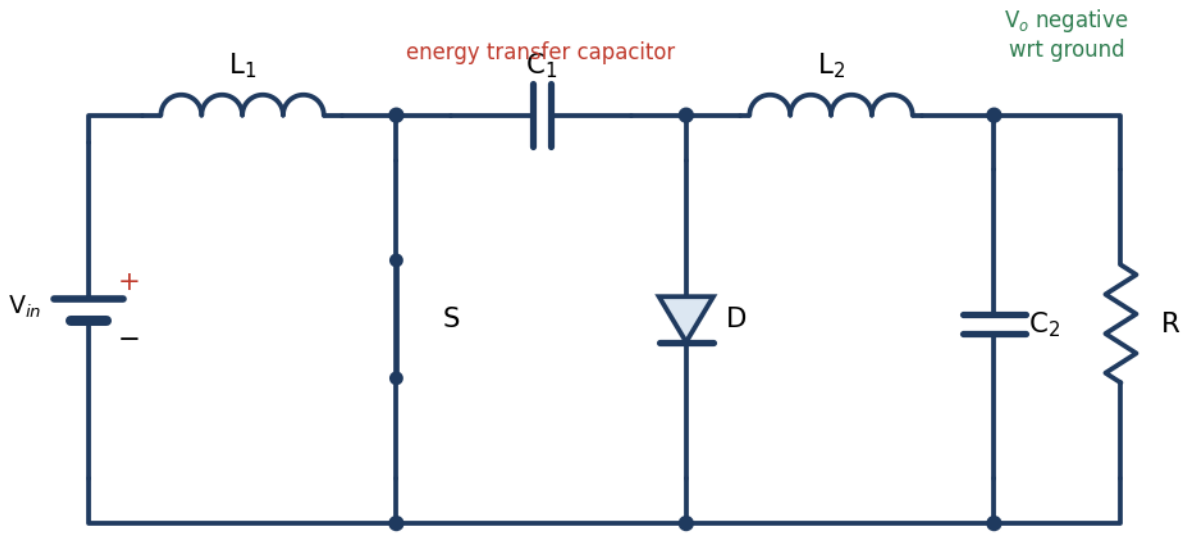


Figure 49. *Cuk converter: L_1 and the source form the input stage; capacitor C_1 carries energy to the output stage with L_2, C_2, load R.*

1. **Switch ON:** S shorts node X to ground. L_1 is connected across V_{in} and its current rises. C_1 is connected through S to the output stage: C_1 discharges into L_2 and the load, and the diode D is reverse biased (its anode is at $-V_{C1}$).
2. **Switch OFF:** L_1 current now flows through C_1 and D: C_1 is *charged* by the input (energy stored). At the same time L_2 discharges to the load through D.

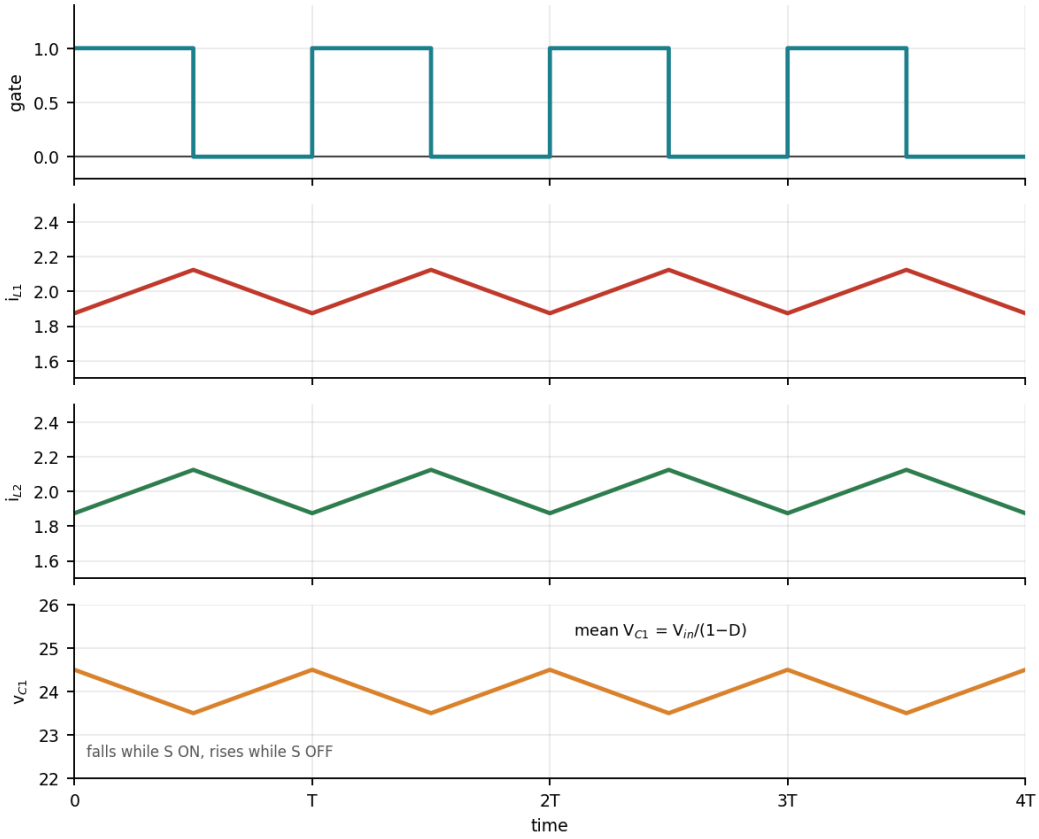


Figure 50. *Cuk waveforms for D = 0.5 (schematic): gate, input inductor current i_{L1}, output inductor current i_{L2}, and coupling-capacitor voltage v_{C1}.*

Derivation. Average voltage across L_1 is zero: $V_{in}\cdot DT + (V_{in} - V_{C1})(1 - D)T = 0 \Rightarrow V_{C1} = V_{in}/(1 - D)$. Average voltage across L_2 is zero (ON: $V_{C1} - |V_o|$; OFF: $-|V_o|$): $(V_{C1} - |V_o|)D - |V_o|(1 - D) = 0 \Rightarrow |V_o| = D\cdot V_{C1}$. Combining:

$$V_{C1} = \frac{V_{in}}{1-D} \qquad V_o = -\frac{D}{1-D}V_{in} \qquad I_{L1} = \frac{D}{1-D}I_o$$

The last relation follows from power balance ($V_{in}\, I_{L1} = |V_o|\, I_o$ for an ideal converter).

Worked example: Cuk numbers
V_{in} = 12 V, D = 0.5, R = 6 Ω.
V_{C1} = 12/0.5 = 24 V; V_o = −0.5/0.5 × 12 = **−12 V**; I_o = 12/6 = 2 A; I_{L1} = (0.5/0.5) × 2 = 2 A; input power = 12 × 2 = 24 W = output power 12 × 2 ✔.
With D = 0.75 the same circuit gives V_o = −36 V (step-up); with D = 0.25, −4 V (step-down).

- **Advantages:** continuous (nearly ripple-free) input and output currents ⇒ low EMI and small filters; wide voltage range; high efficiency.
- **Disadvantages:** inverted output; more components (two inductors, two capacitors); the coupling capacitor C_1 carries large ripple current and must be low-ESR; the switch must withstand $V_{in} + |V_o|$; more complex control.
- **Applications:** battery-powered systems, telecom power supplies, noise-sensitive instrumentation, solar MPPT.

Unit 2, Question 9: Three-phase controlled half-wave rectifier (resistive load)

Exam question – Unit 2 Q9
Explain the working of three-phase controlled half wave rectifier with resistive load controlled with waveforms.

A–D. Purpose and circuit

Industrial DC-motor drives and electrolysis plants need larger DC power than a single-phase bridge can supply smoothly. A **three-phase half-wave (three-pulse) controlled rectifier** uses a three-phase star-connected supply with **one SCR per phase** connected to a common load; the load's return is the supply neutral. The output has three pulses per supply cycle, so the ripple is smaller than a single-phase rectifier's.

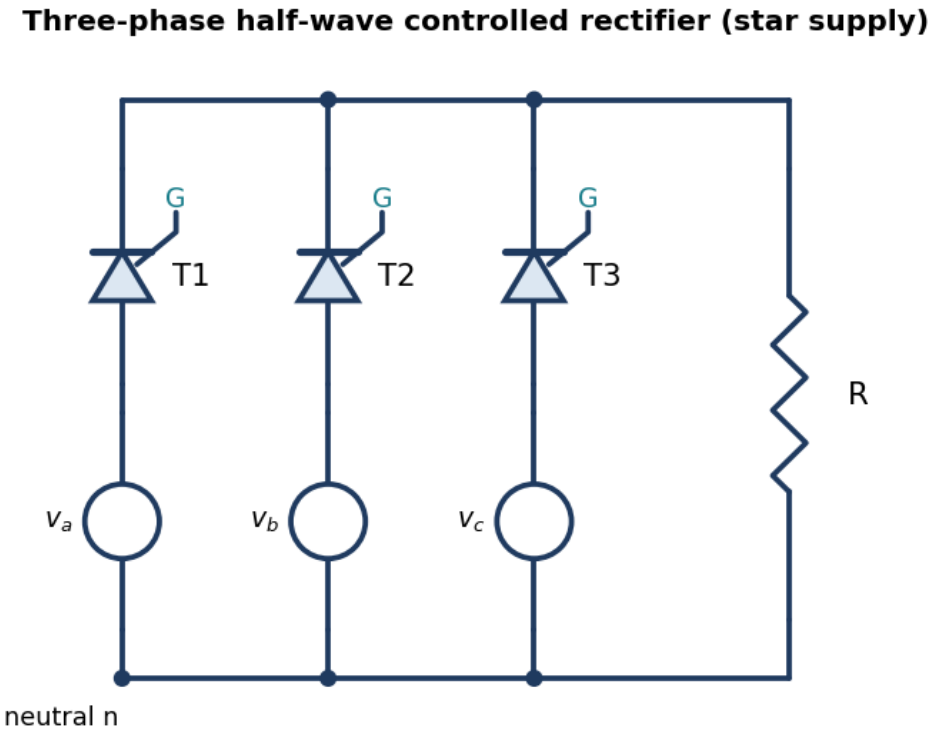


Figure 51. *Three-phase half-wave controlled rectifier: SCRs T1, T2, T3 in phases a, b, c feed a common resistive load, returning to the neutral n.*

F. Working, step by step

Define phase voltages $v_a = V_m \sin \omega t$, $v_b = V_m \sin(\omega t - 120°)$, $v_c = V_m \sin(\omega t - 240°)$, where V_m is the **peak phase voltage**. The firing angle α is measured from the **natural commutation point**: the instant (30° after the positive zero crossing of a phase) at which that phase's voltage first becomes larger than the others.

1. Between 30° and 150° phase a is the highest. If T1 is triggered at $\omega t = 30° + \alpha$, it conducts and $v_o = v_a$.
2. The next SCR T2 is triggered 120° later, at 150° + α. If phase b is now higher than phase a, T2 conducts and T1 is reverse biased and turns off (natural commutation). Similarly T3 at 270° + α.
3. **For α ≤ 30°** the output is never zero: each SCR conducts for 120°. **For α > 30°** with a resistive load, the SCR current falls to zero when its phase voltage reaches zero (at 180° for T1) *before* the next SCR is fired, so the output is zero for part of the cycle (the figure shows α = 60°).
4. For α > 150° there is no conduction: the output is zero.

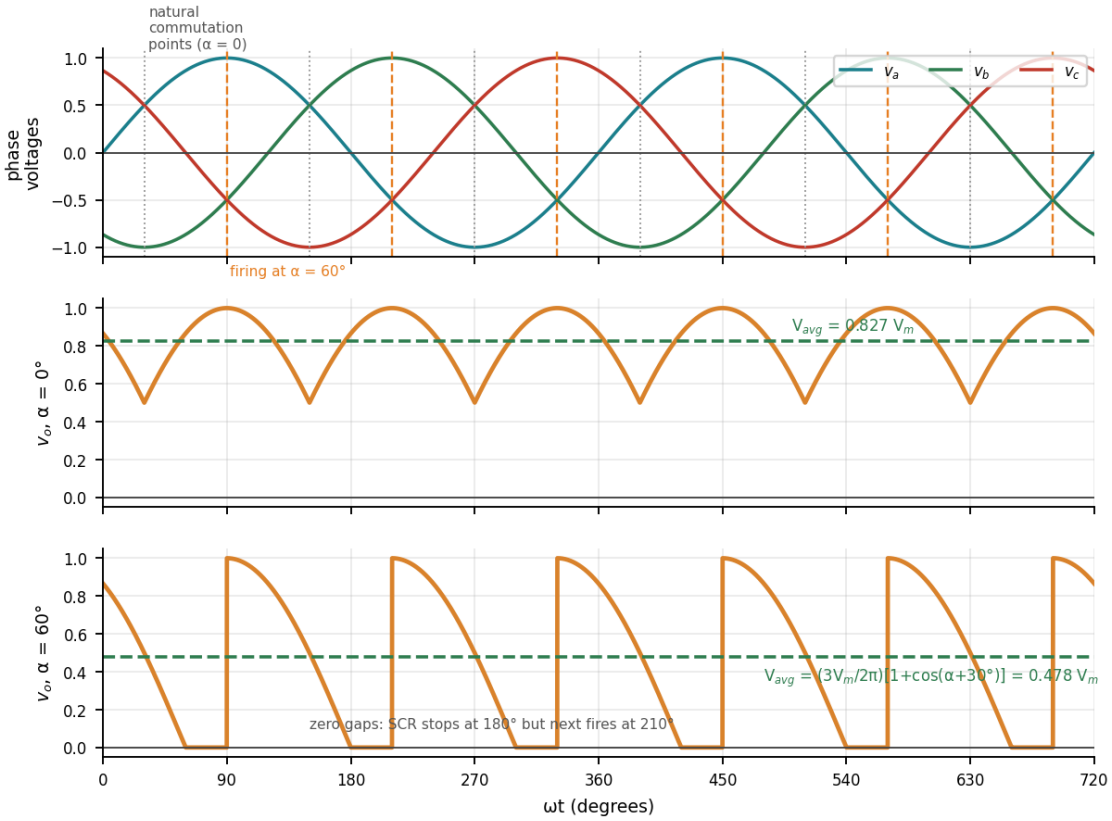


Figure 52. *Waveforms: phase voltages with natural commutation points and firing at α = 60°; output for α = 0° (top envelope) and α = 60° (discontinuous output).*

G. Derivation

Case 1: 0 ≤ α ≤ 30° (continuous output). Each SCR conducts for 120° (from 30° + α to 150° + α). Average over one pulse (3 pulses per cycle, so the factor is 3/2π):

$$V_{avg} = \frac{3}{2\pi}\int_{30^\circ+\alpha}^{150^\circ+\alpha} V_m \sin\theta\, d\theta = \frac{3V_m}{2\pi}[\cos(30^\circ+\alpha) - \cos(150^\circ+\alpha)]$$

Use cos A − cos B = −2 sin[(A+B)/2] sin[(A−B)/2]. Here (A+B)/2 = 90° + α and (A−B)/2 = −60°. So the bracket = −2 sin(90° + α) sin(−60°) = 2 cos α × (√3/2) = √3 cos α. Therefore:

$$V_{avg} = \frac{3\sqrt{3}}{2\pi} V_m \cos\alpha = 0.827\, V_m \cos\alpha = 1.17\, V_{ph,rms} \cos\alpha$$

Case 2: 30° < α ≤ 150° (resistive load, discontinuous). Each SCR conducts from 30° + α up to 180°:

$$V_{avg} = \frac{3}{2\pi}\int_{30^\circ+\alpha}^{180^\circ} V_m \sin\theta\, d\theta = \frac{3V_m}{2\pi}[1 + \cos(\alpha+30^\circ)]$$

Check at α = 30°: Case 1 gives 0.827 × 0.866 = 0.716 V_m; Case 2 gives (3/2π)(1 + cos 60°) = 0.4775 × 1.5 = 0.716 V_m ✔.

Worked example: three-pulse rectifier
Phase voltage 230 V RMS ⇒ V_m = 325.3 V; R = 10 Ω.
α = 0°: V_{avg} = 0.827 × 325.3 = **269.0 V**; I_{avg} = 26.9 A.
α = 60° (discontinuous): V_{avg} = (3 × 325.3 / 2π)[1 + cos 90°] = 155.3 × 1 = **155.3 V**; I_{avg} = 15.5 A. (As V_m = 325.3: 3 × 325.3 = 975.9; ÷ 6.2832 = 155.3 ✔.)
α = 120°: V_{avg} = 155.3 × [1 + cos 150°] = 155.3 × 0.134 = **20.8 V**: the output falls steadily to zero at 150°.

- **Characteristics:** ripple frequency = 3f (150 Hz at 50 Hz); each SCR conducts at most 120°; peak inverse voltage = $\sqrt{3}\ V_m$ (line peak).
- **Limitations:** the transformer or supply has a DC component in each phase (unbalanced), so the neutral is required and the transformer may saturate; therefore, practical high-power rectifiers use the six-pulse full bridge.
- **Applications:** medium-power DC-motor drives, battery chargers, electrochemical processes (historically).

Common misconception
Measuring α from the zero crossing of the phase voltage. In three-phase half-wave rectifiers α is measured from the *natural commutation point* (30° after the zero crossing).

Unit 2, Question 10: Cycloconverter, operation and applications

Exam question – Unit 2 Q10
Explain the operation of a cyclo converter with applications.

A–D. Purpose and definition

A **cycloconverter** is a **direct frequency changer**: it converts AC of fixed frequency and voltage (for example 50 Hz) *directly* into AC of a lower, adjustable frequency and adjustable voltage, without an intermediate DC link. Large slow-speed motors need low frequency (a motor's speed is N = 120f/P), and the cycloconverter delivers it using only phase-controlled thyristors.

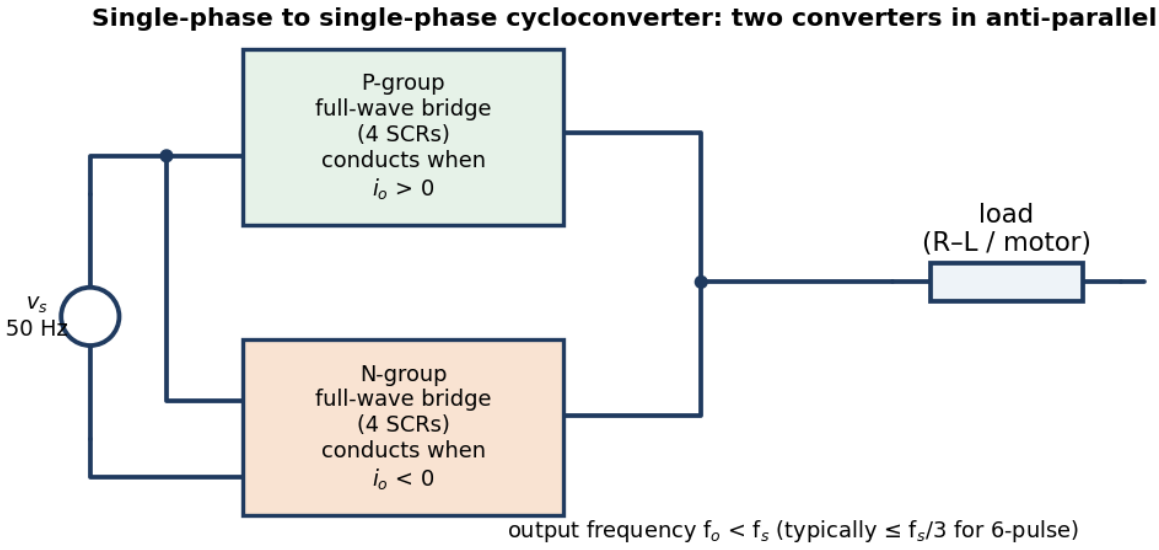


Figure 53. *Single-phase to single-phase cycloconverter: two controlled bridge rectifiers connected back-to-back (anti-parallel) between the supply and the load. (Return wires omitted for clarity.)*

F. Working, step by step

1. **P-group (positive converter)** conducts when the load current is positive and **N-group (negative converter)** when it is negative. Never both at once (otherwise a short circuit between supply phases), unless a circulating-current reactor is provided.
2. During the positive half of the *output* cycle the P-group operates; its firing angle α is changed gradually, for example from 90° → 0° → 90°, so that the average of the output pulses follows a rising and falling sine envelope (**cosine-wave crossing**: $\alpha = \cos^{-1}(r \sin \omega_o t)$, where r is the voltage-control ratio).

3. During the negative half, the N-group operates with the same modulation, producing the negative half of the output.
4. The output is therefore built from ‘slices’ of the supply waveform; its fundamental component has frequency f_o, lower than f_s.

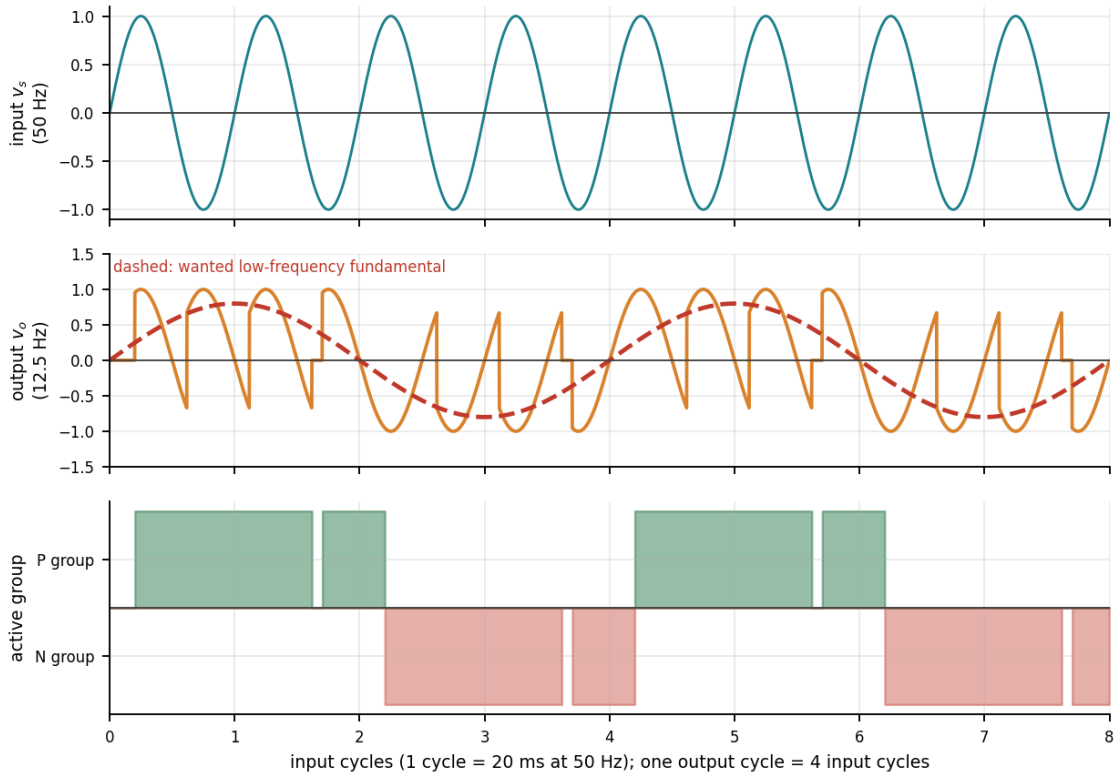


Figure 54. *Illustrative waveforms (computed with the cosine-wave firing rule for r = 0.8): a 50 Hz input becomes a 12.5 Hz output (one-quarter). The dashed line is the wanted fundamental; the thick curve is built from slices of the input. The ripple around the dashed curve is the harmonic content typical of cycloconverters.*

$$f_o < f_s, \quad f_o \leq \frac{f_s}{3}\ \text{(practical, 6-pulse)} \qquad N = \frac{120\, f_o}{P}$$

Three-phase cycloconverters: the 3-pulse (half-wave) type needs 18 SCRs (two groups of three SCRs per output phase); the 6-pulse (bridge) type needs 36. **Operating modes:** *non-circulating-current* (the idle group is blocked; needs blanking time) and *circulating-current* (both groups fire, with a reactor limiting the circulating current; smoother, but lower efficiency).

Worked example: cycloconverter speed range
A 6-pole (P = 6) synchronous motor is fed from a 50 Hz supply through a 6-pulse cycloconverter with $f_{o,max} \approx f_s/3 = 16.7$ Hz.
$N = 120 \times f_o/P = 120 \times 16.7/6 =$ **333 rpm** at maximum, e.g. at $f_o = 10$ Hz, N = 200 rpm.
This low-speed, high-torque range is just what mill and mine hoist drives need, with no gearbox.

I–J. Applications, advantages, limitations

- **Applications:** gearless cement and ore-grinding mill drives, rolling-mill and mine-hoist drives, ship propulsion, large wind-tunnel and test-rig drives, variable-speed constant-frequency generators (historically).
- **Advantages:** a single conversion stage; natural commutation (no commutation circuits); power can flow back to the source (regeneration); smooth, low-frequency waveforms; high power (MW) capability with ordinary thyristors.
- **Limitations:** output frequency is limited to a fraction of the input; poor input power factor (lagging, depends on α); complex control; harmonics on both input and output; many thyristors. In modern systems, IGBT-based VFDs have replaced many cycloconverters, but they remain in the multi-MW

range.

Common misconception
‘A cycloconverter can produce higher frequency than the supply.’ Not in the practical thyristor version: the output is assembled from slices of the supply, so f_o must be a fraction of f_s. (Matrix converters with fully controlled switches are a different technology.)

Unit 2: End-of-Unit Revision Pack

Unit summary

- Inverter: DC → AC. H-bridge = two legs; diagonal pairs conduct alternately; $v_o = \pm V_{dc}$; $V_{o,rms} = V_{dc}$; fundamental $0.9V_{dc}$; same-leg switches never ON together; anti-parallel diodes carry reactive current.
- AC voltage controller: SCR/TRIAC phase control; RMS formula; same frequency; ON–OFF control $\sqrt{(n/(n+m))}$.
- Buck $V_o = DV_{in}$; boost $V_o = V_{in}/(1-D)$; buck-boost $V_o = -DV_{in}/(1-D)$; Cuk similar to buck-boost with continuous input and output currents. All derived from zero average inductor voltage.
- Single-phase fully controlled bridge: $V_{avg} = (V_m/\pi)(1+\cos\alpha)$ for resistive load.
- Three-phase inverter: six switches; 180° mode; six-step phase voltage; $V_{LL} = 0.816V_{dc}$.
- Three-phase half-wave rectifier: $V_{avg} = 0.827V_m \cos\alpha$ for $\alpha \le 30°$; $(3V_m/2\pi)[1+\cos(\alpha+30°)]$ beyond.
- Cycloconverter: direct AC → lower-frequency AC with P and N groups.

Formula and equation sheet

H-bridge output

$$V_{o,rms} = V_{dc}, \quad V_{1,rms} = \frac{4V_{dc}}{\sqrt{2}\pi} = 0.9V_{dc}$$

AC controller (R load)

$$V_{o,rms} = V_s\sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

ON–OFF control

$$V_{o,rms} = V_s\sqrt{\frac{n}{n+m}}$$

Buck

$$V_o = DV_{in},\ \Delta I_L = \frac{V_o(1-D)}{fL},\ \Delta V_o = \frac{\Delta I_L}{8fC}$$

Boost

$$V_o = \frac{V_{in}}{1-D},\ \Delta I_L = \frac{V_{in}D}{fL},\ \Delta V_o = \frac{I_oD}{fC}$$

Buck-boost

$$V_o = -\frac{D}{1-D}V_{in}$$

Cuk

$$V_o = -\frac{D}{1-D}V_{in},\ V_{C1} = \frac{V_{in}}{1-D}$$

1-phase bridge (R load)

$$V_{avg} = \frac{V_m}{\pi}(1+\cos\alpha)$$

3-phase inverter (180°)

$$V_{LL,rms} = \sqrt{\frac{2}{3}} V_{dc},\ V_{ph,rms} = \frac{\sqrt{2}}{3} V_{dc}$$

3-phase half-wave rectifier

$$V_{avg} = \frac{3\sqrt{3}}{2\pi} V_m \cos\alpha \ (\alpha \le 30^\circ)$$

Cycloconverter speed

$$N = \frac{120 f_o}{P}, \quad f_o \le f_s/3$$

Diagram-revision sheet

Practise drawing each of these from memory, labelling every component, then compare with the figure.

Figure	What to draw and label
Fig. 34	Q1: H-bridge circuit with both current paths; waveforms of gates, v_o, i_o
Fig. 36	Q2: AC voltage controller circuit + α waveform
Q3	Buck and boost circuits (ON and OFF paths) and waveforms
Q4	Fully controlled bridge circuit + α waveform
Q5	Buck-boost circuit and waveforms
Q6	AC–AC types; R–L waveform with extinction angle
Q7	Three-phase inverter circuit, switching table, six-step waveforms
Q8	Cuk circuit and waveforms
Q9	Three-phase half-wave circuit; waveforms for α = 0 and 60°
Q10	Cycloconverter circuit and waveform

Glossary of important terms

Term	Meaning
Inverter	DC to AC converter
Shoot-through	Both switches of one leg ON: supply short circuit
Dead time	Brief interval with both switches of a leg OFF
PWM	Pulse-width modulation
Duty cycle D	t_{ON}/T of the switch
CCM	Continuous conduction mode: inductor current never zero
Volt-second balance	Average inductor voltage over a period is zero
Firing angle α	Delay of the SCR trigger
Natural commutation	Turn-off by supply current reaching zero
Phase-angle control	Adjust RMS by firing delay
Extinction angle β	Angle where SCR current becomes zero in R–L load
Cycloconverter	Direct AC–AC frequency changer
P and N groups	Positive and negative converters in a cycloconverter
Cuk converter	DC–DC with inductors at input and output

Conceptual and numerical review questions

1. A buck converter converts 48 V to 12 V at 4 A. Find D and the inductor for ΔI_L = 1 A at 50 kHz.
2. A boost converter must produce 36 V from 12 V. Find D and the input current if the output is 1 A.
3. Single-phase controlled bridge, V_m = 311 V, α = 30°, R-load. Find V_{avg}.
4. An AC controller (R-load) fed from 230 V is operated with 4 cycles ON and 1 cycle OFF. Find $V_{o,rms}$.
5. A 3-phase inverter has V_{dc} = 600 V. Find the six-step line RMS and the fundamental line RMS.
6. Three-phase half-wave rectifier, V_m = 311 V, α = 0°. Find V_{avg}.
7. Why does a Cuk converter have lower EMI than a buck-boost?

Fully explained answers
1. D = 12/48 = 0.25. $L = V_o(1-D)/(f\Delta I_L) = 12 \times 0.75/(50\,000 \times 1) = 9/50\,000 =$ **180 µH**.
2. D = 1 − 12/36 = 0.667. $I_{in} = I_o/(1-D) = 1/0.333 =$ **3 A** (power 36 W).
3. $(311/\pi)(1 + \cos 30°) = 99.0 \times 1.866 =$ **184.7 V**.
4. $230 \times \sqrt{(4/5)} = 230 \times 0.894 =$ **205.7 V**.
5. 0.816 × 600 = **490 V**; 0.78 × 600 = **468 V**.
6. 0.827 × 311 = **257 V**.
7. Its input current (through L_1) and output current (through L_2) are both continuous and nearly ripple-free, whereas in the buck-boost both are pulsating.

Examination-answer guide for the original questions

This guide complements the teaching chapters. It shows how to organise each original question as a complete exam answer.

Question	Structure of a strong answer
Q1	Definition → why needed → H-bridge circuit (four switches + diodes) → two states with current paths → waveforms (gate, v_o, i_o) → $V_{o,rms}$, fundamental → dead time → PWM note → applications.
Q2	Need (three bullets) → circuit with two SCRs or TRIAC → α waveform → derive $V_{o,rms}$ → ON–OFF control → regulation by feedback → applications.
Q3	One page per converter: circuit → ON and OFF states → waveforms → derive V_o by volt-second balance → ripple formulas → example → comparison table.
Q4	Circuit (4 SCRs) → conduction paths → waveforms (v_s, gate, v_o, i_o) → derive V_{avg} (and V_{rms}) → example.
Q5	Define DC–DC → circuit → two modes → waveforms → $V_o = -DV_{in}/(1-D)$ → advantages and limits.
Q6	Three types in a diagram → AC controller circuit and waveform (R and R–L) → formulas → applications.
Q7	Circuit of 6 switches → 180° conduction table → waveforms (gates, v_{AN}, v_{AB}) → formulas for RMS → dead time, PWM, applications.
Q8	Table of ≥ 4 differences (pick any two) → Cuk: circuit, ON/OFF states, waveform, $V_o = -DV_{in}/(1-D)$, advantages.
Q9	Circuit → natural commutation point → waveforms (α = 0 and α > 30°) → V_{avg} in both regions → example → limitations.
Q10	Definition → P and N group circuit → working with α modulation → waveform for $f_o < f_s$ → frequency limit → applications and drawbacks.

UNIT 3: PLCs and Industrial Control

Unit 3 moves from power to **decision-making**. We first need the language machines use to decide: logic. Revisit Chapter E.5 if binary and Boolean algebra feel new. Then we meet the PLC, the controllers (P, PI, PID), the supervisory systems (SCADA, DCS) and the VFD.

Unit 3, Question 1: Requirement of PLC; logic gates and PLC ladder logic

Exam question – Unit 3 Q1
What are the requirement of PLC? Develop PLC ladder logic for the following gates: (Truth table, Boolean expression, Gate symbol & PLC ladder diagram) (a) NAND (b) XNOR (c) NOR (d) XOR (e) AND (f) OR (g) NOT

Note on the source wording
'Requirement of PLC' can mean (i) *why industry needs PLCs* or (ii) *what a PLC must satisfy*. Both are covered below. The gates are listed in the source in the order NAND, XNOR, NOR, XOR, AND, OR, NOT; for learning, we teach AND, OR, NOT first and present the final answers in the source order.

A. The problem: from relay panels to PLCs

Before the 1970s, machine logic was built from hundreds of electromechanical relays wired in cabinets. Changing the process meant *rewiring*, and a single failed relay was hard to find. A **PLC (Programmable Logic Controller)** is a rugged industrial computer that reads inputs (switches, sensors), executes a stored program, and drives outputs (motors, valves, lamps), so that *changing the logic means changing the program, not the wiring*.

Requirements of a PLC

Requirement	Meaning / why industry needs it
Ruggedness	operates in heat, dust, vibration, electrical noise (EMI, Unit 1 Q4); no moving parts or fans in many models
Reliability and long life	continuous 24 × 7 operation; self-diagnostics; battery-backed or non-volatile memory
Reprogrammability / flexibility	logic changed by software; same hardware for many machines
Easy programming by technicians	ladder logic looks like relay diagrams (IEC 61131-3 also defines FBD, ST, SFC)
Direct industrial I/O	24 V DC, 230 V AC, 4–20 mA analogue; opto-isolation between field wiring and electronics
Real-time, deterministic response	the scan is repeated every few milliseconds
Modularity and expandability	add input/output modules; remote I/O
Communication	Ethernet, RS485, Profibus, Modbus for SCADA, HMI and other PLCs (Unit 5)
Troubleshooting and monitoring	online monitoring of the program, forced I/O, fault flags
Cost-effectiveness	cheaper than relay panels beyond about a dozen relays; reduces wiring and downtime
Safety	fail-safe programming, safety PLCs for emergency stops

B–D. Logic basics and how a PLC reads them

Physical input device: a real push-button, limit switch or sensor wired to an input terminal of the PLC, e.g. terminal I0.0. **Input bit**: the PLC converts the voltage at the terminal into a 1 (voltage present) or 0, stored in memory. **Ladder contact**: a symbol in the program that *examines* a bit. **Normally open contact (NO)** ┤ ├ is TRUE (passes 'power flow') when its bit = 1. **Normally closed contact (NC)** ┤/├ is

TRUE when its bit = 0. **Coil ( )**: sets an output bit to 1 when the rung is TRUE; the output module then switches the physical load (e.g. Q0.0).

> **Common misconception**
> The NC *contact in the program* has nothing to do with whether the physical push-button is wired NO or NC. The contact tests the bit; the bit depends on the wiring and the button state. A student wrongly reasons 'I used a NC button, so I must use an NC contact'; the correct reasoning is 'what logic condition do I want?'.

Ladder logic is drawn between two vertical power rails (left = supply, right = return). A **rung** is one horizontal line of contacts leading to a coil. 'Power' flows left to right if there is a continuous path of TRUE contacts. **Series** contacts mean AND; **parallel** branches mean OR. Throughout this answer: A = input I0.0, B = input I0.1, Y = output Q0.0.

E. Gate symbols and Boolean expressions

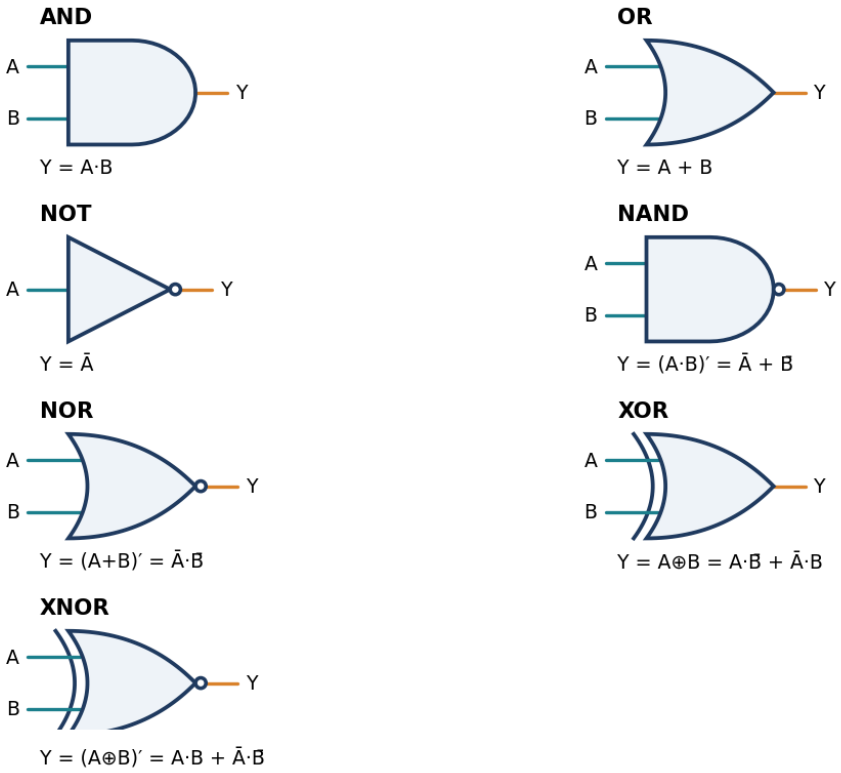


Figure 55. *Standard logic-gate symbols with Boolean expressions. A bubble on the output means NOT (inversion).*

Intuition and truth tables. A **truth table** lists every possible combination of the inputs and the output. With two inputs there are $2^2 = 4$ rows; with one input, 2 rows.

(e) AND gate: Y = A·B

Intuition: two switches in series; the lamp lights only if both are closed.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(f) OR gate: Y = A + B

Intuition: two switches in parallel; the lamp lights if either (or both) is closed.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(g) NOT gate: Y = Ā

Intuition: an inverter; the output is the opposite of the input.

A	Y
0	1
1	0

(a) NAND gate: Y = (A·B)′ = Ā + B̄

Intuition: AND followed by NOT; the output is OFF only when both inputs are ON. (De Morgan's law gives the equivalent form: OR of the complements.)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(c) NOR gate: Y = (A + B)′ = Ā·B̄

Intuition: OR followed by NOT; the output is ON only when both inputs are OFF.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(d) XOR (exclusive OR) gate: Y = A ⊕ B = A·B̄ + Ā·B

Intuition: output ON when the inputs are **different** (e.g. a two-way staircase switch).

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(b) XNOR (exclusive NOR) gate: Y = (A ⊕ B)′ = A·B + Ā·B̄

Intuition: output ON when the inputs are the **same** (an equality detector).

A	B	Y
0	0	1
0	1	0

A	B	Y
1	0	0
1	1	1

F–H. PLC ladder diagrams, rung by rung

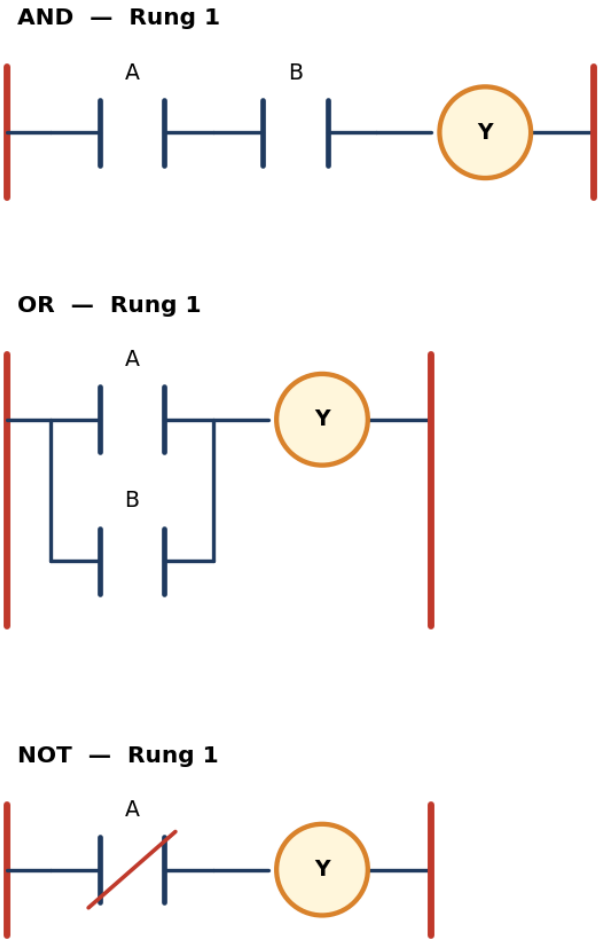


Figure 56. Ladder diagrams for AND, OR and NOT. Red slashed contacts are normally-closed (NC) contacts. The rung is TRUE when a path of TRUE contacts connects the left rail to the coil Y.

- **AND:** A and B contacts in **series** (both NO). Y energises only if both bits = 1.
- **OR:** A and B in **parallel** branches (both NO). Y energises if either is 1.
- **NOT:** a single **NC** contact of A: when A = 0 the contact passes power, so Y = 1.

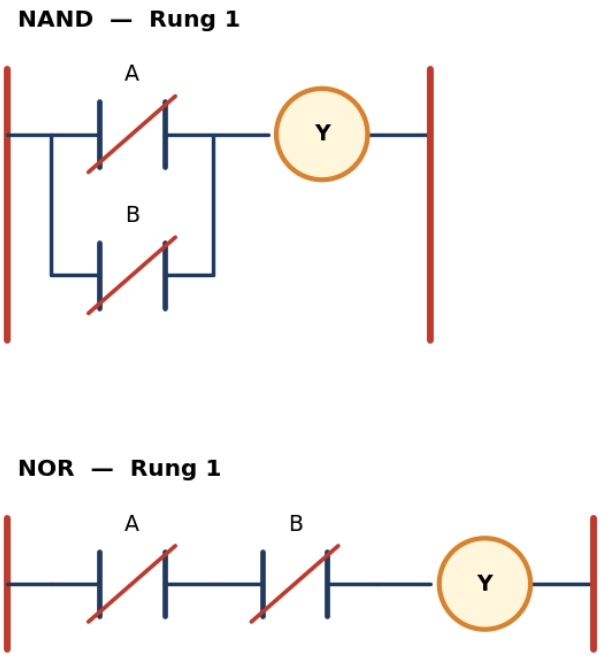


Figure 57. *Ladder diagrams for NAND and NOR.*

- **NAND:** Y = Ā + B̄, so two **NC** contacts in **parallel**. Y is ON unless both A and B are ON (then both NC contacts are open).
- **NOR:** Y = Ā·B̄, so two **NC** contacts in **series**. Y is ON only if both are OFF.

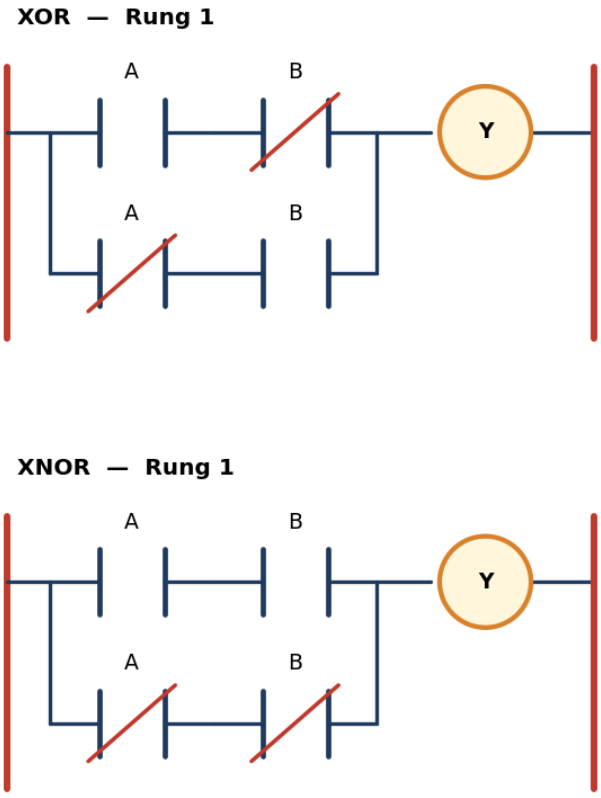


Figure 58. *Ladder diagrams for XOR and XNOR.*

- **XOR:** Y = A·B̄ + Ā·B: two parallel branches. Branch 1: A (NO) in series with B (NC). Branch 2: A (NC) in series with B (NO). Exactly one branch conducts when A ≠ B.

- **XNOR:** Y = A·B + Ā·B̄: Branch 1: A (NO) in series with B (NO); Branch 2: A (NC) in series with B (NC). A branch conducts when A = B.

Check by tracing XOR: A = 1, B = 0 ⇒ branch 1: A(NO) true, B(NC) true ⇒ power flows ⇒ Y = 1 ✔. A = 1, B = 1 ⇒ branch 1: B(NC) false; branch 2: A(NC) false ⇒ no path ⇒ Y = 0 ✔.

Worked example: practical use
A conveyor must run only when the guard is closed (A) AND the start latch is on (B): AND rung. An alarm must sound if either high-temperature (A) OR high-pressure (B): OR rung. A two-way staircase light, where either switch toggles the lamp, is an XOR of the two switch positions.

Summary of the seven gates in the order of the question

Gate	Boolean	Ladder form
(a) NAND	Y = (A·B)′ = Ā + B̄	two NC contacts in parallel
(b) XNOR	Y = A·B + Ā·B̄	(A NO · B NO) parallel (A NC · B NC)
(c) NOR	Y = Ā·B̄	two NC contacts in series
(d) XOR	Y = A·B̄ + Ā·B	(A NO · B NC) parallel (A NC · B NO)
(e) AND	Y = A·B	two NO contacts in series
(f) OR	Y = A + B	two NO contacts in parallel
(g) NOT	Y = Ā	one NC contact

Unit 3, Question 2: Comparison of P, PI and PID controllers

Exam question – Unit 3 Q2
Compare between P, PI, and PID controller (at least 7 parameters)

Terms: **error** e = setpoint − measured value. The controller output u is the command to the actuator (valve opening, VFD speed, heater power). The three controllers differ in how they compute u from e. Full intuition and equations are in Question 5; the comparison is below. A simulated comparison on one process follows.

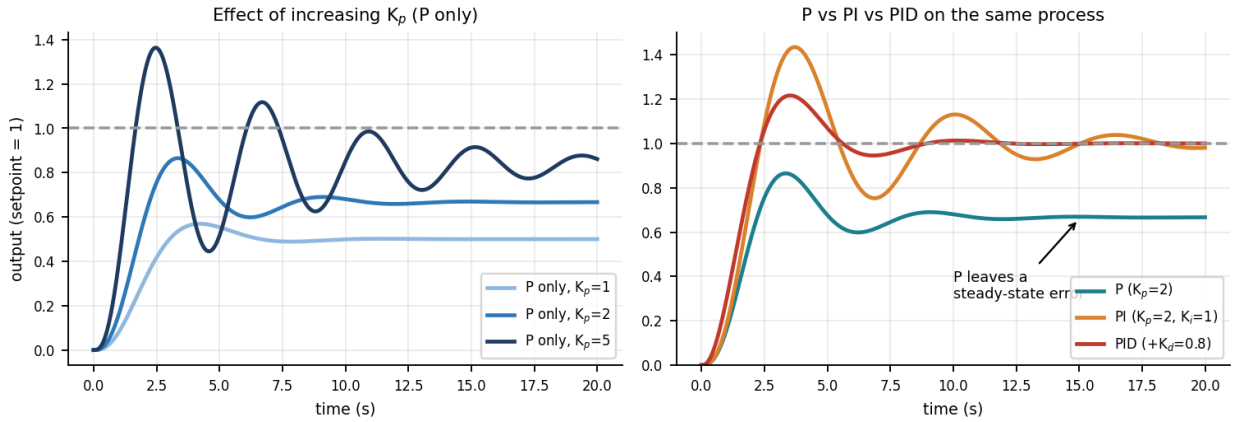


Figure 59. Simulation of a three-stage lag process (transfer function 1/(s+1)³ with a unit step in setpoint). Left: increasing K_p reduces the error but adds oscillation. Right: P leaves a permanent error; PI removes it; PID adds damping. (Computed numerically; illustrative.)

Parameter	P	PI	PID
Control law	$u = K_p e$	$u = K_p e + K_i \int e\,dt$	$u = K_p e + K_i \int e\,dt + K_d\,de/dt$
Transfer function	K_p	$K_p + K_i/s$	$K_p + K_i/s + K_d s$
Action used	present error	present + accumulated past error	present + past + predicted (rate of change)
Steady-state error	non-zero (offset) for most processes	eliminated (for step setpoint)	eliminated

Parameter	P	PI	PID
Speed of response	faster as K_p rises	slower than P at equal K_p, but accurate	fast and well damped
Overshoot / oscillation	grows with K_p	more overshoot than P (integral lag)	least overshoot of the three if well tuned
Stability margin	good for small K_p	reduced by integral action	improved by derivative action
Sensitivity to noise	low	low	high (differentiates noise); needs filtering
Number of tuning parameters	1 (K_p)	2 (K_p, K_i or T_i)	3 (K_p, K_i, K_d)
Special issue	offset	integral windup	noise and derivative kick
Typical use	level in a tank where offset is acceptable, simple on-board loops	flow, pressure, speed, most industrial loops	temperature, slow lagging processes, positioning, drives

Unit 3, Question 3: SCADA versus DCS; DCS architecture and example

Exam question – Unit 3 Q3
Write down the key difference between SCADA & DCS (at least 7 parameters). Explain DCS architecture with suitable block diagram & example.

SCADA (Supervisory Control And Data Acquisition): software and hardware that gather data from remote equipment, display it to operators, log it, raise alarms, and let operators send supervisory commands. **DCS (Distributed Control System)**: a plant-wide control system where the control functions are *distributed* over multiple controllers, tightly integrated with operator stations, to run continuous processes (a refinery, a power plant).

Parameter	SCADA	DCS
Main purpose	data acquisition and supervisory control (monitoring)	continuous, closed-loop process control (regulation) with supervision
Control location	local control in PLC/RTUs; SCADA server only supervises	control loops run in the DCS controllers (built into the system)
Geographic area	wide area (pipelines, grids, water networks, 100s of km)	localised plant or site
Communication	WAN: radio, cellular, satellite, leased lines; slower; may be intermittent	fast, deterministic, redundant plant LAN (usually proprietary or industrial Ethernet)
Data acquisition style	event/poll-driven, tag-based; can tolerate comms delays	process-driven, continuous real-time scanning, loop time of ms–100 ms
Architecture	multi-vendor, open; PLCs/RTUs + SCADA server + HMI	vendor-integrated: controllers, I/O, networks, HMI and engineering tools as one system
Redundancy and availability	optional	built in at the controller, network and power-supply levels
Process complexity	simple to moderate loops, discrete and batch events	complex, interacting continuous loops (cascade, ratio, advanced control)
Database and configuration	separate databases for PLC and SCADA	single integrated database for all tags
Cost and implementation	lower cost, easier to extend	higher cost, long engineering, strong lifecycle support
Typical applications	water supply, oil & gas pipelines, electrical grids, railways, building automation	refineries, chemical plants, power stations, cement and paper mills, pharmaceuticals

DCS architecture

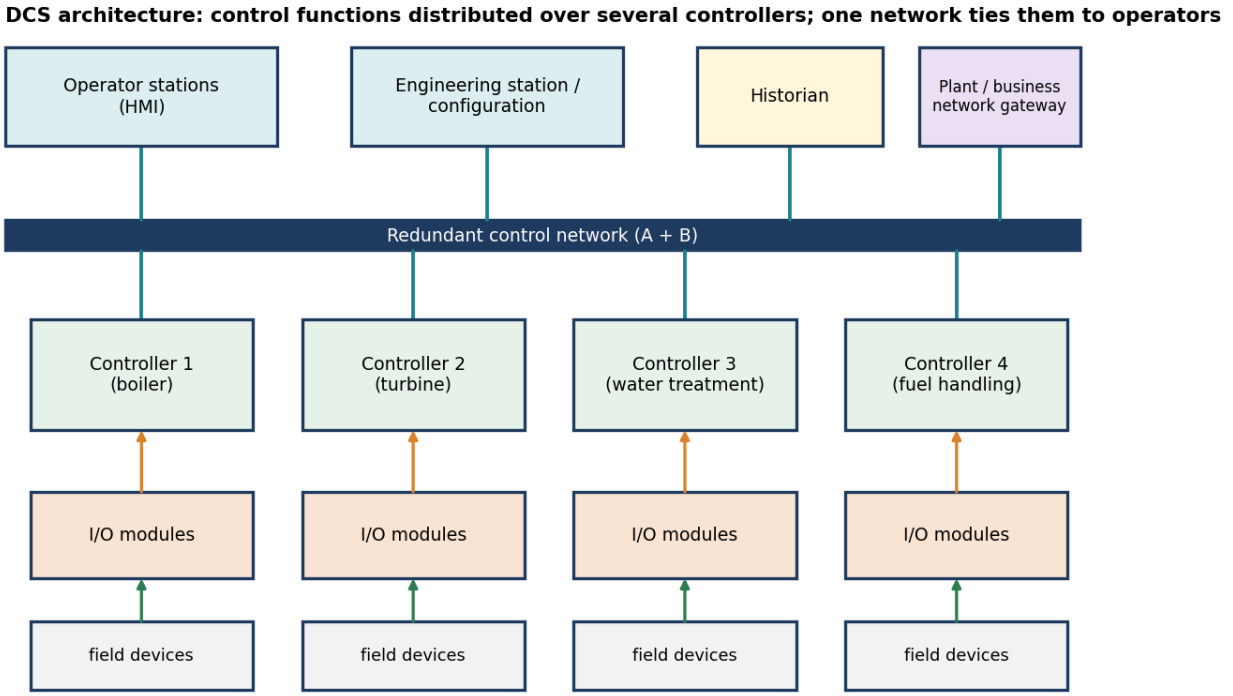


Figure 60. *DCS architecture. Field devices connect to I/O modules; several controllers each run the loops of one plant area; a redundant control network connects them to operator, engineering and historian stations and to the business network through a gateway.*

Layer	Function
Field level	sensors, transmitters, valves, drives, analyser
I/O modules	convert field signals (4–20 mA, RTD, 24 V) to digital data, with isolation
Controllers	execute PID, interlock, sequence and calculation logic for one area; redundant pairs
Control network	redundant, deterministic network (ring or dual bus)
Operator stations (HMI)	process graphics, alarms, trends, set-point changes
Engineering station	configuration and downloading of control strategies
Historian / servers	long-term data storage, reports, links to plant information systems

Worked example: example: a thermal power plant
The boiler, turbine, water-treatment and fuel-handling areas each have a controller (Figure above). In the boiler controller a PID loop holds drum level by adjusting the feed-water valve; another PID loop adjusts fuel to keep steam pressure; interlocks trip the fuel if the drum level or pressure goes out of range.
If controller 2 (turbine) fails, its redundant partner takes over; the boiler controller keeps running (this is the benefit of *distributed* control: a failure affects only one part).
Operators in the control room see all loops, acknowledge alarms, and change set-points from the HMI.

Unit 3, Question 4: Open-loop and closed-loop control; closed-loop transfer function

Exam question – Unit 3 Q4
Define and differentiate between open-loop and closed-loop control systems with examples. Derive the closed loop transfer function.

A–D. Basic vocabulary

Term	Meaning	Water-tank example
System / process (plant)	the thing being controlled	tank and pump
Input / reference / setpoint	desired value	level = 2 m
Output / process variable	actual value	measured level
Disturbance	unwanted influence	outflow valve opened by a user
Sensor	measures the output	level transmitter
Controller	decides the action	PLC
Actuator	carries out the action	VFD + pump

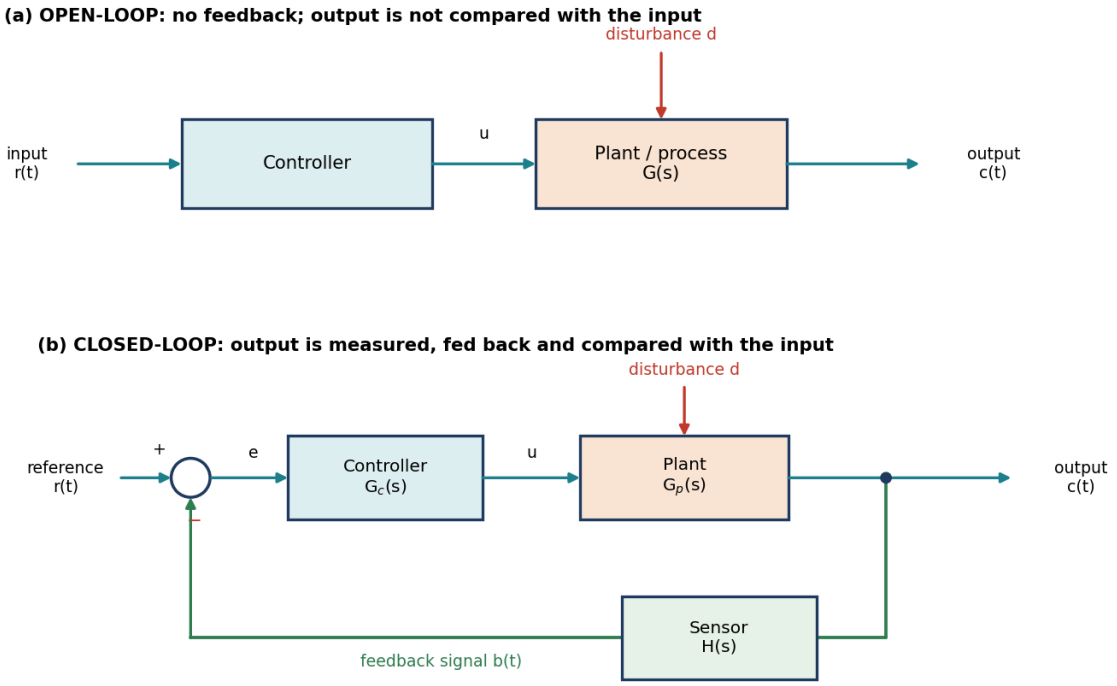


Figure 61. *(a) Open-loop and (b) closed-loop control systems. In (b) the sensor feeds the output back to the summing junction, where it is subtracted from the reference.*

Point	Open-loop	Closed-loop (feedback)
Feedback	none	output is measured and fed back
Action depends on	input only	error between reference and actual output
Accuracy	poor unless the process is well known	high; corrects errors
Effect of disturbances	not compensated	automatically compensated
Stability	no stability problem	may oscillate if gain is too high
Complexity and cost	simple, cheap	more components (sensor, comparator)
Examples	washing machine timer cycle, toaster, traffic-light timer, a fan with a regulator set manually, VFD run at fixed frequency	room thermostat, cruise control, tank level control, motor speed control with tachometer feedback, PID temperature control

G. Derivation of the closed-loop transfer function

A **transfer function** is the ratio of the output to the input in the **Laplace (s) domain** with zero initial conditions; it describes a block's behaviour. It lets us multiply blocks instead of solving differential equations. Define (see Figure above, taking $G = G_c G_p$ as the forward path):

Symbol	Meaning
R(s)	reference input

Symbol	Meaning
C(s)	output
E(s)	actuating error signal at the summing junction
B(s)	feedback signal
G(s)	forward-path transfer function (controller × plant)
H(s)	feedback-path transfer function (sensor)

1. Each block: output = (transfer function) × input. Forward path: **C(s) = G(s) E(s)**.
2. Feedback path: **B(s) = H(s) C(s)**.
3. Summing junction (negative feedback): **E(s) = R(s) − B(s)**.
4. Substitute step 2 into step 3: E(s) = R(s) − H(s)C(s).
5. Substitute step 4 into step 1: C(s) = G(s)[R(s) − H(s)C(s)] = G(s)R(s) − G(s)H(s)C(s).
6. Collect C(s) terms on the left: C(s) + G(s)H(s)C(s) = G(s)R(s), i.e. C(s)[1 + G(s)H(s)] = G(s)R(s).
7. Divide by R(s) and by [1 + G(s)H(s)]:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad \text{(negative feedback)} \qquad \frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)} \quad \text{(positive feedback)}$$

For **unity feedback** (H = 1): C/R = G/(1 + G). The product GH is the **open-loop transfer function**; the denominator 1 + GH = 0 is the **characteristic equation** whose roots determine stability. **With a disturbance** D(s) entering at the plant input (superposition): C = [$G_c G_p$/(1+$G_c G_p$H)]R + [G_p/(1+$G_c G_p$H)]D; a large loop gain $G_c G_p$H shrinks the disturbance's effect.

Worked example: numerical check
G(s) = 10/(s + 2), H = 1.
T(s) = G/(1+G) = [10/(s+2)] / [1 + 10/(s+2)] = [10/(s+2)] / [(s+2+10)/(s+2)] = **10/(s + 12)**.
DC gain: open loop G(0) = 10/2 = 5; closed loop T(0) = 10/12 = 0.83. Time constant: open loop 1/2 = 0.5 s; closed loop 1/12 = 0.083 s, 6 times faster.
So feedback reduced the gain but made the response much faster and less sensitive to changes of G (Figure below). A controller with integral action restores the DC gain to 1.

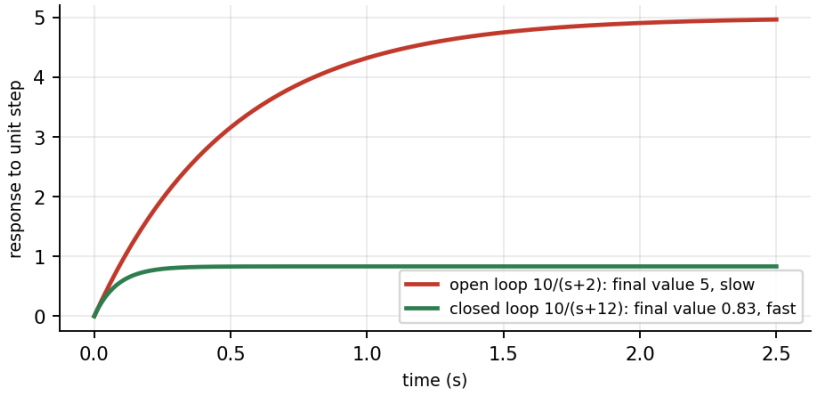


Figure 62. *Step response of G(s) = 10/(s+2) in open loop and with unity negative feedback.*

Common misconception
'Feedback always makes a system better.' Feedback reduces sensitivity and disturbances, but too much gain can make the system oscillate or become unstable (when 1 + GH = 0 has roots in the right half of the s-plane).

Unit 3, Question 5: The PID controller in detail

Exam question – Unit 3 Q5
Explain the working of a PID controller in detail.

A–C. Intuition

Imagine driving a car at 80 km/h on a hill. The speed drops to 70 km/h. A **proportional** driver presses the pedal in proportion to the shortfall (10 km/h). But at a steady hill, pressing proportionally to the shortfall settles at a speed slightly below 80 (the shortfall that produces just enough pedal). An **integral** driver notices that the speed has been below target for a long time and keeps pressing harder until the shortfall disappears. A **derivative** driver sees how *fast* the speed is changing and eases off before the car overshoots. A **PID controller** does all three automatically.

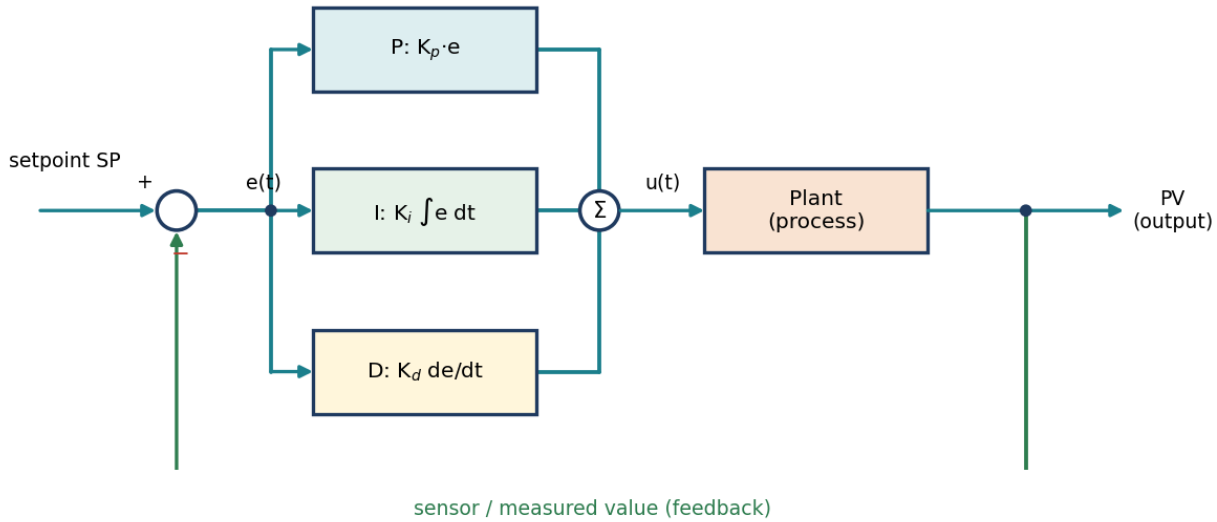


Figure 63. *PID controller in a feedback loop. The error e = SP − PV is processed by three parallel paths whose outputs are summed to produce the control signal u.*

D–G. Equations

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt} \quad \Longleftrightarrow \quad G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

Symbol	Meaning	Unit (example: temperature loop)
e(t)	error = setpoint − measurement	°C
u(t)	controller output	% heater power
K_p	proportional gain	% per °C
K_i (= K_p/T_i)	integral gain: strength of the accumulated-error correction	% per (°C·s)
K_d (= $K_p T_d$)	derivative gain: strength of the rate-of-change correction	% per (°C/s)

Meaning of each term. P reacts to the *present* error: large error ⇒ large correction. **I** sums the error over time (area under the error curve, Chapter E.4): any small persistent error grows the integral until the correction removes it, giving zero steady-state error. **D** is the slope of the error: if the error is falling quickly, D reduces the output (braking); if rising quickly, it adds a push.

Increase…	Rise time	Overshoot	Settling time	Steady-state error
K_p	decreases	increases	small change	decreases
K_i	decreases	increases	increases	eliminated
K_d	small change	decreases	decreases	no effect (in theory)

H. Worked example (discrete PID in a PLC)

A PLC computes PID every sample time T_s (e.g. 1 s) using the rectangular approximation: $u[k] = K_p e[k] + K_i T_s \Sigma e + K_d(e[k] - e[k-1])/T_s$.

Worked example: one PID calculation
$K_p = 2$, $K_i = 0.5\ s^{-1}$, $K_d = 1$ s, $T_s = 1$ s. Errors: $e[0] = 4$, $e[1] = 3$, $e[2] = 1$.
P $= 2 \times 1 = 2$. **I** $= 0.5 \times 1 \times (4 + 3 + 1) = 4$. **D** $= 1 \times (1 - 3)/1 = -2$ (the error is falling, so D reduces the output).
$u[2] = 2 + 4 - 2 =$ **4** (in engineering units of the output). Unit check: each term has the units of u by the definition of K_p, K_i, K_d.

I–J. Tuning, problems and applications

Ziegler–Nichols (closed-loop) tuning: set I and D to zero, raise K_p until the loop oscillates steadily with period P_u; that K_p is the ultimate gain K_u. Then:

Controller	K_p	T_i	T_d
P	$0.5\ K_u$	–	–
PI	$0.45\ K_u$	$P_u/1.2$	–
PID	$0.6\ K_u$	$P_u/2$	$P_u/8$

These are starting values, normally refined by hand. For example, $K_u = 5$ and $P_u = 4$ s gives a PID of $K_p = 3$, $T_i = 2$ s, $T_d = 0.5$ s ($K_i = K_p/T_i = 1.5$, $K_d = K_p T_d = 1.5$).

- **Integral windup:** if the actuator saturates (valve fully open), the integral keeps growing and causes large overshoot afterwards; use anti-windup (clamp or back-calculation).
- **Derivative noise and kick:** D amplifies measurement noise; use a filter and apply D to the measurement rather than the error to avoid ‘kick’ at setpoint changes.
- **Dead time** limits achievable gains; very noisy or highly non-linear loops need other strategies (cascade, feed-forward, gain scheduling).
- **Applications:** furnace and oven temperature, boiler pressure, flow and level, motor speed in VFDs, robot joints, pH control. PID blocks are built into PLCs and VFDs.

Common misconception
Setting K_i very large to remove error quickly. Integral action is slow by nature; too much causes overshoot and oscillation. Increase gradually.

Unit 3, Question 6: PLC architecture

Exam question – Unit 3 Q6
Describe the architecture of a PLC with a neat block diagram.

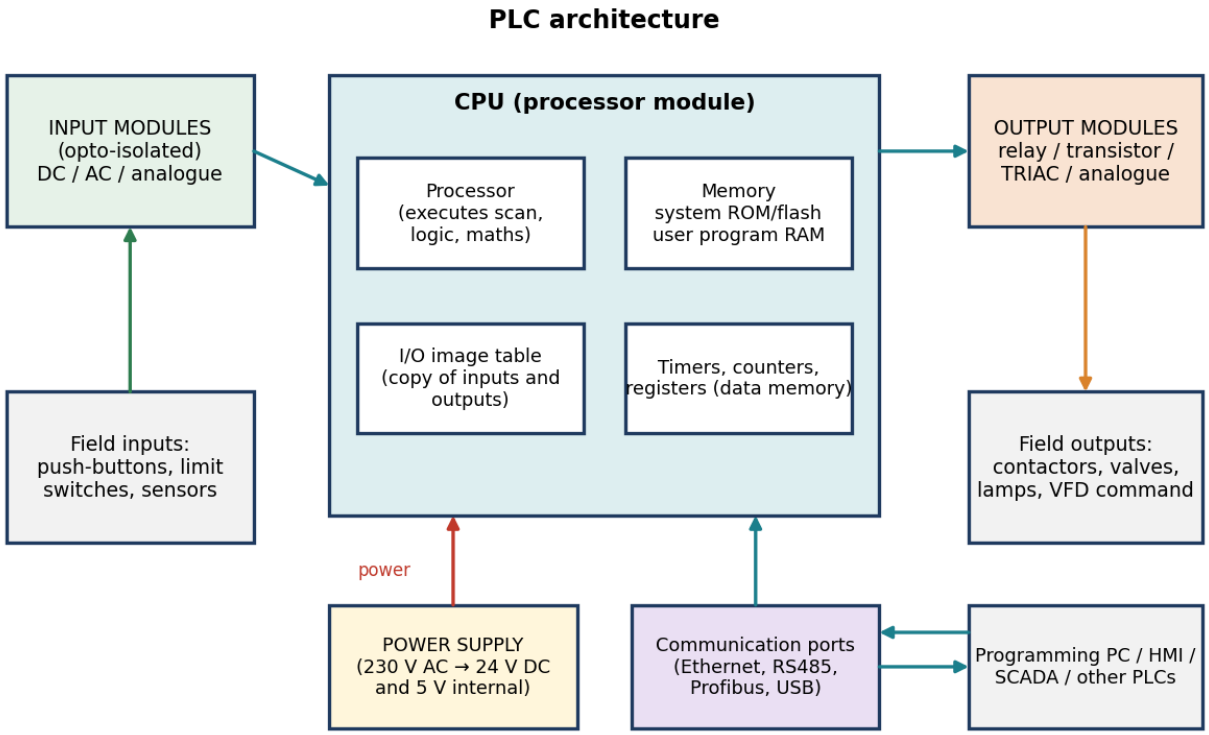


Figure 64. *PLC architecture: power supply, CPU with memory, input and output modules, communication interface and programming device.*

Block	Function
Power supply	converts 230 V AC to 24 V DC (and 5 V internal) for the CPU and modules; protects against surges; may have battery back-up
CPU (processor)	runs the operating system and the scan; executes the user program; handles diagnostics
Memory	**system memory** (ROM/flash, holds the operating system); **user program memory** (flash or battery-backed RAM); **data memory** (I/O image table, timers, counters, registers)
Input modules	accept signals from push-buttons, switches and sensors (digital 24 V DC / 230 V AC, analogue 4–20 mA or 0–10 V). Use **opto-isolators** to separate field voltages from the CPU
Output modules	drive actuators: **relay** outputs (AC/DC, slow, high current), **transistor** (DC, fast), **TRIAC** (AC); analogue outputs for VFD speed reference or valve position
Communication interface	Ethernet, RS232/RS485 (Modbus), Profibus, CAN etc. to connect to SCADA, HMI, drives and other PLCs (Unit 5)
Programming device	PC software to write, download and monitor the program in ladder logic, FBD, ST or SFC (IEC 61131-3)
Rack / backplane	bus that links the CPU and modules (modular PLCs)

The scan cycle

A PLC runs a continuous **scan cycle**: (1) it copies the status of all inputs into the input image table; (2) it executes the program from top to bottom, left to right, rung by rung, using the image table values; (3) it writes the results to the output image table and then to the output modules; (4) it performs housekeeping (communications, diagnostics). The scan time is typically 1–50 ms, so very short input pulses (shorter than the scan) can be missed unless special fast-input features are used.

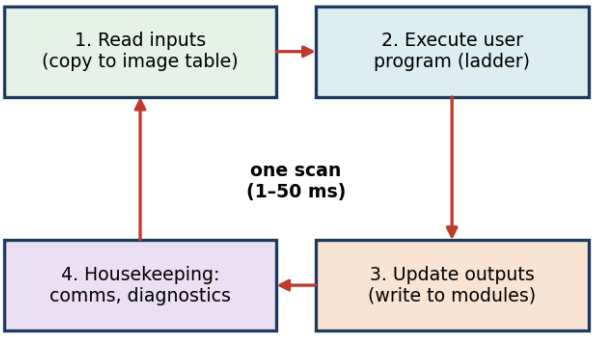


Figure 65. *The PLC scan cycle.*

- **Types:** *compact (fixed)* PLC with built-in I/O for small machines, and *modular* PLC with racks for large plants.
- **Digital input sinking/sourcing, and relay outputs** must match the field device voltage and current; relays handle larger loads but wear out.

Worked example: tracing a rung
Start/stop motor rung: Start (NO, I0.1) in parallel with the seal-in contact Q0.0, in series with Stop (NC, I0.2), driving coil Q0.0.
1. All inputs off: Q0.0 = 0. 2. Press Start: I0.1 = 1 ⇒ path true ⇒ Q0.0 = 1 (motor contactor ON). 3. Release Start: the seal-in contact Q0.0 (now 1) keeps the path true. 4. Press Stop: the NC contact opens ⇒ Q0.0 = 0. This ‘latch’ is the most common PLC rung in industry.

Unit 3, Question 7: SCADA in industrial process control

Exam question – Unit 3 Q7
Explain the role of SCADA in industrial process control with examples and hierarchical block diagram.

SCADA sits above the PLCs and RTUs (Remote Terminal Units). The PLCs do fast local control; SCADA gives the operator a *window into the whole process* and the ability to adjust it. It does not usually close fast loops itself.

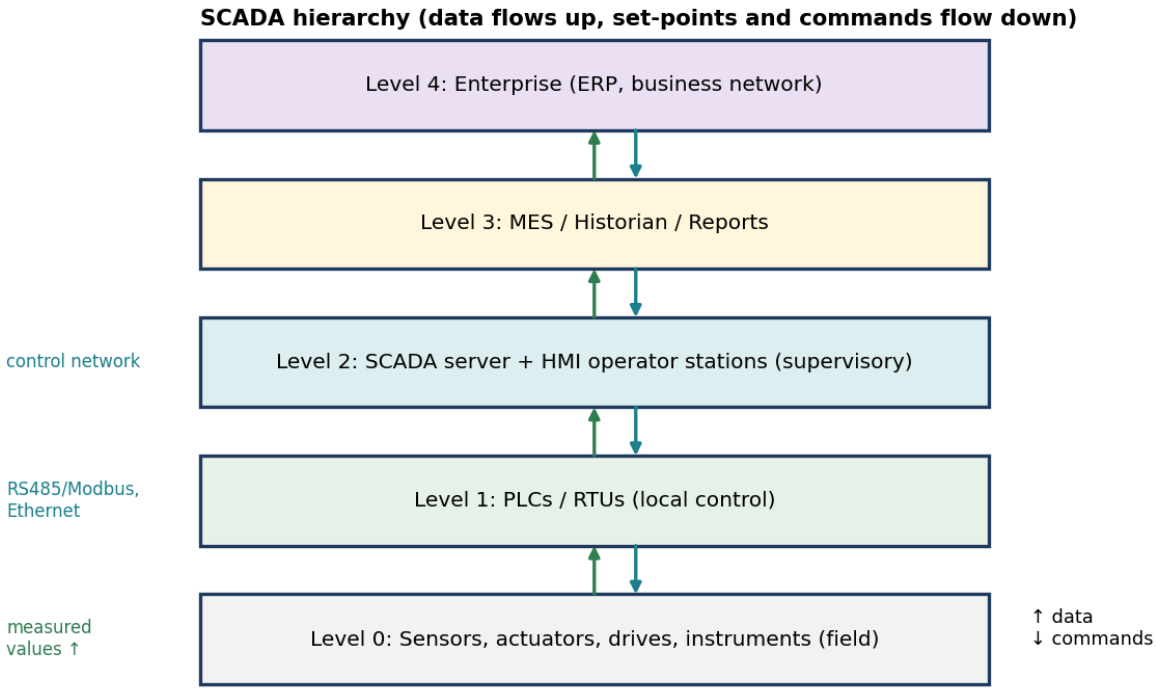


Figure 66. *Hierarchical block diagram of an industrial control system with SCADA at supervisory level 2.*

Function of SCADA	Description
Data acquisition	polls PLCs/RTUs for tag values (levels, flows, states) via Modbus, Ethernet, etc.
Visualisation (HMI)	graphical mimic diagrams, trend charts, status colours
Supervisory control	operator changes setpoints, starts/stops equipment remotely
Alarm management	detects out-of-range values and equipment faults, prioritises, logs and acknowledges alarms
Data logging and historian	stores time-stamped data for reports, audits, analytics
Reporting and integration	production reports, links to MES/ERP, remote access with security

- **Example 1: municipal water supply.** RTUs at pumping stations and reservoirs send tank level and pump status to a central SCADA server by radio/GSM; the operator starts a pump remotely when the level is low and receives an alarm for a low chlorine reading.
- **Example 2: factory conveyor line.** The PLC controls motors; SCADA displays production count, motor currents and faults, and logs downtime.
- **Example 3: electrical substation.** SCADA monitors breakers and voltages and trips feeders on command.
- **Security:** SCADA connected to the internet must use firewalls, segmentation and authentication, because it controls physical processes.

SCADA versus DAS. A data acquisition system (Unit 4) collects and digitises sensor data; SCADA adds wide-area communication, visualisation, alarms, logging and *command capability*. SCADA includes data acquisition, but it is more than a DAS.

Unit 3, Question 8: Role of VFD in variable-speed motor drives

Exam question – Unit 3 Q8
Analyze the role of VFD in variable speed motor drives with suitable block diagram.

A–D. The problem and the idea

An AC induction motor runs at nearly fixed speed set by the supply frequency. Throttling a pump with a valve (or using dampers on fans) wastes energy. A **VFD (Variable Frequency Drive)**, also called an AC drive or inverter drive, changes the **frequency and voltage** supplied to the motor so that its speed can be set anywhere over a wide range, efficiently.

$$N_s = \frac{120\,f}{P} \qquad s = \frac{N_s - N}{N_s} \qquad N = N_s(1 - s)$$

N_s = synchronous speed (rpm), f = supply frequency (Hz), P = number of poles, s = slip (typically 2–5%), N = rotor speed. Speed is controlled through f.

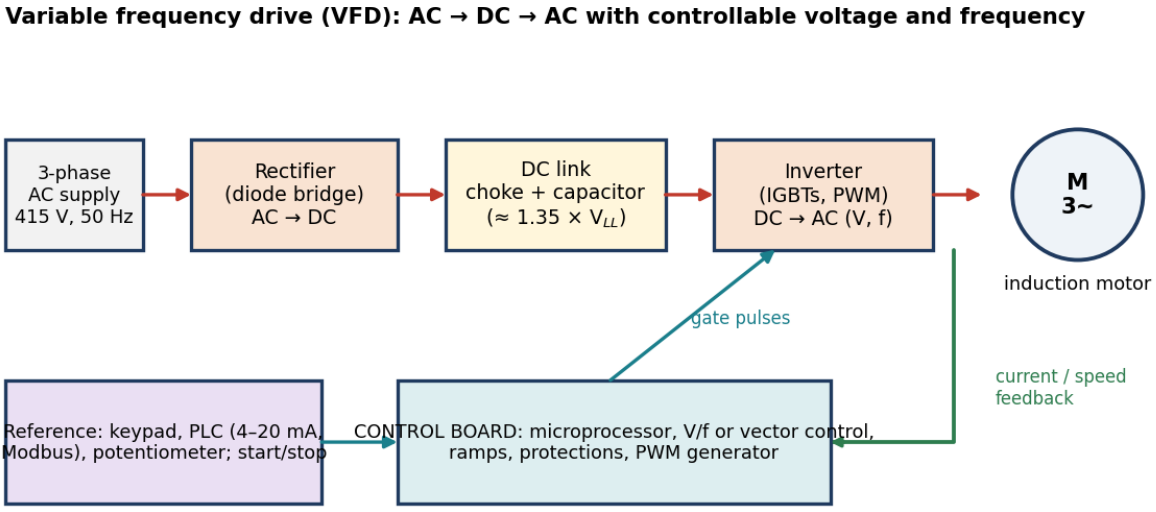


Figure 67. *VFD block diagram: rectifier → DC link → PWM inverter → motor, controlled by a microprocessor board that receives commands and feedback.*

1. **Rectifier:** converts the 3-phase AC input into DC (diode bridge: DC ≈ 1.35 × line voltage, e.g. 560 V from 415 V).
2. **DC link:** capacitor (and sometimes choke) smooths the DC and stores energy.
3. **Inverter:** six IGBTs switch with PWM (Unit 2 Q7) to produce a three-phase AC of chosen frequency and voltage.
4. **Control board:** receives the speed reference (keypad, 4–20 mA, PLC), applies V/f or vector control, ramps, current limiting and protection (overload, over/under-voltage, over-temperature).
5. **Feedback** (current sensors, encoder in closed-loop vector control) improves accuracy of speed and torque.

G. V/f control and the affinity laws

The motor's magnetic flux is proportional to V/f. To keep the flux (and hence torque capability) constant, the VFD keeps **V/f constant**: if the frequency is halved, the voltage is halved. At and above the base frequency the voltage can no longer rise (it is at its rated value) so the drive enters the field-weakening region: constant power, falling torque.

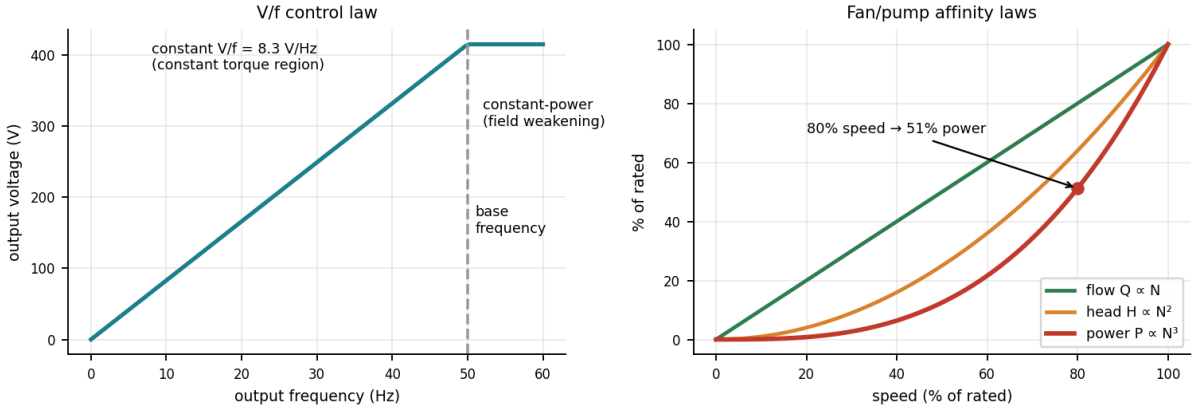


Figure 68. *Left: V/f control. Right: affinity laws for centrifugal pumps and fans; a small speed reduction gives a large power saving.*

$$\frac{Q_2}{Q_1} = \frac{N_2}{N_1} \qquad \frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2 \qquad \frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$$

Worked example: speed and energy
Speed: 4-pole motor, f = 30 Hz: N_s = 120 × 30/4 = 900 rpm; with slip 3%, N = 900 × 0.97 = **873 rpm**.
V/f: a 415 V / 50 Hz motor: 415/50 = 8.3 V/Hz; at 30 Hz the VFD supplies 8.3 × 30 = **249 V**.
Energy saving (illustrative, assumed data): a 15 kW pump motor runs 6000 h a year. At 80% speed: P_2 = 15 × 0.8³ = 15 × 0.512 = **7.68 kW**. Saving = 15 − 7.68 = 7.32 kW × 6000 h = **43 920 kWh/year** (about 49% less energy), compared with the theoretical full-speed-with-throttling case; real savings depend on the pump curve and the system.

I–J. Advantages, issues and links to other topics

- **Advantages:** energy saving on fans/pumps, **soft start** (reduces starting current from 6–7 × I_{rated} to about 1–1.5 ×; reduces voltage sags, Unit 1 Q5), controlled acceleration/deceleration (ramps), speed and torque control, braking and regeneration options, built-in protection and PLC/Modbus communication.
- **Issues:** harmonics injected into the supply (Unit 1 Q5); EMI and high dV/dt on the motor cable (use shielded cable, filters; Unit 1 Q4); bearing currents; need to derate the motor at very low speed (cooling fan slows).
- **Role in the system:** the PLC decides the required speed (from a pressure or level sensor via a PID loop) and sends the reference to the VFD; SCADA displays speed, current, faults. The VFD is therefore the **actuator's power stage** in the closed loop of Chapter A.

Common misconception
'A VFD changes only the frequency.' It changes frequency *and* voltage together (V/f), otherwise the motor flux saturates or weakens.

Unit 3: End-of-Unit Revision Pack

Unit summary

- PLC: rugged programmable controller; reads inputs, executes program, updates outputs in a scan.
- Ladder contacts test bits: NO true when bit = 1; NC true when bit = 0; series = AND, parallel = OR.
- Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR, with truth tables and ladder equivalents.
- P handles present error, I removes steady-state error, D damps; PID: u = K_pe + K_i∫e + $K_d$de/dt.
- Closed-loop transfer function: C/R = G/(1+GH).
- SCADA: supervisory, wide-area; DCS: distributed, plant-level continuous control.
- PLC architecture: power supply, CPU + memory, I/O modules, communications, programming device.
- VFD: rectifier + DC link + inverter + control; V/f constant; affinity laws give energy savings.

Formula and equation sheet

Closed-loop transfer function

$$\frac{C}{R} = \frac{G}{1 + GH}$$

PID (time domain)

$$u = K_p e + K_i \int e\,dt + K_d \frac{de}{dt}$$

PID (s domain)

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

Ziegler–Nichols PID

$$K_p = 0.6K_u,\ T_i = \frac{P_u}{2},\ T_d = \frac{P_u}{8}$$

Synchronous speed

$$N_s = \frac{120f}{P}$$

Slip

$$s = \frac{N_s - N}{N_s}$$

V/f constant

$$\frac{V}{f} = \text{constant}$$

Affinity law (power)

$$\frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$$

Diagram-revision sheet

Practise drawing each of these from memory, labelling every component, then compare with the figure.

Figure	What to draw and label
Q1	Seven gate symbols; seven ladder rungs
Q2	Step responses P vs PI vs PID
Q3	DCS architecture
Q4	Open-loop and closed-loop block diagrams
Q5	PID block diagram
Q6	PLC architecture and scan cycle
Q7	SCADA hierarchy
Q8	VFD block diagram; V/f curve

Glossary of important terms

Term	Meaning
PLC	Programmable logic controller
Rung	One line of ladder logic
NO / NC contact	Contact true when bit = 1 / bit = 0
Coil	Ladder output element
Truth table	Output for every input combination
Setpoint	Desired value
Error	Setpoint − measured
Feedback	Measured output returned to the input
Transfer function	Output/input in the s-domain
Windup	Integral growing while the actuator saturates
SCADA	Supervisory control and data acquisition
DCS	Distributed control system
RTU	Remote terminal unit
HMI	Human–machine interface
VFD	Variable frequency drive
Slip	Relative speed difference, rotor vs rotating field
Scan cycle	Repeated read–execute–write loop

Conceptual and numerical review questions

1. A motor must start only if the guard is closed (A) and the E-stop is not pressed (B, 1 = pressed). Write Y and draw the rung.
2. $G(s) = 20/(s+4)$, $H = 1$. Find the closed-loop transfer function and DC gain.
3. $e = 5$, $\int e = 8$, $de/dt = -2$, $K_p = 1.5$, $K_i = 0.25$, $K_d = 2$. Find u.
4. A 6-pole motor runs from a VFD at 40 Hz. Find N_s and the V/f voltage for a 415 V/50 Hz motor.
5. Speed of a fan is reduced to 70%. By how much does power fall?
6. State two differences between SCADA and DCS.

Fully explained answers
1. Y = A·B: A (NO contact) in series with B (NC contact).
2. T = 20/(s+4+20) = **20/(s+24)**; DC gain 20/24 = 0.83.
3. u = 1.5×5 + 0.25×8 + 2×(−2) = 7.5 + 2 − 4 = **5.5**.
4. N_s = 120×40/6 = **800 rpm**; V = 8.3×40 = **332 V**.
5. 0.7^3 = 0.343, so power falls to 34% (a 66% saving).
6. SCADA is wide-area supervisory with control in PLCs/RTUs; DCS is local, integrated, redundant continuous-process control.

Examination-answer guide for the original questions

This guide complements the teaching chapters. It shows how to organise each original question as a complete exam answer.

Question	Structure of a strong answer
Q1	Need for PLC (table of requirements) → logic and ladder basics → for each gate: symbol, Boolean expression, truth table, ladder rung (in the order NAND, XNOR, NOR, XOR, AND, OR, NOT).
Q2	Definition of P, PI, PID (one line) → table with ≥ 7 parameters (law, transfer function, error, speed, overshoot, noise, tuning, application) → one response graph.
Q3	Definitions → table with ≥ 7 parameters → DCS block diagram → layer table → example (power plant, refinery).
Q4	Definitions → open vs closed block diagrams → comparison table → derivation C/R = G/(1+GH) in 7 steps → numerical example.
Q5	Block diagram → equation → effect of P, I, D → tuning → problems (windup) → applications.
Q6	Block diagram → function of each block → scan cycle → I/O types.
Q7	Hierarchical diagram → functions → two examples → relation to PLC and DAS.
Q8	Block diagram → function of each block → V/f → speed formula → advantages (soft start, saving) → example.

UNIT 4: Sensors, Signal Conditioning and Data Acquisition

Note on the source wording
The question bank numbers Unit 4 as 1, 2, 3, **3**, 4, 5, 6, 7, 8, 9: the number 3 appears twice (temperature DAS, then accelerometers/gyroscopes). To keep every item and avoid renumbering your source, this book labels them **Unit 4 Q3 (first)** for temperature and **Unit 4 Q3 (second)** for accelerometers and gyroscopes; the rest keep the printed numbers 4–9. That makes ten items in total.

The measurement chain. Every question in this unit is a part of one chain. Learn it once, and every answer becomes an explanation of one or more blocks.

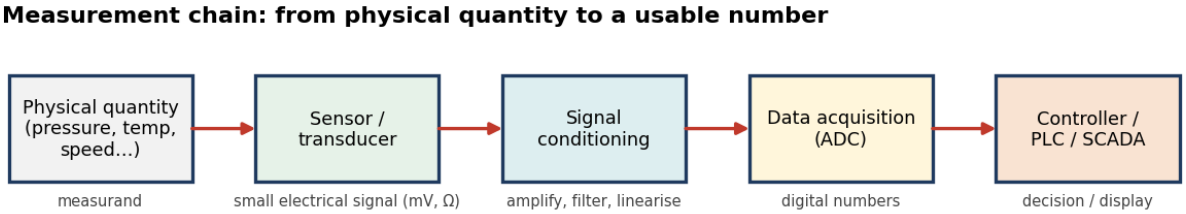


Figure 69. The measurement chain used throughout Unit 4.

Vocabulary: the **measurand** is the physical quantity being measured; the **sensor** detects it; **signal conditioning** makes the weak or awkward electrical signal suitable for conversion; the **ADC (Analogue-to-Digital Converter)** turns it into numbers; the **DAS** combines these and the processor; the **controller** (PLC) uses the numbers (Unit 3).

Unit 4, Question 1: Differences between sensor and transducer

Exam question – Unit 4 Q1
Write down differences between sensor and transducer.

Beginner view. A **sensor** is the ‘sense organ’: it detects something. A **transducer** is any device that converts one form of energy into another. In practice the two words overlap strongly, so textbooks draw the line slightly differently; the commonly accepted distinctions are:

Point	Sensor	Transducer
Definition	detects a physical, chemical or biological quantity and produces a measurable response (usually electrical)	converts energy from one form to another (e.g. pressure → voltage, current → sound)
Function	sensing / detecting	energy conversion (sensing element + converting stage)
Relationship	the sensing element is the front end	a complete transducer may include a sensor plus extra circuitry (conditioning, standardised output)
Output	often a small, raw signal (mV, change of resistance)	often a standardised signal (0–10 V, 4–20 mA, digital) when a ‘transmitter’
Direction	input is always a quantity to be measured	can convert either way: sensing (measurement) or actuating (e.g. loudspeaker, motor, heater)
Examples	thermocouple, LDR, strain gauge, proximity switch	pressure transmitter, LVDT with demodulator, microphone, loudspeaker, piezoelectric buzzer
Typical statement	every sensor is a kind of transducer	not every transducer is a sensor (an actuator is a transducer but does not sense)

Worked example: same device, two names
A thermocouple converts temperature (heat energy) into a small voltage. As a **sensor**, it detects temperature; as a **transducer**, it converts thermal to electrical energy. A **loudspeaker** converts electrical to sound energy: a transducer, but not a sensor. A **pressure transmitter** (diaphragm sensor + bridge + amplifier + 4–20 mA output) is a transducer package built around a sensor.

Common misconception
Insisting that the terms are always different. Many manufacturers use them interchangeably; in an exam, state the definitions, give the difference table, and add that 'a sensor is a transducer when its output is electrical'.

Unit 4, Question 2: Signal conditioning and DAS for pressure sensors

Exam question – Unit 4 Q2
Explain signal conditioning requirements and DAS (Data acquisition System) for pressure sensors with proper block diagram.

A–D. Pressure and its sensors

Pressure = force per unit area (pascal, Pa; 1 bar = 10^5 Pa). **Gauge** pressure is relative to the atmosphere; **absolute** pressure is relative to vacuum; **differential** pressure is the difference between two ports (used for flow and level). Common sensors: **strain-gauge/piezoresistive** (most common), **capacitive**, **piezoelectric** (dynamic pressure only), **Bourdon tube** with LVDT or potentiometer.

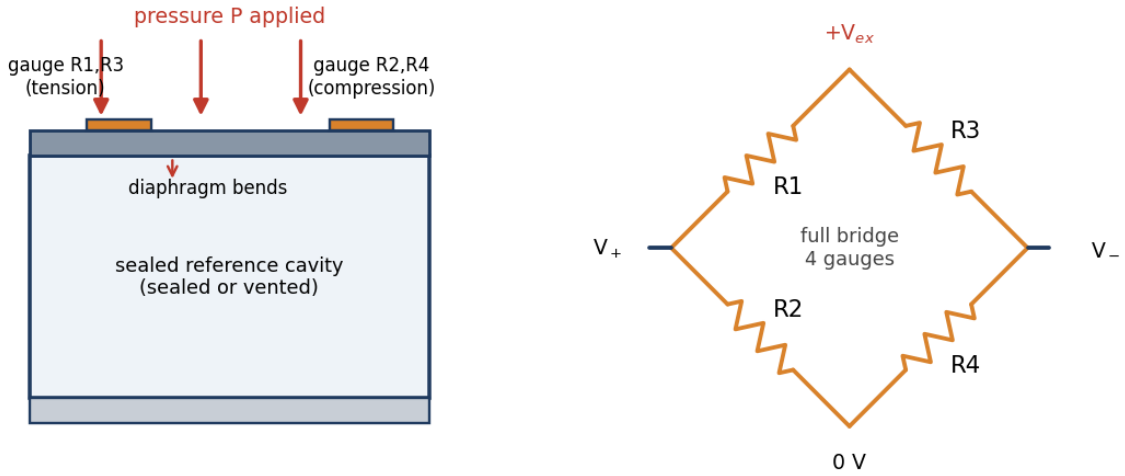


Figure 70. Left: a diaphragm pressure sensor. Pressure bends the diaphragm; strain gauges on it change resistance (two in tension, two in compression). Right: the four gauges form a full Wheatstone bridge.

Working. A strain gauge's resistance changes with strain ε as $\Delta R/R = GF \times \varepsilon$ (GF = gauge factor, about 2 for metal foil). The diaphragm strain is proportional to pressure, so $\Delta R/R \propto P$. The gauges are connected in a bridge (see Question 9), which converts the tiny $\Delta R/R$ into a voltage proportional to the excitation.

E–G. Signal conditioning requirements

Requirement	Why it is needed (pressure sensor)
Excitation	the bridge needs a stable DC supply (5–10 V); output scales with it (ratiometric)
Amplification	full-scale bridge output is only ≈ 10–100 mV; amplify (instrumentation amplifier, high CMRR) to match the ADC input (e.g. 0–5 V)
Zero (offset) and span adjustment	remove the bridge's offset at zero pressure; set full-scale gain

Requirement	Why it is needed (pressure sensor)
Temperature compensation	gauge resistance and sensitivity drift with temperature
Filtering	low-pass filter to remove pump pulsation, vibration and EMI pick-up (Unit 1 Q4)
Linearisation	correct small non-linearity of sensor or amplifier
Isolation	galvanic isolation avoids ground loops and protects the DAS
Current-loop conversion	industrial transmitters convert to 4–20 mA, which is immune to noise and long cables (2-wire loop powered; live zero at 4 mA detects a broken wire)
Overpressure / surge protection	snubber, protective diodes

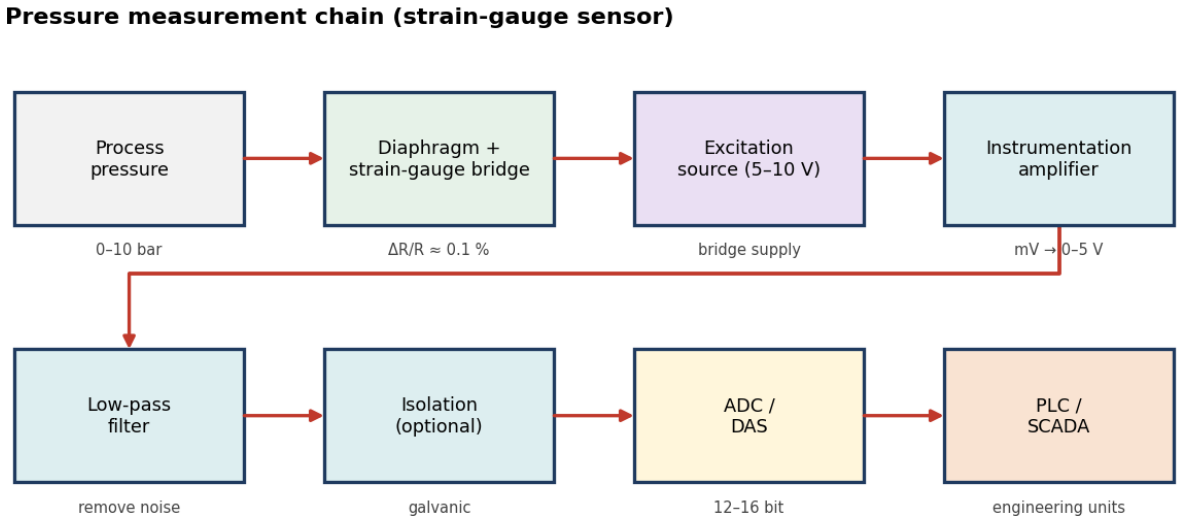


Figure 71. *Block diagram: pressure measurement chain and data acquisition.*

Equations and worked example

$$V_{out} = S \cdot V_{ex} \cdot \frac{P}{P_{FS}} \qquad G = \frac{V_{ADC,FS}}{V_{out,FS}} \qquad LSB = \frac{V_{ADC,FS}}{2^N}$$

S = sensitivity in mV per volt of excitation at full scale (e.g. 2 mV/V), V_{ex} = excitation, P_{FS} = full-scale pressure, G = amplifier gain, N = ADC bits, LSB = least significant bit (smallest ADC step).

Worked example: pressure DAS
Sensor: 0–10 bar, S = 2 mV/V, V_{ex} = 10 V ⇒ full-scale output = 2 × 10 = **20 mV**.
ADC input range 0–5 V ⇒ gain G = 5 V/20 mV = **250**.
At P = 6 bar: V_{out} = 20 mV × 6/10 = 12 mV; amplified = 12 mV × 250 = **3.0 V**.
12-bit ADC: LSB = 5/4096 = 1.22 mV ⇒ pressure resolution = 10 bar/4096 = **0.0024 bar**. The ADC output code at 3.0 V = 3.0/5 × 4096 = **2458**.
4–20 mA alternative: a transmitter outputs I = 4 + 16 × (P/10) mA ⇒ at 6 bar: 4 + 9.6 = **13.6 mA**. A 250 Ω resistor in the PLC input converts it to 1–5 V (3.4 V here). At exactly 12 mA, P = 5 bar.

Data acquisition: after conditioning, a multiplexer selects channels, a sample-and-hold circuit freezes the value, the ADC digitises it, and the processor scales it to bar, logs it and sends it to the PLC/SCADA (Question 6 explains DAS in general). **Calibration:** apply two known pressures (zero and span) and set offset and gain; periodically repeat because drift occurs.

- **Industrial uses:** hydraulic and pneumatic systems, boiler and steam lines, tank level by hydrostatic pressure, flow by differential pressure, compressors, process plants.

Unit 4, Question 3 (first): Signal conditioning and DAS for temperature sensors

Exam question – Unit 4 Q3 (first)
Explain signal conditioning requirements and DAS (Data acquisition System) for temperature sensors with proper block diagram.

A–D. The temperature sensors

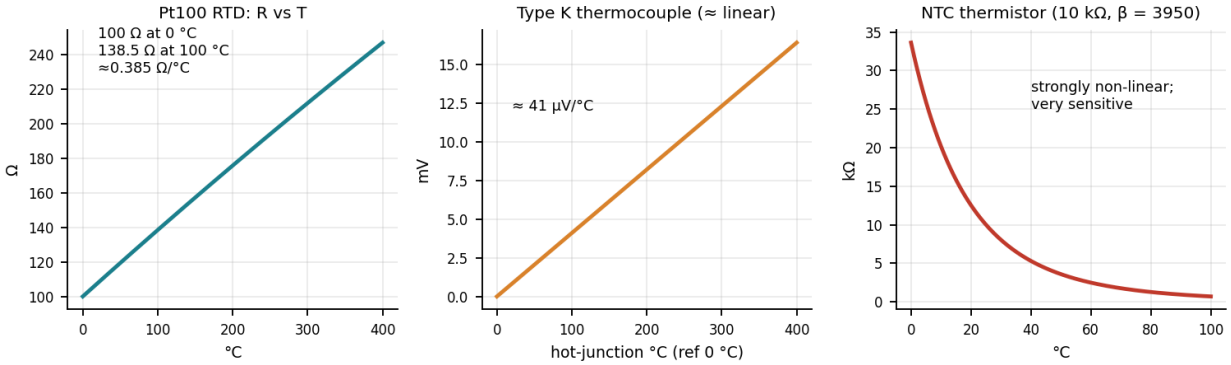


Figure 72. Characteristics of the three main electrical temperature sensors (schematic values).

Sensor	Principle	Range (typical)	Output	Conditioning needs
Thermocouple	Seebeck effect: two dissimilar metals joined; voltage ∝ $T_{hot} - T_{ref}$	−200 to 1300 °C (type K)	≈ 41 µV/°C (K)	high-gain low-noise amplifier; **cold-junction compensation**; linearisation; open-circuit detection
RTD (Pt100)	resistance of platinum rises with T: $R = R_0(1 + AT + BT^2)$	−200 to 650 °C	0.385 Ω/°C	constant excitation current (≈ 1 mA) or bridge; **lead-resistance compensation** (3- or 4-wire); avoid self-heating
Thermistor (NTC)	semiconductor resistance falls steeply with T	−50 to 150 °C	large, non-linear	voltage divider or bridge; strong linearisation (Steinhart–Hart)
IC sensor (e.g. 10 mV/°C)	silicon junction voltage ∝ T	−55 to 150 °C	linear voltage	little needed; limited range

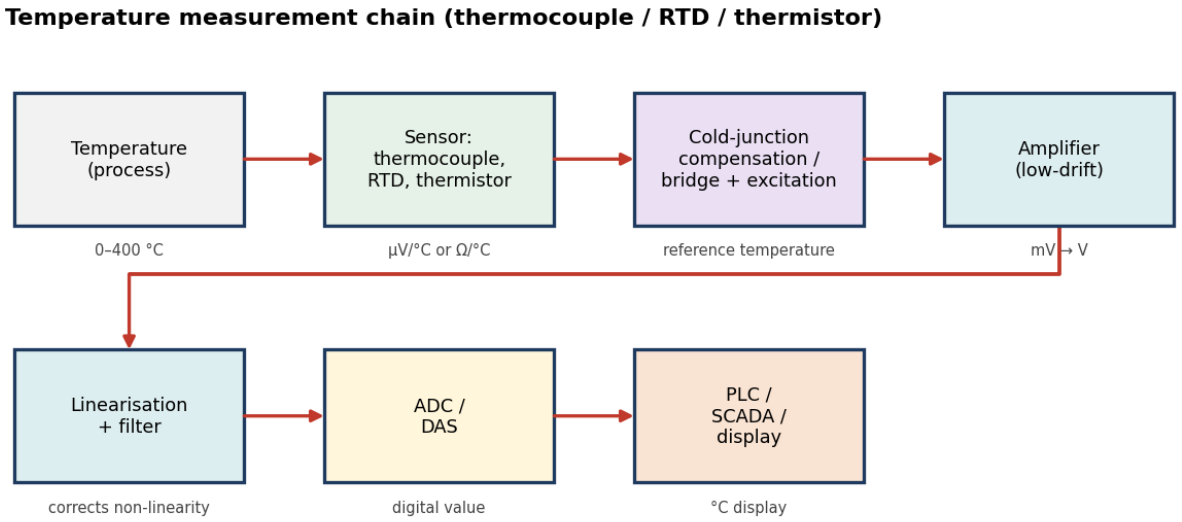


Figure 73. Block diagram: temperature measurement chain and data acquisition.

E–H. Signal-conditioning requirements, with reasons and numbers

- **Amplification:** a thermocouple at 400 °C gives only 16.4 mV; ADC inputs are 0–5 V ⇒ gain ≈ 300.
- **Cold-junction compensation (CJC):** a thermocouple measures the *difference* between the hot and the reference (terminal) junction; the terminal block temperature is measured (by an IC sensor or RTD) and its equivalent voltage added electronically.
- **Excitation and bridge (RTD, thermistor):** they are resistances, so they need current or a bridge to give a voltage.
- **Lead-wire compensation:** in a 2-wire RTD the lead resistance adds to the sensor. With 0.5 Ω per lead, 1 Ω extra means 1/0.385 = 2.6 °C error. A 3-wire connection (in a bridge) or 4-wire (separate current and voltage leads) cancels it.
- **Linearisation:** by polynomial or look-up table in software.
- **Filtering, isolation, noise protection:** thermocouple leads are long and pick up EMI; use shielded twisted pair, a low-pass filter (temperatures change slowly) and isolation.
- **Burn-out detection:** pull the amplifier input high so that a broken thermocouple reads full scale and raises an alarm.
- **Temperature transmitters:** a head-mounted transmitter does all this and outputs 4–20 mA.

Worked example: thermocouple DAS
Type K, S ≈ 41 µV/°C (approximation; real tables are non-linear). Cold-junction at 25 °C. Measured voltage = 10.0 mV. Find the hot-junction temperature.
$V = S(T_{hot} - T_{ref}) \Rightarrow T_{hot} - 25 = 10.0\ mV/0.041\ mV/°C = 243.9 \Rightarrow T_{hot} =$ **268.9 °C**. (Ignoring CJC would give only 243.9 °C: a 25 °C error.)
Range 0–400 °C ⇒ full-scale voltage = 400 × 0.041 = 16.4 mV; gain for 0–5 V = 5/0.0164 = **305**. 12-bit ADC ⇒ 400/4096 = **0.098 °C per count**.
RTD: Pt100 at 100 °C = 138.5 Ω; with 1 mA excitation, V = 138.5 mV; each 1 °C changes it by 0.385 mV.

Common misconception
Reading the thermocouple voltage as if it were an absolute temperature. It depends on both junctions; without CJC the temperature is wrong by about the ambient temperature.

Unit 4, Question 3 (second): Accelerometers and gyroscopes

Exam question – Unit 4 Q3 (second)
Discuss the operating principles and applications of accelerometers and gyroscopes using diagrams for explanation.

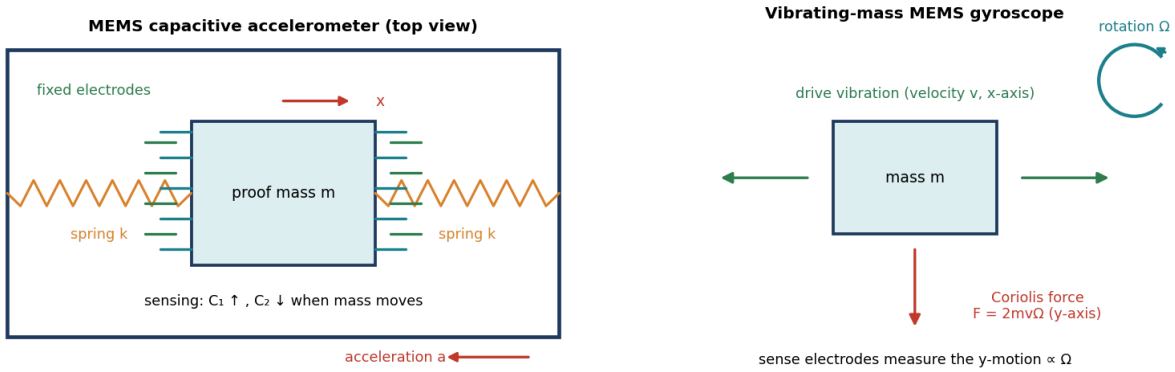


Figure 74. *Left: MEMS capacitive accelerometer. Right: vibrating-mass gyroscope using the Coriolis effect.*

Accelerometer: principle

Acceleration is the rate of change of velocity (m/s²; 1 g = 9.81 m/s²). A **proof mass** m is held by springs (stiffness k) inside a frame. When the frame accelerates with a, the mass lags behind (inertia): the spring stretches until its force balances. By Newton's law F = ma and Hooke's law F = kx:

$$x = \frac{ma}{k} \qquad f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad C = \frac{\varepsilon A}{d} \Rightarrow \Delta C \approx C_0 \frac{x}{d_0}$$

The displacement x is proportional to a (below the natural frequency f_n). In a **MEMS capacitive** accelerometer, the movable mass carries comb fingers between fixed electrodes; the gaps change, so the capacitances change (C_1 rises, C_2 falls), and the electronics convert ΔC to a voltage. A **piezoelectric** accelerometer instead uses a crystal that generates charge when squeezed by the mass: it responds to fast vibration but not to constant acceleration (no DC response).

Worked example: MEMS accelerometer numbers (assumed values)
m = 2 μg = 2×10^{-9} kg, k = 2 N/m, a = 10 g = 98.1 m/s².
x = ma/k = $(2\times10^{-9} \times 98.1)/2 = 9.8\times10^{-8}$ m = **98 nm**. $f_n = (1/2\pi)\sqrt{(2/2\times10^{-9})} = (1/2\pi)(31\,623) \approx$ **5.0 kHz**.
With C_0 = 1 pF and gap d_0 = 2 μm: ΔC ≈ 1 pF × 0.098/2 = **0.049 pF** (very small, hence on-chip electronics).
Tilt measurement: at rest the sensor reads gravity; if the sensitive axis reads 0.5 g, tilt = $\sin^{-1}(0.5)$ = **30°**.

Gyroscope: principle

A **gyroscope** measures **angular rate** (rotation speed, in °/s). A MEMS gyro keeps a mass vibrating at velocity v (drive mode). If the frame rotates at Ω about the perpendicular axis, a **Coriolis force** F = 2mvΩ pushes the mass sideways (sense mode); the sideways motion, measured capacitively, is proportional to Ω. The angle is obtained by integrating Ω over time: θ = ∫Ω dt, so small rate errors accumulate as **drift**. Older types are spinning-rotor (mechanical) and optical (ring-laser, fibre-optic) gyros.

Worked example: Coriolis force
m = 2 μg, v = 0.1 m/s, Ω = 100 °/s = 1.745 rad/s: F = 2 × 2×10^{-9} × 0.1 × 1.745 = **7×10^{-10} N** (nanonewton level, hence sensitive electronics).

	Accelerometer	Gyroscope
Measures	linear acceleration (and tilt via gravity)	angular rate of rotation
Principle	inertia of a proof mass (F = ma)	Coriolis force on a vibrating mass; or optical Sagnac effect
Output	g, m/s²	°/s
Drift issue	bias, vibration rectification	integration drift of angle
Typical sensors	MEMS capacitive, piezoelectric, piezoresistive	MEMS vibratory, fibre-optic gyro

Applications

- **Accelerometers:** machine vibration and bearing condition monitoring (predictive maintenance, Unit 5 Q5), tilt and levelling, airbag and crash sensing, shock recorders, phone orientation, smart-city structural health monitoring.
- **Gyroscopes:** drones and robots (stabilisation), vehicle stability control, camera stabilisation, navigation when GPS is lost, motion control of cranes and robots.
- **IMU (inertial measurement unit):** accelerometer + gyroscope (+ magnetometer) with sensor fusion (complementary or Kalman filter) to give a stable estimate of orientation: gyro for fast changes, accelerometer to correct long-term drift.

Unit 4, Question 4: Wireless sensor networks in Industry 4.0

Exam question – Unit 4 Q4
Explain the use of wireless sensor networks in Industry 4.0 (industrial automation) with proper block diagram.

A **wireless sensor network (WSN)** is a group of spatially distributed **sensor nodes** that sense physical quantities and send the data by radio, usually through several hops, to a **gateway**. Each node has: a sensor, a microcontroller, a radio, and a power source (battery or energy harvester).

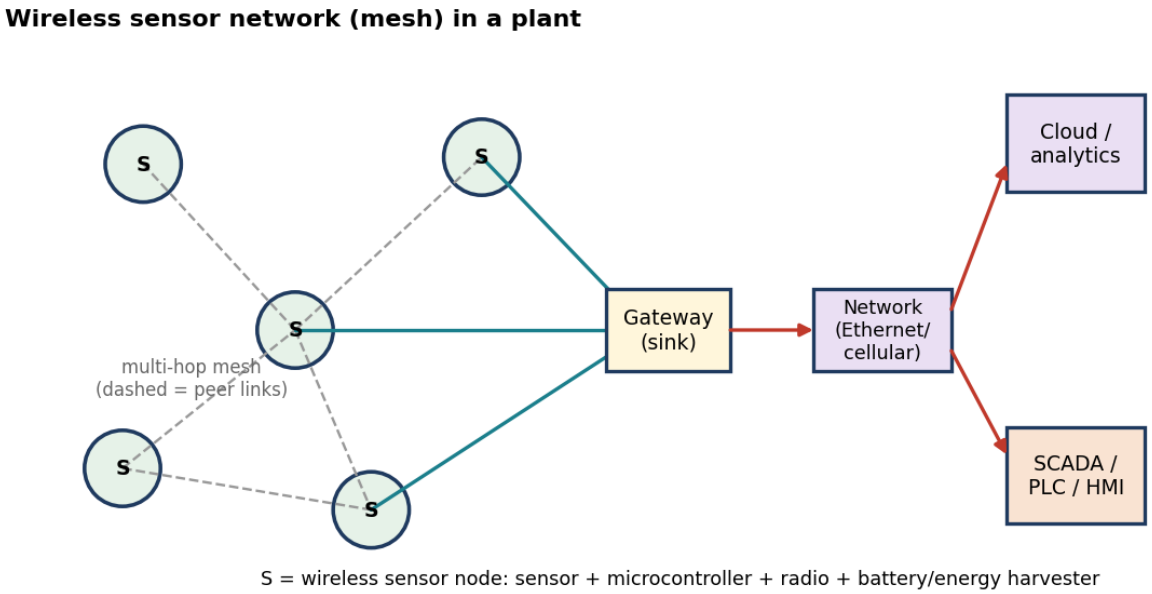


Figure 75. *Wireless sensor network feeding a gateway, a network and the cloud / SCADA in an Industry 4.0 plant.*

Element	Role
Sensor node	measures (temperature, vibration, pressure, humidity, current) and does simple local processing
Mesh/star topology	nodes forward each other's data (mesh) for coverage and resilience; star nodes connect directly to the gateway
Gateway (sink)	collects data, translates the wireless protocol to Ethernet/Modbus/OPC UA/MQTT, adds security
Backbone network	wired Ethernet, Wi-Fi or cellular to the plant network or cloud
Analytics, SCADA, PLC	display, store, alarm, and run predictive-maintenance models

Technologies (examples): IEEE 802.15.4-based protocols such as **WirelessHART** and ISA100.11a (industrial process monitoring), ZigBee/6LoWPAN (mesh), Bluetooth Low Energy (short range), Wi-Fi (high data rate, higher power), LoRaWAN (long range, low power, low data rate), and 4G/5G for cellular links.

- **Why use them in Industry 4.0:** retrofit monitoring on old machines without running cables, access to rotating or moving parts, fast deployment and low cost per point, flexible reconfiguration, large-scale data for analytics and predictive maintenance.
- **Applications:** condition monitoring of motors and pumps (vibration, temperature), tank-level and leak monitoring, environment and energy monitoring, asset tracking, worker safety.
- **Challenges:** limited battery energy, interference and reliability in metal-rich plants, latency (not suited to fast control loops), data security, standard interoperability.

Worked example: battery-life estimate (assumed values)
Node with a 2400 mAh battery. Sleep current 10 µA; active (measure + transmit) 20 mA for 0.1 s every 60 s.
Average current = 10 µA + 20 mA × (0.1/60) = 10 µA + 33.3 µA = **43.3 µA**.
Life ≈ 2400 mAh / 0.0433 mA ≈ 55 400 h ≈ **6.3 years** (ignoring battery self-discharge and temperature effects, which will shorten this in practice). The calculation shows why nodes sleep almost all the time.

Unit 4, Question 5: Proximity sensors

Exam question – Unit 4 Q5
Explain the working principles and applications of proximity sensors in automation industry.

A **proximity sensor** detects the presence (or absence) of an object **without physical contact**, within a defined **sensing distance (S_n)**. The output is normally a switching signal (ON/OFF, 24 V DC) wired to a PLC input.

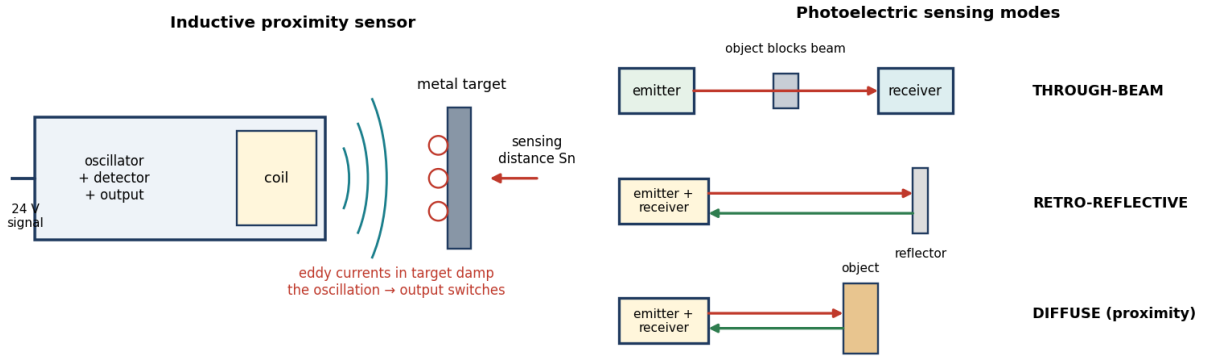


Figure 76. *Left: inductive proximity sensor; eddy currents in a metal target damp the oscillator. Right: the three photoelectric sensing modes.*

Type	Working principle	Detects	Typical range	Notes
Inductive	oscillator coil produces a high-frequency field; a metal target induces **eddy currents** that absorb energy; the amplitude drops and the output switches	metals only	1–40 mm	robust, immune to dirt and oil; range depends on metal
Capacitive	object changes the capacitance of the sensing electrodes (dielectric constant)	metals, plastics, liquids, powders, wood	3–30 mm	detects through thin non-metal walls (level through tank wall); sensitive to humidity
Photoelectric (optical)	emitter (LED/laser) and receiver; the object interrupts or reflects light (through-beam, retro-reflective, diffuse)	almost any opaque object	mm to tens of m	long range; affected by dirt and ambient light
Ultrasonic	transmits sound pulses; measures echo time: distance = ½ v t	solid, liquid, transparent objects	cm to several m	not affected by colour or transparency; slow
Magnetic (reed switch, Hall effect)	magnetic field of a magnet closes a reed switch or changes the Hall voltage	magnet-bearing parts	a few mm–cm	pneumatic cylinder position sensing

Worked example: effect of target material
An inductive sensor has rated S_n = 4 mm for mild steel. Typical reduction factors (indicative; check the datasheet): stainless steel ≈ 0.8, brass ≈ 0.4, aluminium ≈ 0.4, copper ≈ 0.3.
For aluminium: effective range ≈ 4 × 0.4 = **1.6 mm**. Mounting must therefore leave a safety margin (often set the working distance at about 80% of S_n).
Electrical output: 3-wire DC sensors are **PNP** (sourcing) or **NPN** (sinking), and NO or NC; this must match the PLC input type.

- **Applications in automation:** counting parts on conveyors, detecting a cylinder piston position, checking the presence of a cap or label, detecting bottle level through a wall (capacitive), robot gripper presence, machine-tool turret position, safety-gate switches, vehicle detection.
- **Advantages over limit switches (Unit 1 Q7):** no wear, high switching speed (kHz), sealed (IP67), long life.
- **Selection:** object material, distance, environment, speed, output type and supply voltage.

Unit 4, Question 6: Signal conditioning and data acquisition systems (SCADA) in brief

Exam question – Unit 4 Q6
Explain signal conditioning and data acquisition systems(SCADA) in brief.

Note on the source wording
The question writes ‘data acquisition systems (SCADA)’ as if they were identical. A **DAS** and **SCADA** are related but different: a DAS collects and digitises measurements; SCADA is a supervisory layer that adds communications, visualisation, alarms, logging and remote commands (Unit 3 Q7). Both are explained below, with a table of differences.

Signal conditioning

Signal conditioning is the set of circuits that convert a raw sensor signal into a signal that suits the next stage (usually an ADC).

Function	What it does	Example
Amplification	raises small signals to the ADC range	thermocouple 16 mV → 5 V
Attenuation	reduces large signals	divider for 0–100 V
Excitation	supplies current/voltage the sensor needs	bridge, RTD
Filtering	removes noise and aliasing frequencies	50 Hz notch, low-pass
Linearisation	corrects non-linear characteristic	thermocouple tables
Isolation	galvanic separation (opto/transformer)	safety, ground loops
Conversion	voltage ↔ current, frequency ↔ voltage	4–20 mA to 1–5 V
Compensation	corrects for temperature, lead resistance	cold junction, 3-wire RTD

Data acquisition system (DAS)

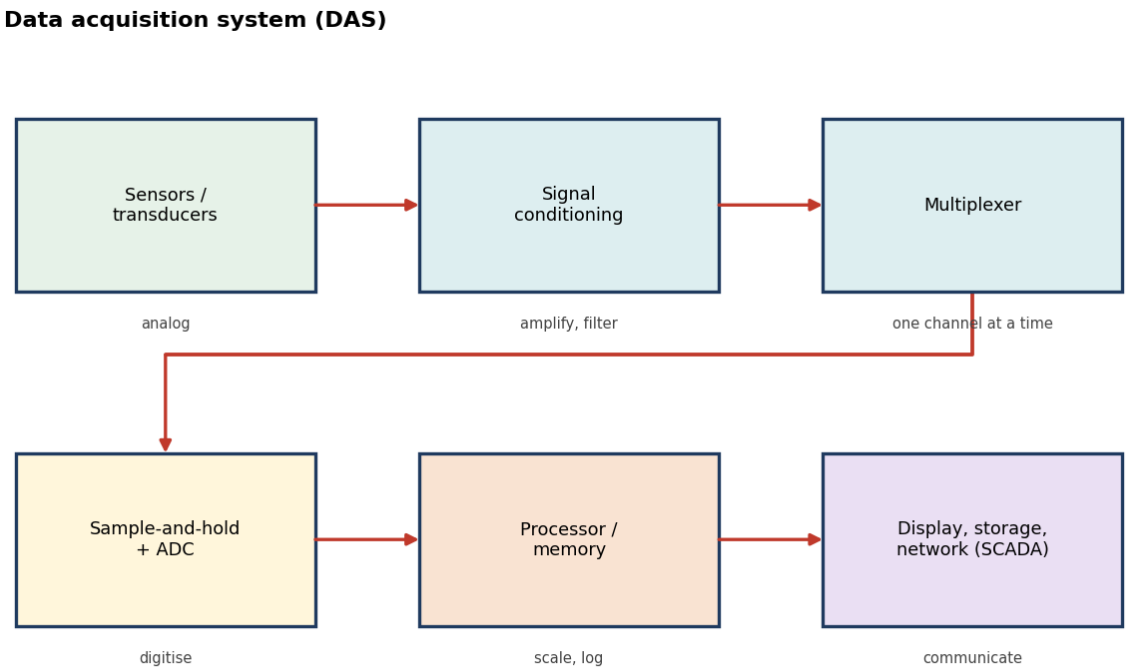


Figure 77. *Block diagram of a data acquisition system.*

A **DAS** measures several signals, digitises them, and processes, stores or sends the results. Main blocks: sensors → signal conditioning → **multiplexer** (selects one of many channels in turn) → **sample-and-hold** (holds the value constant during conversion) → **ADC** → processor/memory → display, storage or network.

$$LSB = \frac{V_{FS}}{2^N} \qquad \text{quantisation error} = \pm\frac{LSB}{2} \qquad f_s \geq 2f_{max}\ \text{(Nyquist)}$$

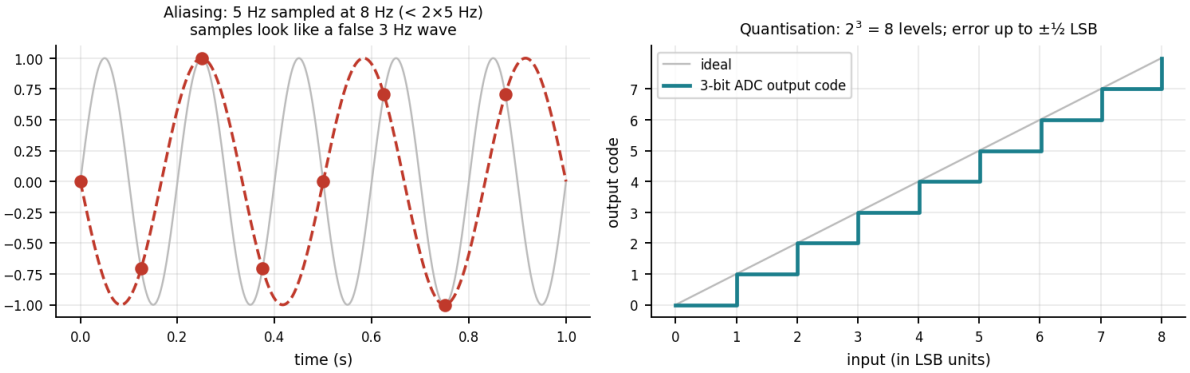


Figure 78. *Left: sampling below the Nyquist rate creates a false (aliased) low-frequency signal. Right: quantisation by a 3-bit ADC.*

Worked example: ADC and sampling
N = 12, V_{FS} = 10 V ⇒ 2^{12} = 4096 levels; LSB = 10/4096 = **2.44 mV**; error up to ±1.22 mV.
A machine vibration has components up to 1 kHz ⇒ minimum sampling rate 2 kHz (Nyquist); in practice 5–10 kHz, with an **anti-aliasing low-pass filter** before the ADC.
If the multiplexer scans 8 channels and each conversion takes 20 µs, one scan takes 160 µs, so each channel can be sampled up to 1/160 µs = 6.25 kHz.

Point	DAS	SCADA
Main purpose	measure, digitise, store data	supervise and control a process or site

Point	DAS	SCADA
Scope	a laboratory, machine or local measurement point	plant-wide or wide-area (water network, pipeline)
Control	normally none (monitoring)	supervisory commands, setpoints, alarms
Communication	local bus, USB, Ethernet	industrial protocols, radio, WAN
Software	acquisition and logging software	HMI, alarm server, historian, database
Relation	a DAS can be a part of SCADA at the data-collection level	SCADA uses PLC/RTU/DAS as its field data source

Unit 4, Question 7: Encoders, construction and applications in robotics

Exam question – Unit 4 Q7
Describe the construction and applications of encoders in robotic industry.

An **encoder** is a sensor that converts **angular (rotary) or linear position** into an electrical signal (digital pulses or a digital code). In robots, every joint needs to know its angle; the encoder provides it.

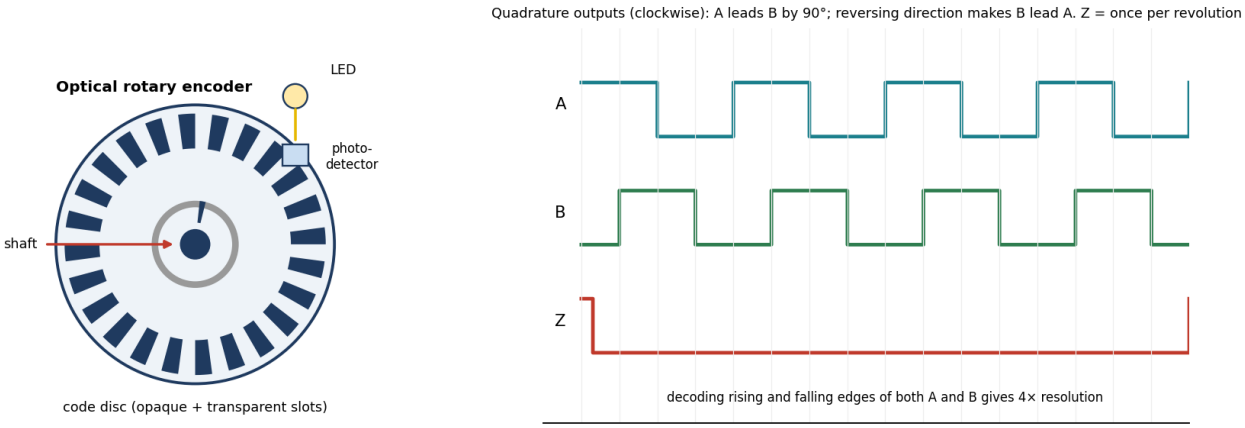


Figure 79. Left: construction of an optical rotary encoder. Right: incremental encoder outputs A, B (quadrature) and index Z.

Construction

- **Light source** (LED), **code disc** (glass, plastic or metal with alternating opaque and transparent segments), **photodetectors** (with a mask), and **electronics** (amplifier, comparator, line driver) in a housing with a shaft and bearings. As the shaft turns, the disc interrupts the light; the detectors produce pulses.
- **Magnetic encoders** use a magnetised ring and Hall sensors (more robust to dirt and vibration), **capacitive** and **inductive** types also exist.

Types

Type	Output	Information	Disadvantage
Incremental	pulse trains A, B (90° apart) and index Z (once per turn)	position found by **counting** pulses; direction from which channel leads; speed from the pulse frequency	loses position when power is off; needs a homing run to Z
Absolute (single-turn)	a unique digital code for each angle (Gray or binary code), e.g. 12–19 bit	position known immediately at power-up	more complex and costly
Absolute multi-turn	also counts the number of revolutions (gears or battery-backed counter)	full range of the robot axis	cost

Why Gray code? In ordinary binary code several bits can change at once (011 → 100), so a slightly misaligned reading can give a large wrong value. In Gray code only **one** bit changes between adjacent positions (000, 001, 011, 010, 110, 111, 101, 100), so the error is at most one step.

$$\text{counts per rev} = 4 \times PPR \text{ (quadrature)}, \qquad \Delta\theta = \frac{360^\circ}{4\,PPR} \qquad n = \frac{f_{pulse}}{PPR} \times 60 \text{ (rpm)}$$

Worked example: encoder numbers
Incremental encoder with PPR = 1024 (pulses per revolution on channel A).
Quadrature decoding counts every edge of A and B ⇒ 4 × 1024 = **4096 counts/rev** ⇒ resolution Δθ = 360°/4096 = **0.088°**.
Speed: if A has f = 5120 Hz, n = 5120/1024 × 60 = **300 rpm**.
Absolute 12-bit encoder: 2^{12} = 4096 positions per turn: same 0.088° resolution, but with no homing.

Applications in robotics and industry

- **Robot arms and servo drives:** joint angle feedback in closed-loop position and speed control (PID, Unit 3 Q5).
- **Mobile robots and AGVs:** wheel odometry (distance and heading).
- **CNC machines** (Unit 5 Q2): spindle and axis position; **conveyors**: length and speed; **cranes, lifts, packaging machines**; **pick-and-place** positioning.
- **Selection:** resolution, absolute or incremental, interface (A/B/Z, SSI, fieldbus), mechanical mounting, environment (IP rating, vibration).

Unit 4, Question 8: Classification of industrial sensors

Exam question – Unit 4 Q8
Classify different types of industrial sensors with examples (temperature, flow, level, pressure, optical)

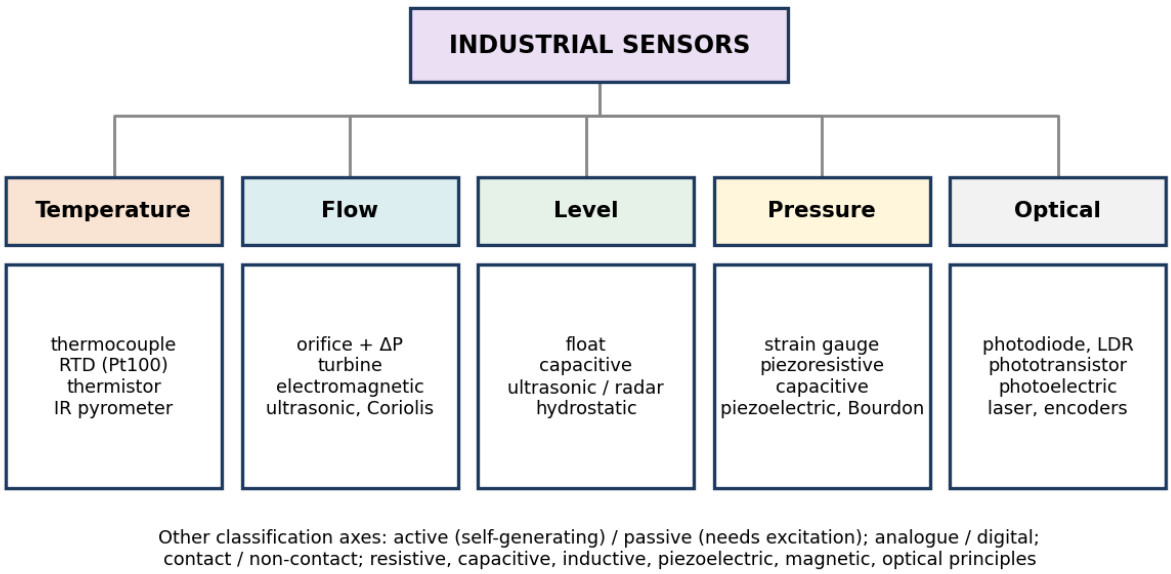


Figure 80. *Classification of industrial sensors by measured quantity, with examples.*

Quantity	Sensor	Working principle	Typical use
Temperature	Thermocouple	Seebeck voltage	furnaces, kilns, engines
	RTD (Pt100)	resistance rises with T	process temperature, precision

Quantity	Sensor	Working principle	Typical use
	Thermistor	resistance changes steeply with T	motors, battery packs, HVAC
	IR pyrometer	thermal radiation (non-contact)	moving or hot objects
Flow	Orifice plate + differential pressure	$\Delta P \propto Q^2$	liquids, gases, steam
	Turbine meter	blade speed ∝ flow	clean fluids, fuel
	Electromagnetic flowmeter	Faraday's law (conductive liquids)	water, slurries
	Ultrasonic flowmeter	transit-time or Doppler	non-invasive (clamp-on)
	Coriolis meter	mass-flow from tube deflection	high accuracy, mass flow
Level	Float	buoyancy moves a switch/potentiometer	tanks
	Capacitive	capacitance changes with level	liquids, solids
	Ultrasonic / radar	echo time	open tanks, silos
	Hydrostatic (pressure)	$P = \rho g h$	deep tanks, wells
Pressure	Strain gauge / piezoresistive	ΔR with diaphragm strain	hydraulics, process
	Capacitive	diaphragm changes capacitance	low pressure, differential
	Piezoelectric	charge on crystal (dynamic)	combustion, shock
	Bourdon tube	mechanical deflection (gauge)	local indication
Optical	Photodiode / phototransistor	light → current	light detection, communications
	LDR	resistance falls with light	street lights, simple switching
	Photoelectric switch	beam interruption/reflection	object detection (Q5)
	Laser sensor, encoder	distance/position by light	positioning, measurement (Q7)

Other classification criteria: *active* sensors generate their own signal (thermocouple, piezoelectric); *passive* sensors need excitation (RTD, strain gauge, LDR); *analogue* vs *digital* output; *contact* vs *non-contact*; by the physical effect used (resistive, capacitive, inductive, piezoelectric, magnetic, optical).

Selection criteria: measurement range, accuracy and repeatability, response time, linearity, environment (temperature, dust, explosion hazard), output type and signal standard, size, cost, maintenance and calibration needs.

Unit 4, Question 9: The Wheatstone bridge, balance condition and applications

> **Exam question – Unit 4 Q9**
> What is a Wheatstone bridge? Derive its balance condition and explain its applications.

A **Wheatstone bridge** is a circuit of four resistors arranged in a diamond, with a supply across one diagonal and a detector (galvanometer or amplifier) across the other diagonal. It is used to measure an unknown resistance, or small changes of resistance, very accurately by **comparison** with known resistors.

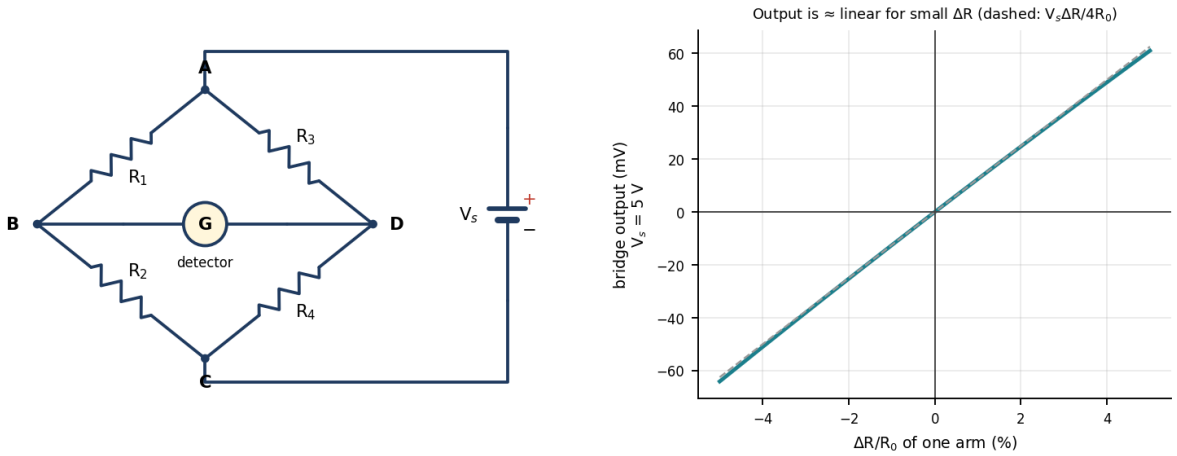


Figure 81. *Left: Wheatstone bridge. Right: for small changes ΔR the output voltage is nearly linear in $\Delta R/R_0$.*

Derivation of the balance condition (step by step)

The supply V_s is applied between A (top) and C (bottom). Two resistor chains lie between A and C: the left chain R_1 (A–B) and R_2 (B–C), and the right chain R_3 (A–D) and R_4 (D–C). The detector is between B and D. Take C as the 0 V reference.

1. The left chain is a **voltage divider**: current $I_1 = V_s/(R_1 + R_2)$, so the voltage at B is $V_B = I_1 R_2 = V_s R_2/(R_1 + R_2)$.
2. The right chain is also a voltage divider: $V_D = V_s R_4/(R_3 + R_4)$.
3. The detector voltage is $V_{BD} = V_B - V_D = V_s[R_2/(R_1 + R_2) - R_4/(R_3 + R_4)]$.
4. **Balance** means the detector shows zero: $V_{BD} = 0$, i.e. $V_B = V_D$ (no current flows through the detector).
5. So $R_2/(R_1 + R_2) = R_4/(R_3 + R_4)$. Cross-multiply: $R_2(R_3 + R_4) = R_4(R_1 + R_2)$.
6. Expand: $R_2R_3 + R_2R_4 = R_4R_1 + R_4R_2$.
7. The term R_2R_4 appears on both sides; cancel it: $R_2R_3 = R_1R_4$.
8. Rearranged:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \qquad \Longleftrightarrow \qquad R_1R_4 = R_2R_3$$

Physical meaning: at balance, the two voltage dividers give the same fraction of V_s, so points B and D are at the same potential and no current flows through the detector. The balance does not depend on V_s (a stable supply is not needed for a balance measurement) or on the detector's resistance. To find an unknown R_4, adjust a known R_3 (or the ratio R_2/R_1) until the detector reads zero, then $R_4 = R_2R_3/R_1$.

> **Worked example: measuring an unknown resistance**
> $R_1 = R_2 = 100\ \Omega$; the variable R_3 is adjusted until the galvanometer reads zero at $R_3 = 315\ \Omega$. Then $R_4 = R_2R_3/R_1 = 100 \times 315/100 =$ **315 Ω**.
> If $R_1 = 1\ k\Omega$ and $R_2 = 100\ \Omega$, $R_3 = 2.2\ k\Omega$ at balance: $R_4 = 100 \times 2200/1000 =$ **220 Ω**; the ratio arm lets us measure resistances 10 times smaller than the standard.

Deflection (unbalanced) mode: sensors

In sensors the bridge is balanced at the reference condition and a change ΔR in one arm unbalances it. For a **quarter bridge** ($R_1 = R_2 = R_3 = R_0$, $R_4 = R_0 + \Delta R$):

$$V_{BD} = V_s\left[\frac{1}{2} - \frac{R_0 + \Delta R}{2R_0 + \Delta R}\right] = -V_s\frac{\Delta R}{2(2R_0 + \Delta R)} \approx -\frac{V_s}{4}\frac{\Delta R}{R_0}$$

(Steps: bring to a common denominator: $[(2R_0 + \Delta R) - 2(R_0 + \Delta R)] / [2(2R_0 + \Delta R)] = -\Delta R/[2(2R_0 + \Delta R)]$; for small ΔR the denominator $\approx 4R_0$.) The sign shows the polarity of the output; the magnitude is

$V_s\Delta R/(4R_0)$. A **full bridge** (four active gauges, two in tension and two in compression) gives $V_s\Delta R/R_0$, four times more output, with temperature effects cancelling.

Worked example: strain-gauge bridge
Gauge $R_0 = 350\ \Omega$, GF = 2, strain $\varepsilon = 1000\ \mu\varepsilon = 0.001$, $V_s = 5$ V, quarter bridge.
$\Delta R/R_0 = GF \times \varepsilon = 2 \times 0.001 = 0.002 \Rightarrow \Delta R = 0.7\ \Omega$.
$V_{out} \approx (V_s/4)(\Delta R/R_0) = (5/4)(0.002) =$ **2.5 mV**. Exact: $5 \times 0.7/(2 \times (700 + 0.7)) = 3.5/1401.4 = 2.498$ mV ✔.
A full bridge would give $5 \times 0.002 =$ **10 mV** for the same strain.

Applications

- **Strain gauges and load cells** (weighing, force), **pressure transducers** (Question 2), torque sensors.
- **RTD and thermistor temperature measurement** (Question 3 first), with 3-wire lead compensation: one lead in each of two adjacent arms, so their resistances cancel.
- **Resistance measurement** in the laboratory (the original use); **gas sensors, humidity sensors, light sensors** with resistive elements.
- **AC bridges** for measuring inductance and capacitance (Maxwell, Wien and Schering bridges).
- **Sensor interfaces:** the output is amplified by an instrumentation amplifier and digitised in the DAS (Question 6).

Common misconception
‘The galvanometer reading is proportional to the unknown resistance.’ At balance the reading is zero, whatever the supply; the unknown is found from the resistor values, not the reading. In deflection mode the reading is proportional to the *change*, not to the resistance itself.

Unit 4: End-of-Unit Revision Pack

Unit summary

- Sensor detects; transducer converts energy; the electrical-output sensor is a transducer, but an actuator is a transducer and not a sensor.
- Measurement chain: sensor → signal conditioning → ADC/DAS → controller.
- Pressure: strain-gauge bridge, amplified; 4–20 mA is the industrial standard.
- Temperature: thermocouple (CJC), RTD (3-wire, excitation), thermistor (non-linear).
- Accelerometer: proof mass and spring, x = ma/k. Gyroscope: Coriolis force F = 2mvΩ.
- WSN: nodes, mesh, gateway, cloud; low power; used for retrofit monitoring.
- Proximity sensors: inductive (metals), capacitive (any), photoelectric, ultrasonic, magnetic.
- DAS: multiplexer, S/H, ADC; Nyquist $f_s \geq 2f_{max}$; SCADA is a supervisory layer.
- Encoders: incremental (A, B, Z) and absolute (Gray code); 4× decoding.
- Wheatstone bridge balance: $R_1/R_2 = R_3/R_4$.

Formula and equation sheet

Strain-gauge sensitivity

$$\frac{\Delta R}{R} = GF \cdot \varepsilon$$

Wheatstone balance

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Quarter bridge output

$$V_{out} \approx \frac{V_s}{4}\frac{\Delta R}{R_0}$$

ADC resolution

$$LSB = \frac{V_{FS}}{2^N}$$

Nyquist

$$f_s \geq 2f_{max}$$

4–20 mA scaling

$$I = 4 + 16\frac{P}{P_{FS}}\ \text{mA}$$

Accelerometer

$$x = \frac{ma}{k}, \quad f_n = \frac{1}{2\pi}\sqrt{k/m}$$

Coriolis force

$$F = 2mv\Omega$$

Encoder resolution

$$\Delta\theta = \frac{360^\circ}{4\,PPR}$$

Thermocouple

$$V = S\,(T_{hot} - T_{ref})$$

Diagram-revision sheet

Practise drawing each of these from memory, labelling every component, then compare with the figure.

Figure	What to draw and label
Q2	Pressure DAS block diagram
Q3 first	Temperature DAS block diagram
Q3 second	MEMS accelerometer and gyroscope
Q4	WSN block diagram
Q5	Inductive sensor; photoelectric modes
Q6	DAS block diagram; aliasing/quantisation
Q7	Optical encoder; A, B, Z waveforms
Q8	Sensor classification tree
Q9	Wheatstone bridge

Glossary of important terms

Term	Meaning
Sensor	Detects a physical quantity
Transducer	Converts one form of energy to another
Measurand	Quantity being measured
Signal conditioning	Amplify, filter, linearise, isolate
DAS	Data acquisition system
ADC	Analogue-to-digital converter
Nyquist	Sampling at least twice the highest frequency
Aliasing	False low frequency from under-sampling
Gauge factor	(ΔR/R)/strain
Cold-junction compensation	Corrects thermocouple reference temperature
Coriolis force	Apparent force on a moving mass in a rotating frame
WSN	Wireless sensor network
Gateway	Bridge between sensor network and plant network
Proximity sensor	Non-contact object detector
Encoder	Converts position to digital signals
PPR	Pulses per revolution
Gray code	Code in which one bit changes per step
Wheatstone bridge	Four-resistor circuit for precise resistance comparison

Conceptual and numerical review questions

1. A load cell has $R_0 = 120\ \Omega$, GF = 2.0, strain 500 με, full bridge, $V_s = 10$ V. Find the output.
2. A thermocouple reads 8.2 mV with the reference at 20 °C (S = 41 μV/°C). Find the hot-junction temperature.
3. A 14-bit ADC has a 5 V range. Find the LSB and the maximum quantisation error.
4. An encoder has PPR = 2500 and is read in quadrature. Find the resolution in degrees and the counts per revolution.
5. Balance: $R_1 = 50\ \Omega$, $R_2 = 200\ \Omega$, $R_3 = 120\ \Omega$. Find R_4.
6. Which proximity sensor detects a plastic bottle's liquid level through the wall?

Fully explained answers

1. $\Delta R/R_0 = 2 \times 0.0005 = 0.001$; full-bridge output = $V_s \Delta R/R_0 = 10 \times 0.001 =$ **10 mV**.
2. $T_{hot} = 20 + 8.2/0.041 = 20 + 200 =$ **220 °C**.
3. LSB = 5/16384 = 0.305 mV; error = ±0.153 mV.
4. 10 000 counts; 360/10 000 = **0.036°**.
5. $R_4 = R_2R_3/R_1 = 200 \times 120/50 =$ **480 Ω**.
6. A capacitive sensor (it senses dielectric changes through non-metallic walls).

Examination-answer guide for the original questions

This guide complements the teaching chapters. It shows how to organise each original question as a complete exam answer.

Question	Structure of a strong answer
Q1	Definitions → table of ≥ 6 differences → examples (thermocouple, speaker) → note on overlap.
Q2	Sensor types → pressure chain block diagram → signal-conditioning requirements → DAS → worked numbers → 4–20 mA.
Q3 first	Sensor types table (thermocouple, RTD, thermistor) → block diagram → CJC, lead compensation, linearisation → ADC → example.
Q3 second	Accelerometer diagram + F = ma, x = ma/k → gyroscope diagram + Coriolis → applications → IMU.
Q4	Definition → block diagram (nodes, mesh, gateway, cloud) → why in Industry 4.0 → applications and challenges.
Q5	Table of types with principles → inductive and photoelectric diagrams → applications in automation.
Q6	Signal conditioning table → DAS block diagram → Nyquist and ADC equations → DAS vs SCADA.
Q7	Construction diagram → incremental vs absolute → quadrature waveform → resolution → robot applications.
Q8	Classification tree with five categories → table of principle and use for each example.
Q9	Circuit → derivation in steps → balance condition → worked example → applications (strain gauge, RTD).

UNIT 5: Industrial Automation, Communication and Industry 4.0

Unit 5 zooms out. We have power (Units 1–2), decisions (Unit 3) and measurements (Unit 4). Unit 5 asks: how are they organised into whole automated factories, how do machines exchange data, and how is the data used to make plants smarter?

Unit 5, Question 1: Industrial automation, advantages and the role of robotics

> **Exam question – Unit 5 Q1**
> Define industrial automation and list its advantages. Also explain the role of robotics in smart manufacturing & modern industries.

A–D. Definition

Industrial automation is the use of control systems (PLCs, DCS, computers), sensors, actuators, robots and information technology to operate machines and processes with minimal human intervention. A human defines *what* must be done (the recipe or program); the automation system does *how*, repeatedly and consistently.

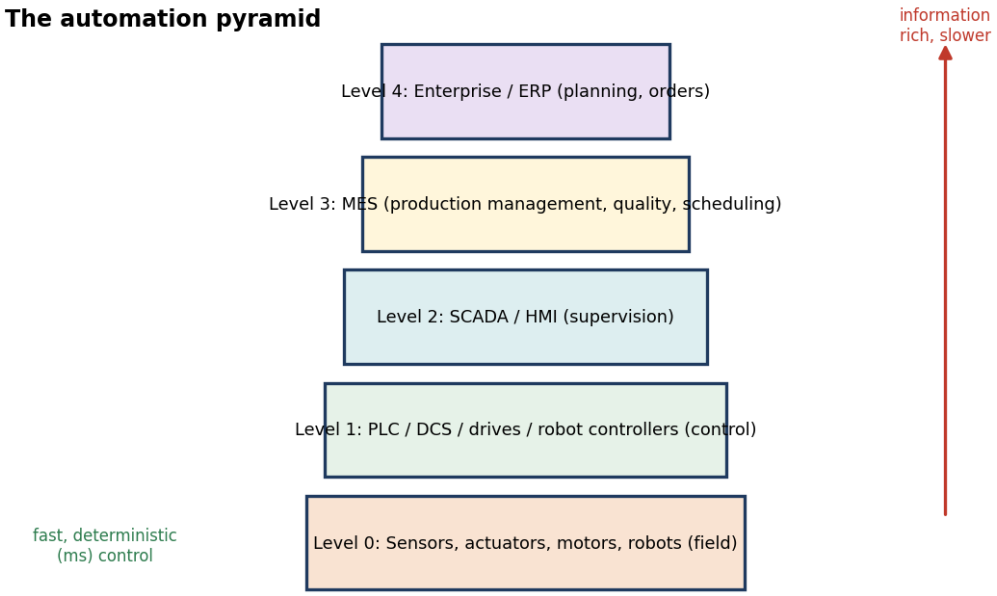


Figure 82. *The automation pyramid shows where each technology studied in this book sits, from field devices at the bottom to enterprise planning at the top.*

Type of automation	Meaning	Example
Fixed (hard) automation	sequence fixed by the equipment; very high volume, little flexibility	bottle-filling line, transfer lines
Programmable automation	sequence changed by reprogramming; batch production	CNC machines, PLC-controlled batch plants
Flexible automation	quick changeover without long stoppages	robotic assembly cells, flexible manufacturing systems
Process automation	continuous control of variables	refinery, power plant, chemical plant (DCS)

Advantages

Advantage	Explanation
Higher productivity	machines run 24 × 7 at constant speed
Consistent quality	repeatable operations reduce defects; in-line inspection
Lower operating cost	less labour per unit, lower scrap, energy optimisation (Unit 3 Q8)
Safety	people are removed from hazardous work (high temperature, toxic, heavy loads)
Flexibility and fast changeover	programs replace rewiring
Data and traceability	every batch is logged for quality, audit and optimisation
Reduced downtime	diagnostics and predictive maintenance (Question 5)

Disadvantages and limits (balanced view): high initial investment, need for skilled maintenance, less flexibility of fixed automation, cybersecurity risks, and the social need to retrain workers.

Role of robotics in smart manufacturing

A **robot** is a reprogrammable, multi-functional machine that moves tools or materials through programmed motions. An industrial robot is built from: a **manipulator** (links and joints), **actuators** (electric servo motors with gearboxes), **sensors** (encoders on every joint, force, vision), an **end effector** (gripper, welding torch, spray gun, vacuum cup) and a **controller** (computes the motion and closes the position loop; Unit 3).

Articulated (6-axis) industrial robot arm: each joint = motor + gearbox + encoder

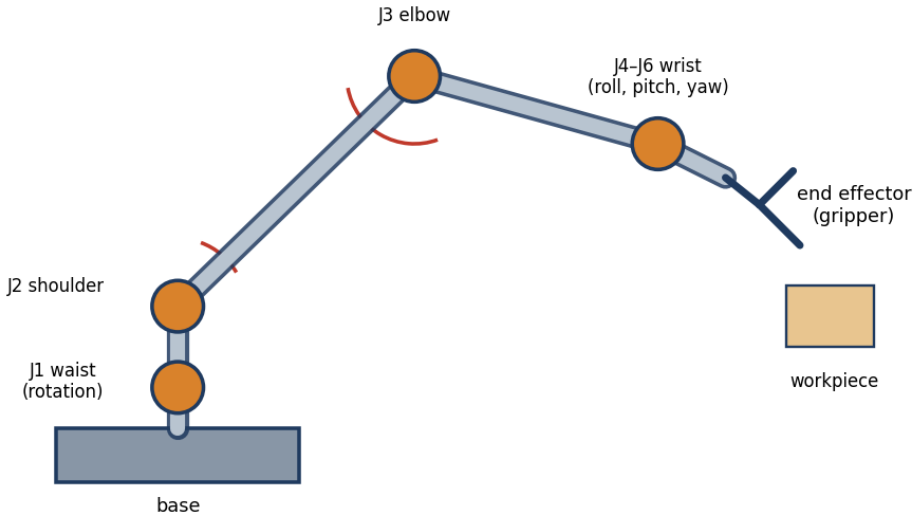


Figure 83. *A six-axis articulated robot. Each joint has a servo motor, gear reducer and encoder (Unit 4 Q7).*

Robot type	Description	Typical use
Articulated (6-axis)	rotary joints; large workspace	welding, painting, machine tending
SCARA	rigid vertical, compliant horizontal; fast and precise	electronics assembly, pick-and-place
Delta (parallel)	three arms from a fixed base; very fast	packaging, food sorting
Cartesian / gantry	three linear axes	palletising, CNC loading, 3D printing
Collaborative robot (cobot)	force-limited; works near humans without a cage (with risk assessment)	small-batch assembly, inspection
AMR / AGV	autonomous mobile robots	material transport in factories

- **Applications:** arc and spot welding (automotive), painting, assembly, pick-and-place, packaging and palletising, material handling, inspection, machine tending (loading CNC machines).
- **In smart manufacturing:** robots are networked devices that exchange data (production counts, motor currents, faults) with the PLC and SCADA; vision-guided robots adapt to part variation; AI and digital twins plan paths and detect faults; cobots allow flexible, low-volume production (Question 6).
- **Benefits:** repeatability (typically ±0.02–0.1 mm for industrial arms), speed, endurance, safe handling of heavy or hazardous tasks, high quality. **Limits:** cost, programming effort, limited dexterity compared with humans, safety rules (guarding, emergency stops, ISO 10218 and related standards).

Worked example: robot productivity
A robot welds a part in 40 s and loads/unloads in 10 s: cycle time 50 s ⇒ parts per hour = 3600/50 = **72**. A manual cell with a 90 s cycle gives 3600/90 = 40/h. With 90% availability, the robot makes 72 × 0.9 = **64.8 parts/hour** effectively.

Unit 5, Question 2: CNC machine, architecture and working

Exam question – Unit 5 Q2
Describe the architecture and working of a CNC machine using basic block diagram

CNC (Computer Numerical Control): a machine tool (milling, turning, drilling, laser, router) whose movements are controlled by a computer that follows a numerical **part program** instead of by manual handwheels or cams. It converts a digital drawing into precise motion.

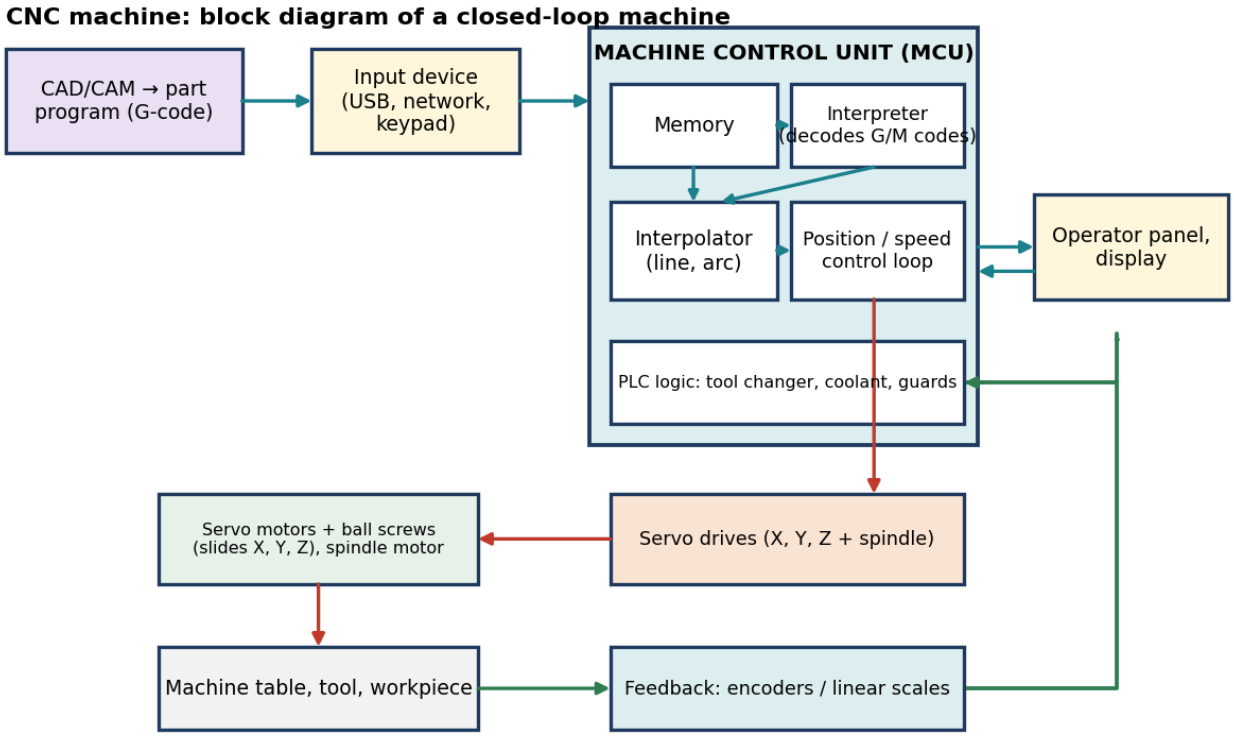


Figure 84. *CNC machine architecture: program → MCU → drives → motors and slides, with encoders closing the position loop.*

Block	Function
Part program (G-code)	list of instructions: G-codes for motion (G00 rapid, G01 linear feed, G02/G03 circular), M-codes for auxiliary functions (M03 spindle on, M08 coolant), plus X, Y, Z coordinates, F feed, S spindle speed. Produced by CAM software from the CAD model
Input device	loads the program (USB, network, memory card, keyboard)
MCU (machine control unit)	*interpreter* decodes each block; *interpolator* computes the intermediate points between the start and end of each move (linear, circular) so that axes move in coordination; *position/speed loops* compare the commanded and actual positions

Block	Function
Servo drives and motors	amplify the command and turn the slide motors (and spindle motor) through ball screws, producing linear motion
Feedback devices	encoders (Unit 4 Q7) or linear scales on the axes give actual position; **closed-loop** control corrects errors (Unit 3 Q4)
PLC (inside or beside the MCU)	handles tool changer, coolant, door interlocks, lubrication, safety (Unit 3 Q6)
Operator panel and display	program editing, jog, simulation, alarms
Machine structure	bed, slides, table, spindle, tool; accuracy depends on rigidity and thermal stability

Working, step by step

1. The designer creates the part in CAD; CAM software generates the tool path and G-code.
2. The operator loads the program, sets the work zero and tool offsets, clamps the workpiece.
3. The MCU reads one block at a time, interprets it, and the interpolator splits each move into small steps (many per second).
4. Position commands go to the servo drives; motors turn ball screws; slides move the tool/table.
5. Encoders report the real position; the control loop corrects the difference (following error) every few milliseconds.
6. The spindle turns at speed S; coolant and the tool changer operate on M-codes; at the end of the program the machine stops.

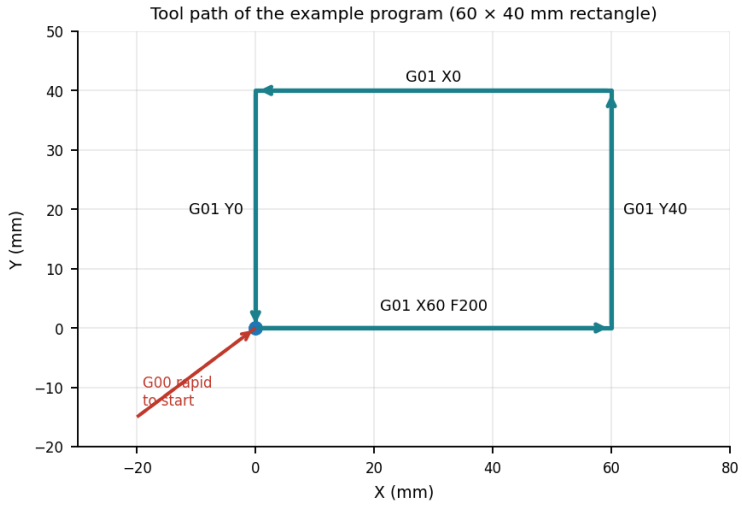


Figure 85. *Tool path for the example program below.*

Example G-code (illustrative): `N10 G21 G90 (mm, absolute)  N20 G00 X0 Y0 Z5  N30 M03 S1600  N40 G01 Z-2 F100  N50 G01 X60 F200  N60 G01 Y40  N70 G01 X0  N80 G01 Y0  N90 G00 Z5  N100 M05  N110 M30`. It starts the spindle, plunges 2 mm and mills a 60 × 40 mm rectangle.

Worked example: CNC calculations (assumed data)
Resolution (BLU, basic length unit): ball-screw pitch 5 mm per revolution; encoder 2500 PPR with 4× decoding = 10 000 counts/rev ⇒ BLU = 5 mm/10 000 = **0.0005 mm = 0.5 µm**.
Spindle speed: cutter D = 20 mm, cutting speed V_c = 100 m/min ⇒ N = $1000V_c/(\pi D)$ = 100 000/(3.1416 × 20) = **1592 rpm**.
Table feed: 4-tooth cutter, feed per tooth 0.05 mm ⇒ F = 0.05 × 4 × 1592 = **318 mm/min**.
Linear interpolation: move (0,0) → (40,30) at F = 300 mm/min: path length = $\sqrt{(40^2 + 30^2)}$ = 50 mm ⇒ time = 50/300 = 0.167 min = 10 s. Axis speeds: V_x = 300 × 40/50 = **240 mm/min**; V_y = 300 × 30/50 = **180 mm/min**, so both axes finish together.

- **Types:** open-loop (stepper motors, no feedback; cheaper, may lose steps) and closed-loop (servo with feedback; used in industry). By axes: 2-axis (lathe), 3-axis, 5-axis machining centres.

- **Advantages:** accuracy, repeatability, complex shapes, quick changeover, reduced human error.
  Limitations: cost, programming skill, tool wear, thermal errors.

Unit 5, Question 3: Comparison of Modbus, Profibus, CAN, RS485 and Ethernet

Exam question – Unit 5 Q3
Compare between Modbus, Profibus, CAN, RS485 & Ethernet communication protocols. (At least 7 parameters)

Note on the source wording
Not all five are 'protocols' in the same sense. **Modbus** is an application-layer messaging protocol. **Profibus** and **CAN** are complete fieldbus systems (protocol + physical layer). **RS485** is only an electrical (physical-layer) standard. **Ethernet** is a network technology (IEEE 802.3 physical and data-link layers) on which other protocols run. The table compares them as the question asks, and labels each one's category.

Why machines must exchange data

A PLC must read sensors that are 100 m away, command drives, report to SCADA and share data with other PLCs. Running one pair of wires per signal (the old 4–20 mA way) is expensive. A **communication network** lets many devices share one cable and carries diagnostics as well as values. Three ideas must be separated: the **physical layer** (cable and voltage levels: e.g. RS485), the **protocol** (rules and message format: e.g. Modbus), and the **network technology** (how many devices access the medium: e.g. Ethernet).

Parameter	Modbus	Profibus (DP)	CAN	RS485	Ethernet
Category	application-layer protocol (master/slave)	fieldbus protocol (Siemens-origin, IEC 61158)	fieldbus: data-link + physical, ISO 11898	physical-layer electrical standard (TIA/EIA-485) only	network technology (IEEE 802.3); carries TCP/IP, Modbus TCP, Profinet, EtherNet/IP
Physical medium	RS485 (RTU) or RS232; or Ethernet (TCP)	RS485 twisted pair (DP) or fibre; MBP for PA	twisted pair, differential	twisted pair, differential	twisted pair (Cat5e/6), fibre
Topology	bus (RTU) or star (TCP)	linear bus, segments with repeaters	linear bus (backbone + short stubs)	multi-drop bus	star with switches (also ring)
Typical speed	9.6–115.2 kbps (RTU); 10/100 Mbps (TCP)	9.6 kbps – 12 Mbps	up to 1 Mbps (CAN FD data phase up to ~8 Mbps)	up to 10 Mbps (short); 100 kbps at 1.2 km	10/100/1000 Mbps (and above)
Max. distance	≈ 1200 m (RTU on RS485); network-limited (TCP)	≈ 1200 m at 93.75 kbps or less; ≈ 100 m at 12 Mbps per segment	≈ 40 m at 1 Mbps; ≈ 500 m at 125 kbps; ≈ 6 km at 10 kbps	≈ 1200 m (at ≤ 100 kbps)	100 m per copper segment; km with fibre/switches
Max. nodes	247 slaves (RTU)	32 per segment, 126 addresses total	≈ 110 (transceiver limited)	32 unit loads (up to 256 with low-load drivers)	thousands (IP addressing)
Access method	master polls slaves; one at a time	token passing between masters + master/slave	CSMA with non-destructive priority arbitration (multi-master)	defined by the protocol above it	CSMA/CD on shared; full-duplex switched (no collisions)
Error checking	CRC-16 (RTU), LRC (ASCII)	parity + checksum/FCS (HD = 4)	15-bit CRC, bit stuffing, ACK	none (done by protocol)	CRC-32 frame check

Parameter	Modbus	Profibus (DP)	CAN	RS485	Ethernet
Determinism	moderate (poll-time dependent)	good, cyclic data exchange	good (priority-based latency)	depends on protocol	not inherently deterministic; industrial variants (Profinet, EtherCAT) add real-time
Typical applications	PLC–instrument, VFD, meter, SCADA links	factory automation with PLC and remote I/O, drives	automotive, machinery, mobile equipment, CANopen/DeviceNet	wiring layer for Modbus RTU, Profibus DP, BACnet MS/TP	plant networks, SCADA, MES, IoT, cloud links
Cost / complexity	very low, simple	medium	low	very low	medium, widespread hardware

Values are typical figures from the standards and manufacturer practice; actual limits depend on cable, number of nodes and device capability. Always check device datasheets.

Worked example: which one to choose?
Reading 20 energy meters over 800 m in a plant: Modbus RTU over RS485 (cheap, long distance, meters already speak Modbus).
Linking PLCs to SCADA and the cloud in a building: Ethernet (Modbus TCP or OPC UA).
Fast I/O in a machine with many nodes: Profibus DP or an Ethernet-based real-time protocol.
Controllers inside a vehicle or mobile machine: CAN (robust, priority messages).

Unit 5, Question 4: RS232, RS485 and Ethernet communication in automation

Exam question – Unit 5 Q4
Write short notes on RS232, RS485, and Ethernet communication in automation.

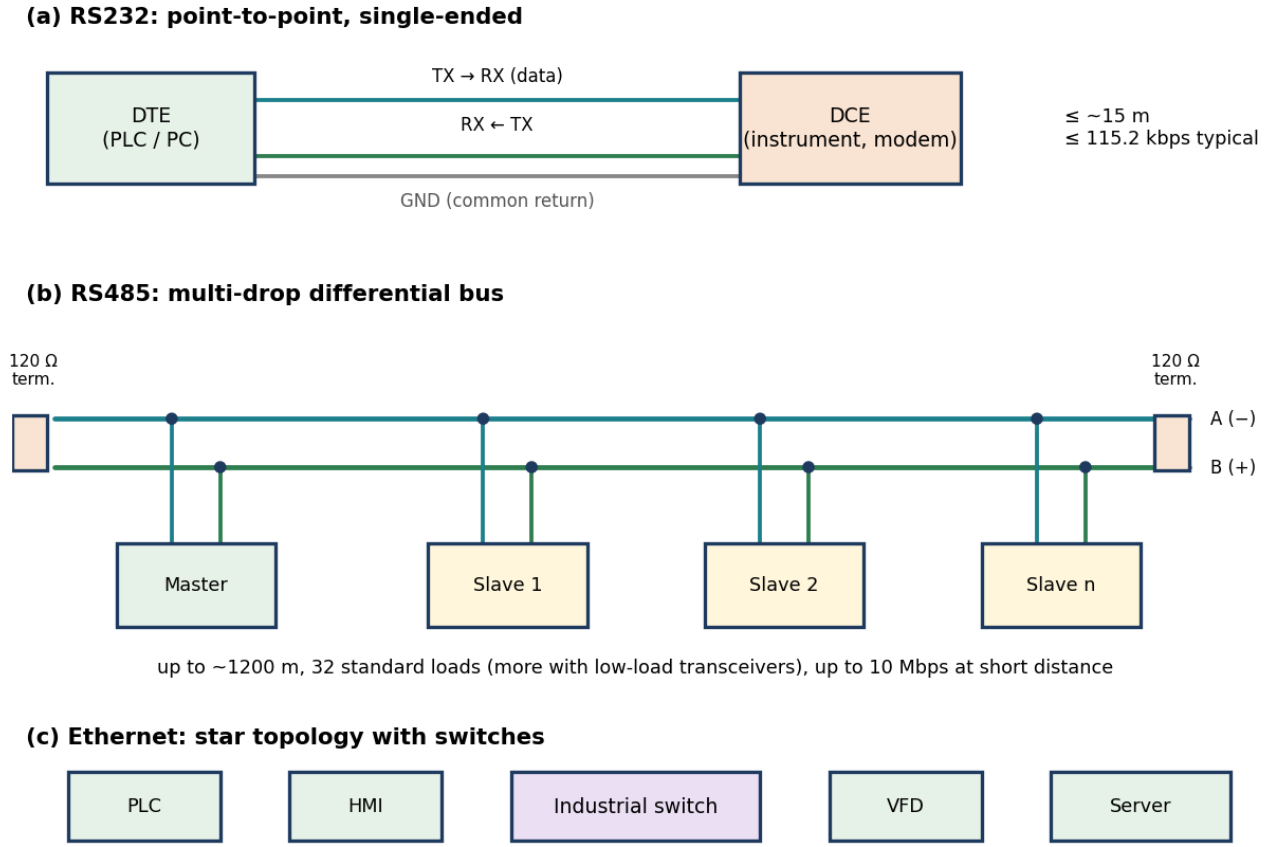


Figure 86. *(a) RS232 point-to-point link. (b) RS485 multi-drop bus with 120 Ω terminators at both ends. (c) Ethernet star topology with a switch.*

RS232

RS232 (Recommended Standard 232) is a serial standard for point-to-point connection between two devices, a **DTE** (data terminal equipment, e.g. a PC or PLC) and a **DCE** (data communication equipment, e.g. a modem or instrument). **Single-ended signalling**: each signal is a voltage relative to a common ground; logic 1 (mark) is −3 to −15 V and logic 0 (space) is +3 to +15 V. Main lines: TX, RX, GND (and handshake lines RTS/CTS). Full duplex. Typical distance up to about 15 m at 19.2 kbps (shorter at higher rates; up to 115.2 kbps is common). Connectors: DB9 or DB25.

- **Uses:** PLC programming ports, barcode scanners, weighing scales, CNC part-program transfer, laboratory instruments, GSM modems.
- **Limits:** only two devices, short distance, ground-potential differences and noise (single-ended), relatively slow.

RS485

RS485 (TIA/EIA-485) is a **differential** serial standard: a bit is the voltage difference between two wires A and B (≥ ±200 mV at the receiver; typically ±1.5 to ±5 V from the driver). Noise appears equally on both wires and is cancelled in the receiver, hence high noise immunity (Unit 1 Q4). Features: **multi-drop** bus with up to 32 unit loads (more with newer low-load transceivers), cable lengths up to about 1200 m, speeds up to 10 Mbps (at ≈ 12 m), half-duplex on 2 wires (one talks at a time) or full duplex on 4 wires. Needs **120 Ω termination** at both ends of the cable to prevent reflections, and biasing to hold the idle line state.

- **Uses:** physical layer for Modbus RTU, Profibus DP, BACnet MS/TP, DMX lighting; VFD, meters, sensors, RTU networks.
- **Limits:** it defines only the electrical layer; the higher-level protocol must manage who may talk (master/slave); modest data rate over long distance; no built-in addressing.

Worked example: Modbus RTU timing on RS485
Baud rate 9600, 8 data bits + even parity + 1 stop + 1 start = 11 bits per byte ⇒ 1 byte = 11/9600 = **1.146 ms**.
A request 'read 2 holding registers' is 8 bytes (address, function, start (2), count (2), CRC (2)); the reply is 9 bytes (address, function, byte count, 4 data bytes, CRC (2)). Total 17 bytes ⇒ 17 × 1.146 = **19.5 ms**, plus silent gaps of 3.5 characters (≈ 4 ms each).
Polling 20 slaves sequentially needs roughly 20 × 25 ms = **0.5 s**, which explains why Modbus RTU suits slow supervisory data but not fast control.

Ethernet

Ethernet (IEEE 802.3) is the dominant LAN technology. Devices are connected in a **star** through **switches** using twisted pair (Cat5e/6, 100 m per segment) or fibre. Speeds: 10 Mbps, 100 Mbps (Fast Ethernet), 1 Gbps and higher. Each device has a unique MAC address; with TCP/IP it also has an IP address. Switched, full-duplex Ethernet has no collisions. Frames carry a 32-bit CRC.

- **Industrial Ethernet:** uses rugged connectors (M12), managed switches, VLANs, redundancy (ring protocols), and real-time extensions: Profinet, EtherNet/IP, EtherCAT, Modbus TCP, OPC UA (the interoperable data model).
- **Role in automation:** connects PLCs, HMIs, drives, robots, SCADA servers and the IT network and the cloud (IoT, Question 5, Industry 4.0, Question 6).
- **Advantages:** high speed, standard cheap hardware, long reach with fibre, easy integration with IT, remote diagnostics. **Limits:** standard Ethernet is not deterministic; needs cybersecurity (firewalls, segmentation) because it is exposed to IT threats; cable length per segment limited to 100 m.

Feature	RS232	RS485	Ethernet
Signalling	single-ended	differential	differential (twisted pair)
Devices	2 (point-to-point)	up to 32+ (multi-drop)	many (switched network)
Distance	≈ 15 m	≈ 1200 m	100 m per copper link
Speed	≤ ~115 kbps (typical)	≤ 10 Mbps	10 Mbps – 10 Gbps
Noise immunity	poor	very good	good
Typical use	programming port, instruments	fieldbus wiring, meters, VFDs	plant and enterprise networks

Unit 5, Question 5: IoT-based predictive maintenance; IoT devices and gateways

Exam question – Unit 5 Q5 (a)
Explain how predictive maintenance is achieved using IoT in industrial systems with a suitable example.

Exam question – Unit 5 Q5 (b)
Describe the role of IoT devices and gateways in industrial monitoring and predictive maintenance.

A–D. Maintenance strategies

Strategy	Rule	Drawback
Reactive (run-to-failure)	repair after it breaks	unplanned downtime, secondary damage
Preventive (time-based)	service at fixed intervals	unnecessary work; failures between intervals
Predictive (condition-based)	service when the measured condition shows a developing fault	needs sensors, data and analysis
Prescriptive	predicts and recommends the best action and timing	needs mature models

Predictive maintenance (PdM) monitors the actual condition of equipment and uses the data to detect faults early and to estimate when a failure may occur, so that maintenance is done just in time.

Part (a): how IoT achieves predictive maintenance

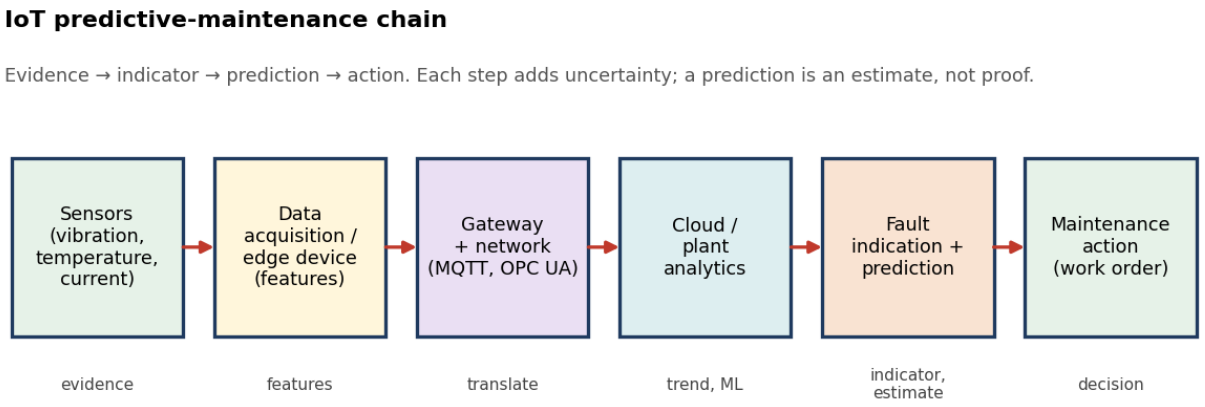


Figure 87. *The predictive-maintenance chain: sensor → edge/data acquisition → gateway and network → analytics → fault indication and prediction → maintenance action.*

1. **Sensing (measured evidence):** accelerometers on bearings (vibration), thermocouples/RTDs (temperature), current sensors (motor current), pressure, oil debris, acoustic sensors (Unit 4).
2. **Data acquisition / edge device:** samples the signal (respecting Nyquist, Unit 4 Q6), filters it, and computes *features* such as RMS velocity, peak, kurtosis, FFT peaks.
3. **Gateway and network:** sends the features (and occasional raw waveforms) by MQTT or OPC UA over Ethernet/Wi-Fi/cellular to the plant server or cloud, securely.
4. **Analytics:** compares features with a healthy baseline and thresholds; rule-based or machine-learning models trend the data and classify the type of fault.
5. **Fault indication and prediction:** an alert (e.g. ‘outer-race bearing defect likely’) and an estimate of remaining useful life with uncertainty.
6. **Maintenance action:** the system generates a work order in the maintenance system (CMMS); an engineer inspects and repairs at a convenient shutdown.

Measured evidence, diagnostic indicator and prediction are different things. A rising vibration level is *evidence*; a peak at the bearing’s characteristic frequency is a *diagnostic indicator*; ‘the bearing will reach the alarm level in about 3 weeks’ is a *prediction* (an estimate). None of these *proves* that the bearing will fail; sensor faults, a change of load or a loose mounting can give similar readings, so an engineer confirms before acting.

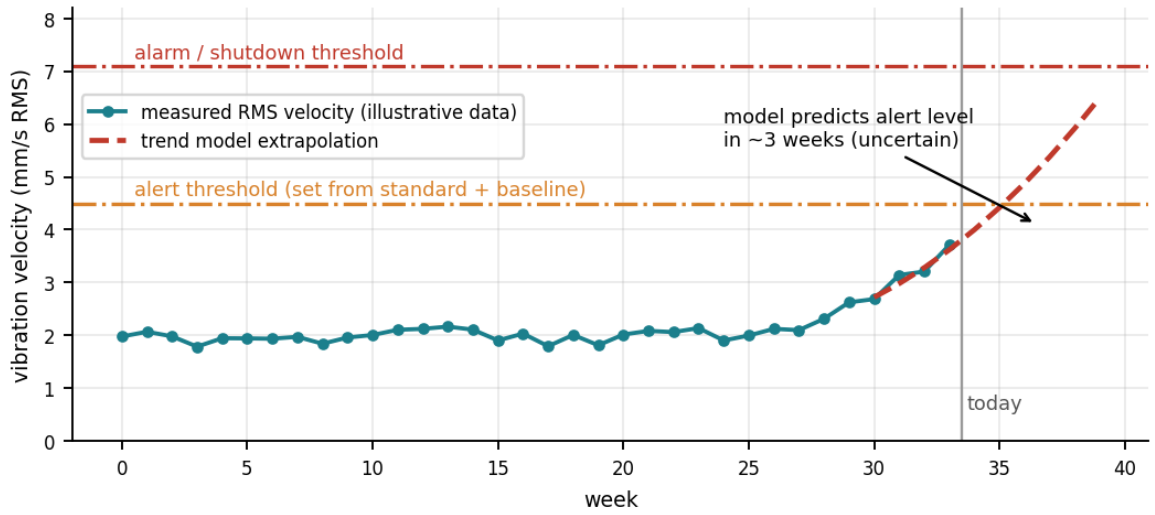


Figure 88. *Illustrative vibration trend (synthetic data to show the idea, not a real measurement). The model extrapolation crosses the alert level in a few weeks.*

$$f_{BPFO} = \frac{n}{2} f_r \left(1 - \frac{d}{D}\cos\phi\right)$$

f_{BPFO} = ball-pass frequency of the outer race (Hz) where a defect in the outer race of a bearing produces a vibration; n = number of rolling elements, f_r = shaft rotation frequency (Hz), d = rolling-element diameter, D = pitch diameter, φ = contact angle.

Worked example: example: pump motor bearing (assumed data)
Motor 1500 rpm ⇒ f_r = 1500/60 = 25 Hz. Bearing: n = 9 balls, d/D = 0.2, φ = 0°.
f_{BPFO} = (9/2) × 25 × (1 − 0.2 × cos 0°) = 4.5 × 25 × 0.8 = **90 Hz**.
The vibration sensor's spectrum shows a growing peak at 90 Hz (and its harmonics 180, 270 Hz) while overall RMS rises from 2.0 to 4.5 mm/s over several weeks. This is a **diagnostic indicator** of an outer-race defect. The analytics system extrapolates the trend and warns that the alert threshold will be reached in about three weeks; maintenance is scheduled at the next planned stop. Temperature and current data are checked for consistency.

Part (b): role of IoT devices and gateways

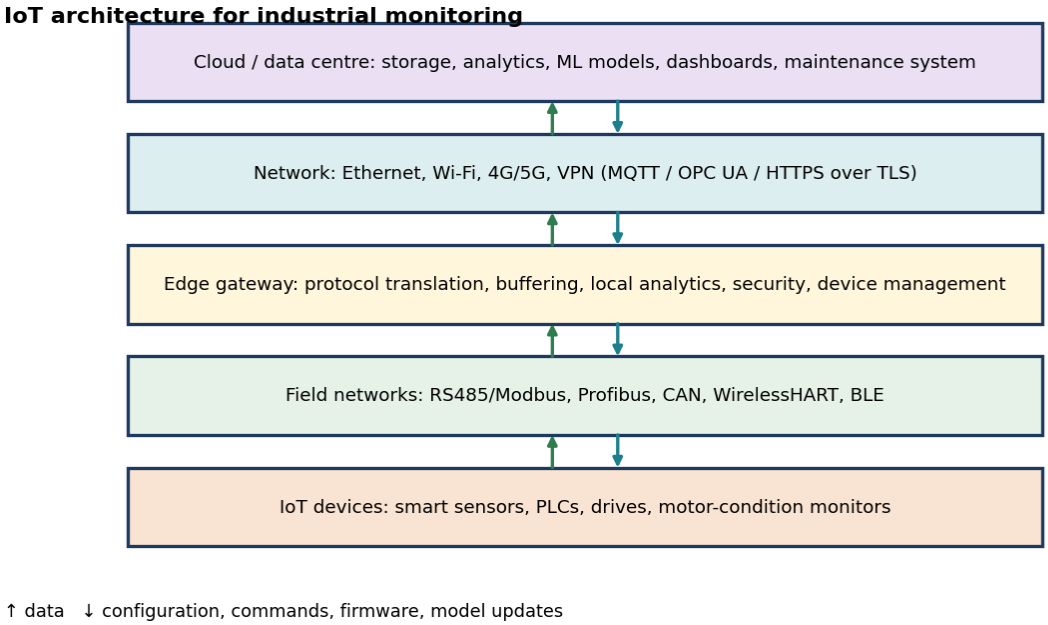


Figure 89. *Industrial IoT architecture: devices, field networks, edge gateway, network, cloud.*

Element	Role in monitoring and predictive maintenance
IoT devices (smart sensors, PLCs, drives, condition monitors)	measure vibration, temperature, current, pressure; many have built-in processing and a digital interface (IO-Link, Modbus, wireless)
Field networks	link devices to the gateway (RS485/Modbus, Profibus, CAN, WirelessHART)
Gateway	(1) **protocol translation** between field protocols and IT protocols (Modbus → MQTT/OPC UA); (2) **aggregation and buffering** (store-and-forward when the link is down); (3) **edge analytics**: compute features, detect anomalies locally with low delay; (4) **security**: encryption (TLS), certificates, firewall, isolation of OT from IT; (5) **device management**: configuration, firmware updates
Network	carries data to the cloud or data centre; bandwidth and latency matter
Cloud / analytics	long-term storage, machine learning, dashboards, fleet comparison, work-order integration

Worked example: why edge processing matters (assumed numbers)
A vibration sensor samples at 10 kHz, 16 bits (2 bytes): 10 000 × 2 = 20 000 bytes/s ⇒ × 86 400 s/day = 1.73 × 10^9 bytes ≈ **1.73 GB per day per sensor**.
If the edge device sends 10 features × 4 bytes once per minute: 40 × 1440 = 57 600 bytes ≈ **58 kB per day** ⇒ about **30 000 times less** data, enough for trending; raw waveforms are uploaded only on alarms for deeper analysis.

- **MQTT** (publish/subscribe): the gateway publishes values to a broker under topics such as `plant1/pump3/vibration_rms`; analytics and dashboards subscribe. It is light and works over unreliable links. **OPC UA** gives a standard, secure information model for industrial data.
- **Security:** OT networks must be segmented from the internet; use authenticated, encrypted channels.

Unit 5, Question 6: Industry 4.0 and its impact on smart manufacturing

Exam question – Unit 5 Q6
What is Industry 4.0? Explain Industry 4.0 and its impact on smart manufacturing with a suitable example.

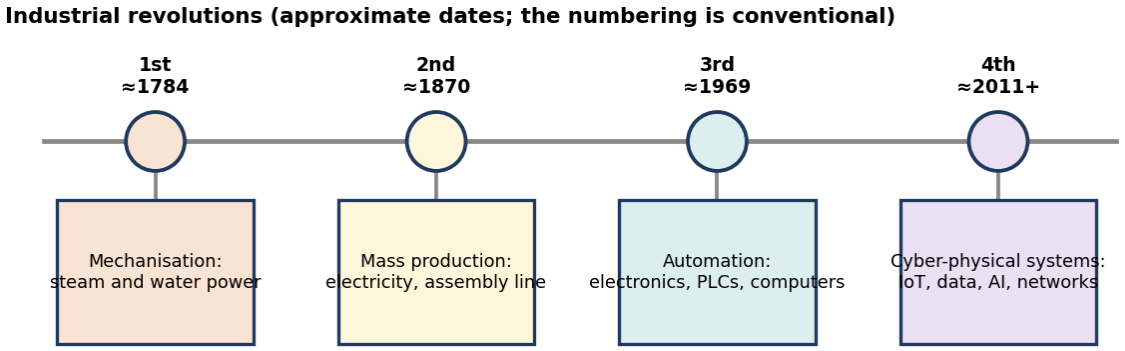


Figure 90. *The four industrial revolutions (conventional numbering).*

Industry 4.0 (the term comes from a German government initiative announced around 2011) is the digital transformation of manufacturing in which machines, products and people are **connected**, produce **data**, and use **analytics and automation** to make factories flexible, efficient and self-optimising. Its core is the **cyber-physical system**: a physical machine tightly coupled with computing, sensing and networking.

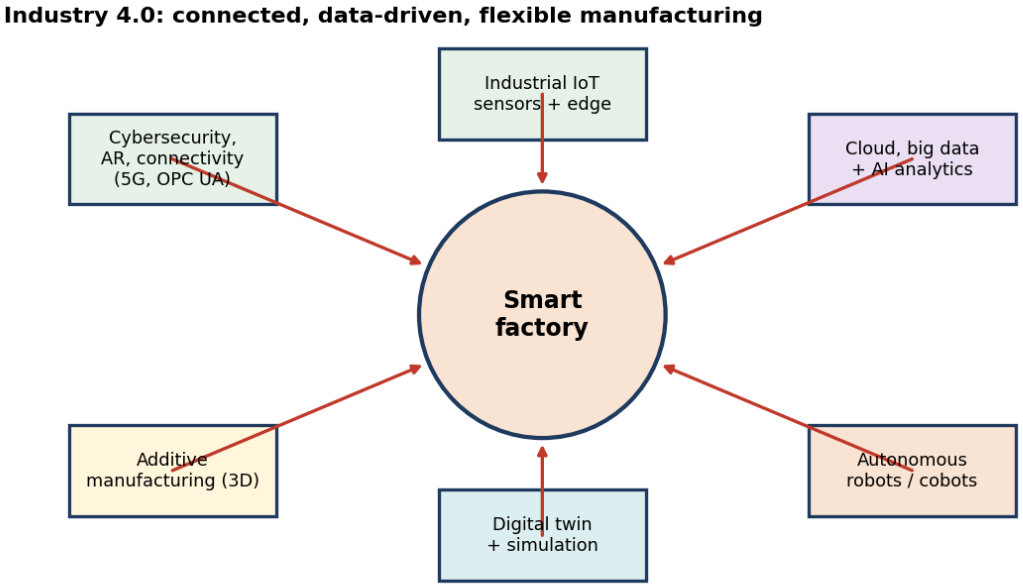


Figure 91. *Industry 4.0 connects the technologies of this book into one coherent smart factory.*

Pillar / technology	Contribution	Link to this book
Industrial IoT sensors	real-time data from every machine	Unit 4, Unit 5 Q5
Industrial networks, 5G, OPC UA	reliable connectivity and interoperability	Unit 5 Q3–Q4
Cloud and big data	storage and analytics of production data	Unit 5 Q5
AI and analytics	predictive maintenance, quality prediction, optimisation	Unit 5 Q5
Autonomous robots / cobots	flexible production and handling	Unit 5 Q1
Digital twin and simulation	virtual model for planning and testing	—
Horizontal and vertical integration	connecting machines to MES/ERP and suppliers to customers	Unit 3 Q7, pyramid
Additive manufacturing	customised parts, rapid tooling	—
Cybersecurity	protection of connected plants	Unit 1 Q4 (analogy: EMC), Unit 5

Pillar / technology	Contribution	Link to this book
AR/VR	maintenance guidance, training	—

Impact on smart manufacturing

- **Flexibility and mass customisation:** lot-size-one production by reprogramming robots and CNC machines automatically from orders.
- **Higher efficiency (OEE):** real-time monitoring and removal of bottlenecks.
- **Quality:** in-line inspection with vision, traceability of every product.
- **Reduced downtime:** predictive maintenance (Question 5).
- **Energy and resource efficiency:** VFDs, monitoring of energy per product.
- **Safety and working conditions:** cobots and AR support.
- **Challenges:** cybersecurity, integrating old machines (retrofit with sensors and gateways), skills shortage, investment, data standards.

$$OEE = A \times P \times Q = \text{Availability} \times \text{Performance} \times \text{Quality}$$

Worked example: example: a smart bottling plant (illustrative)
A beverage plant fits bottle-line machines with vibration and temperature sensors, a vision system, and PLCs on Ethernet. Data flows through gateways to SCADA/MES and a cloud analytics service.
What happens: the MES sends an order for a new flavour; the PLC changes recipe, the robot changes the gripper format; vision rejects mis-capped bottles and tags the batch; the analytics detect a rising bearing temperature on the filler and raise a work order before it fails; the energy dashboard compares kWh per 1000 bottles.
Metric: OEE = A × P × Q = 0.90 × 0.95 × 0.98 = **0.838 = 83.8%**. If predictive maintenance raises availability from 90% to 94%, OEE becomes 0.94 × 0.95 × 0.98 = **87.5%**, i.e. 3.7 percentage points more output from the same machines.

Common misconception
‘Industry 4.0 means replacing all workers with robots.’ It is about connected, data-driven production; many gains come from information (monitoring, predictive maintenance, flexibility), not only robots.

Unit 5: End-of-Unit Revision Pack

Unit summary

- Automation uses controllers, sensors, actuators and robots to run processes with minimal human effort; benefits are productivity, quality, safety, flexibility and data.
- Robots: manipulator + actuators + sensors + end effector + controller; types: articulated, SCARA, delta, Cartesian, cobots.
- CNC: G-code → MCU (interpreter, interpolator, loops) → drives → servo motors, with encoder feedback.
- Modbus (application protocol), Profibus and CAN (fieldbus), RS485 (physical layer), Ethernet (network technology).
- RS232: point-to-point single-ended; RS485: multi-drop differential 1200 m; Ethernet: switched star.
- Predictive maintenance: sensor → edge → gateway → analytics → indicator/prediction → action.
- Gateways translate protocols, buffer, run edge analytics and provide security.
- Industry 4.0: connected cyber-physical production; OEE = A × P × Q.

Formula and equation sheet

Spindle speed

$$N = \frac{1000\,V_c}{\pi D}$$

Table feed

$$F = f_z \cdot z \cdot N$$

Encoder BLU

$$BLU = \frac{\text{pitch}}{4 \times PPR}$$

Bearing outer-race frequency

$$f_{BPFO} = \frac{n}{2} f_r \left(1 - \frac{d}{D}\cos\phi\right)$$

Serial byte time

$$t_{byte} = \frac{\text{bits per frame}}{\text{baud}}$$

OEE

$$OEE = A \times P \times Q$$

Parts per hour

$$\frac{3600}{\text{cycle time (s)}}$$

Diagram-revision sheet

Practise drawing each of these from memory, labelling every component, then compare with the figure.

Figure	What to draw and label
Q1	Automation pyramid; robot arm
Q2	CNC block diagram; tool path
Q3	Comparison table of five networks (≥ 7 rows)
Q4	RS232, RS485, Ethernet connection diagrams
Q5a	PdM chain; trend plot
Q5b	IoT architecture
Q6	Industry 4.0 smart factory

Glossary of important terms

Term	Meaning
Automation	Operation of processes with minimal human intervention
Robot	Reprogrammable multi-axis manipulator
Cobot	Collaborative robot
CNC	Computer numerical control
G-code	Programming language of CNC motions
Interpolation	Computing intermediate points between two positions
Modbus	Master/slave application protocol
Profibus	Fieldbus standard (DP/PA)
CAN	Controller area network, multi-master priority bus
RS232 / RS485	Serial electrical standards
Ethernet	IEEE 802.3 network technology
OPC UA	Standard industrial data model and protocol
MQTT	Publish/subscribe messaging protocol for IoT
Edge computing	Processing data near the machine
Gateway	Device that links networks and translates protocols
PdM	Predictive maintenance
OEE	Overall equipment effectiveness
Cyber-physical system	Physical machine tightly coupled with computing and networking
Digital twin	Virtual model of a physical asset

Conceptual and numerical review questions

1. A robot cell has a 36 s cycle. How many parts per hour at 95% availability?
2. A cutter D = 16 mm runs at V_c = 80 m/min. Find N.
3. Encoder 1000 PPR, ball-screw pitch 10 mm, 4× decoding. Find BLU.
4. Serial link at 19200 baud, 11 bits/byte. Time for 25 bytes?
5. OEE with A = 85%, P = 90%, Q = 99%.
6. Why can RS485 be used over 1 km but RS232 cannot?
7. List two roles of an IoT gateway.

Fully explained answers
1. 3600/36 = 100/h; × 0.95 = **95 parts/h**.
2. N = 80 000/(π×16) = **1592 rpm**.
3. 10/(4×1000) = **0.0025 mm = 2.5 µm**.
4. 25 × 11/19200 = 14.3 ms.
5. 0.85 × 0.90 × 0.99 = **75.7%**.
6. RS485 is differential (noise cancels, higher swing, terminated line); RS232 is single-ended with ground-potential differences and capacitive limits (~15 m).
7. Protocol translation (e.g. Modbus → MQTT) and edge analytics/security.

Examination-answer guide for the original questions

This guide complements the teaching chapters. It shows how to organise each original question as a complete exam answer.

Question	Structure of a strong answer
Q1	Definition → types → advantages table → robot types/diagram → role of robots in smart manufacturing → limits.
Q2	Block diagram (program, MCU, drives, motors, feedback) → function of each block → step-wise working → example G-code → advantages.
Q3	Note on categories (protocol vs physical vs network) → table of ≥ 7 parameters (category, medium, topology, speed, distance, nodes, access, errors, use) → selection.
Q4	Three short notes: RS232 (single-ended, ≤15 m), RS485 (differential, multi-drop, termination), Ethernet (switched star, industrial variants) → comparison table.
Q5 (a)	Maintenance strategies → chain diagram → six steps → example (bearing) → evidence vs indicator vs prediction.
Q5 (b)	Architecture diagram → role table for devices and gateways → edge benefit → MQTT/OPC UA, security.
Q6	Definition and history → pillars → impact (flexibility, quality, uptime) → example with OEE → challenges.

REFERENCE SECTION: Master Glossary

Term	Meaning
AC / DC	Alternating / direct current
ADC	Analogue-to-digital converter
Aliasing	False low frequency from under-sampling
APFC	Automatic power factor correction panel
Automation	Operation of processes with minimal human intervention
Binary / Boolean	Base-2 numbers; algebra of true/false
CAN	Controller area network, multi-master priority bus
CCM	Continuous conduction mode: inductor current never zero
Charge	Property of matter; unit coulomb (C)
CNC	Computer numerical control
Cobot	Collaborative robot
Coil	Ladder output element
Cold-junction compensation	Corrects thermocouple reference temperature
Conduction loss	Loss V_{ON}×I while the switch is ON
Coriolis force	Apparent force on a moving mass in a rotating frame
Cuk converter	DC–DC with inductors at input and output
Current	Rate of flow of charge; ampere (A)
Cyber-physical system	Physical machine tightly coupled with computing and networking
Cycloconverter	Direct AC–AC frequency changer
DAS	Data acquisition system
DCS	Distributed control system
Dead time	Brief interval with both switches of a leg OFF
Digital twin	Virtual model of a physical asset
Diode	One-way semiconductor device
Duty cycle	Fraction of period a switch is ON
Duty cycle D	t_{ON}/T of the switch
Edge computing	Processing data near the machine
EMI / EMC	Electromagnetic interference / compatibility
Encoder	Converts position to digital signals
Error	Setpoint − measured
Ethernet	IEEE 802.3 network technology
Extinction angle β	Angle where SCR current becomes zero in R–L load
Feedback	Measured output returned to the input
Firing angle α	Delay of gate pulse after zero crossing
Frequency	Cycles per second; hertz (Hz)
G-code	Programming language of CNC motions
Gateway	Bridge between sensor network and plant network
Gauge factor	(ΔR/R)/strain
Gray code	Code in which one bit changes per step
Harmonic	Component at an integer multiple of fundamental frequency
HMI	Human–machine interface

Term	Meaning
Holding current	Minimum anode current to keep SCR ON
IGBT	Insulated-gate bipolar transistor
Interpolation	Computing intermediate points between two positions
Inverter	DC to AC converter
Latching current	Minimum anode current at end of gate pulse to keep SCR on
Limit switch	Mechanically actuated position switch
Measurand	Quantity being measured
Modbus	Master/slave application protocol
MQTT	Publish/subscribe messaging protocol for IoT
Natural commutation	SCR turns off when AC current falls to zero
NO / NC contact	Contact true when bit = 1 / bit = 0
Nyquist	Sampling at least twice the highest frequency
OEE	Overall equipment effectiveness
Ohm’s law	V = IR
OPC UA	Standard industrial data model and protocol
P and N groups	Positive and negative converters in a cycloconverter
PdM	Predictive maintenance
Phase-angle control	Adjust RMS by firing delay
PLC	Programmable logic controller
Power	Rate of energy conversion; watt (W)
Power factor	P/S = cos φ
Power switch	Semiconductor operated ON/OFF to control power
PPR	Pulses per revolution
Profibus	Fieldbus standard (DP/PA)
Proximity sensor	Non-contact object detector
PWM	Pulse-width modulation
Reactive power	Q = V I sin φ, in VAr
Relay	Electrically operated mechanical switch
Resistance	Opposition to current; ohm (Ω)
RMS	Value giving the same heating as DC
Robot	Reprogrammable multi-axis manipulator
RS232 / RS485	Serial electrical standards
RTU	Remote terminal unit
Rung	One line of ladder logic
SCADA	Supervisory control and data acquisition
Scan cycle	Repeated read–execute–write loop
SCR / thyristor	Four-layer latching switch triggered by gate
Sensor	Detects a physical quantity
Setpoint	Desired value
Shoot-through	Both switches of one leg ON: supply short circuit
Signal conditioning	Amplify, filter, linearise, isolate
Slip	Relative speed difference, rotor vs rotating field
Switching loss	Loss during turn-on/off transitions

Term	Meaning
THD	Total harmonic distortion
Transducer	Converts one form of energy to another
Transfer function	Output/input in the s-domain
Transistor	Three-terminal semiconductor switch/amplifier
Truth table	Output for every input combination
VFD	Variable frequency drive
Volt-second balance	Average inductor voltage over a period is zero
Voltage	Electrical push between two points; volt (V)
Voltage sag	Short fall of RMS voltage to 10–90%
Wheatstone bridge	Four-resistor circuit for precise resistance comparison
Windup	Integral growing while the actuator saturates
WSN	Wireless sensor network

Abbreviations: ADC analogue-to-digital converter · APFC automatic power factor correction · CAN controller area network · CNC computer numerical control · DAS data acquisition system · DCS distributed control system · DIAC diode for AC · EMC/EMI electromagnetic compatibility/interference · HMI human–machine interface · IGBT insulated-gate bipolar transistor · IoT Internet of Things · MOSFET metal-oxide-semiconductor FET · OEE overall equipment effectiveness · PID proportional–integral–derivative · PLC programmable logic controller · PWM pulse-width modulation · RTD resistance temperature detector · RTU remote terminal unit · SCADA supervisory control and data acquisition · SCR silicon-controlled rectifier · TRIAC triode for AC · VFD variable frequency drive · THD total harmonic distortion.

REFERENCE SECTION: Consolidated Formula Sheet

Foundations

Ohm’s law

$$V = IR$$

Power and energy

$$P = VI = I^2R = \frac{V^2}{R}, \quad E = Pt$$

Series / parallel resistance

$$R_T = R_1 + R_2, \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Sine wave

$$v = V_p \sin(2\pi f t), \; T = \frac{1}{f}, \; V_{rms} = \frac{V_p}{\sqrt{2}}$$

Rectified averages

$$V_{avg} = \frac{V_p}{\pi} \text{ (half wave)}, \quad \frac{2V_p}{\pi} \text{ (full wave)}$$

Unit 1

Power triangle and PF

$$S^2 = P^2 + Q^2, \quad PF = \cos\varphi = \frac{P}{S}$$

Line loss versus PF

$$P_{loss} \propto \frac{1}{\cos^2\varphi}$$

SCR half-wave average (resistive)

$$V_{avg} = \frac{V_m}{2\pi}(1 + \cos\alpha)$$

THD

$$THD = \frac{\sqrt{\sum_{n \geq 2} V_n^2}}{V_1} \times 100\%$$

Capacitor reactive power

$$Q_C = P(\tan\varphi_1 - \tan\varphi_2)$$

Capacitance for PF correction

$$C = \frac{Q_C}{2\pi f V^2}$$

Unit 2

H-bridge output

$$V_{o,rms} = V_{dc}, \quad V_{1,rms} = \frac{4V_{dc}}{\sqrt{2}\pi} = 0.9V_{dc}$$

AC controller (R load)

$$V_{o,rms} = V_s\sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

ON–OFF control

$$V_{o,rms} = V_s\sqrt{\frac{n}{n+m}}$$

Buck

$$V_o = DV_{in},\ \Delta I_L = \frac{V_o(1-D)}{fL},\ \Delta V_o = \frac{\Delta I_L}{8fC}$$

Boost

$$V_o = \frac{V_{in}}{1-D},\ \Delta I_L = \frac{V_{in}D}{fL},\ \Delta V_o = \frac{I_oD}{fC}$$

Buck-boost

$$V_o = -\frac{D}{1-D}V_{in}$$

Cuk

$$V_o = -\frac{D}{1-D}V_{in},\ V_{C1} = \frac{V_{in}}{1-D}$$

1-phase bridge (R load)

$$V_{avg} = \frac{V_m}{\pi}(1+\cos\alpha)$$

3-phase inverter (180°)

$$V_{LL,rms} = \sqrt{\frac{2}{3}}V_{dc},\ V_{ph,rms} = \frac{\sqrt{2}}{3}V_{dc}$$

3-phase half-wave rectifier

$$V_{avg} = \frac{3\sqrt{3}}{2\pi}V_m\cos\alpha\ (\alpha \le 30^\circ)$$

Cycloconverter speed

$$N = \frac{120f_o}{P}, \quad f_o \le f_s/3$$

Unit 3

Closed-loop transfer function

$$\frac{C}{R} = \frac{G}{1+GH}$$

PID (time domain)

$$u = K_p e + K_i \int e\,dt + K_d \frac{de}{dt}$$

PID (s domain)

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

Ziegler–Nichols PID

$$K_p = 0.6K_u,\ T_i = \frac{P_u}{2},\ T_d = \frac{P_u}{8}$$

Synchronous speed

$$N_s = \frac{120f}{P}$$

Slip

$$s = \frac{N_s - N}{N_s}$$

V/f constant

$$\frac{V}{f} = \text{constant}$$

Affinity law (power)

$$\frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$$

Unit 4

Strain-gauge sensitivity

$$\frac{\Delta R}{R} = GF \cdot \varepsilon$$

Wheatstone balance

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Quarter bridge output

$$V_{out} \approx \frac{V_s}{4}\frac{\Delta R}{R_0}$$

ADC resolution

$$LSB = \frac{V_{FS}}{2^N}$$

Nyquist

$$f_s \geq 2f_{max}$$

4–20 mA scaling

$$I = 4 + 16\,\frac{P}{P_{FS}}\ \text{mA}$$

Accelerometer

$$x = \frac{ma}{k}, \quad f_n = \frac{1}{2\pi}\sqrt{k/m}$$

Coriolis force

$$F = 2mv\Omega$$

Encoder resolution

$$\Delta\theta = \frac{360^\circ}{4\,PPR}$$

Thermocouple

$$V = S\,(T_{hot} - T_{ref})$$

Unit 5

Spindle speed

$$N = \frac{1000\,V_c}{\pi D}$$

Table feed

$$F = f_z \cdot z \cdot N$$

Encoder BLU

$$BLU = \frac{\text{pitch}}{4 \times PPR}$$

Bearing outer-race frequency

$$f_{BPFO} = \frac{n}{2} f_r \left(1 - \frac{d}{D}\cos\phi\right)$$

Serial byte time

$$t_{byte} = \frac{\text{bits per frame}}{\text{baud}}$$

OEE

$$OEE = A \times P \times Q$$

Parts per hour

$$\frac{3600}{\text{cycle time (s)}}$$

REFERENCE SECTION: Master Diagram Index

Fig.	Caption
1	Block diagram of the automatic water tank, colour-coded by the Unit that studies each block. Arrows show power (orange path) and information (blue/green paths).
2	The water analogy (left) and the matching electric circuit (right). Height difference ↔ voltage; flow of water ↔ current; narrow pipe ↔ resistance.
3	A simple complete circuit. Conventional current leaves the battery's (+) terminal, passes the closed switch and the lamp, and returns to (−).
4	Three states of the same circuit: (a) open circuit, switch open, no current; (b) closed circuit, normal current; (c) short circuit, a wire bypasses the load, current becomes dangerously large.
5	Left: memory triangle for Ohm's law. Right: for a resistor, the V–I graph is a straight line through the origin, and its slope equals R.
6	(a) Series: components share one path. (b) Parallel: components share the same two nodes (connection points).
7	Common circuit symbols used throughout this book. Component names and reference labels (R1, S1, D1…) are written beside each symbol.
8	Left: DC is a flat line. Right: AC swings above and below zero; the '−' half means the polarity is reversed.
9	A 50 Hz mains sine wave with 230 V RMS: peak 325 V, peak-to-peak 650 V, period 20 ms.
10	Left: the average of a plain sine over a full cycle is zero. Right: squaring makes it non-negative; its mean is ½, so RMS = Vp×√½.
11	Current lagging voltage by 60°. The two waves have the same frequency but the red wave reaches each peak 60° later.
12	A 12 V supply switched ON for 0.4 ms of each 1 ms period (duty cycle D = 0.4). Average = 12 × 0.4 = 4.8 V.
13	Diode symbol with the allowed current direction (left) and its V–I characteristic (right, schematic, not to scale).
14	Half-wave rectifier (ideal diode): current flows only while the source is positive, so the load voltage is the positive half-waves only.
15	An NPN transistor used as a switch. With no base current the collector–emitter path is open and the load is off (left). Base current closes it and load current flows (right).
16	Output characteristics of a BJT. Cut-off and saturation are the OFF and ON states used when switching; the active region is used for amplification.
17	Switching gives the same average voltage as a dropping resistor without the wasted heat.
18	Symbols of the six power switches in your question bank. Each is studied in detail in Unit 1.
19	Left: key sine values. Right: the area under half a sine (integral) and the slope (derivative) at a point.
20	Schematic V–I characteristics of the six devices. Curves are drawn for understanding, not to scale.
21	Left: the power triangle. Right: with a 60° lag (PF = 0.5), instantaneous power p = v·i goes negative for part of each cycle: energy is returned to the source rather than used.
22	Left: four-layer PNPN structure with three junctions J1, J2, J3. Right: the same device behaves like a PNP and an NPN transistor, each supplying the other's base current.
23	V–I characteristics of an SCR (schematic). Increasing gate current reduces the break-over voltage; the dashed segment is the unstable negative-resistance transition.
24	Half-wave SCR circuit with resistive load RL, and its waveforms. The SCR conducts from α to π; by changing α the average output voltage is adjusted.
25	EMI chain: every EMI problem has a source, a coupling path and a victim; EMC engineering acts on one or more of them.
26	A fundamental plus a 3rd and a 5th harmonic produces a distorted wave. The harmonics are the 'extra' sine waves hidden in the distortion.
27	A voltage sag: the amplitude falls to 0.5 per unit for four cycles, then recovers. Depth and duration characterise the sag.
28	IGBT: simplified cross-section (left) and equivalent circuit of a MOSFET driving a PNP transistor (right).

Fig.	Caption
29	IGBT characteristics (schematic): output characteristic with VGE steps, transfer characteristic showing the threshold, and a switching sketch with turn-off tail.
30	Electromechanical relay: the coil circuit (left) is electrically separate from the load circuit (right).
31	Limit switch at the end of a conveyor (left) and the contact states of NO and NC types (right).
32	Left: capacitor connected in parallel with an inductive load. Right: phasor diagram; IC cancels part of the lagging reactive current so the supply current I and angle φ2 become smaller.
33	The four basic kinds of power conversion. The word **converter** is the general name; ‘rectifier’, ‘inverter’, ‘chopper/DC–DC converter’ and ‘AC controller/cycloconverter’ are specific kinds.
34	Single-phase full-bridge (H-bridge) inverter. Each switch (S1–S4: IGBT or MOSFET) has an anti-parallel diode. The red path shows the current in each state.
35	Waveforms of the H-bridge: gate signals, output voltage (square wave; dashed = its fundamental), load current for an R–L load, and intervals in which the feedback diodes conduct.
36	Single-phase AC voltage controller with resistive load: two SCRs T1, T2 in inverse parallel (or one TRIAC). Right: phase-angle waveforms (middle) and integral-cycle (ON–OFF) control (bottom).
37	Buck converter in its two states. Red = current path. In the OFF interval the inductor current continues through the diode.
38	Buck converter waveforms for D = 0.5 (schematic ripple exaggerated): gate signal, switch-node voltage vX, inductor current iL and output voltage.
39	Boost converter in its two states.
40	Boost converter waveforms for D = 0.5: gate, switch-node voltage (0 while ON, Vo while OFF), inductor (input) current, and diode current.
41	Single-phase fully controlled bridge: SCRs T1–T4 between the AC source and a resistive load R. Current direction: T1, T2 conduct in the positive half-cycle; T3, T4 conduct in the negative half-cycle.
42	Waveforms for α = 60°: supply, gate pulses, load voltage (shaded region between α and π, then π+α to 2π), and load current.
43	Buck-boost converter (inverting). Inductor L is across the source when S is ON; when S is OFF, L pushes its current into C and the load through D.
44	Buck-boost waveforms for D = 2/3 (Vo = −2Vin): gate, inductor voltage, inductor current, output voltage.
45	Three ways to convert single-phase AC to AC.
46	AC voltage controller with R–L load: α = 75°, φ = 50°. The SCR keeps conducting beyond 180° until the current reaches zero (extinction angle β). During conduction vo = vs.
47	Three-phase bridge inverter: six switches S1–S6 (IGBT with anti-parallel diode) in three legs (A: S1/S4, B: S3/S6, C: S5/S2), supplied from Vdc. The star-connected load (motor windings) is connected to A, B, C.
48	Three-phase 180° inverter waveforms: the six gate signals, phase voltages vAN and vBN (six-step, levels ±Vdc/3 and ±2Vdc/3), and line voltage vAB (quasi-square, ±Vdc and 0).
49	Cuk converter: L1 and the source form the input stage; capacitor C1 carries energy to the output stage with L2, C2, load R.
50	Cuk waveforms for D = 0.5 (schematic): gate, input inductor current iL1, output inductor current iL2, and coupling-capacitor voltage vC1.
51	Three-phase half-wave controlled rectifier: SCRs T1, T2, T3 in phases a, b, c feed a common resistive load, returning to the neutral n.
52	Waveforms: phase voltages with natural commutation points and firing at α = 60°; output for α = 0° (top envelope) and α = 60° (discontinuous output).
53	Single-phase to single-phase cycloconverter: two controlled bridge rectifiers connected back-to-back (anti-parallel) between the supply and the load. (Return wires omitted for clarity.)
54	Illustrative waveforms (computed with the cosine-wave firing rule for r = 0.8): a 50 Hz input becomes a 12.5 Hz output (one-quarter). The dashed line is the wanted fundamental; the thick curve is built from slices of the input. The ripple around the dashed curve is the harmonic content typical of cycloconverters.
55	Standard logic-gate symbols with Boolean expressions. A bubble on the output means NOT (inversion).
56	Ladder diagrams for AND, OR and NOT. Red slashed contacts are normally-closed (NC) contacts. The rung is TRUE when a path of TRUE contacts connects the left rail to the coil Y.

Fig.	Caption
57	Ladder diagrams for NAND and NOR.
58	Ladder diagrams for XOR and XNOR.
59	Simulation of a three-stage lag process (transfer function $1/(s+1)^3$ with a unit step in setpoint). Left: increasing Kp reduces the error but adds oscillation. Right: P leaves a permanent error; PI removes it; PID adds damping. (Computed numerically; illustrative.)
60	DCS architecture. Field devices connect to I/O modules; several controllers each run the loops of one plant area; a redundant control network connects them to operator, engineering and historian stations and to the business network through a gateway.
61	(a) Open-loop and (b) closed-loop control systems. In (b) the sensor feeds the output back to the summing junction, where it is subtracted from the reference.
62	Step response of $G(s) = 10/(s+2)$ in open loop and with unity negative feedback.
63	PID controller in a feedback loop. The error $e = SP - PV$ is processed by three parallel paths whose outputs are summed to produce the control signal u.
64	PLC architecture: power supply, CPU with memory, input and output modules, communication interface and programming device.
65	The PLC scan cycle.
66	Hierarchical block diagram of an industrial control system with SCADA at supervisory level 2.
67	VFD block diagram: rectifier → DC link → PWM inverter → motor, controlled by a microprocessor board that receives commands and feedback.
68	Left: V/f control. Right: affinity laws for centrifugal pumps and fans; a small speed reduction gives a large power saving.
69	The measurement chain used throughout Unit 4.
70	Left: a diaphragm pressure sensor. Pressure bends the diaphragm; strain gauges on it change resistance (two in tension, two in compression). Right: the four gauges form a full Wheatstone bridge.
71	Block diagram: pressure measurement chain and data acquisition.
72	Characteristics of the three main electrical temperature sensors (schematic values).
73	Block diagram: temperature measurement chain and data acquisition.
74	Left: MEMS capacitive accelerometer. Right: vibrating-mass gyroscope using the Coriolis effect.
75	Wireless sensor network feeding a gateway, a network and the cloud / SCADA in an Industry 4.0 plant.
76	Left: inductive proximity sensor; eddy currents in a metal target damp the oscillator. Right: the three photoelectric sensing modes.
77	Block diagram of a data acquisition system.
78	Left: sampling below the Nyquist rate creates a false (aliased) low-frequency signal. Right: quantisation by a 3-bit ADC.
79	Left: construction of an optical rotary encoder. Right: incremental encoder outputs A, B (quadrature) and index Z.
80	Classification of industrial sensors by measured quantity, with examples.
81	Left: Wheatstone bridge. Right: for small changes ΔR the output voltage is nearly linear in $\Delta R/R0$.
82	The automation pyramid shows where each technology studied in this book sits, from field devices at the bottom to enterprise planning at the top.
83	A six-axis articulated robot. Each joint has a servo motor, gear reducer and encoder (Unit 4 Q7).
84	CNC machine architecture: program → MCU → drives → motors and slides, with encoders closing the position loop.
85	Tool path for the example program below.
86	(a) RS232 point-to-point link. (b) RS485 multi-drop bus with 120 Ω terminators at both ends. (c) Ethernet star topology with a switch.
87	The predictive-maintenance chain: sensor → edge/data acquisition → gateway and network → analytics → fault indication and prediction → maintenance action.
88	Illustrative vibration trend (synthetic data to show the idea, not a real measurement). The model extrapolation crosses the alert level in a few weeks.

Fig.	Caption
89	Industrial IoT architecture: devices, field networks, edge gateway, network, cloud.
90	The four industrial revolutions (conventional numbering).
91	Industry 4.0 connects the technologies of this book into one coherent smart factory.

REFERENCE SECTION: Comparison Tables of Major Devices and Control Systems

Power semiconductor devices (summary of Unit 1 Q1)

Device	Control	Turn-off by gate?	Speed	Voltage/current class	Typical use
BJT	base current	yes	medium	medium	legacy, low cost
MOSFET	gate voltage	yes	very fast	low–medium V	SMPS, DC–DC
SCR	gate pulse	no	slow	very high	rectifiers, AC control, HVDC
IGBT	gate voltage	yes	medium	medium–high	VFD, inverter, EV
DIAC	none	–	–	trigger	TRIAC triggering
TRIAC	gate pulse	no (current zero)	line frequency	low–medium	dimmers, heaters

Converter families (Unit 2)

Converter	Input → output	Key relation	Where studied
Single-phase controlled bridge	AC → DC	$V_{avg} = (V_m/\pi)(1+\cos\alpha)$	Q4
3-phase half-wave	AC → DC	$0.827\ V_m \cos\alpha$ ($\alpha \le 30°$)	Q9
Buck	DC → lower DC	DV_{in}	Q3
Boost	DC → higher DC	$V_{in}/(1-D)$	Q3
Buck-boost / Cuk	DC → DC (inverted)	$-DV_{in}/(1-D)$	Q5, Q8
H-bridge / 3-phase inverter	DC → AC	square or six-step; PWM for sine	Q1, Q7
AC voltage controller	AC → AC (same f)	$V_s\sqrt{[(\pi-\alpha+\sin 2\alpha/2)/\pi]}$	Q2, Q6
Cycloconverter	AC → AC (lower f)	$f_o \le f_s/3$	Q10

Control and supervisory systems (Unit 3)

System	Role	Key point
PLC	local discrete/sequence control	scan cycle, ladder logic
P / PI / PID	feedback regulation	P: present, I: past, D: future trend
SCADA	wide-area supervision, data logging	above PLC/RTU
DCS	integrated, redundant plant control	distributed controllers
VFD	motor speed control	V/f constant, PWM inverter

Industrial communication (Unit 5)

Standard	Layer / type	Strength	Limit
RS232	physical, point-to-point	simple	15 m, 2 devices
RS485	physical, multi-drop differential	1200 m, noise-immune	needs a protocol and termination
Modbus	application protocol	simple, universal	slow polling
Profibus	fieldbus	fast, deterministic	vendor ecosystem
CAN	fieldbus	robust, priority-based	short at high speed
Ethernet	network technology	fast, IT-compatible	not inherently deterministic

REFERENCE SECTION: Consolidated List of Original Questions

The questions are reproduced exactly as printed in the question bank (including its spelling and grammar). In Unit 4 the number 3 appears twice in the source; they are labelled below as 3 (first) and 3 (second).

Unit 1

No.	Question
1.1	Define Power Switches. Compare the working and characteristics BJTs, MOSFETs, SCR, IGBTs, DIAC, and TRIAC based on switching characteristics, operating parameters, applications etc. with neat diagrams.
1.2	What impact does a poor power factor have on industrial power system performance, and which techniques are commonly applied to correct it explain in details.
1.3	Explain the working principle of SCR/Thyristor with VI characteristics.
1.4	Explain EMI/EMC considerations in industrial circuits with suitable examples.
1.5	What are the major power quality issues in industrial systems? Explain harmonics and voltage sags in detail
1.6	Explain the working principle of IGBT with V-I characteristics.
1.7	Explain the operation of electromechanical relays and limit switches used in automation with suitable diagram.
1.8	Design and explain a simple circuit for power factor improvement using capacitors.

Unit 2

No.	Question
2.1	What is an inverter? Explain single-phase H-bridge inverter working principle with neat diagram and waveforms.
2.2	What is the need of AC voltage controllers? Explain the principle of voltage regulation in industrial systems.
2.3	Describe buck converter & boost converter (separately) with circuit diagram and waveform.
2.4	Explain the working of single-phase fully controlled full wave bridge rectifier with resistive load controlled with waveforms.
2.5	What are DC-DC converters? Describe buck-boost DC–DC converters with circuit diagram and waveform.
2.6	Write short note on single phase AC to AC converters with neat diagram and waveforms.
2.7	Explain three-phase inverter circuit with neat diagram and waveforms.
2.8	Differentiate between inverters and converters (any two). Write short notes on Cuk converter.(with circuit diagram and waveform)
2.9	Explain the working of three-phase controlled half wave rectifier with resistive load controlled with waveforms.
2.10	Explain the operation of a cyclo converter with applications.

Unit 3

No.	Question
3.1	What are the requirement of PLC? Develop PLC ladder logic for the following gates: (Truth table, Boolean expression, Gate symbol & PLC ladder diagram) (a) NAND (b) XNOR (c) NOR (d) XOR (e) AND (f) OR (g) NOT
3.2	Compare between P, PI, and PID controller (at least 7 parameters)
3.3	Write down the key difference between SCADA & DCS (at least 7 parameters). Explain DCS architecture with suitable block diagram & example.
3.4	Define and differentiate between open-loop and closed-loop control systems with examples. Derive the closed loop transfer function.
3.5	Explain the working of a PID controller in detail.

No.	Question
3.6	Describe the architecture of a PLC with a neat block diagram.
3.7	Explain the role of SCADA in industrial process control with examples and hierarchical block diagram.
3.8	Analyze the role of VFD in variable speed motor drives with suitable block diagram.

Unit 4

No.	Question
4.1	Write down differences between sensor and transducer.
4.2	Explain signal conditioning requirements and DAS (Data acquisition System) for pressure sensors with proper block diagram.
4.3 (first)	Explain signal conditioning requirements and DAS (Data acquisition System) for temperature sensors with proper block diagram. [printed as the first ‘3’]
4.3 (second)	Discuss the operating principles and applications of accelerometers and gyroscopes using diagrams for explanation. [printed as the second ‘3’]
4.4	Explain the use of wireless sensor networks in Industry 4.0 (industrial automation) with proper block diagram.
4.5	Explain the working principles and applications of proximity sensors in automation industry.
4.6	Explain signal conditioning and data acquisition systems(SCADA) in brief.
4.7	Describe the construction and applications of encoders in robotic industry.
4.8	Classify different types of industrial sensors with examples (temperature, flow, level, pressure, optical)
4.9	What is a Wheatstone bridge? Derive its balance condition and explain its applications.

Unit 5

No.	Question
5.1	Define industrial automation and list its advantages. Also explain the role of robotics in smart manufacturing & modern industries.
5.2	Describe the architecture and working of a CNC machine using basic block diagram
5.3	Compare between Modbus, Profibus, CAN, RS485 & Ethernet communication protocols. (At least 7 parameters)
5.4	Write short notes on RS232, RS485, and Ethernet communication in automation.
5.5	(a) Explain how predictive maintenance is achieved using IoT in industrial systems with a suitable example. (b) Describe the role of IoT devices and gateways in industrial monitoring and predictive maintenance.
5.6	What is Industry 4.0? Explain Industry 4.0 and its impact on smart manufacturing with a suitable example.

REFERENCE SECTION: Final Revision Guide

Priority checklist before the examination

1. Draw from memory: the power-semiconductor symbols; the SCR V–I curve; the H-bridge with both current paths and its waveforms; the buck, boost and buck-boost circuits; the controlled-bridge waveform with α; the three-phase inverter table.
2. Know the derivations: C/R = G/(1+GH); AC controller V_{rms}; bridge V_{avg}; buck/boost V_o by volt-second balance; Wheatstone balance; capacitor for PF correction.
3. Know the seven gates: expression, truth table, ladder rung.
4. Know the comparison tables: power devices (Unit 1 Q1), P/PI/PID (Unit 3 Q2), SCADA vs DCS (Unit 3 Q3), five communication standards (Unit 5 Q3), sensor vs transducer (Unit 4 Q1).
5. Practise one numerical in each style: PF correction, SCR average voltage, buck/boost design, ADC resolution, encoder resolution, V/f and speed, strain-gauge bridge output.

Common exam traps

- Forgetting units, or mixing kW/kVA/kVAr.
- Wrong firing-angle reference (three-phase rectifiers measure α from the natural commutation point).
- Calling a SCR ‘turned off by the gate’.
- Mixing up protocol, physical standard and network technology in Unit 5 Q3.
- Writing a block diagram without labelling the signal flowing between blocks.
- Giving a derivation as a ‘from the above, we get’ jump: write each step.

How to turn knowledge into marks

A strong answer follows the same shape each time: **definition → diagram → working in numbered steps → equations (with symbols defined) → table of advantages/limits/applications → short conclusion**. For a comparison, a table with at least the number of parameters requested; for a derivation, every intermediate step; for a design, assumptions stated first.

REFERENCE SECTION: Coverage Checklist

Every question and sub-part of the uploaded question bank, with the section of this book that answers it. A tick (✔) means the section contains the explanation, the required diagram(s)/waveform(s)/table(s) and the worked example for that item.

Unit / Q	Sub-requirements in the question	Section in this book	Done
1.1	Define power switches; compare BJT, MOSFET, SCR, IGBT, DIAC, TRIAC (switching, parameters, applications) with diagrams	Unit 1, Question 1	✔
1.2	Impact of poor power factor; correction techniques in detail	Unit 1, Question 2	✔
1.3	SCR working principle with V–I characteristics	Unit 1, Question 3	✔
1.4	EMI/EMC with examples	Unit 1, Question 4	✔
1.5	Power quality issues; harmonics and voltage sags in detail	Unit 1, Question 5	✔
1.6	IGBT working with V–I characteristics	Unit 1, Question 6	✔
1.7	Relays and limit switches with diagram	Unit 1, Question 7	✔
1.8	Design a capacitor power-factor improvement circuit	Unit 1, Question 8	✔
2.1	Inverter; single-phase H-bridge with diagram and waveforms	Unit 2, Question 1	✔
2.2	Need for AC voltage controllers; voltage regulation principle	Unit 2, Question 2	✔
2.3	Buck and boost converters separately, circuits and waveforms	Unit 2, Question 3	✔
2.4	Single-phase fully controlled full-wave bridge rectifier, resistive load, waveforms	Unit 2, Question 4	✔
2.5	DC–DC converters; buck-boost with circuit and waveform	Unit 2, Question 5	✔
2.6	Single-phase AC–AC converters with diagram and waveforms	Unit 2, Question 6	✔
2.7	Three-phase inverter with diagram and waveforms	Unit 2, Question 7	✔
2.8	Differentiate inverters and converters (any two); Cuk converter with circuit and waveform	Unit 2, Question 8	✔
2.9	Three-phase controlled half-wave rectifier, resistive load, waveforms	Unit 2, Question 9	✔
2.10	Cycloconverter operation and applications	Unit 2, Question 10	✔
3.1	PLC requirements; seven gates (NAND, XNOR, NOR, XOR, AND, OR, NOT) with truth table, Boolean expression, symbol and ladder	Unit 3, Question 1	✔
3.2	P, PI, PID comparison (≥ 7 parameters)	Unit 3, Question 2	✔
3.3	SCADA vs DCS (≥ 7 parameters); DCS architecture with block diagram and example	Unit 3, Question 3	✔
3.4	Open vs closed loop with examples; derive closed-loop transfer function	Unit 3, Question 4	✔
3.5	PID controller in detail	Unit 3, Question 5	✔
3.6	PLC architecture with block diagram	Unit 3, Question 6	✔
3.7	SCADA role with examples and hierarchical block diagram	Unit 3, Question 7	✔
3.8	VFD role with block diagram	Unit 3, Question 8	✔
4.1	Sensor vs transducer	Unit 4, Question 1	✔
4.2	Pressure sensors: signal conditioning, DAS, block diagram	Unit 4, Question 2	✔
4.3 (first)	Temperature sensors: signal conditioning, DAS, block diagram	Unit 4, Question 3 (first)	✔
4.3 (second)	Accelerometers and gyroscopes with diagrams	Unit 4, Question 3 (second)	✔
4.4	Wireless sensor networks in Industry 4.0 with block diagram	Unit 4, Question 4	✔
4.5	Proximity sensors: principles and applications	Unit 4, Question 5	✔
4.6	Signal conditioning and data acquisition (SCADA) in brief	Unit 4, Question 6	✔

Unit / Q	Sub-requirements in the question	Section in this book	Done
4.7	Encoders: construction and applications in robotics	Unit 4, Question 7	✔
4.8	Classification of industrial sensors (temperature, flow, level, pressure, optical)	Unit 4, Question 8	✔
4.9	Wheatstone bridge: definition, balance derivation, applications	Unit 4, Question 9	✔
5.1	Industrial automation, advantages, role of robotics	Unit 5, Question 1	✔
5.2	CNC architecture and working with block diagram	Unit 5, Question 2	✔
5.3	Modbus, Profibus, CAN, RS485, Ethernet (≥ 7 parameters)	Unit 5, Question 3	✔
5.4	Short notes: RS232, RS485, Ethernet	Unit 5, Question 4	✔
5.5 (a)	Predictive maintenance using IoT with example	Unit 5, Question 5 (a)	✔
5.5 (b)	Role of IoT devices and gateways	Unit 5, Question 5 (b)	✔
5.6	Industry 4.0, impact on smart manufacturing with example	Unit 5, Question 6	✔

Totals: 8 + 10 + 8 + 10 + 6 = 42 questions as printed (Unit 4’s duplicated ‘3’ counted as two). Unit 5 Q5 has parts (a) and (b), so 43 rows are listed above. Foundations A–E precede Unit 1.